

PISGAH UNITED METHODIST CHURCH

FELLOWSHIP HALL

Released For Construction Bids

November 5, 2021

CODE SUMMARY	
APPLICABLE CODES:	Florida Bldg. Code, 7th Ed. (2020) – Building
	Florida Fire Prevention Code (effective 12/31/2020)
	Florida Bldg. Code, 7th Ed. (2020) – Accessibility
	Florida Bldg. Code, 7th Ed. (2020) – Mechanical
	Florida Bldg. Code, 7th Ed. (2020) – Plumbing
	National Electrical Code (NFPA 70–2017)
OCCUPANCY TYPE:	National Fire Alarm and Signaling Code (NFPA 72–2016)
CONSTRUCTION TYPE:	Small Assembly A–3 Religious Worship
TOTAL OCCUPANCY:	Type V–B Combustible One Story
	Fellowship Hall – 248 Persons
	Classroom – 18 Persons
NUMBER OF EXITS:	Two (2) Required; Three (3) Provided
EXIT WIDTH:	34" Clear Minimum
TRAVEL DISTANCE:	75' Limit Required; 50' Provided
FIRE PROTECTION:	Not Fire Sprinklered
EXIT SIGNS:	Signs to be readily visible as indicated on drawings
BUILDING HEIGHT:	Allowable: 40'–0" (Table 504.3)
TBD	Actual: 18'–0" to Roof Mid–Point
NUMBER OF STORIES:	One
TOTAL BUILDING AREA:	Fellowship Hall Building is 54'x60' = 3,240 SF
	Porte–Co–Chere is 16'x24' = 384 SF
	Heated & Cooled Space is 3,240 SF
	Allowable Unsprinklered Area Type 5–B is 6,000 SF

INTERIOR FINISHES	
Group A–3 Interior Finishes	Nonsprinklered
Rooms & Enclosed Spaces	Class B
Corridors / Exit Access	Class A

- NOTES:
- Acoustical ceiling system shall be installed in accordance with manufacturers' written instructions and in accordance with ASTM C635 and ASTM C636.
 - Paint exterior with 2 coats Sherwin–Williams Duration Extra White Gloss.

ARCHITECT:

Jerry L. Hicks, AIA
AR. No. 8326

CIVIL SITE ENGINEERING:

Inovia Consulting Group
James F. Waddell, P.E.
P.E. No. 46019

Structural Engineering:

Structural Engineering Group, LLC
Carson D. Hicks, P.E.
P.E. No. 68195

Mechanical (HVAC) and Plumbing Engineering:

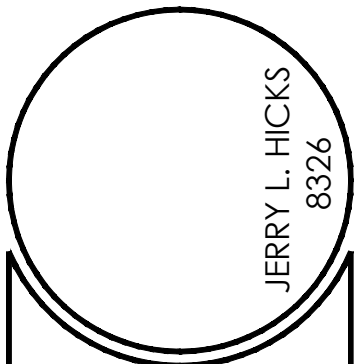
r.e. Walsh Engineering, Inc.
Roger E. Walsh, P.E.
P.E. No. 36997

Electrical Engineering:

Ooten and Associates, LLC
Homer A. Ooten, P.E.
P.E. No. 13168

INDEX TO DRAWINGS	
NUMBER	TITLE / SUBJECT
Cover Sheet	(this sheet)
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C2.0	Existing Conditions, and Erosion Control Plan
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S3.2	Wood Details, Cont'd.
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A1	Floor Plan and Notes; Door Schedule
A2	Enlarged Toilet Room Plan
	Toilet Room Accessories Schedule
A3	Door Hardware Schedule
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M1	Mechanical Specifications & Schedules
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P2	Plumbing Plan
P3	Plumbing Risers, Details and Fixture Schedule
E101	Electrical Specifications and Scope of Work
E102	Electrical Notes, Legend and Fixture Schedule
E103	Partial Electrical Site Plan
E201	Electrical Lighting Plan
E202	Dimming System Schematic
E203	Electrical Power Plan
E301	Electrical Riser Diagrams
E401	Lightning Protection Plan
E402	Lightning Protection Specifications and Details

NOTE:
Contractor shall provide and submit all required Florida product approval data.



REVISIONS	
NO.	DATE:

COVER SHEET

FELLOWSHIP HALL
PISGAH UNITED METHODIST CHURCH
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

JERRY L. HICKS, ARCHITECT
AR NO. 8326 (FLORIDA)
5383 PEMBRIDGE PLACE
TALLAHASSEE, FLORIDA 32309
850-544-0540

These drawings are the property of the Engineer and shall not be used in whole or part without written consent of the Engineer.

OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
PE 13168 EB 5043
6700 TREASURE OAKS CIRCLE
TALLAHASSEE, FLORIDA 32309
850-906-9501

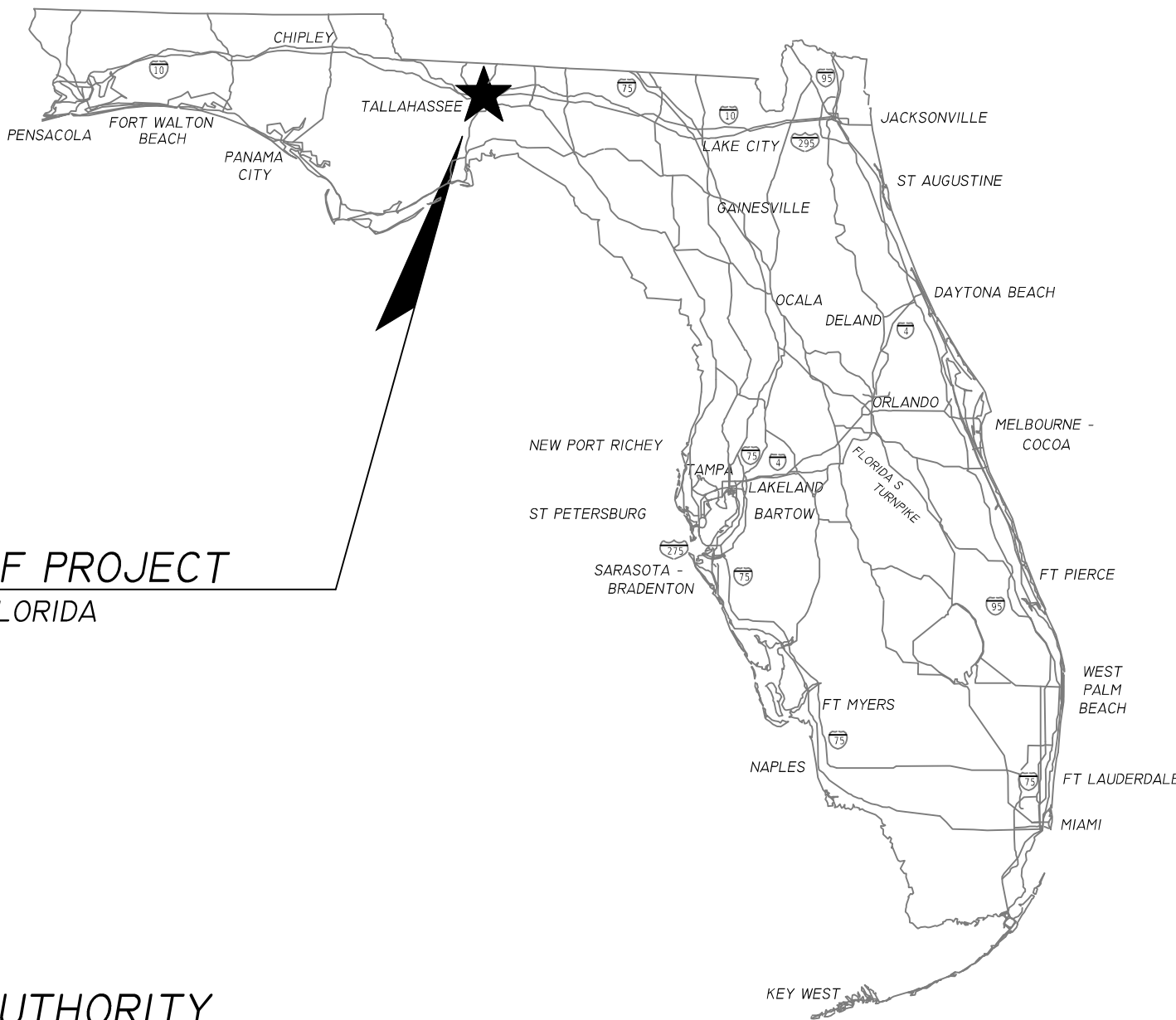
DATE:	11-05-2021
DRAWN BY:	JLH/YMO
CHECKED BY:	HAO
PROJ. NO.	38022

SHEET NO.
COVER SHEET

CONSTRUCTION PLANS
FOR
PISGAH METHODIST
CHURCH FELLOWSHIP HALL
TALLAHASSEE, FLORIDA
FINAL DESIGN PLAN APPROVAL TRACK

PREPARED FOR:
PISGAH METHODIST CHURCH
7000 PISGAH CHURCH RD,
TALLAHASSEE, FLORIDA
32309

LOCATION OF PROJECT
TALLAHASSEE, FLORIDA



INDEX OF PLANS

C0.0	KEY SHEET
C1.0	GENERAL NOTES
C1.1	GENERAL DETAILS
C2.0	EXISTING CONDITIONS AND EROSION CONTROL PLAN
C3.0	DIMENSIONAL SITE, GRADING, AND UTILITY PLAN
C4.0	SWMF DETAILS

OWNER/RESPONSIBLE AUTHORITY

PISGAH METHODIST CHURCH
HOMER A. OOTEN, P.E.
7000 PISGAH CHURCH RD
TALLAHASSEE, FLORIDA 32309
(850) 906-9501

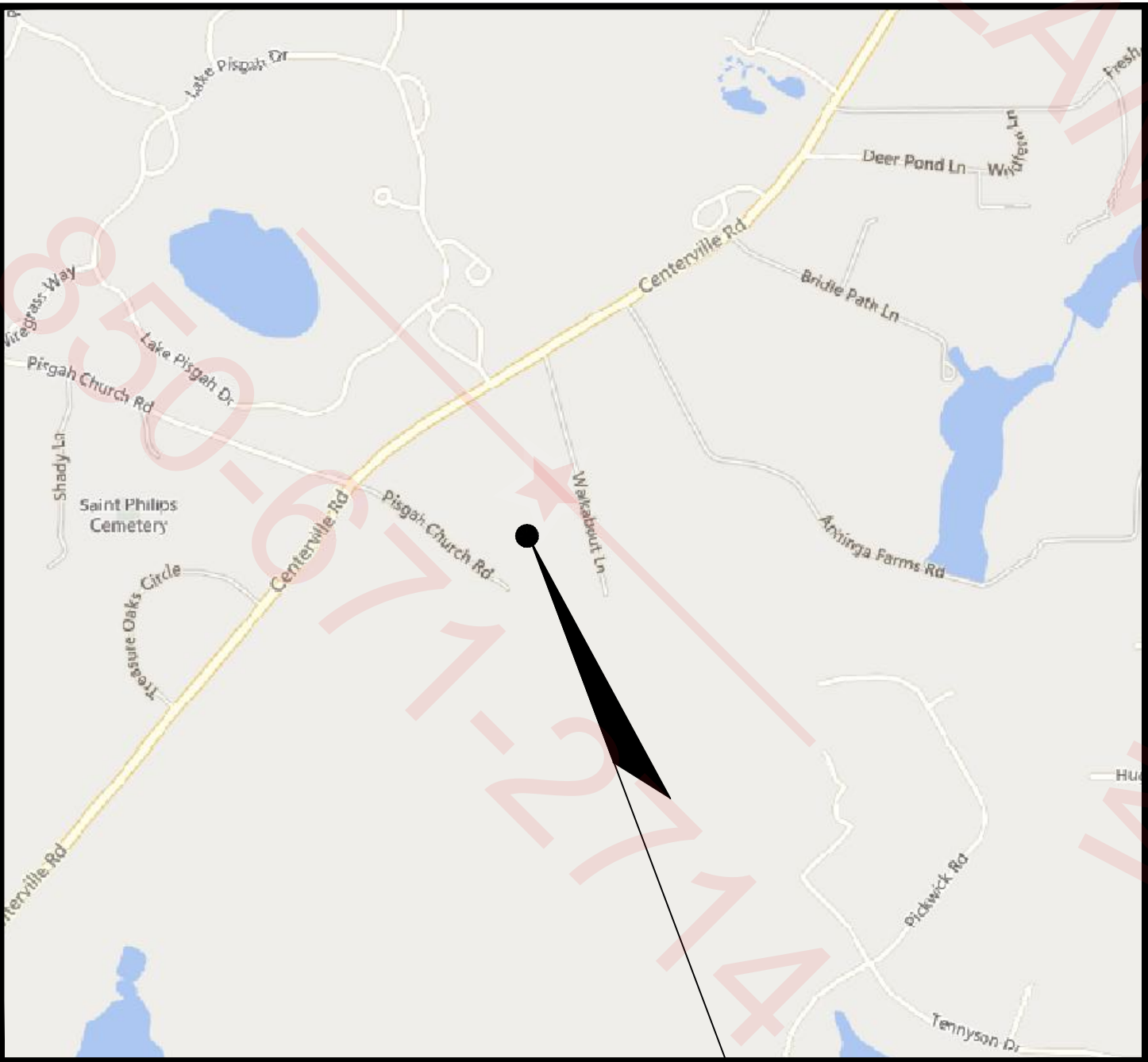
PROJECT TEAM

ENGINEER OF RECORD
INOVIA CONSULTING GROUP
2015 CENTRE POINT BOULEVARD, SUITE 103
TALLAHASSEE, FLORIDA 32308
(850) 298-4213
MR. JAMES F. WADDELL, P.E. No. 46019

SURVEYOR OF RECORD
THURMAN RODDENBERRY & ASSOCIATES, INC
125 SHELTON STREET
SOPCHOPPY, FLORIDA 32358
JAMES T. RODDENBERRY, P.S.M. 4261

PARCEL INFORMATION

TAX ID NO. 15-19-20-634-0000
CURRENT LAND USE: RELIGIOUS / NON-PROFIT (R-RURAL)
FRONTAGE STREET: CENTERVILLE ROAD
STREET MAINTENANCE: LEON COUNTY
WATER SERVICE: N/A
SEWER SERVICE: N/A
ELECTRIC SERVICE: TALQUIN ELECTRIC
BOUNDARY SOURCE: THURMAN RODDENBERRY & ASSOCIATES, INC
TOPOGRAPHIC SOURCE: THURMAN RODDENBERRY & ASSOCIATES, INC
DATUM: NGVD 1988



LOCATION MAP
NTS

LOCATION OF PROJECT

PROJECT NOTES

- GOVERNING SPECIFICATIONS -
 - FDOT STANDARD SPECIFICATION FOR ROAD AND BRIDGE CONSTRUCTION, JANUARY 2021
 - CITY OF TALLAHASSEE TECHNICAL SPECIFICATIONS FOR WATER AND SEWER, LATEST EDITION.
- CONSTRUCTION STANDARDS - FDOT ROADWAY AND TRAFFIC DESIGN STANDARDS, FY 2021-22
- SHOP DRAWINGS SHALL BE SUBMITTED TO INOVIA CONSULTING GROUP.

NO.	BY	REVISIONS	DATE



2015 CENTRE POINTE BLVD, SUITE 103
TALLAHASSEE, FL 32308
PHONE (850) 298-4213
CA8225

PLAN RELEASE STATUS	
☐ PROGRESS PRINT	JAMES F. WADDELL, P.E. PE NO.: 46019 DATE: 11/5/21
☐ PERMIT RELEASE	
☒ BID RELEASE	
☐ CONSTRUCTION RELEASE	
RELEASE DATE: 11/5/21	

1. IT IS THE CONTRACTOR'S RESPONSIBILITY TO DETERMINE THE EXACT LOCATION OF EXISTING UTILITIES AND TO DETERMINE IF OTHER UTILITIES WILL BE ENCOUNTERED DURING THE COURSE OF THE WORK AND TAKE APPROPRIATE STEPS NECESSARY TO PROVIDE FOR THEIR PROTECTION. CONTACT SUNSHINE STATE ONE CALL AT 811.
2. ALL VALVE AND METER BOXES SHALL BE SET FLUSH WITH GRADE.

- CONTRACTOR TO COORDINATE WITH LOCAL UTILITIES FOR REMOVAL AND/OR RELOCATION OF EXISTING UTILITY POLES, AERIAL LINES, BURIED CABLE, AND OTHER UTILITIES.
- PROVISIONS SHALL BE MADE FOR FLOW OF SEWERS, DRAINS, AND WATER COURSES ENCOUNTERED DURING CONSTRUCTION.
- THE CONTRACTOR SHALL PLACE AND MAINTAIN BARRICADES, CONSTRUCTION SIGNS, FLASHING LIGHTS, AND LIGHTS TO BE USED FOR NIGHT CONSTRUCTION AND UNTIL IT IS SAFE FOR BOTH PEDESTRIAN AND VEHICULAR TRAFFIC. TRAFFIC CONTROL SHALL BE IN ACCORDANCE WITH F.O.D.T. STANDARDS.
- CONTRACTOR IS TO RESTORE ALL DISTURBED RIGHTS-OF-WAY TO A CONDITION AS EXISTED PRIOR TO CONSTRUCTION.
- THE CONTRACTOR SHALL VISIT THE SITE AND BE FAMILIAR WITH THE PROJECT PRIOR TO BEGINNING CONSTRUCTION.
- THE CONTRACTOR SHALL NOTIFY THE ENGINEER IMMEDIATELY OF ANY CONFLICTS BETWEEN CONTRACT DOCUMENTS AND EXISTING CONDITIONS.
- THESE DRAWINGS REPRESENT KNOWN STRUCTURES AND UTILITIES LOCATED IN THE PROJECT AREA. THE CONTRACTOR IS CAUTIONED THAT OTHER STRUCTURES AND UTILITIES, ABOVE OR BELOW GROUND, MAY BE ENCOUNTERED DURING THE COURSE OF THE PROJECT. THE CONTRACTOR SHALL NOTIFY THE ENGINEER IMMEDIATELY UPON ENCOUNTERING ANY UNEXPECTED STRUCTURE, UTILITY LINE OR OTHER UNUSUAL CONDITION. VERIFY EXISTING CONDITIONS PRIOR TO COMMENCING WORK.
- IF EXISTING UTILITY LINES ARE DISTURBED DUE TO CONSTRUCTION ACTIVITY THEY WILL BE REPAIRED OR RELOCATED AS PER APPROPRIATE UTILITY REQUIREMENTS.
- CONTRACTOR SHALL SOD ALL DISTURBED AREAS UNLESS NOTED OTHERWISE ON PLANS.
- HATCH NEAR PAVEMENT ELEVATION WITH EXISTING PAVEMENT ELEVATION AT DRIVEWAYS, ETC.
- IN CASES WHERE CONSTRUCTION IS REFERENCED BY FOOT STANDARD INDEXES, ALL CONSTRUCTION MATERIALS AND METHODS SHALL BE IN ACCORDANCE WITH FOOT ROADWAY DESIGN STANDARD PLANS (2000-01) EDITION. MISCELLANEOUS CONSTRUCTION NOT SPECIFICALLY DETAILED SHALL ALSO BE ACCOMPLISHED IN ACCORDANCE WITH THESE STANDARDS.
- DURING ALL CONSTRUCTION PHASES THE CONTRACTOR SHALL POST THE ENTIRE PROPERTY AS A CONSTRUCTION SITE IN ACCORDANCE WITH F.L.A. STATUTE 316.010.01. SIGNING SHALL BE POSTED WHICH READS "THIS AREA IS A DESIGNATED CONSTRUCTION SITE AND ANYONE WHO TRESPASSES ON THIS PROPERTY COMMITS A FELONY"
- UNLESS OTHERWISE NOTED PIPE LENGTHS SHOWN ARE FROM CENTERLINE TO CENTERLINE OF STRUCTURES.
- CONTRACTOR IS RESPONSIBLE FOR STAKEOUT.
- CONTRACTOR TO SECURE SERVICE OF AN APPROVED TESTING LAB FOR ALL REQUIRED TESTING.
- RETAINING WALLS EQUAL TO OR GREATER THAN 2 FEET EXPOSED WALL HEIGHT REQUIRE A SEPARATE BUILDING PERMIT. CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING BUILDING PERMIT.
- EROSION CONTROL MEASURES SHOWN ARE MINIMUM REQUIRED. ADDITIONAL MEASURES MAY BE REQUIRED PER CITY OF TALLAHASSEE ENVIRONMENTAL INSPECTOR.
- THE CONTRACTOR SHALL NOT SUBSTITUTE ANY ARTICLE, DEVICE, PRODUCT, MATERIAL OR FIXTURE, OR ANY FORM OR TYPE OF CONSTRUCTION, FOR THAT WHICH IS INDICATED IN THE APPROVED PLANS WITHOUT THE EXPRESSED WRITTEN APPROVAL OF THE ENGINEER.
- THE CONTRACTOR SHALL ENSURE THAT A FOREMAN OR SUPERVISOR WHO HAS BEEN TRAINED IN THE CITY OF TALLAHASSEE STORMWATER EROSION AND SEDIMENTATION CONTROL INSPECTOR TRAINING PROGRAM IS AVAILABLE IN PERSON OR BY PHONE AT ALL TIMES DURING THE CONSTRUCTION ACTIVITIES.
- ALL DISTURBED AREAS TO BE LEFT LIE LONGER THAN 7 DAYS MUST BE STABILIZED WITH QUICK GROW PLANTS SEED AND SOIL WASTE AND CROSS CONNECTION COORDINATOR.
- BEFORE APPLYING FOR A BUILDING PERMIT, BUILDING PLANS MUST BE REVIEWED AND APPROVED BY THE CITY OF TALLAHASSEE STORMWATER EROSION AND SEDIMENTATION COORDINATOR.
- THE CONTRACTOR SHALL REPLACE ANY SIDEWALK THAT IS DAMAGED DUE TO DEVELOPMENT CONSTRUCTION.
- ALL PAVEMENT MARKINGS WITHIN PUBLIC RIGHT OF WAY SHALL BE THERMOPLASTIC.
- NO TRENCING OR EXCAVATION SHALL BE ALLOWED WITHIN THE CPZ OF PROTECTED TREES, EXCEPT WHERE DEBITS HAVE BEEN NOTED ON PLANS.

The Contractor & Owner will be held accountable during construction for all site improvements. Compliance with Florida Statutes 553.5041 (F.S.), and the 6th Edition (2017 Florida Accessibility Code for Building Construction (FACBC)), is mandatory. If incorrect at final inspection, contractor will be required to modify construction to comply with F.S. and FACBC. The following items take precedence and supersede other site details on drawings:

1. **ACCESSIBLE PARKING** spaces shall be located on an accessible route no less than 44" wide so that users will not be compelled to walk or wheel behind parked vehicles except behind the first or their own vehicle. **#2001 & #502.3, FACBC and S.F. 553.5041.**
 2. **ACCESSIBLE PARKING spaces and access aisles** serving a particular building shall be located on the shortest accessible **ACCESSIBLE PARKING** parking to an accessible entrance. **#200.31 FACBC and S.F. 553.5041a(5)(b).**
 3. **ACCESSIBLE PARKING spaces shall be 12' wide**, and outlined with blue paint. **#1502.2 and #502.6, FACBC.**
 4. **ACCESS AISLES** required adjacent to parking spaces shall be 5' wide with diagonal striping. **#502.6, FACBC.**
 5. **ACCESSIBLE PARKING and access aisles** shall be level (not exceed 1/48") on a stable, firm & slip resistant surface. **#300.21, #502.3, FACBC.**
 6. **ACCESSIBLE PARKING signs** shall be **FOOT** approved and shall read **"PARKING BY DISABLED"** and shall include the **International Symbol of Access** (a stylized figure, with a minimum depth (inches) from the ground to the bottom of the symbol). **#1502.2 and S.F. 553.5041.**
 7. **CURB RAMPS** shall not exceed 1:12 slope and curb ramp flares shall not exceed 1:10 slope. **#406.4, FACBC.** Curb ramps shall not be provided on a sloped surface, such as a driveway, sidewalk, or vehicular traffic lanes. The counter slope of adjacent road surfaces & gutters shall not exceed 1:20. **R.F. #405.2, #406, FACBC.**
 8. **CURB RAMPS** shall have a landing with a minimum clear length of 36" shall be located at the end of the curb ramp and shall be at least as wide as the curb ramp (excluding flared sides) leading to it. Exception: for alterations, where there is no landing, curb ramp flares shall be provided, and shall not be steeper than 1:12 slope. **R.F. #406, FACBC.**
 9. **ALL RAMP** surfaces shall be firm, stable, slip resistant and shall have a maximum slope of 1:40.5 or less. **Ramps** shall have 60" min level landings at the top & bottom of 1:40.5/7, FACBC.
 10. **ALL RAMPS** with a rise greater than 6" shall have handrails on both sides with 12" horizontal extensions at the top & bottom of the ramp. **R.F. #1010.9 FCB-B (Florida Building Code - 2001), #505, FACBC.**
 11. **ACCESSIBLE ROUTES** to "main entry" from an accessible parking space, and from the "public way", shall not exceed 1:20 slope (unless ramps, handrails with proper extensions are provided) with cross slope not in excess of 1:48". **R.F. #206, #402 and #403, FACBC.**
 12. **Street crossings** within a same site with an accessible route which shall not exceed 1:20 slope for ramps and handrails are provided and a maximum cross slope of 1:48. **R.F. #206 FACBC.**
- EXCEPTION:** An accessible route shall not be required between accessible buildings, accessible parking spaces, or accessible transit stops where the only means of access between them is a vehicular way not providing pedestrian access. **R.F. #206.2 FACBC.**

1. SECURE APPROVAL TO COMMENCE CONSTRUCTION FROM ATTENDING A PRE-CONSTRUCTION MEETING WITH THE ENVIRONMENTAL INSPECTOR ON THE PROJECT SITE AND SUBSEQUENT INSPECTION AND APPROVAL OF INSTALLATION OF PERIMETER EROSION CONTROLS.
2. POST ALL PLACARDS. DO NOT NAIL PLACARDS TO TREES.
3. FLAG AND STAKE THE LIMITS OF CONSTRUCTION.
4. INSTALL PERIMETER SEDIMENT AND EROSION CONTROLS.
5. CONSTRUCT TREE PROTECTION BARRICADES.
6. DESIGNATE VEHICLE PATHWAYS, MATERIAL STORAGE, PARKING, STOCKPILING AND FUELING AREAS AWAY FROM CPZ OF PROTECTED TREES, WATERWAYS, AND WATERCOURSES.
7. CLEAR & GRUB REMAINDER OF SITE AND INSTALL IMPROVEMENTS AS INDICATED ON THE SITE PLAN.
8. STABILIZE ALL DISTURBED AREAS.
9. TWO WEEKS PRIOR TO REQUESTING A FINAL ENVIRONMENTAL INSPECTION, PROVIDE ALL CLOSE-OUT DOCUMENT, SUCH AS COMPLIANCE REPORTS AND AS-BUILT DRAWINGS, FOR REVIEW AND COMMENT.
10. PRIOR TO PROJECT CLOSE OUT, A CERTIFIED ARBORIST MUST PROVIDE AN URBAN FOREST AND LANDSCAPING COMPLIANCE REPORT FOR TREES IMPACTED TO REMAIN.

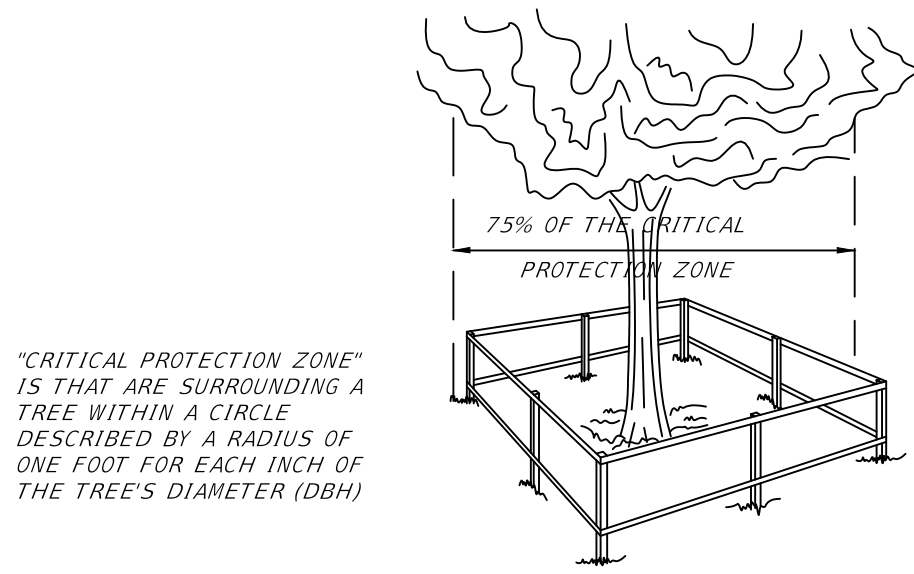


REDEVELOPMENT TABLE:			
	TOTAL AREA OF SITE	SQUARE FEET	PERCENTAGE
		1,226,400	
EXISTING CONDITIONS:	EXISTING IMPERVIOUS AREA	6,899	0.6%
	EXISTING VEHICLE USE AREA	0	0.0%
PROPOSED CONDITIONS:	DEVELOPMENT COMPONENTS		
	PROPOSED IMPERVIOUS AREA	10,259	0.8%
	PROPOSED VEHICLE USE AREA	0	
	TOTAL ADDED IMPERVIOUS AREA	3,360	0.3%
	GREEN SPACE COMPONENTS		
	MINIMUM GREEN SPACE REQUIRED	245,280	20.0%
	MINIMUM GREEN SPACE PROVIDED	1,216,141	99.2%

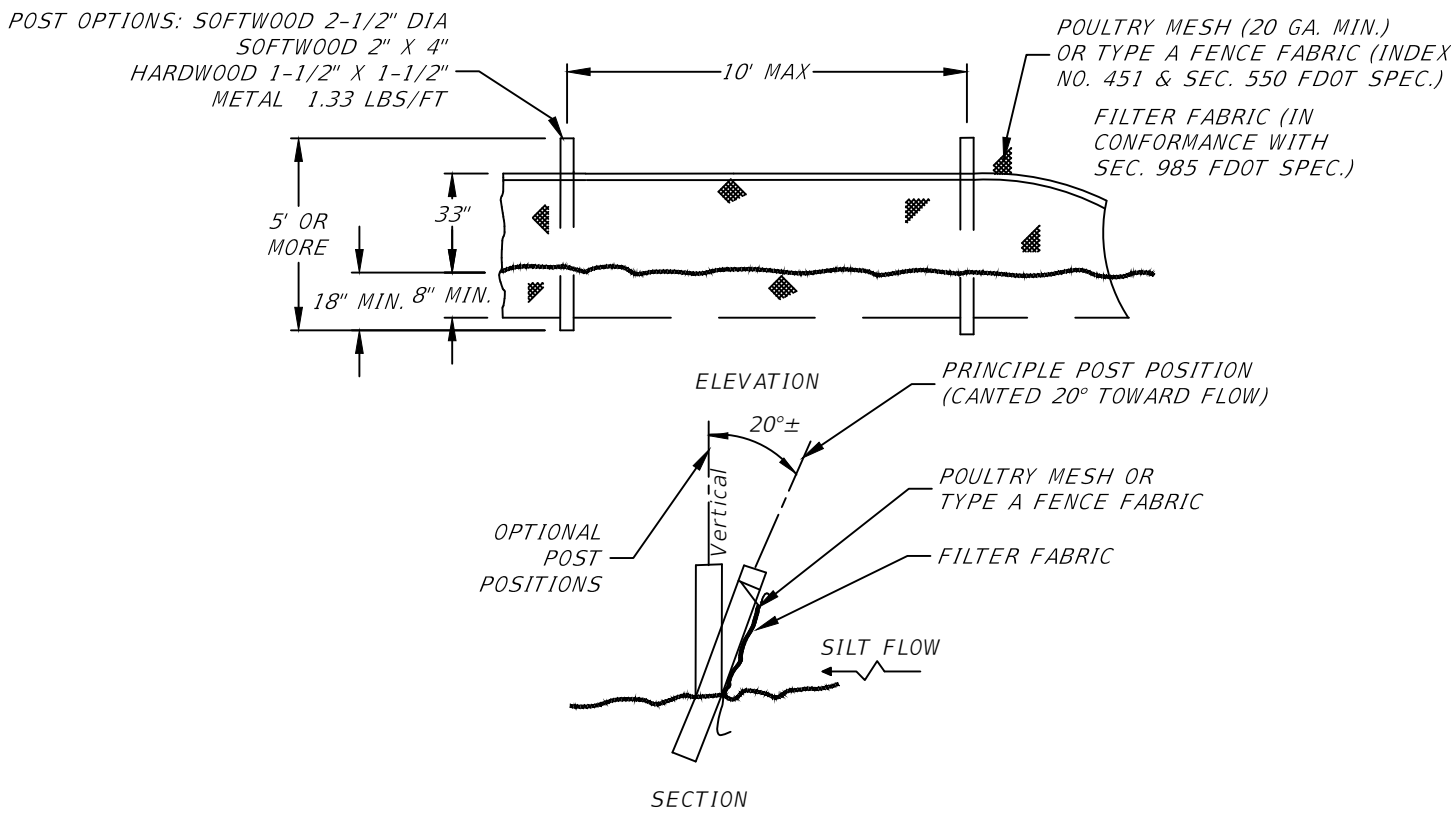
	EXISTING BURIED TELEPHONE WIRE		SIR = SET IRON ROD
	EXISTING OVERHEAD WIRE		FCM = FOUND CONCRETE MONUMENT
	LIMITED ACCESS R/W		FIP = FOUND IRON PIPE
	EXISTING FENCE		FN/C = FOUND NAIL & CAP
	PROPOSED FENCE		R = RADIUS, OR RANGE
	HEAVY DUTY SILT FENCE		SN/C = SET NAIL & CAP
	SILT FENCE		R.O.W. = RIGHT OF WAY
	EXISTING WATER LINE		Δ = DELTA
	PROPOSED WATER LINE		L = ARC LENGTH
	PROPOSED CONTOUR		CH = CHORD
	EXISTING CONTOUR		(P) = per PLAT PROVIDED BY CLIENT
	EXISTING GAS MAIN		(D) = per DEED PROVIDED BY CLIENT
	PROPOSED GAS MAIN		(M) = AS MEASURED
	EXISTING SANITARY SEWER		FF = FINISHED FLOOR ELEVATION
	PROPERTY LINE		INV = INVERT ELEVATION
	RIGHT-OF-WAY		RCP = REINFORCED CONCRETE PIPE
	BUILDING SETBACKS		TCP = TERRA COTTA PIPE
	EASEMENT		EL = ELEVATION
	EXISTING RAILROAD		PROP. = PROPOSED
	PROPOSED SANITARY SEWER		SSMH = SANITARY SEWER MANHOLE
	EXISTING WETLANDS LINE		SSMH = STORMWATER MANHOLE
	EXISTING MEAN HIGH WATER LINE		EXIST. = EXISTING
	EXISTING TREES		CMP = CORRUGATED METAL PIPE
	TREES TO BE REMOVED		UE = UNDERGROUND ELECTRIC
	TREES IMPACTED TO REMAIN		CONC = CONCRETE
	EXISTING SPOT ELEVATION		P.O.C. = POINT OF COMMENCEMENT
	PROPOSED SPOT ELEVATION		P.O.B. = POINT OF BEGINNING
	BM = BENCHMARK AS NOTED		F.A.C. = FLORIDA ADMINISTRATIVE CODE
	WV = WATER VALVE		C.O.T. = CITY OF TALLAHASSEE
	GV = GAS VALVE		(TYP) = TYPICAL
	PP = POWER POLE		SWMF = STORMWATER MANAGEMENT FACILITY
	LP = LIGHT POLE		CMP = CORRUGATED METAL PIPE
	AP = ANCHOR POLE		PROPOSED URBAN FOREST / NATURAL AREA (CONSERVATION EASEMENT)
	WM = WATER METER		EXISTING CONCRETE
	TPED = TELEPHONE PEDESTAL		PROPOSED CONCRETE
	INDICATES DIRECTION OF FLOW		INTERIOR LANDSCAPE ISLANDS

	EXISTING SITE			PROPOSED SITE		
SITE STATISTICS:	1,226,400	SF	28.15 AC	1,226,400	SF	28.15 AC
<u>IMPERVIOUS AREA</u>						
EXISTING / FUTURE BUILDINGS	6,466	SF	1%	9,826	SF	1%
ASPHALT VEHUCULAR USE AREA	0	SF	0%	0	SF	0%
SIDEWALKS AND MISC. IMPERVIOUS AREA	433	SF	0%	433	SF	0%
TOTAL IMPERVIOUS AREA	6,899	SF	1%	10,259	SF	1%
<u>PERVIOUS AREA</u>						
URBAN FOREST PROVIDED	0	SF	0%	0	SF	0%
LANDSCAPE AREA	1,219,501	SF	99%	1,216,141	SF	99%
TOTAL PERVIOUS AREA	1,219,501	SF	99%	1,216,141	SF	99%
<u>INTERIOR LANDSCAPE AREA</u>						
INTERIOR LANDSCAPE AREA REQUIRED (SEE REDEVELOPMENT TABLE)				0	SF	
INTERIOR LANDSCAPE AREA PROVIDED				0	SF	
<u>TOTAL LANDSCAPE/URBAN FOREST AREA</u>						
TOTAL LANDSCAPE AREA	1,219,501	SF	99%	1,216,141	SF	99%
TOTAL NATURAL AREA / CONSERVATION AREA	0	SF	0%	0	SF	0%
TOTAL LANDSCAPE PROVIDED	1,219,501	SF	99%	1,216,141	SF	99%

[illegible]



TREE PROTECTION BARRICADE
N.T.S.



TYPE IV SILT FENCE
N.T.S.

NOTE: SILT FENCE TO BE CONSTRUCTED IN ACCORDANCE WITH FDOT INDEX 102.



2015 CENTRE POINTE BLVD.
SUITE 103
TALLAHASSEE, FLORIDA
32303
PHONE 850.298.4213
WWW.INOVIAGROUP.COM
CA8225

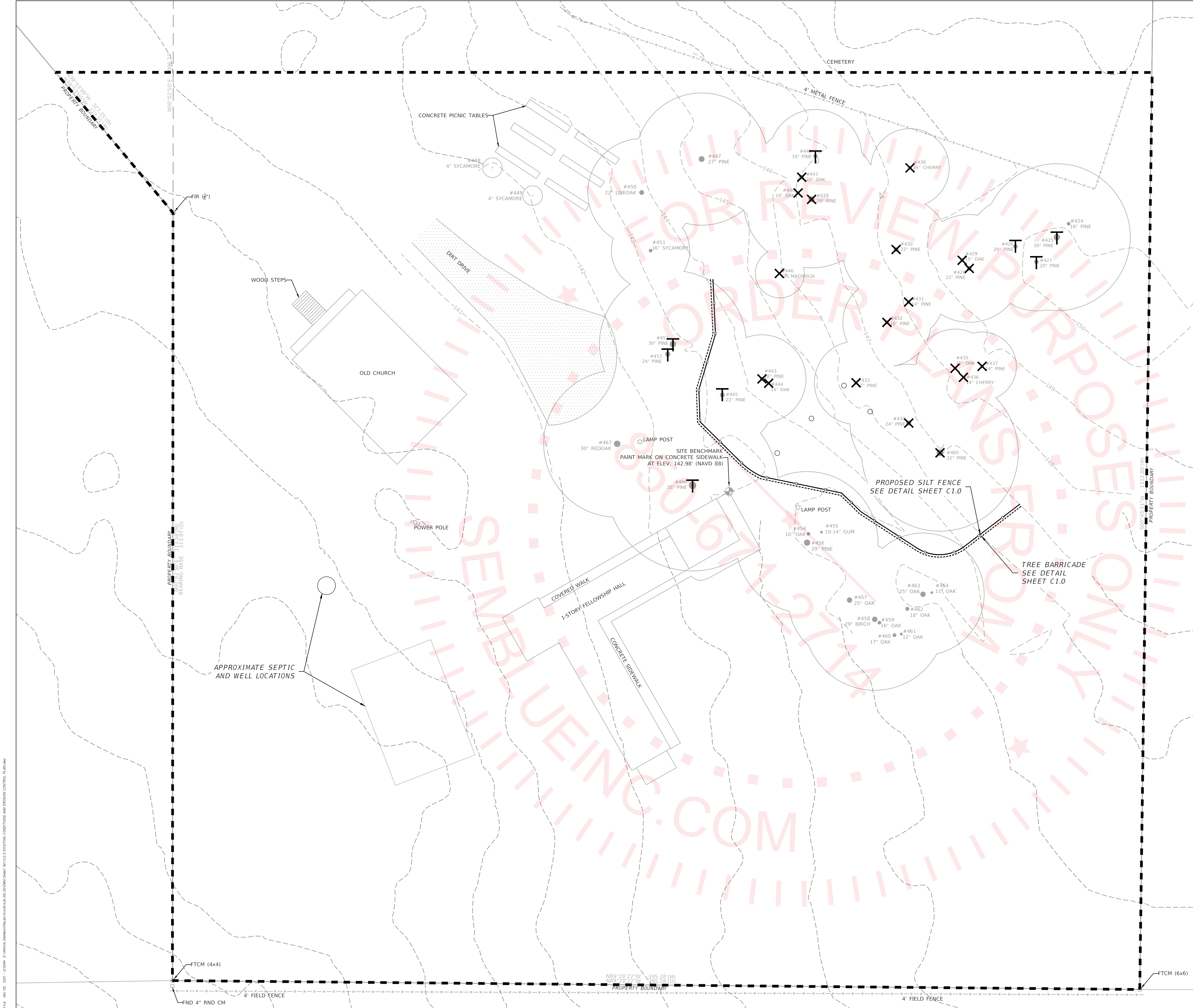
NO.	BY	REVISIONS	DATE
PROJECT NO.: 416.001.00			
DRAWING FILE: C1.0 GENERAL NOTES AND DETAILS.DWG			

CLIENT	PISGAH METHODIST CHURCH
PROJECT	PISGAH METHODIST CHURCH FELLOWSHIP HALL
SHEET	GENERAL DETAILS

PLAN RELEASE STATUS	
☐	PROGRESS PRINT
☐	PERMIT RELEASE
☒	BID RELEASE
☐	CONSTRUCTION RELEASE
RELEASE DATE: 05 NOV 21	
	INITIAL DATE
DRAWN BY	JQV 11/5/21
CHECKED BY	JFW 11/5/21
SCALE: AS SHOWN	

JAMES F. WADDELL, P.E.
PE NO.: 46019
DATE: 11/5/21

SHEET
C1.1



- EROSION CONTROL NOTES:**
- THE CONTRACTOR SHALL ENSURE THAT A FOREMAN OR SUPERVISOR WHO HAS BEEN CERTIFIED UNDER FLORIDA STORMWATER, EROSION AND SEDIMENTATION CONTROL INSPECTOR TRAINING PROGRAM IS AVAILABLE IN PERSON OR BY PHONE AT ALL TIMES DURING THE CONSTRUCTION ACTIVITIES.
 - CONTACT: TO BE DETERMINED
 - UNDER NO CONDITIONS SHALL THE SILT FENCE BE TRENCHED WITHIN THE CRITICAL PROTECTION ZONE (CPZ) OF A TREE TO REMAIN. IN SUCH AREAS WHERE SILT FENCE IS REQUIRED THE CONTRACTOR SHALL PLACE A FABRIC FLAP UNDER HAY BALES AND STAKE THROUGH.
 - ALL DISTURBED AREAS TO BE LEFT IDLE LONGER THAN 14 DAYS MUST BE STABILIZED WITH TEMPORARY SEED AND/OR MULCH.
 - CONTRACTOR SHALL PROVIDE STREET SWEEPING OPERATIONS AS NECESSARY DURING CONSTRUCTION.
 - ALL STAGING AREAS MUST OCCUR WITHIN THE PROPERTY BOUNDARY UNLESS OTHERWISE APPROVED BY THE CITY OF TALLAHASSEE AND ENGINEER PRIOR TO CONSTRUCTION.
 - THE TRANSPORT, STORAGE, AND DISPOSAL OF CONSTRUCTION MATERIALS MUST BE DONE SO IN ACCORDANCE WITH ALL CURRENT STATUTES.
 - ADDITIONAL SEDIMENT AND EROSION CONTROL MEASURES MAY BE REQUIRED, DURING ANY PHASE OF DEVELOPMENT, AT THE DISCRETION OF THE LEON COUNTY ENVIRONMENTAL INSPECTOR.

TREE CREDIT LIST			
Tree #	Species	Diameter	Credits*
TREES TO REMAIN			
424	PINE	16	6
447	PINE	27	10
448	SYCAMORE	4	2
449	SYCAMORE	4	2
450	LIVE OAK	22	8
451	SYCAMORE	16	6
454	OAK	16	6
455	GUM	14	6
456	PINE	29	10
457	OAK	25	10
458	BIRCH	29	10
459	OAK	16	6
460	OAK	17	6
461	OAK	12	4
462	OAK	18	6
463	OAK	25	10
464	OAK	11	4
467	RED OAK	30	10
TOTAL TREE CREDITS			122

*NOTE: Tree Credits only represent the surveyed trees. There are countless other trees included in the wooded area that covers 90% of the 28 acre site.

TREE DEBIT LIST			
Tree #	Species	Diameter	Minimum Tree Replacemet Credits
REMOVED/IMPACTED TREES			
** 425	PINE	30	10
** 426	PINE	20	8
** 427	PINE	20	8
** 428	PINE	22	8
** 429	OAK	15	6
** 430	PINE	22	8
** 431	PINE	14	6
** 432	PINE	18	6
** 433	PINE	17	6
** 434	PINE	24	8
** 435	OAK	16	6
** 436	CHERRY	14	6
** 437	PINE	14	6
** 438	CHERRY	16	6
** 439	PINE	28	10
** 440	OAK	14	6
** 441	OAK	20	8
** 442	PINE	19	8
** 443	PINE	18	6
** 444	OAK	14	6
** 445	PINE	22	8
** 446	MAGNOLIA	10	4
** 452	PINE	30	10
** 453	PINE	34	8
** 465	PINE	32	16
** 466	PINE	35	16
TOTAL MINIMUM TREE REPLACEMENT			204

** On-Site Tree - Impacted to Remain

2015 CENTRE POINTE BLVD.
SUITE 103
TALLAHASSEE, FLORIDA 32303
PHONE 850.298.4213
WWW.INOVIAGROUP.COM
CA8225

CLIENT	PROJECT	SHEET
PISGAH METHODIST CHURCH	PISGAH METHODIST CHURCH FELLOWSHIP HALL	EXISTING CONDITIONS AND EROSION CONTROL PLAN

PLAN RELEASE STATUS

☐ PROGRESS PRINT
☐ PERMIT RELEASE
☒ BID RELEASE
☐ CONSTRUCTION RELEASE

RELEASE DATE: 05 NOV 21

INITIAL DATE

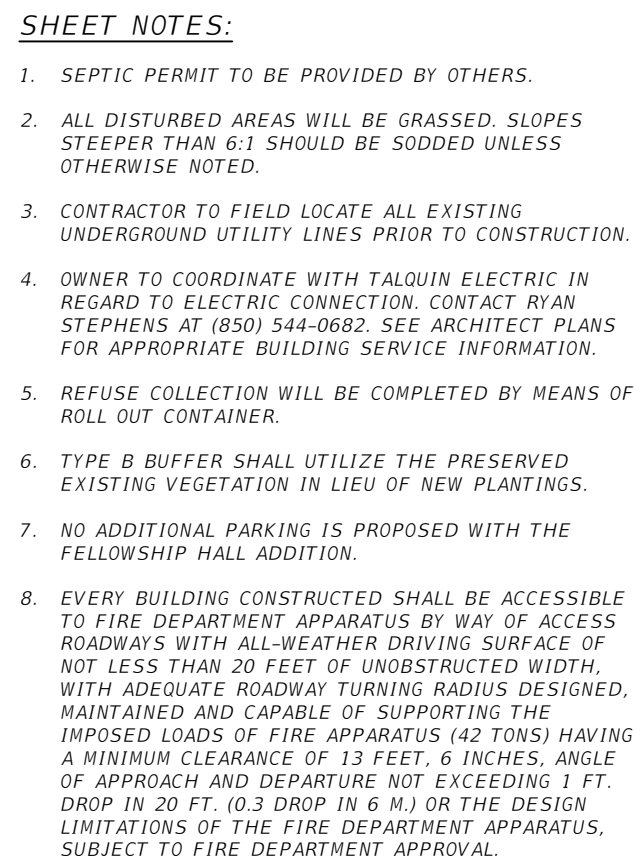
DRAWN BY JOV 11/5/21

CHECKED BY JFW 11/5/21

SCALE: AS SHOWN

JAMES F. WADDELL, P.E.
PE NO.: 46019
DATE: 11/5/21

SHEET
C2.0



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CONSULTING GROUP

2015 CENTRE POINTE BLVD.
SUITE 103
TALLAHASSEE, FLORIDA
32303
PHONE 850.298.4213
WWW.INOVIAGROUP.COM
CA8225

[illegible]

CLIENT	PISGAH METHODIST CHURCH
PROJECT	PISGAH METHODIST CHURCH FELLOWSHIP HALL
SHEET	DIMENSIONAL SITE, GRADING, AND UTILITY PLAN

PLAN RELEASE STATUS		
☐	PROGRESS PRINT	
☐	PERMIT RELEASE	
☒	BID RELEASE	
☐	CONSTRUCTION RELEASE	
RELEASE DATE: 05 NOV 21		
	INITIAL	DATE
DRAWN BY	JQV	11/5/21
CHECKED BY	JFW	11/5/21
SCALE: AS SHOWN		

JAMES F. WADDELL, P.E.
PE NO.: 46019
DATE: 11/5/21

SHEET
C3.0

STRUCTURAL NOTES

GENERAL NOTES

1. The governing Code for this Project is the Florida Building Code, 7th Edition (2020). This Code prescribes which edition of each referenced standard applies to this Project.

2. To the best of our knowledge, the Structural Drawings and comply with the applicable requirements of the governing Building Code.

3. Construction is to comply with the requirements of the governing Building Code and all other applicable Federal, State, and Local Codes, Standards, Regulations and Laws.

4. The Structural Documents are to be used in conjunction with the Architectural Documents. If a conflict exists, the more stringent governs.

5. Details labeled "typical" apply to all situations that are the same or similar to those specifically referenced, whether or not they are keyed in at each location. Questions regarding the applicability of typical details shall be resolved by the Architect.

6. Contractors who discover discrepancies, omissions or variations in the Contract Documents during bidding shall immediately notify the Architect. The Architect will resolve the condition and issue a written clarification.

7. The General Contractor shall coordinate all Contract Documents with field conditions and dimensions and Project Shop Drawings prior to construction. Do not scale drawings; use only printed dimensions. Report any discrepancies in writing to the Architect prior to proceeding with work. Do not change size or location of structural members without written instructions from the Structural Engineer of Record.

8. The Contractor shall protect adjacent property, his own work and the public from harm. The Contractor is solely responsible for construction means and methods, and jobsite safety including all OSHA requirements.

9. The Structure is designed to be structurally sound when completed. Prior to completion, the Contractor is responsible for stability and temporary bracing, including, but not limited to, masonry walls. Wherever the Contractor is unsure of these requirements, the Contractor shall retain a Florida Licensed Engineer to design and inspect the temporary bracing and stability of the structure.
10. Design Superimposed Loads:

OCCUPANCY	LIVE LOAD	DEAD LOAD
Roof	20 psf	15 psf *
Public Areas	100 psf	10 psf
Public Corridors Above 1 st Floor	100 psf	10 psf

* Does not Includes Wood Joists, Beams, Trusses or Decking.
13. Design Wind Loads

Governing Code	Florida Building Code, 7 th Edition / ASCE 7-16
Basic Wind Speed	V _{ult} = 120 mph V _{asd} = 93 mph
Building Risk Category	II
Wind Exposure	B
Internal Pressure Coefficient	GCPi = +/- 0.18
Design Wind Pressure for Components & Cladding	See Drawing S0.2
Directionality Factor	K _d = 0.85
Mean Roof Height	17'-0"

SHOP DRAWINGS AND OTHER SUBMITTALS

1. Incomplete submittals will be returned without review.
2. Submit specific components, such as columns, footings, etc., in a single package. Submit similar floors together.
3. On first submittal, clearly flag and cloud all differences from the Contract Documents. On resubmittals, flag and cloud all changes and additions to previous submittal; only clouded items will be reviewed.
4. Submittals for special structural, load-carrying items that are required by Codes or Standards to resist forces must be prepared by, or under the direct supervision of, a Delegated Engineer. Examples include Prefabricated Wood Components such as Wood Trusses and Exterior Endosure Systems.
5. A Delegated Engineer is defined as a Florida Licensed Engineer who specializes in and undertakes the design of Structural Components or Structural Systems included in a specific submittal prepared for this Project and is an employee or officer of, or consultant to, the Contractor or Fabricator responsible for the submittal. The Delegated Engineer shall sign, seal and date the submittal, including calculations and drawings. See Specifications for more specific criteria.
6. The Trade Contractor is responsible for confirming and correlating dimensions at the job sites, for tolerances, clearances, quantities, fabrication processes and techniques of construction, coordination of the work with other trades and full compliance with the Contract Documents.
7. The General Contractor/Construction Manager shall review and approve submittals and shall sign and date each drawing prior to submitting to the Architect. This approval is to confirm that the Submittal is complete, complies with the Submittal Requirements and is coordinated with field dimensions, other trades, erection sequencing and constructability.
8. The Structural Engineer reviews submittals to confirm that the submittal is in general conformance with the design concept presented in the Contract Documents. Quantities and dimensions are not checked. Notations on submittals do not authorize changes to the contract sum. Checking of the submittal by the Structural Engineer shall not relieve the Contractor of responsibility for deviations from the Contract Documents and from errors or omissions in the submittal.
9. In addition to the above, the Structural Engineer's review of Delegated Engineer submittals is limited to verifying that the specified structural submittal has been furnished, signed and sealed by the Delegated Engineer and that the Delegated Engineer has understood the design intent and used the specified Structural Criteria. No detailed check of calculations will be made. The Delegated Engineer is solely responsible for his/her design, including but not limited to the accuracy of his/her calculations and compliance with the applicable codes and standards.

EXCAVATION, BACKFILL AND DEWATERING

1. The Contractor is solely responsible for all excavation procedures including lagging, shoring, and protection of adjacent property, structures, streets and utilities in accordance with the requirements of the Local Building Department and OSHA regulations. Do not excavate within one foot of the angle of repose of any soil bearing foundation unless the foundation is properly protected against settlement.
2. Do not backfill against walls until 7 days after the walls are braced by the structure or are temporarily braced. Do not backfill until after completion and inspection of any waterproofing.
3. The Contractor is responsible for the disposal of all accumulated water in a manner that does not inconvenience or damage the work.

SHALLOW FOUNDATIONS

1. Foundation design, soil preparation and compaction are based on Geotechnical Investigation, Data and Recommendations in Report # 15-19-20-634-000-0 by Earthworks geotechnical, Inc. and Testing Services, Inc. dated March 9, 2020.
2. Footing sizes and reinforcing are based on an assumed allowable soil bearing capacity of 2,000 psf. All footings shall bear on compacted fill, natural soil or rock prepared per the Geotechnical Report.
3. Subgrade preparation shall be field controlled and tested by a Licensed Soils Engineer in accordance with the Geotechnical Report. At completion, that Engineer shall prepare and submit to the Owner, Architect, Contractor and Structural Engineer a signed and sealed letter indicating that the recommendations of the Geotechnical Report have been followed.
4. Center all footings under their respective columns or walls, u.o.n.

REINFORCED CONCRETE

1. Comply with ACI 301 and 318.
2. Provide structural concrete with a minimum ultimate compressive design strength in 28 days as follows:

Element	Strength
Footings	4,000 psi
Thicken Slabs / Slabs on Grade	4,000 psi
Sidewalks	3,000 psi
3. Contractor shall provide one set of concrete test cylinders for each placement of concrete to verify the concrete strength. The four (4) test cylinders should be provided and tested as follows:

1 Test Cylinder break at 7 days
2 Test Cylinders break at 28 days
1 Test Cylinder break at 54 days or hold

4. Use normal weight concrete for all structural members. u.o.n.
5. Provide ASTM A-615 Grade 60 reinforcing steel. Reinforcing shall be accurately placed, rigidly supported and firmly tied in place, with appropriate bar supports and spacers. Lap continuous reinforcing 48 bar dia. Lap bottom steel over supports and top steel at midspan (u.o.n.). Hook discontinuous ends of all top bars and all bars in walls, u.o.n. provide cover over reinforcing as follows:

Element	Bottom	Top	Sides
Footings	3"	2"	3"
Thicken Slabs / Slabs on Grade	2"	1"	2"
6. Where specified, provide plain, cold-drawn electrically-welded wire reinforcement conforming to ASTM A-185. Supply in flat sheets only. Lap splice one cross wire spacing plus two inches.
7. Utilities shall not penetrate beams or columns but may pass through slabs and walls individually, uon. For openings 24" long or less, cut reinforcing and replace alongside opening with splice bars of equivalent area with 48 bar dia. lap. Prepare and submit shop drawings for openings longer than 24". For rectangular openings 12" long or longer, add 1#5 x 6' mid depth diagonal at all 4 corners.
8. Where reinforcing steel congestion permits, conduit and pipes up to 1" diameter may be embedded in concrete per ACI 318, Section 6.3. Space at 3 diameters o.c. Place between outer layers of reinforcing if conduits are significantly congested, additional reinforcing perpendicular to piping may be required. Requests to embed larger pipes should be accompanied by a detailed description and be submitted to the architect for evaluation.
9. Provide construction joints in accordance with ACI 318, Section 6.4. Provide keyways and adequate dowels. Submit drawings showing location of construction joints and direction of pour for review.

10. Provide 3/4" chamfer for all exposed corners.

11. Provide reinforcing steel placer with a set of Structural Drawings for field reference. Inspect reinforcing steel placing from structural drawings.

CONCRETE ANCHORS

1. Use Simpson Strong-Tie Titen HD anchors or accepted equivalent for anchors to concrete. Follow manufacturer's specifications for use and installation. See details for quantities, sizes and lengths.
2. Confirm the absence of reinforcing steel by drilling a 1/4" diameter pilot hole for each anchor. Do not cut reinforcing steel without approval of the Structural Engineer.
3. Provide anchor embedment, spacing and edge distance as shown on the Drawings.

SLABS ON GRADE

1. Refer to Geotechnical Report for subgrade preparation more than 12" below bottom of slab.
2. Above subgrade, use fill containing not more than 10% passing #200 sieve and maximum 1 inch diameter. Compact to 95% of maximum dry density as determined by modified proctor ASTM D-1557. Each layer of fill shall not exceed 6" loose thickness. Compact prior to placement of the next layer.
3. Fill placement and compaction shall be monitored and accepted by the testing agency. Take a min. of one field density test (ASTM D-1556 or D-2922) for each 2,500 square feet of each layer. The testing agency shall randomly select test locations.
4. For interior slabs place 10 mil polyethylene sheeting between soil and bottom of slab. Do not use any sheeting below exterior concrete slabs.
5. Use 4" thick slabs on grade reinforced with 6 x 6 - W2.0 x W2.0 welded wire reinforcement supplied in flat sheets only unless noted otherwise. Use chairs to support wire fabric in the center of slab.
6. Place concrete in long-strip construction method. Provide crack control joints at 10'-0" feet maximum to limit areas between joints to 100 sq. ft. in all floating slabs on grade. Locate to conform to bay spacing whenever possible, add crack control joints at re-entrant corners which tend to invite cracks.
7. In sidewalks and walkways, locate isolation joints at 20 ft. o.c. maximum score and tool between isolation joints in equal bays of 5 ft. or less.
8. See the Architectural Drawings for slab on grade depressions and other requirements.

WOOD CONSTRUCTION

1. All wood construction and connections shall conform to AITC "American Institute of Timber Construction" Manual, and the "National Design Specifications" for Wood Construction, 2001 Edition, and Florida Building Code, Chapter 23.
2. All member sizes are to be as shown on drawing and provide the following minimum properties:

Member	Species	Fb (PSI)	E (PSI)
Joists	So. Pine No. 2	1500	1,400,000
Studs	So. Pine No. 2	1500	1,400,000
Other	So. Pine No. 2	1500	1,400,000
3. All wood in contact with concrete, masonry or exposed to weather shall be pressure treated.
4. All bolts and bolted connections shall conform to ASTM A307 and shall be galvanized. Use washers between wood and all bolt heads and nuts.
5. All metal wood connectors shall be galvanized and shall be manufactured by Simpson Strong Tie Co., or approved equal. All fasteners exposed to weather shall be galvanized or designed by the manufacturer to resist corrosion.
6. All joists shall be laterally supported at ends by solid blocking.
7. Stud walls shall be 2 X 6 at 16" o.c..
8. Unless noted otherwise on plan, provide double studs at all jambs of openings up to 6'-0". Use triple studs for openings greater than 6'-0" and less than 10'-0".
9. Where beams or columns are formed of two (2) or more members, they shall be full length and fastened together per Chapter 23 of the Florida Building Code or as shown in the structural details.
10. Exterior walls shall have 1/2" CDX or better plywood nailed with 10d ring shank nails at 6" o.c. maximum at all panel edges. Space nails at 12" o.c. maximum along intermediate framing. Block all panel edges.

PRE-ENGINEERED WOOD TRUSSES

1. Design and fabricate all timber in conformance the "National Design Specification For Stress-Grade Lumber And Its Fastenings" and Florida Building Code, 7th Edition (2020), and the National Design Standard for metal plate connected wood truss construction.
2. In accordance with Rule 61G15-31.003 of the Florida Administrative Code, the Truss System Engineer, a Delegated Engineer, shall design the Truss System. The Truss System Engineer shall submit shop drawings and calculations for review to Architect/Engineer for the assemblage of prefabricated, engineered wood trusses and truss girders, together with all bracing, connections and other structural elements and all spacing and location criteria (truss placement plan), that, in combination, function to support the dead, live and wind loads applicable to the roof Truss System. The Truss System does not include walls, or any other structural support systems. These shop drawings and calculations shall be signed and sealed by the Truss System Engineer.
3. In accordance with Rule 61G15-31.003 of the Florida Administrative Code, the Truss Design Engineer, a Delegated Engineer, shall design the individual trusses of the Truss System, but does not design the Truss System. The Truss Design Engineer shall submit shop (piece) drawings and calculations for each different truss of the Truss System such that each truss will function to support the dead, live and wind loads applicable to each truss and truss girder that together comprise the Truss System. These shop drawings and calculations shall be signed and sealed by the Truss Design Engineer.
4. The Truss System Engineer and the Truss Design Engineer shall each be responsible for their own work. However, they may be the same individual providing two separate services.
5. The loads, layouts and connections provided on the structural construction documents are the minimums to be followed by the Truss System Engineer and the Truss Design Engineer. The Truss System Engineer is responsible for all truss to truss connections.
6. Use stress-rated timber for all wood structural members. Do not use wood structural members with a bending stress less than 1200 psi or a modulus of elasticity less than 1,400,000 psi when used at 19% maximum moisture content.
7. Pressure treatment of all structural lumber shall be in accordance with AWPA Standards C1 and C2, latest Editions with a waterborne preservative in accordance with Standard P5, latest Edition. All lumber to be kiln-dried after treatment to a moisture content not to exceed 19% oven-dry basis, per Standard C2. All lumber 4 x 4 in. (nominal size) and larger in dimension to be treated to the soil and fresh water retention and penetration requirements of Standard C2. All lumber less than 4 x 4 (nominal size) to be treated to the above ground requirements of Standard C2.
8. Minimum design loads for roof trusses:

20 psf LL top chord
15 psf DL top chord
15 psf DL bottom chord
Wind pressure based on ASCE-7. See General Notes and Drawing S0.2.
9. The design and erection of wood trusses, including bracing, shall conform to the commentary and recommendations of the truss plate Institute. In addition to continuous lateral bracing of top and bottom chords (designed by the Truss System Engineer, but spaced not more than 10' o.c.) provide diagonal bracing (minimum 2" thick nominal lumber) as follows:

A. In the plane of the top chord: locate between lateral bracing, set at 45 degree angles, see drawings for intervals.

B. In the plane of the web members (perpendicular to the trusses): at each web member requiring continuous lateral bracing but not more than 16' intervals, spacing between sets of diagonals shall not exceed 20' or twice the horizontal run of the diagonal.

C. In the plane of the bottom chord: place between continuous lateral bracing at 45 degree angle at each end of building.
10. Plywood roof sheathing shall be 5/8" thick CDX with exterior glue. In compliance with AITC Standards, connect to supports with 10d ring shank nails at 4" o.c. at panel edges and 6" at intermediate supports.

SHEET NUMBERING SYSTEM

S0.1 Series:	Notes
S0.2 Series:	Wind Loading
S1.0 Series:	Plans
S2.0 Series:	Foundation & Concrete Details
S3.0 Series:	Wood Details

FOR CONSTRUCTION BIDS

REVISIONS

NO.	DATE



STRUCTURAL
ENGINEERING
GROUP, LLC

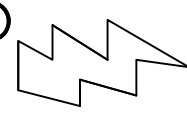
CARSON D. HICKS, P.E.
FL PE No. 68195
1382 Timberlane Road
Suite B
Tallahassee, Florida 32312
(850) 591-5896
SEG Job No. 20T025

To best of my knowledge, the plans & specifications comply with applicable minimum building codes.

PISGAH FELLOWSHIP HALL
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

JERRY L. HICKS, ARCHITECT
5383 PEMBRIDGE PLACE
TALLAHASSEE, FLORIDA 32309
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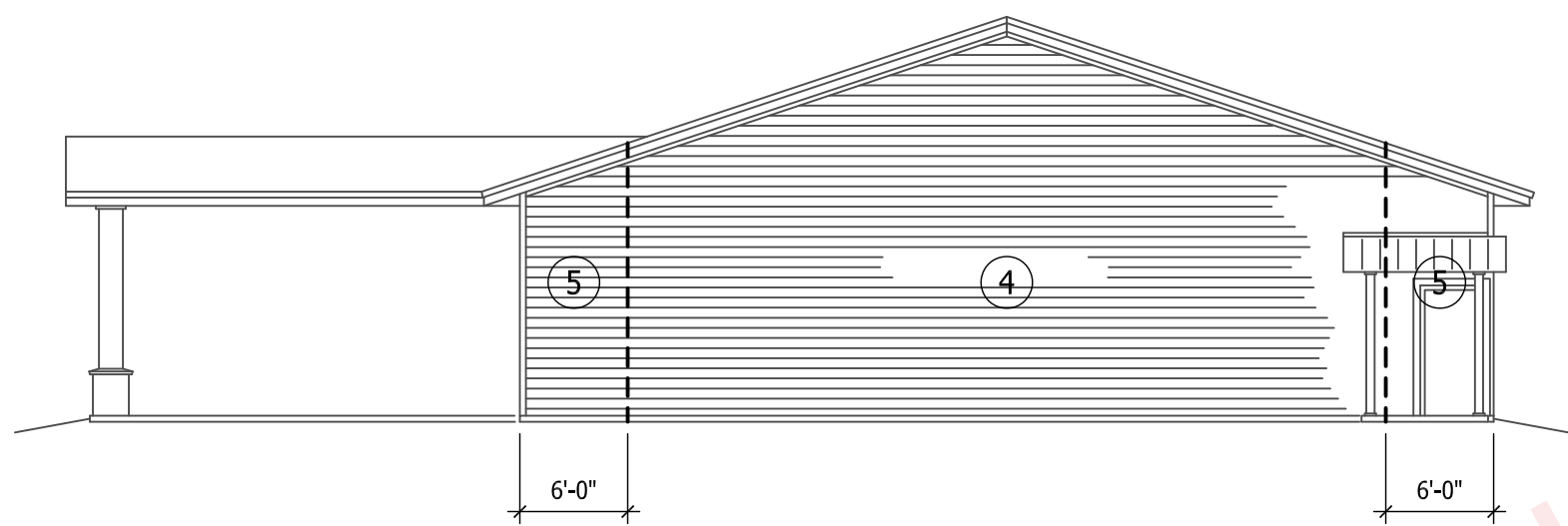
OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
PE 13168 EB 5043
6700 TREASURE OAKS CIRCLE
TALLAHASSEE, FLORIDA 32309
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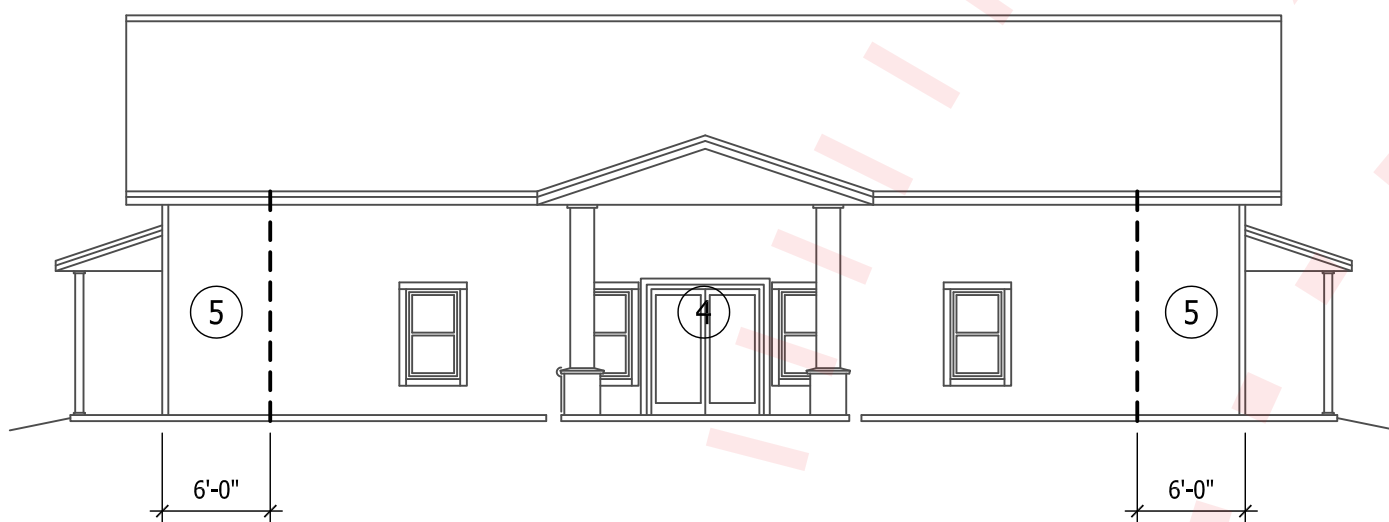
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DRAWN BY: WAG
CHECKED BY: CDH
PROJ. NO. 38022

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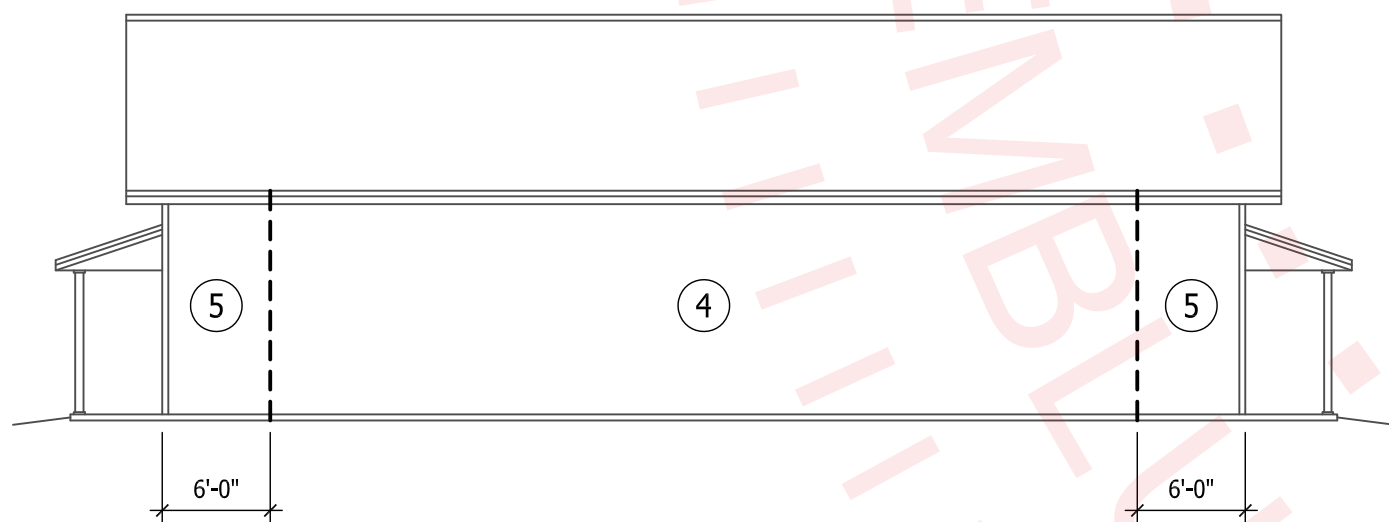
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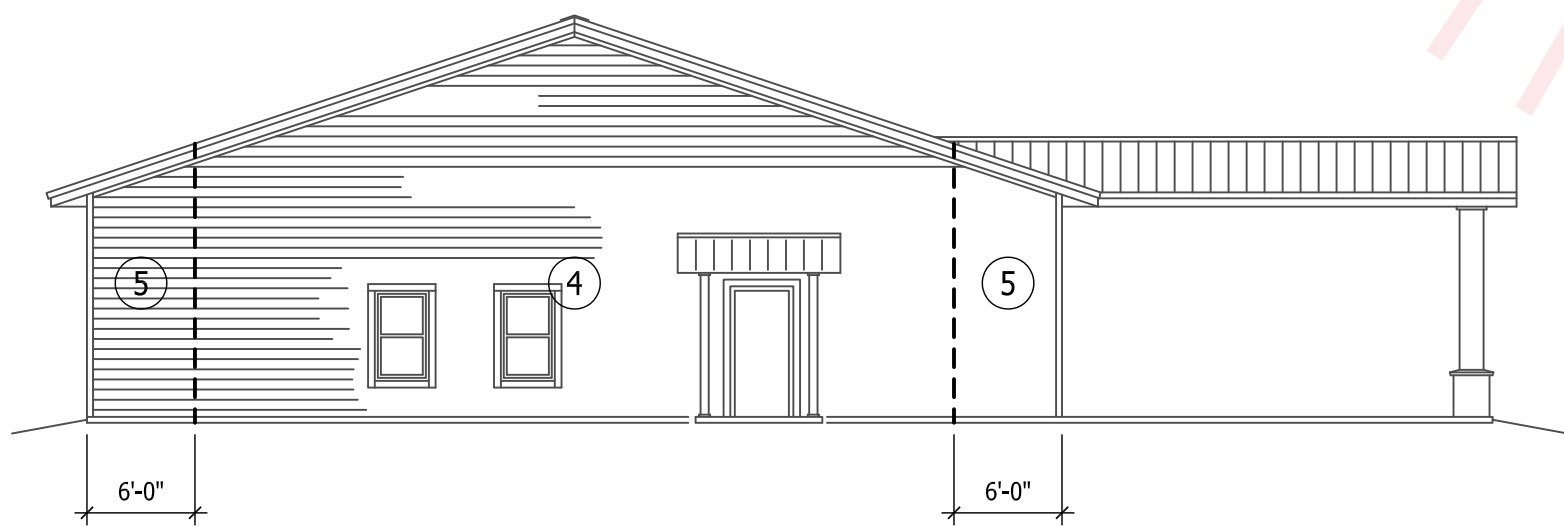
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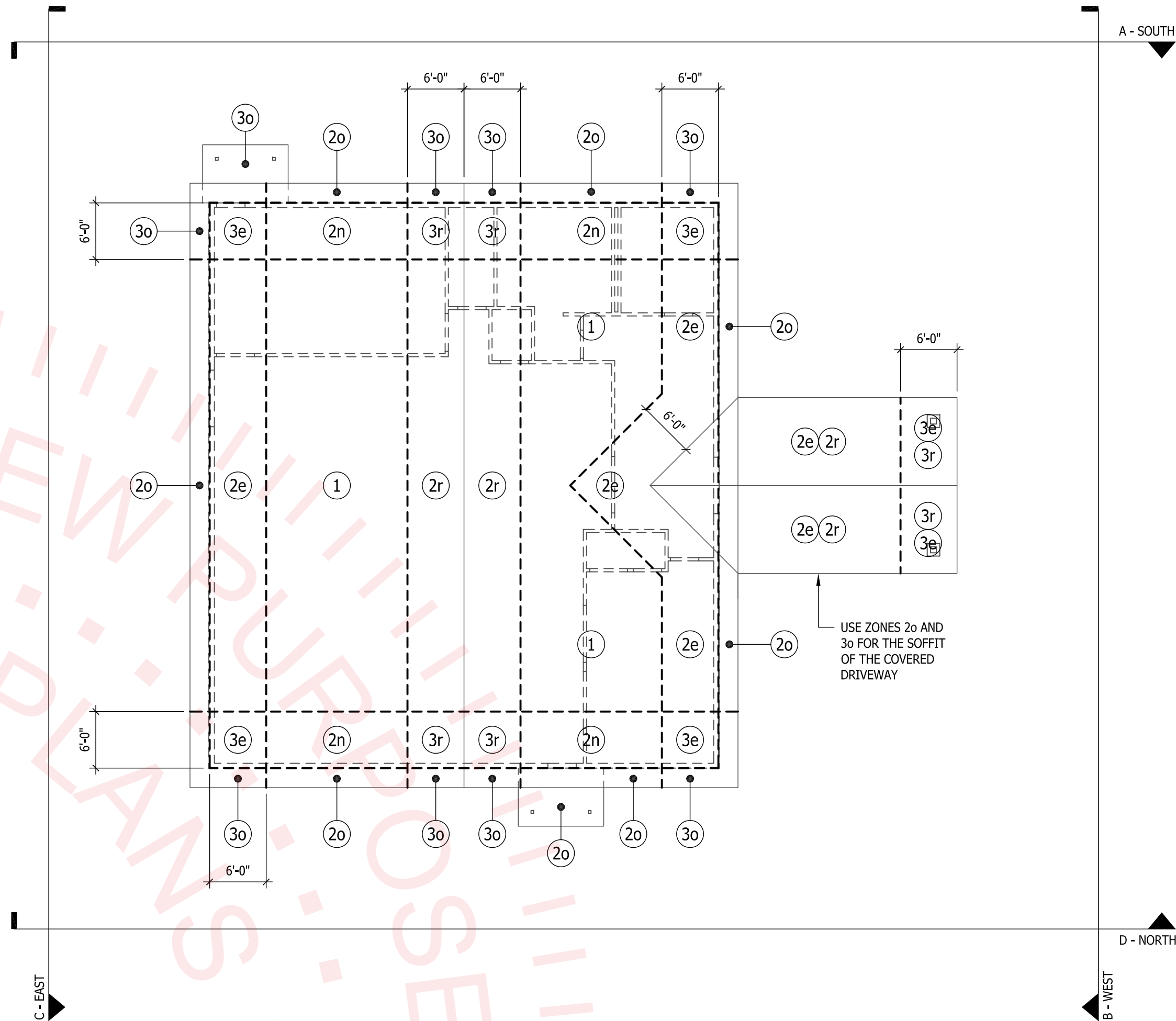
B WEST ELEVATION WIND LOADING



C EAST ELEVATION WIND LOADING



D NORTH ELEVATION WIND LOADING



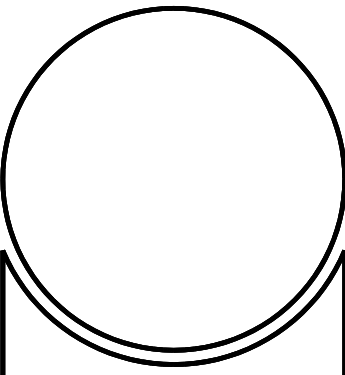
ROOF WIND LOADING DIAGRAM

SCALE: 3/32"=1'-0"

ZONES	COMPONENTS AND CLADDING ROOF AND WALL DESIGN WIND PRESSURES					
	TRIBUTARY AREA (SF)					
	10	20	50	100	200	500
1	-50/20	-50/20	-30/20	-25/20	-25/20	-25/20
2e	-50/20	-50/20	-35/20	-35/20	-35/20	-35/20
2n	-70/20	-65/20	-50/20	-40/20	-35/20	-35/20
2r	-70/20	-65/20	-50/20	-40/20	-35/20	-35/20
2o	-90	-85	-70	-60	-55	-55
3e	-70/20	-65/20	-50/20	-40/20	-35/20	-35/20
3r	-85/20	-75/20	-60/20	-50/20	-50/20	-50/20
3o	-105	-95	-80	-70	-70	-70
4	±30	-30/25	-30/25	±25	±25	-25/20
5	-35/30	-35/25	-30/25	-30/25	-30/25	-25/20

WIND PRESSURE NOTES

- Numbers on this sheet are the components and cladding gross factored load pressures perpendicular to the surface (in P.S.F.) based on tributary area. Multiply factored load pressures by the square root of 0.6 to obtain wind pressures for service loads using allowable stress design (ASCE 7-16 2.4).
- Pressures are derived from ASCE 7-16.
- Directionality factor $K_d = 0.85$
- Negative pressures act away from surface, positive pressures act toward surface.



REVISIONS	
NO.	DATE



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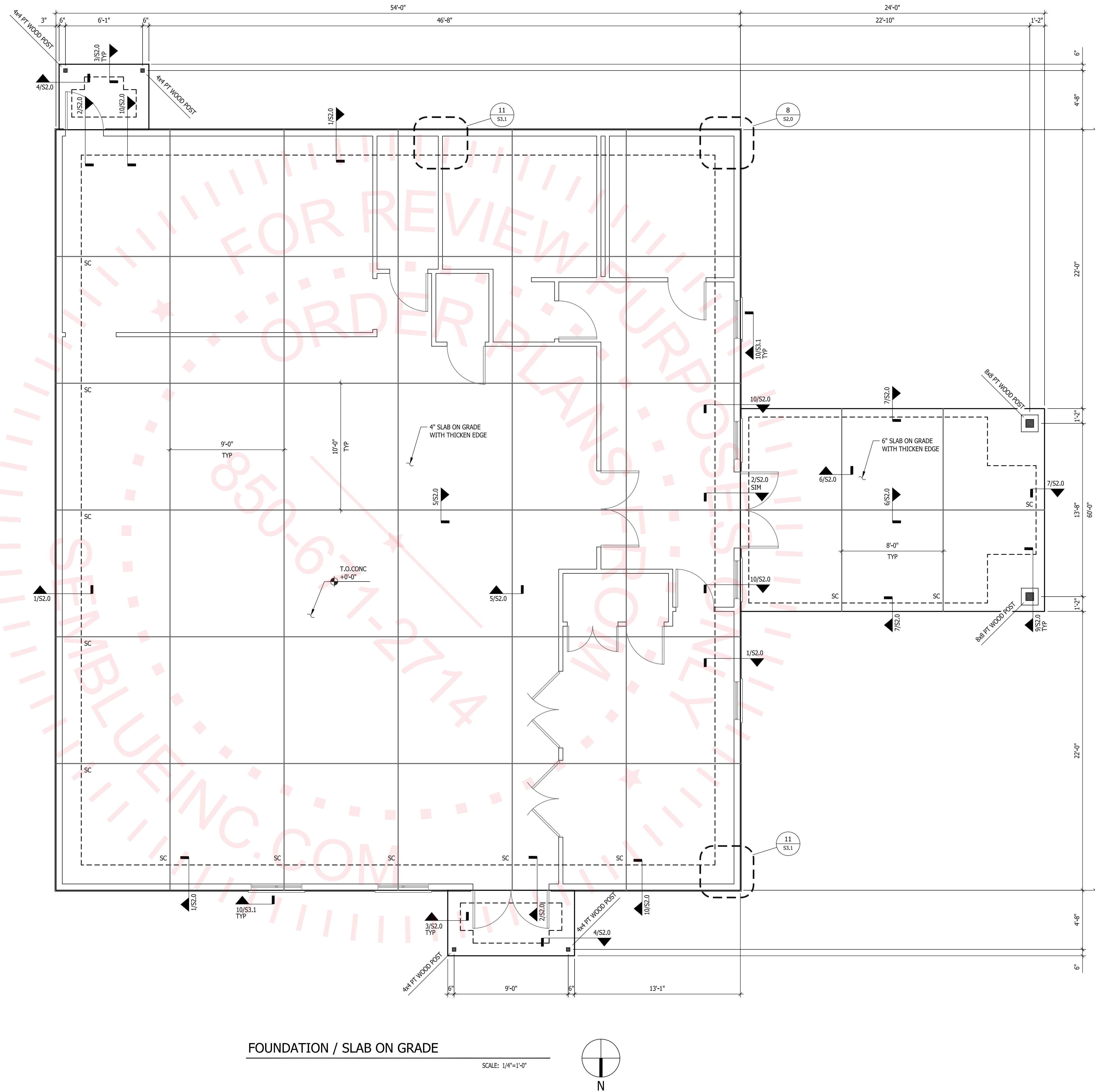
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OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
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DATE:	11-5-2021
DRAWN BY:	WAG
CHECKED BY:	CDH
PROJ. NO.	38022

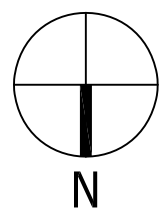


FOR CONSTRUCTION BIDS

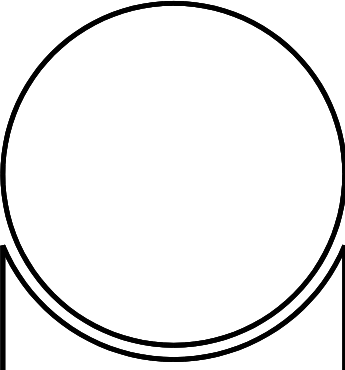


FOUNDATION / SLAB ON GRADE

SCALE: 1/4"=1'-0"



FOR CONSTRUCTION BIDS



REVISIONS	
NO.	DATE



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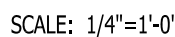
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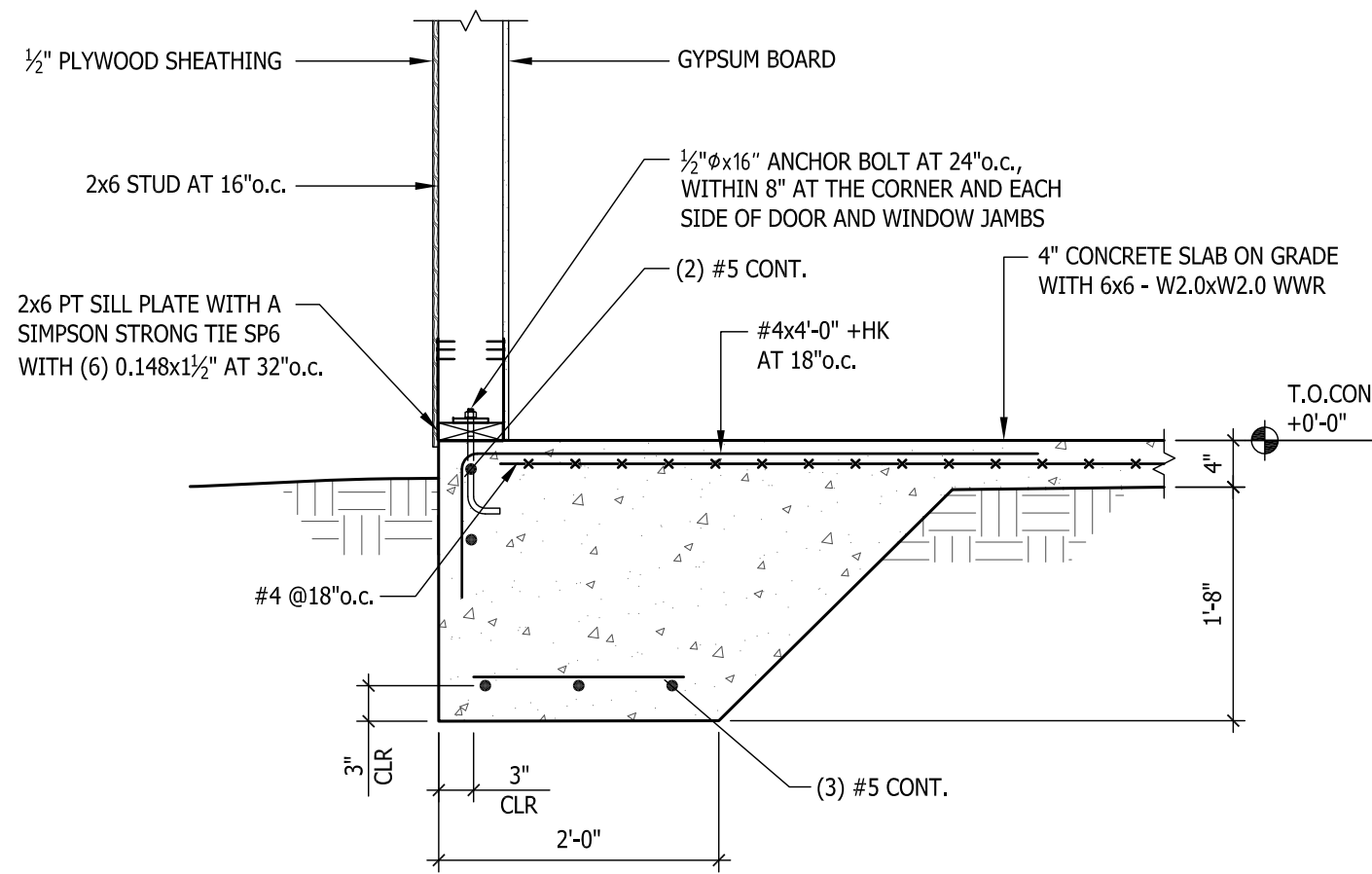
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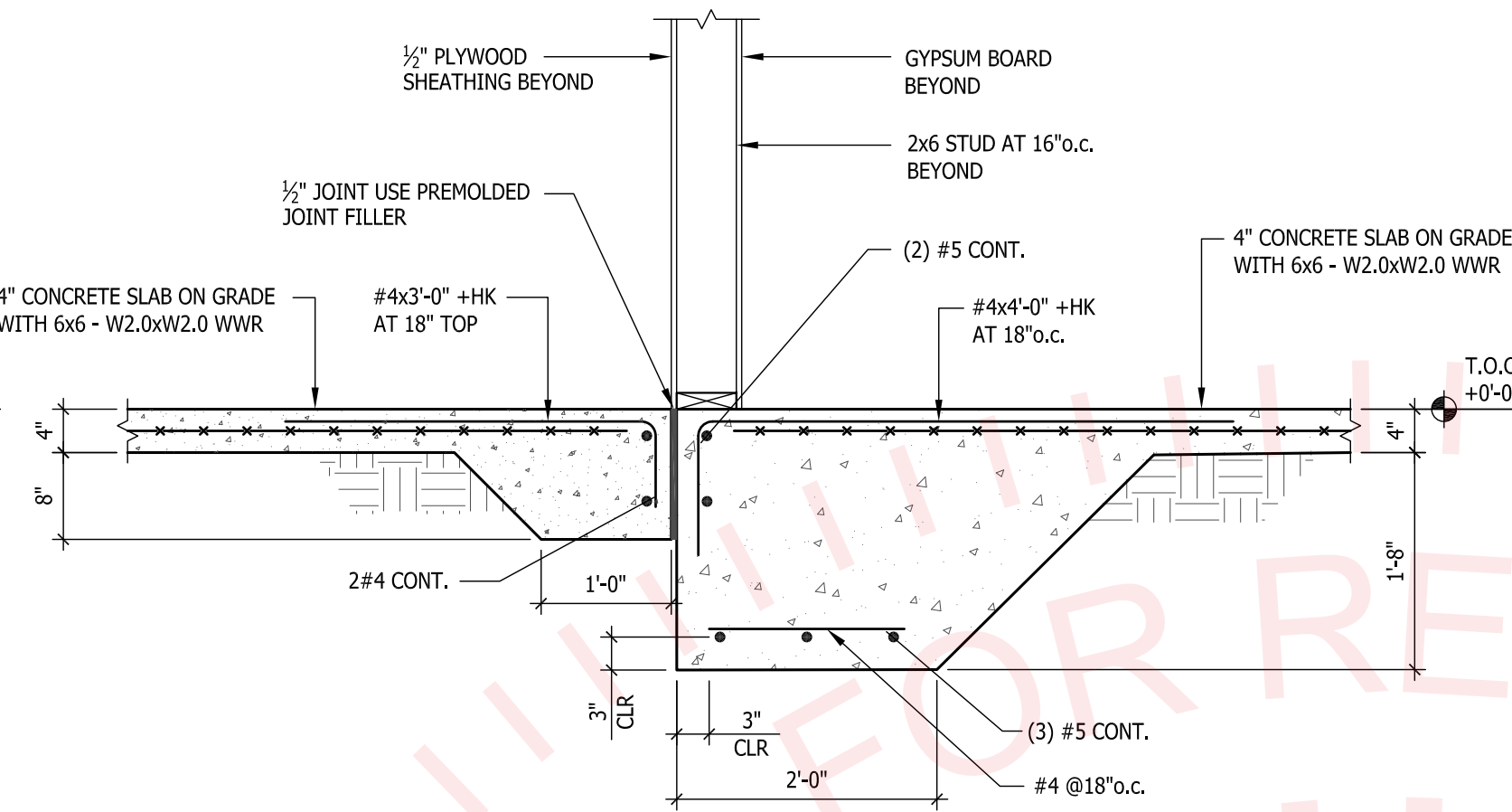


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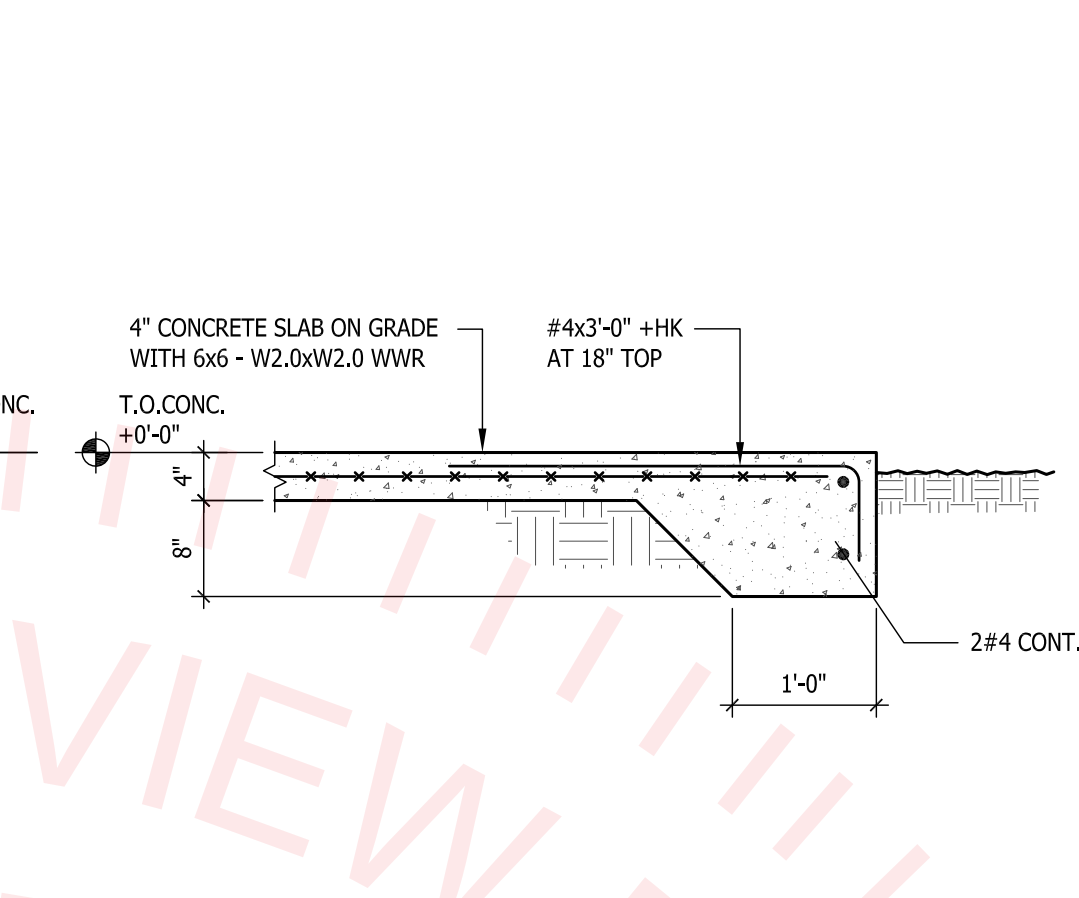
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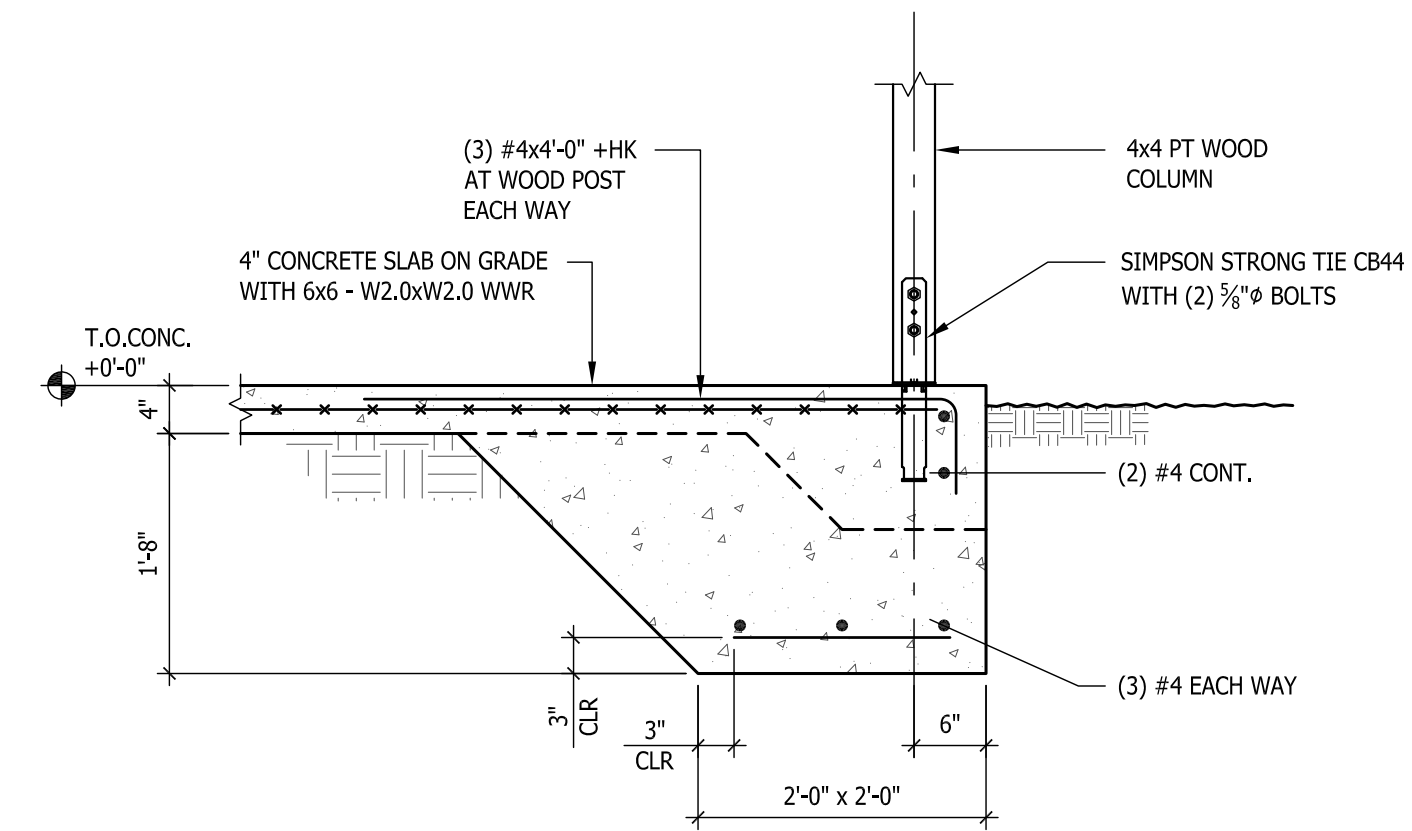
1 THICKEN SLAB EDGE DETAIL
AT WOOD FRAMED WALL



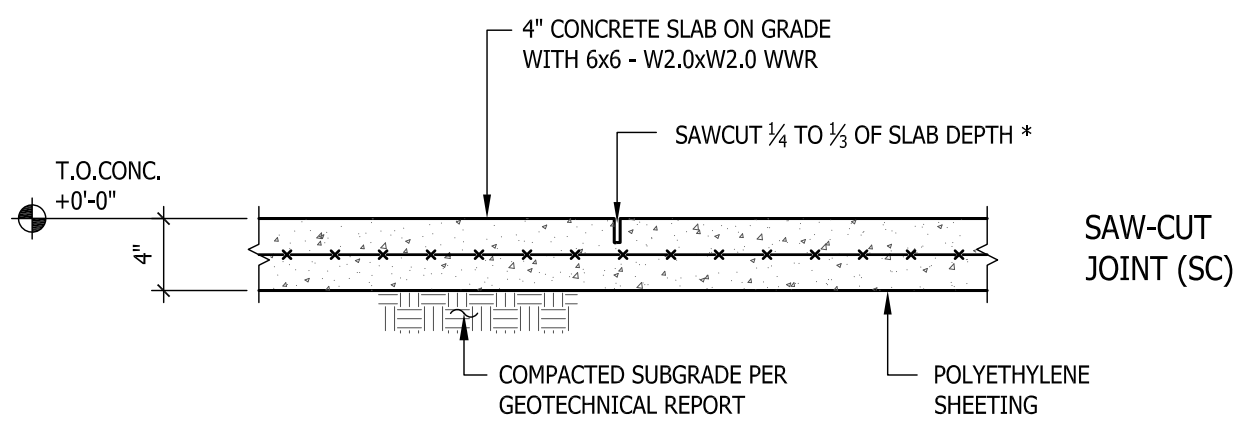
2 THICKEN SLAB EDGE DETAIL
AT DOOR OPENINGS



3 SLAB EDGE DETAIL
AT COVERED PORCH

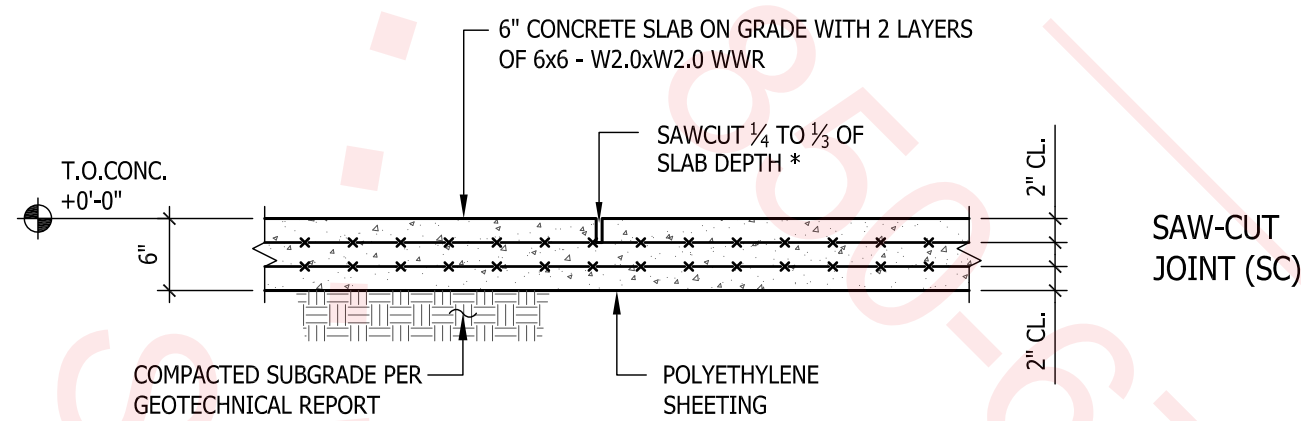


4 FOUNDATION AT WOOD POST
AT COVERED PORCH



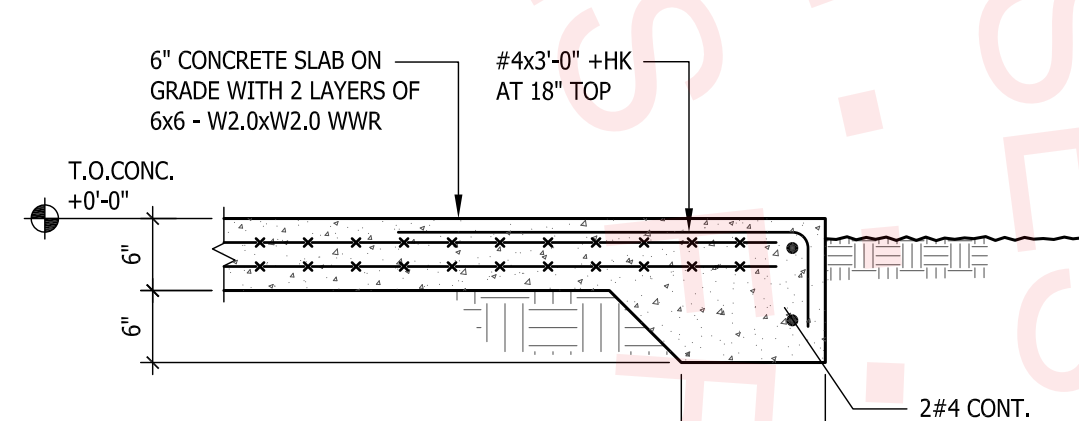
* PROVIDE SEALANT WHERE REQUIRED BY ARCHITECT
SAWCUT AS SOON AS POSSIBLE AFTER CONCRETE HARDENS.
SAW CUTTING MUST BE COMPLETE WITHIN 8 HRS. AFTER
POURING.

5 CONTROL JOINTS
SLAB ON GRADE

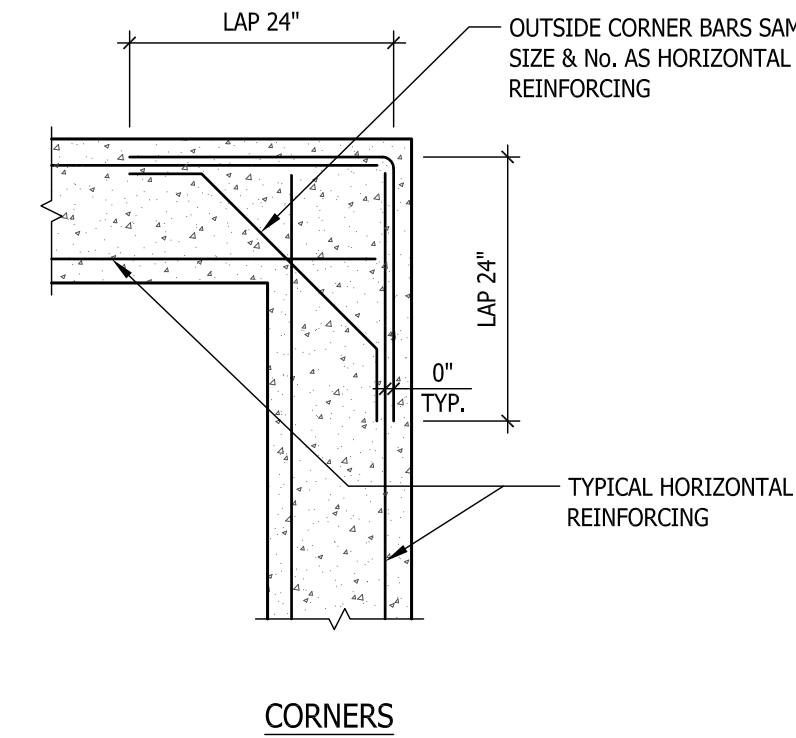


* PROVIDE SEALANT WHERE REQUIRED BY ARCHITECT
SAWCUT AS SOON AS POSSIBLE AFTER CONCRETE HARDENS.
SAW CUTTING MUST BE COMPLETE WITHIN 8 HRS. AFTER
POURING.

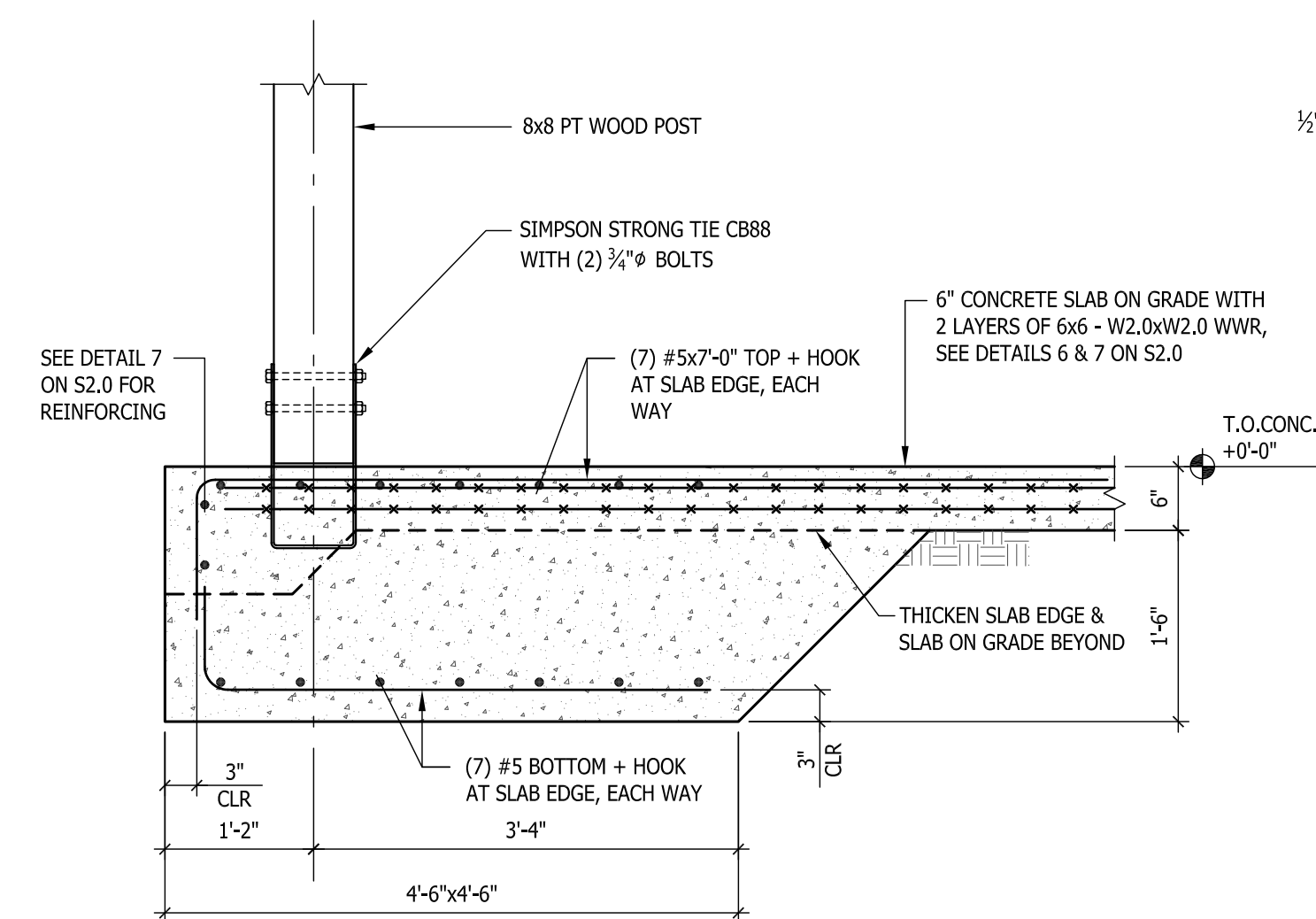
6 TYPICAL SLAB ON GRADE DETAIL
WITH WWR



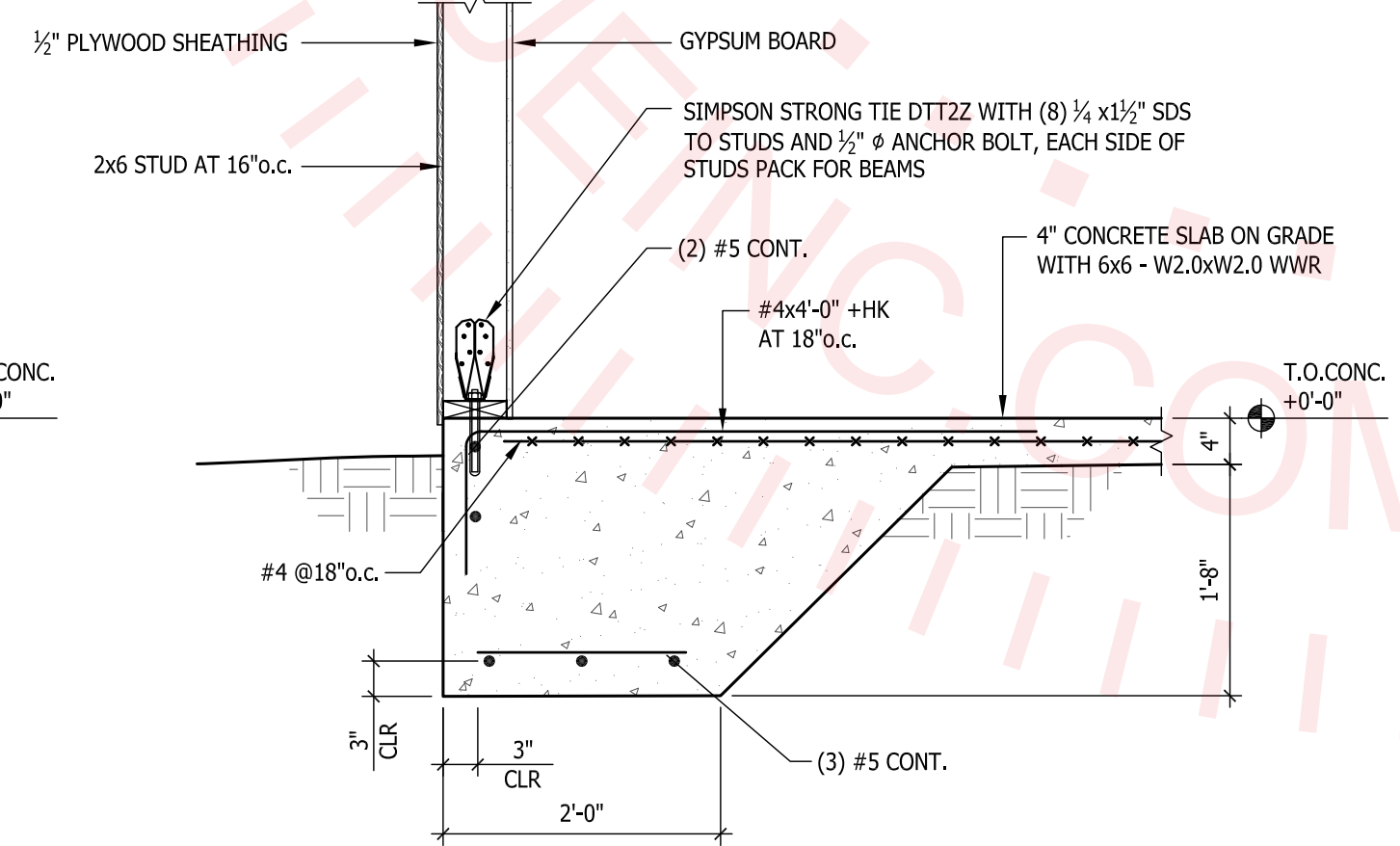
7 SLAB EDGE DETAIL
AT COVERED ENTRANCE / DRIVEWAY



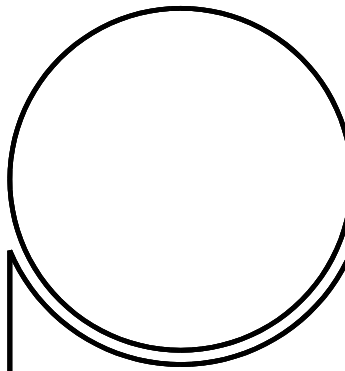
8 TYPICAL HORIZONTAL REINFORCING
CONCRETE SLAB AND FOOTINGS



9 WOOD POST COLUMN / FOOTING DETAIL



10 TYPICAL DETAIL AT STUDS FOR BEAMS
FOUNDATION CONNECTION FOR STUDS AT WOOD BEAMS



REVISIONS	
NO.	DATE



CARSON D. HICKS, P.E.
FL PE No. 68195
1382 Timberlane Road
Suite B
Tallahassee, Florida 32312
(850) 591-5896
SEG Job No. 20T025

To best of my knowledge, the
plans & specifications comply
with applicable minimum
building codes.

PISGAH FELLOWSHIP HALL
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

JERRY L. HICKS, ARCHITECT
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850-544-0540

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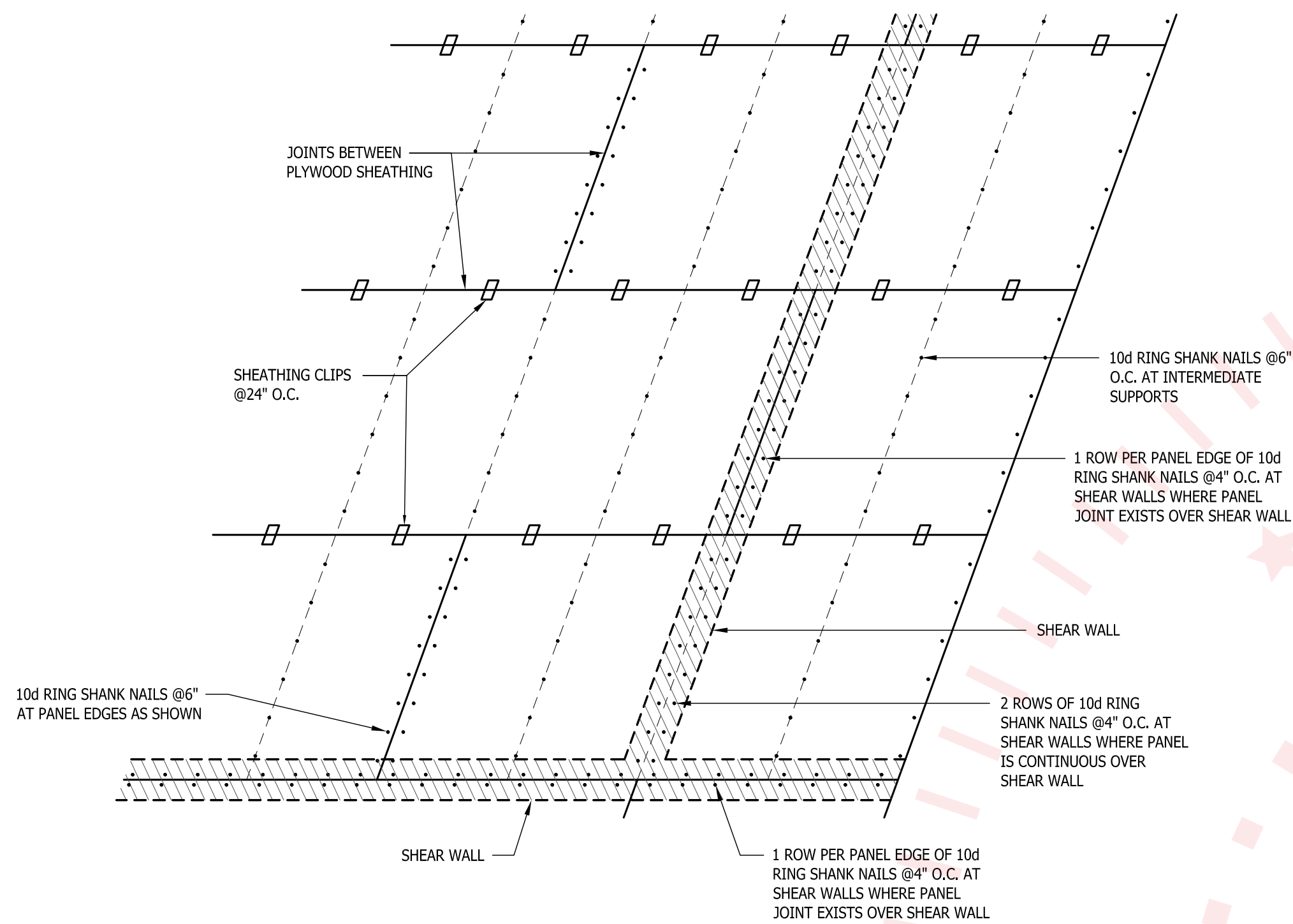
OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
PE 13168 EB 5043
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TALLAHASSEE, FLORIDA 32309
850-906-9501

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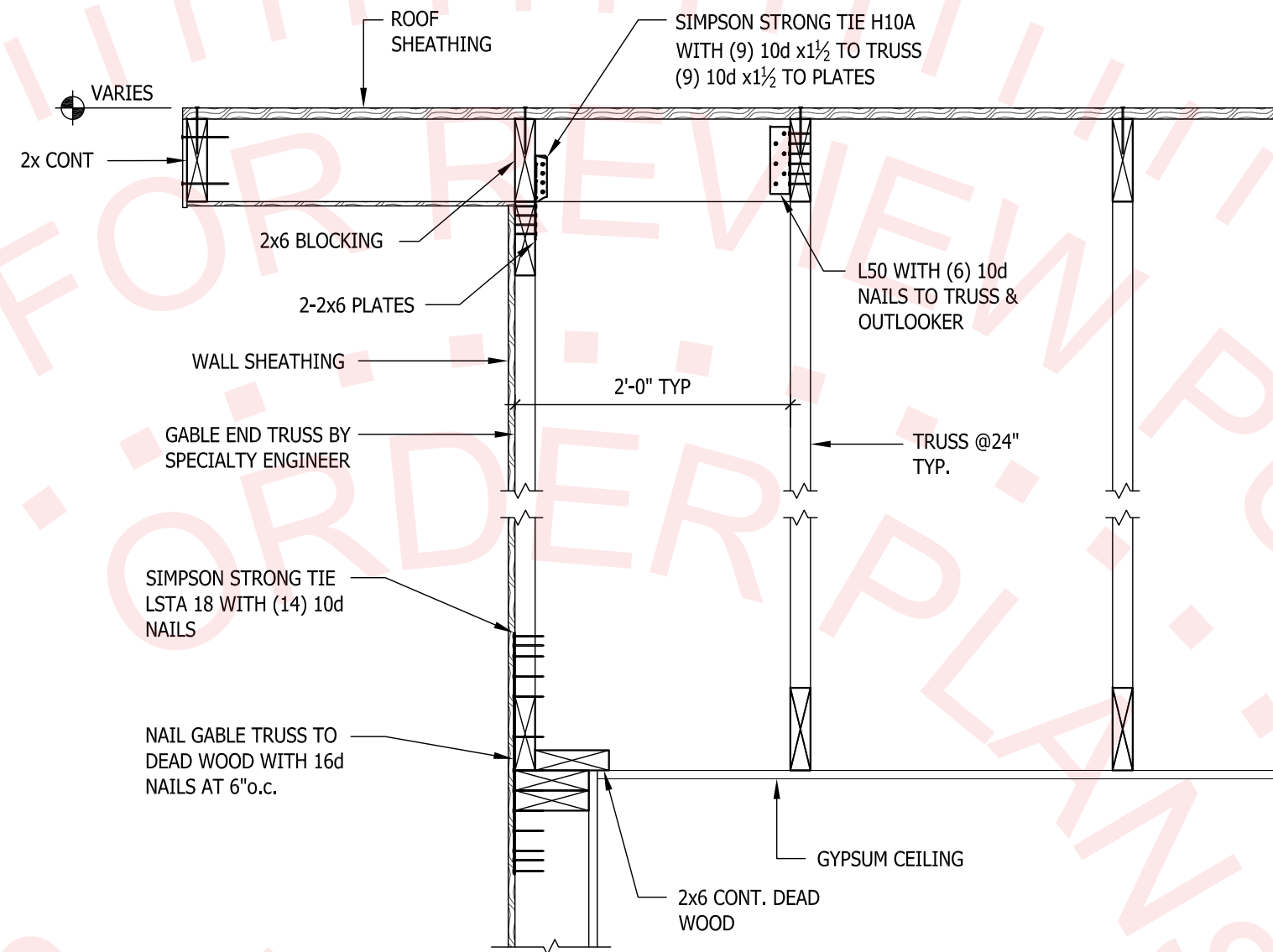
SHEET NO.
S2.0

FOR CONSTRUCTION BIDS

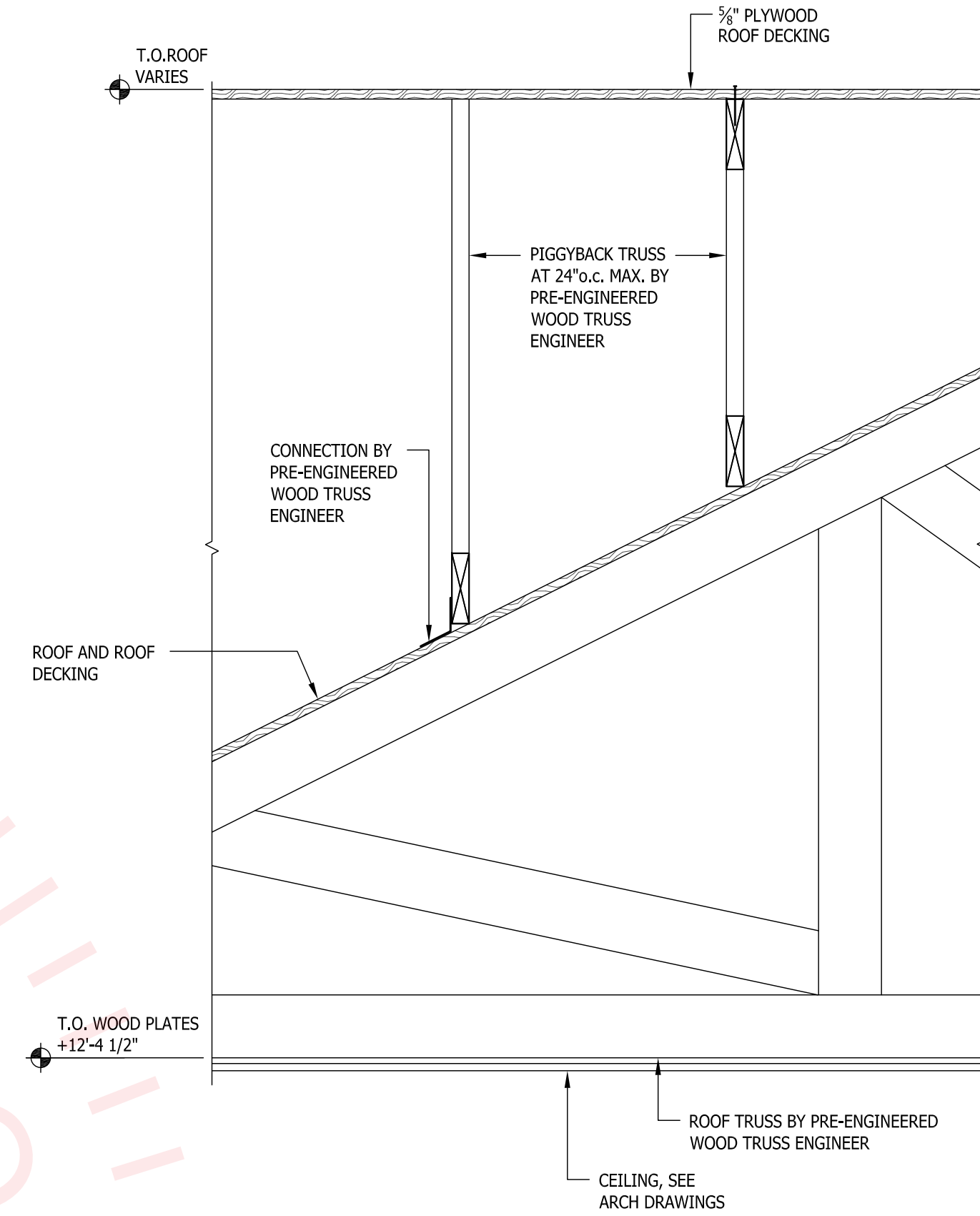
DRAWING SCALE: 3/4" = 1'-0"



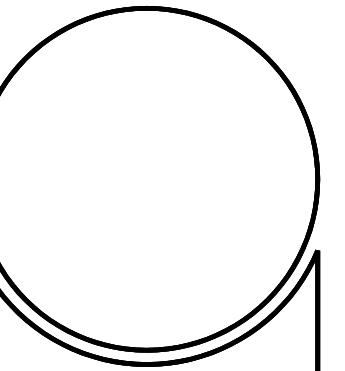
1 TYPICAL ROOF NAIL PATTERN
UNBLOCKED DIAPHRAGM



2 GABLE END WALL



3 TYPICAL DETAIL AT PIGGYBACK TRUSSES



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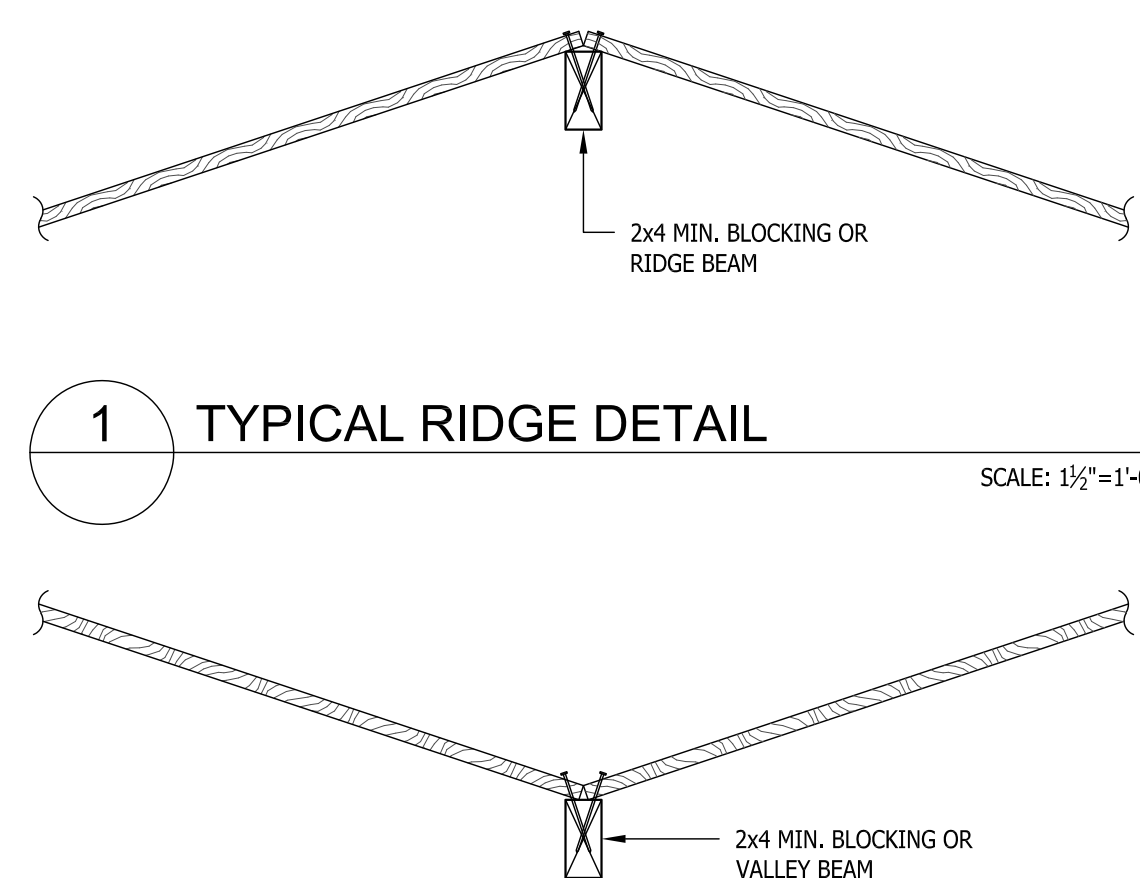
OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
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PE 13168 EB 5043
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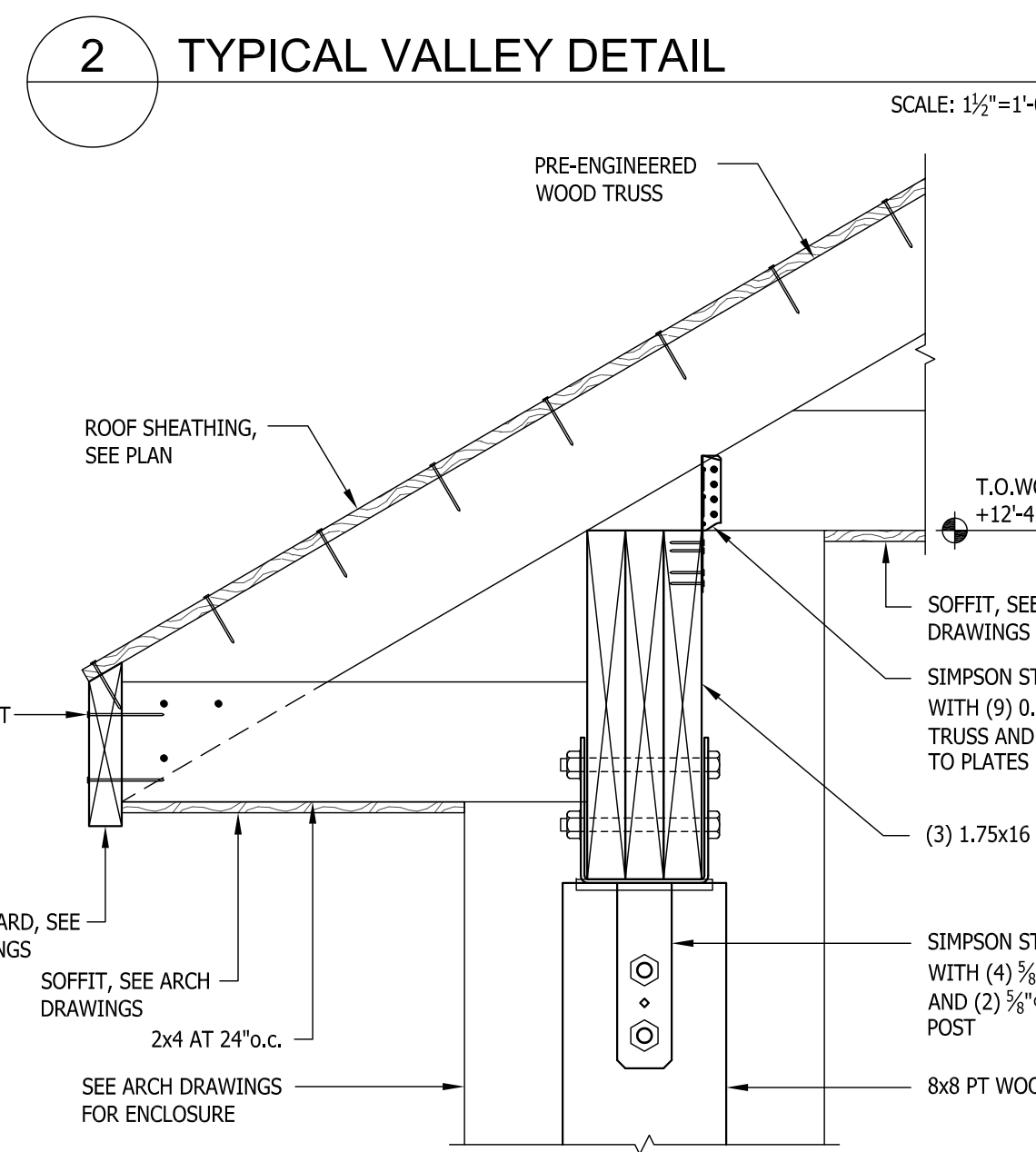
SHEET NO.
S3.0

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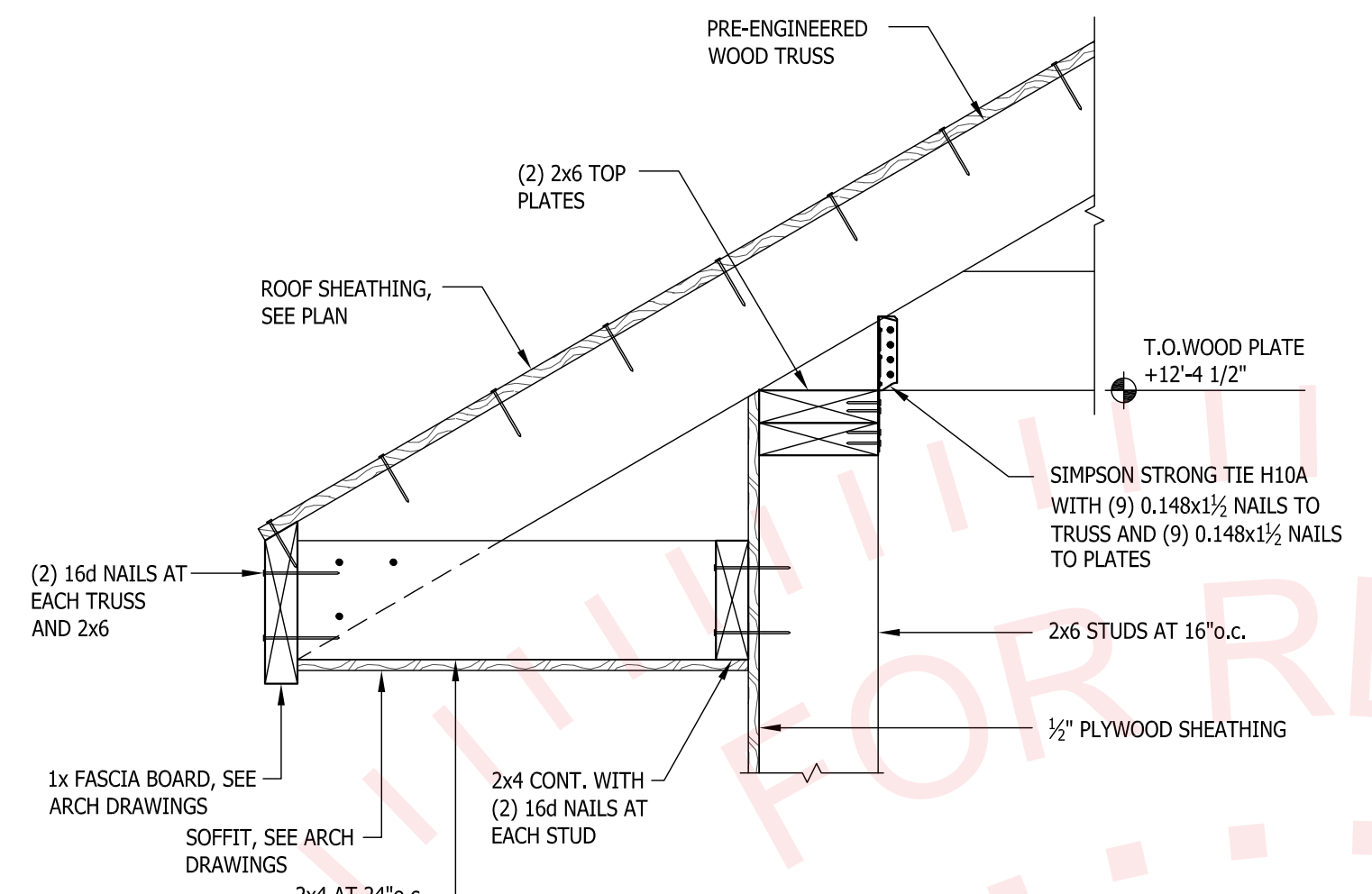
DRAWING SCALE: 1" = 1'-0"



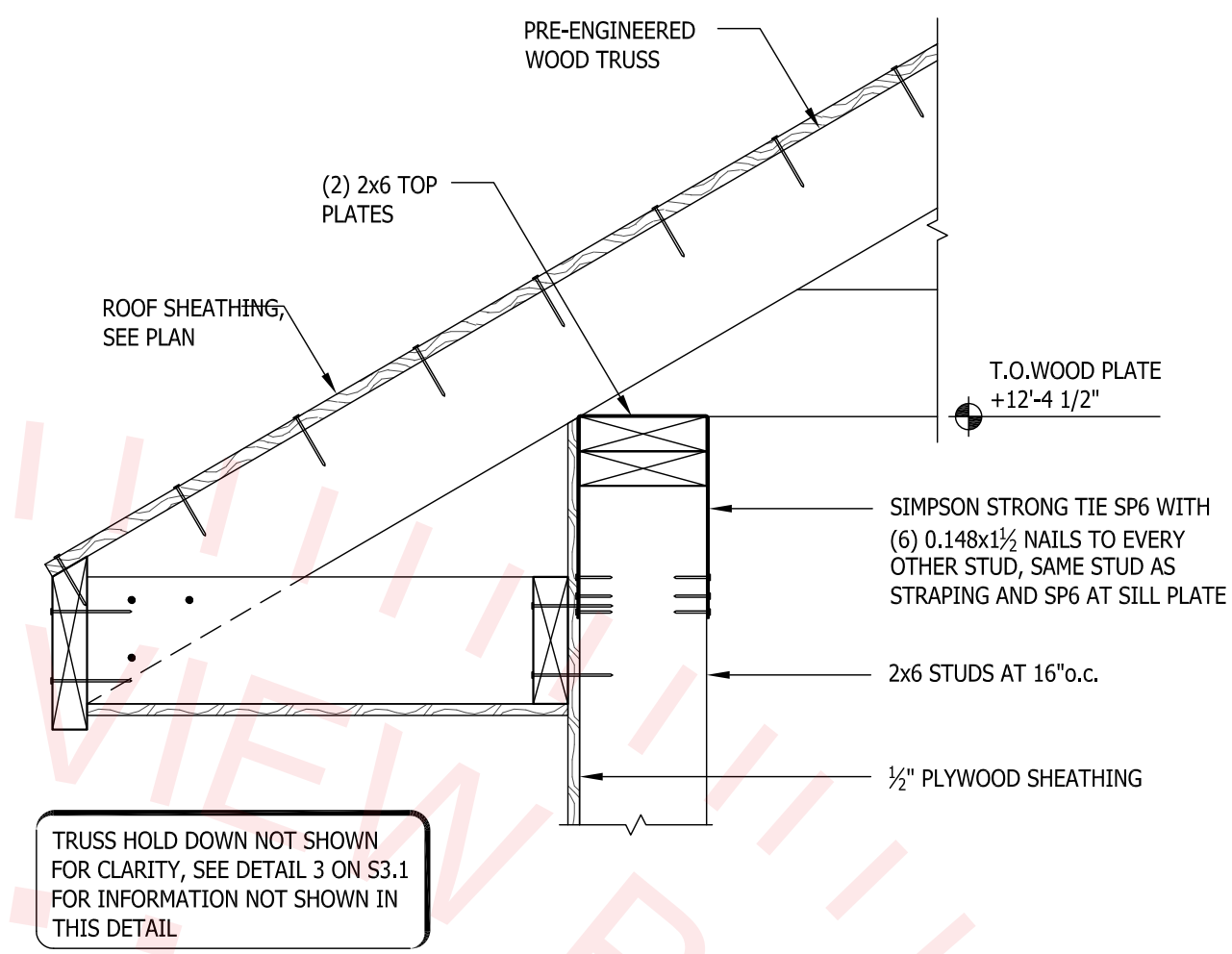
1 TYPICAL RIDGE DETAIL
SCALE: 1 1/2" = 1'-0"



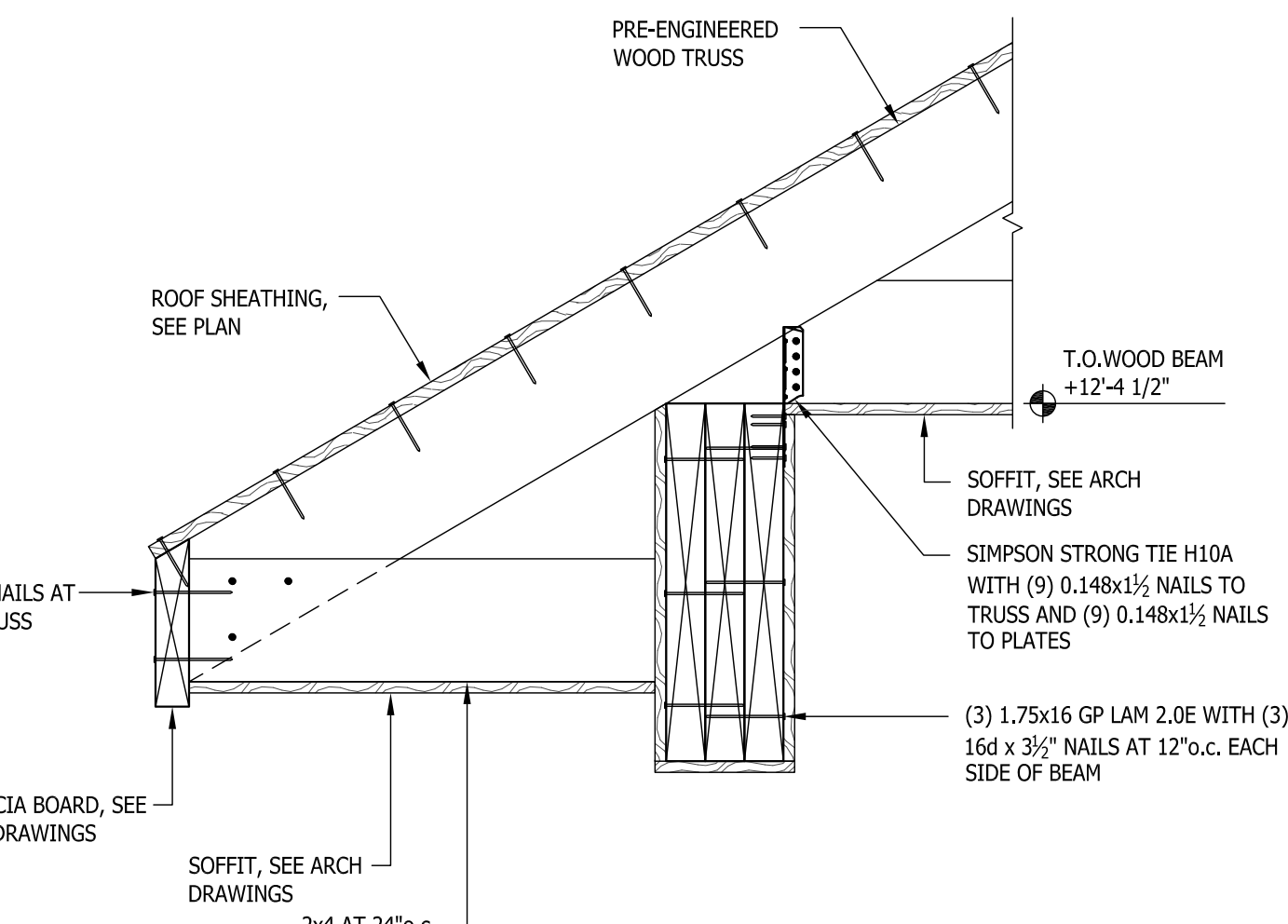
2 TYPICAL VALLEY DETAIL
SCALE: 1 1/2" = 1'-0"



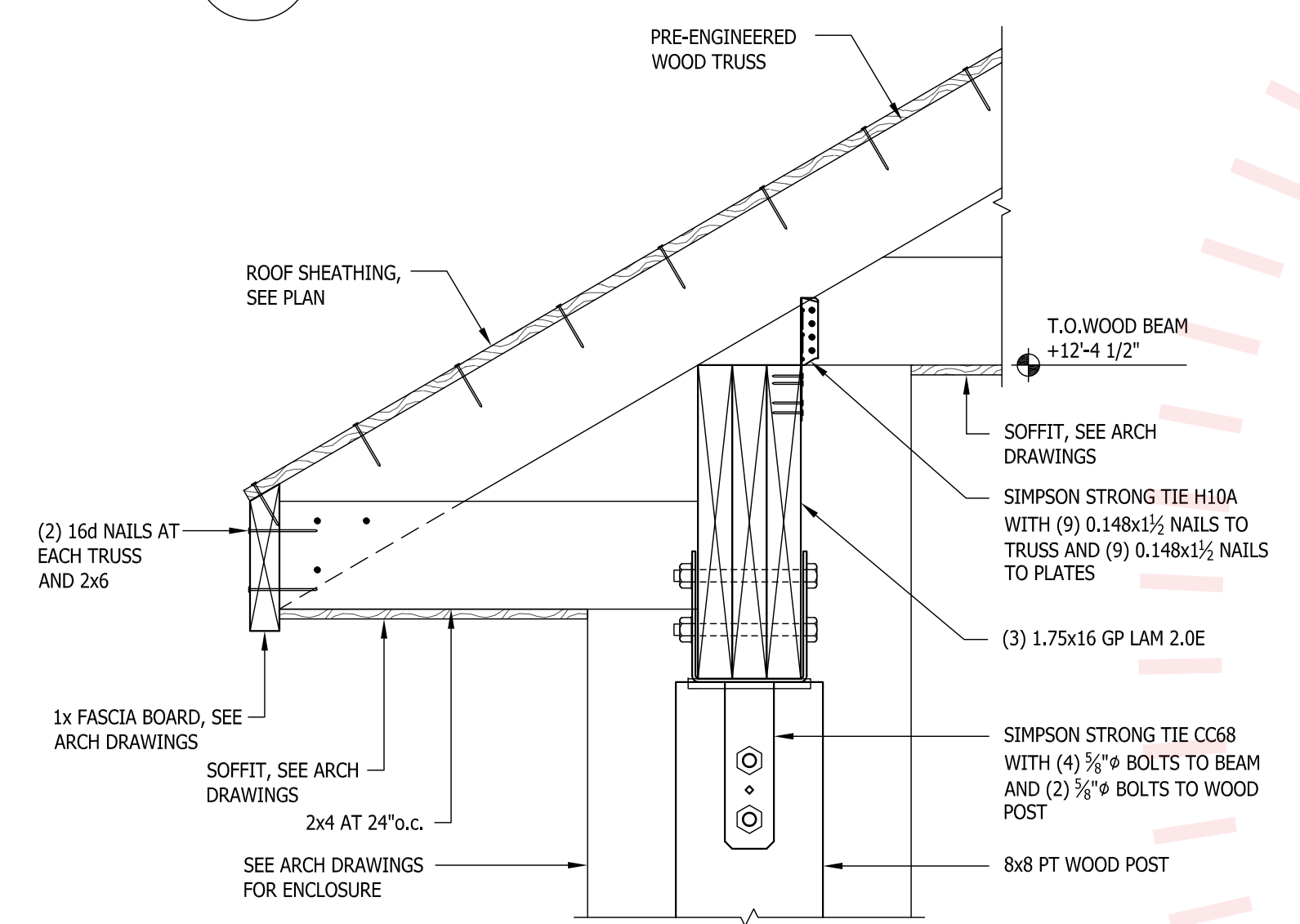
3 TYPICAL ROOF TRUSS CONNECTION
AT STUD WALL



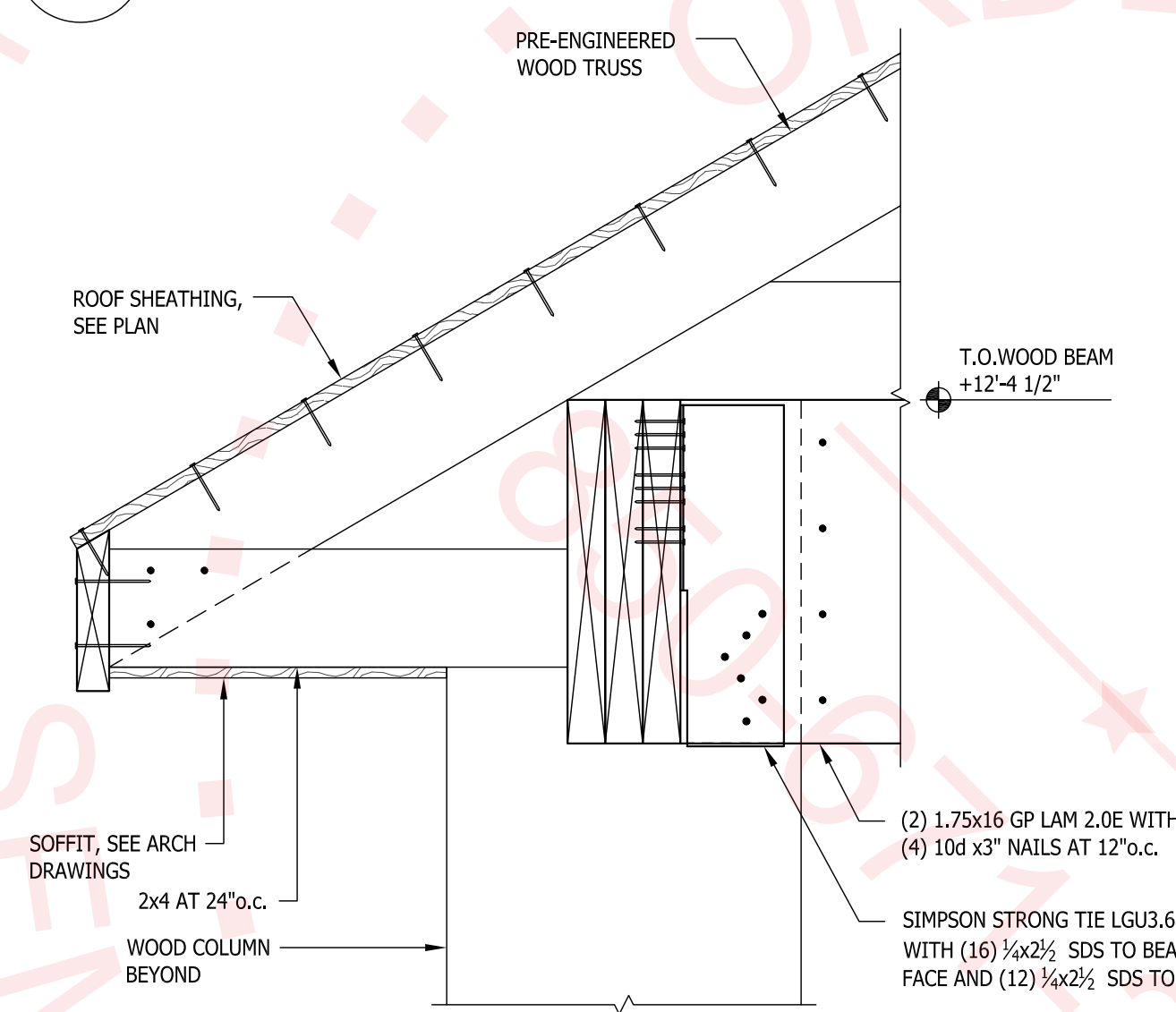
4 TYPICAL ROOF TOP PLATE CONNECTION
AT STUD WALL



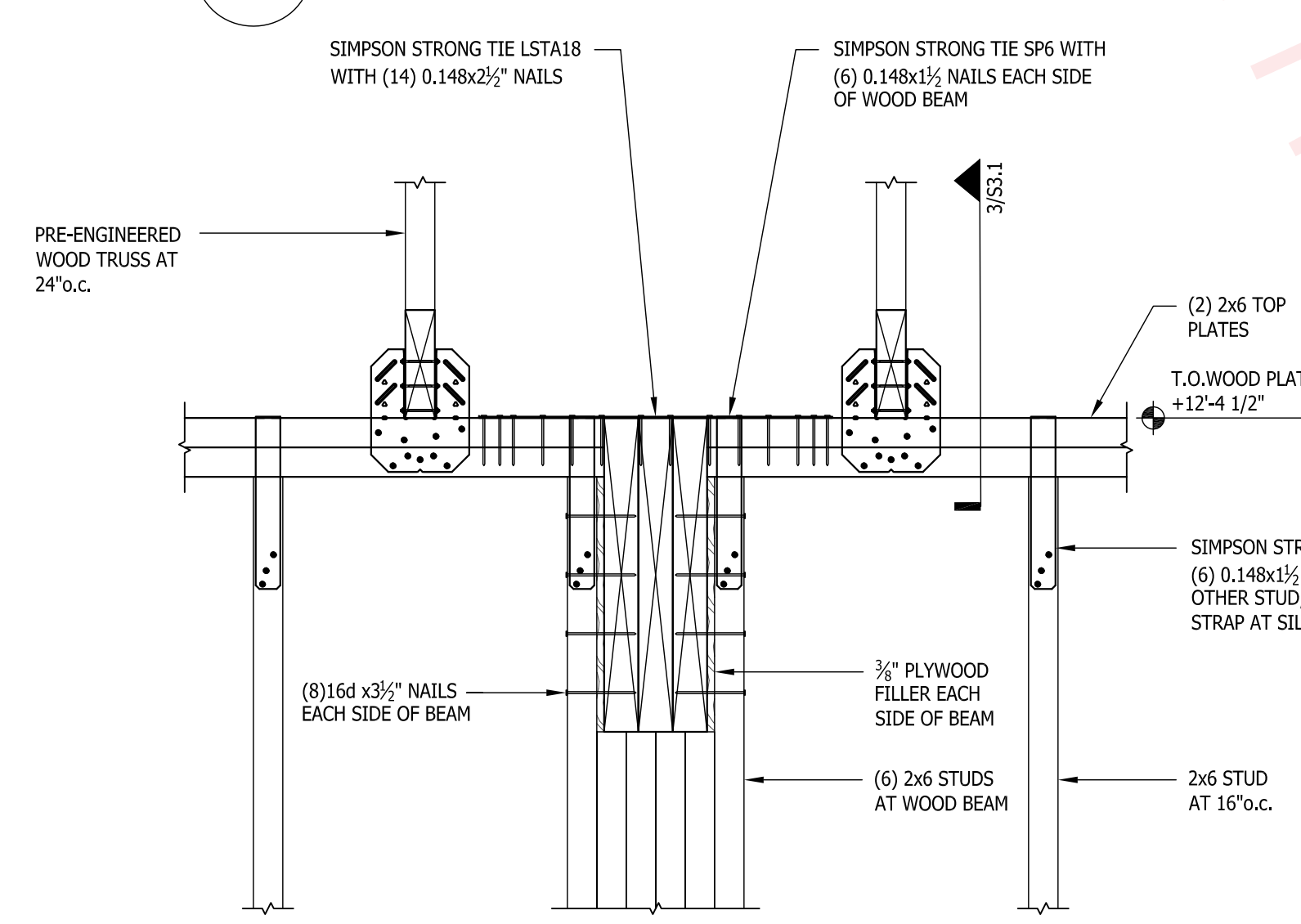
5 TYPICAL ROOF TRUSS CONNECTION
AT COVERED DRIVEWAY



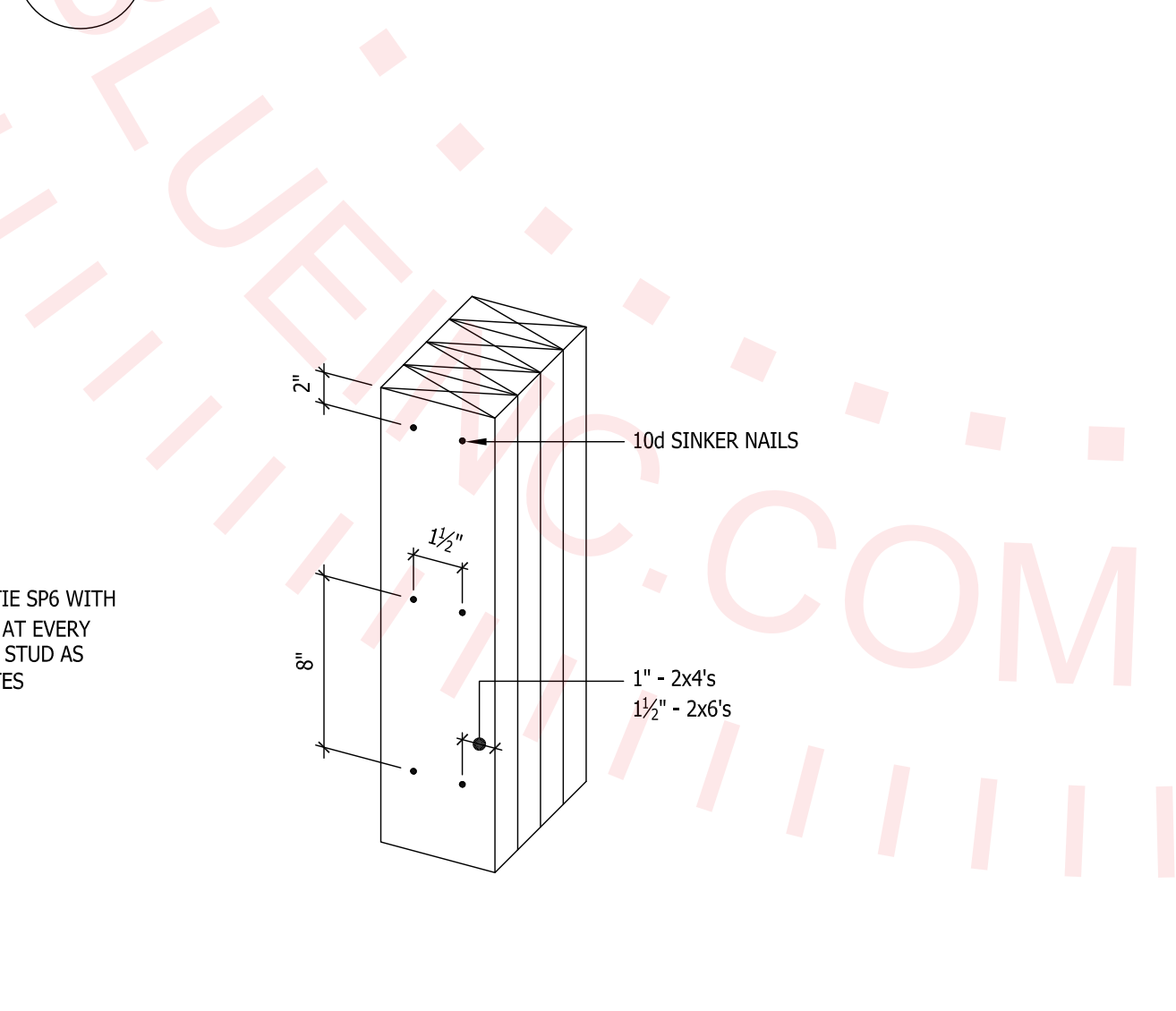
6 TYPICAL WOOD POST CONNECTION
AT COVERED DRIVEWAY



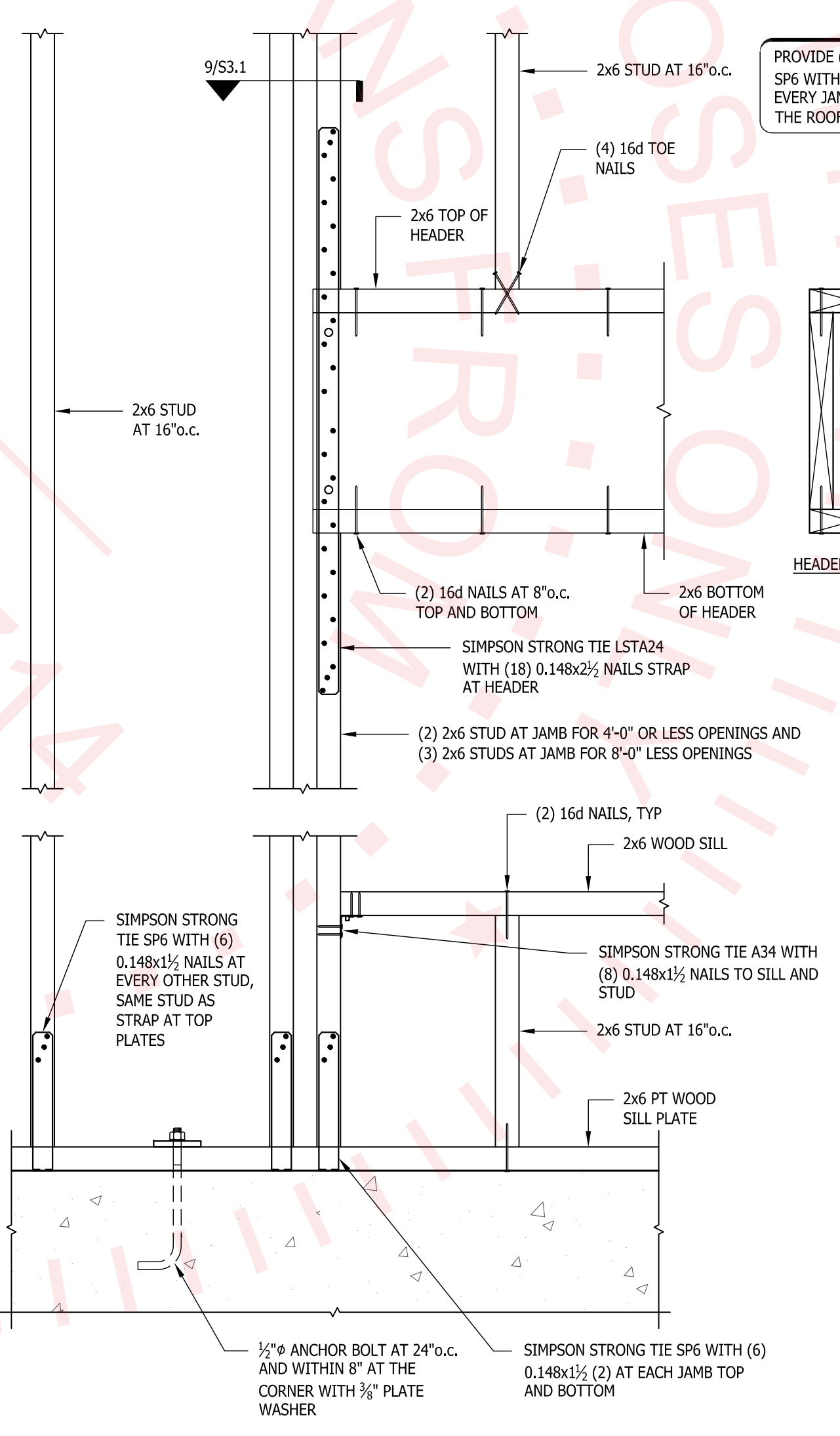
7 TYPICAL WOOD BEAM TO BEAM CONN.
AT COVERED DRIVEWAY



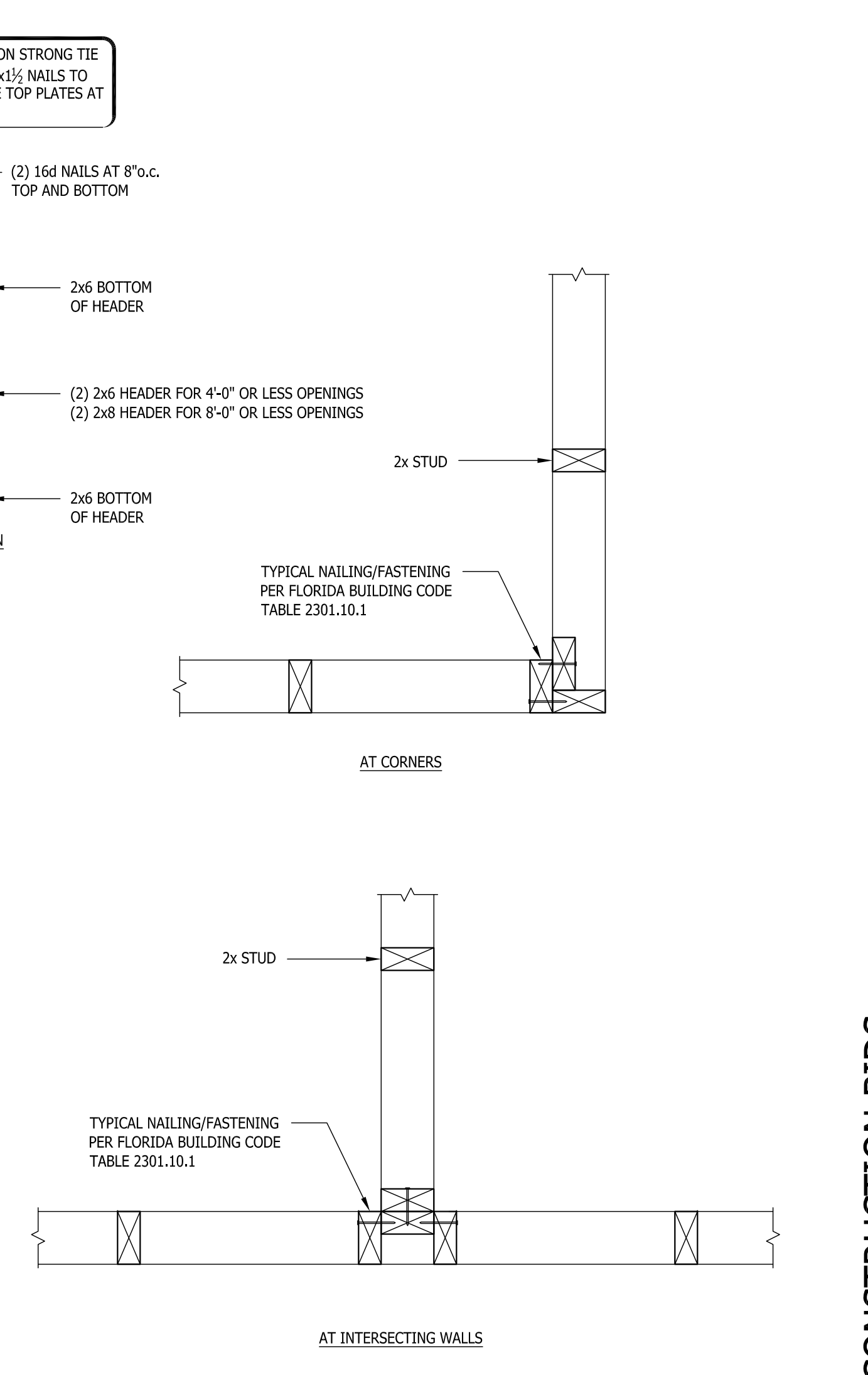
8 TYPICAL WOOD BEAM TO WALL CONN.
AT COVERED DRIVEWAY



9 BUILT-UP COLUMN FASTENING AT JAMBS
CONNECTION REQ'D PER LAYER OF BUILD-UP



10 TYPICAL OPENING DETAIL
DOOR AND WINDOW OPENINGS



11 TYPICAL PARTITION WALL FRAMING

REVISIONS

NO.	DATE

**STRUCTURAL
ENGINEERING
GROUP, LLC**

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(850) 591-5896
SEG Job No. 20T025

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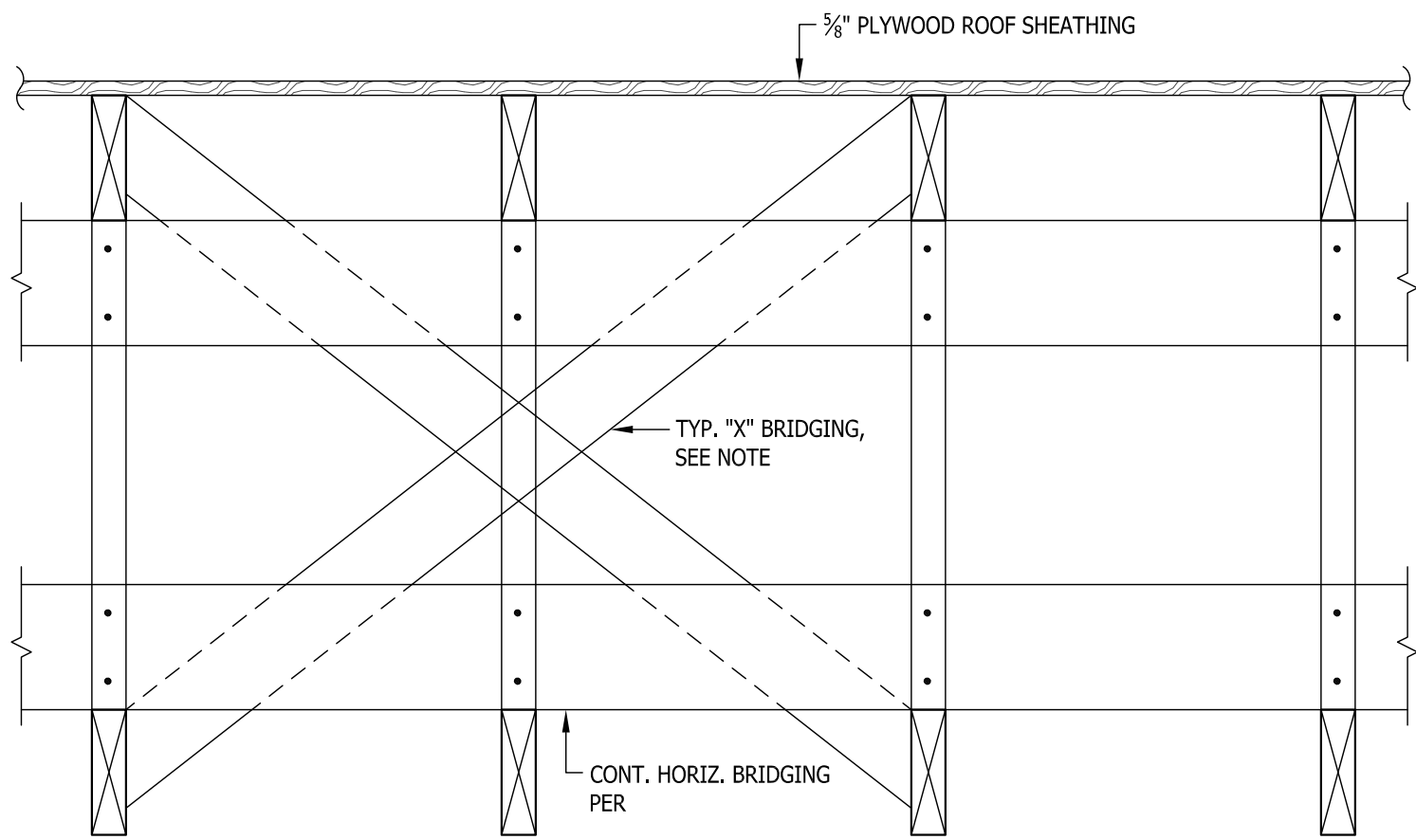
JERRY L. HICKS, ARCHITECT
5383 PEMBRIDGE PLACE
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850-544-0540

FOR CONSTRUCTION BIDS

OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
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6700 TREASURE OAKS CIRCLE
TALLAHASSEE, FLORIDA 32309
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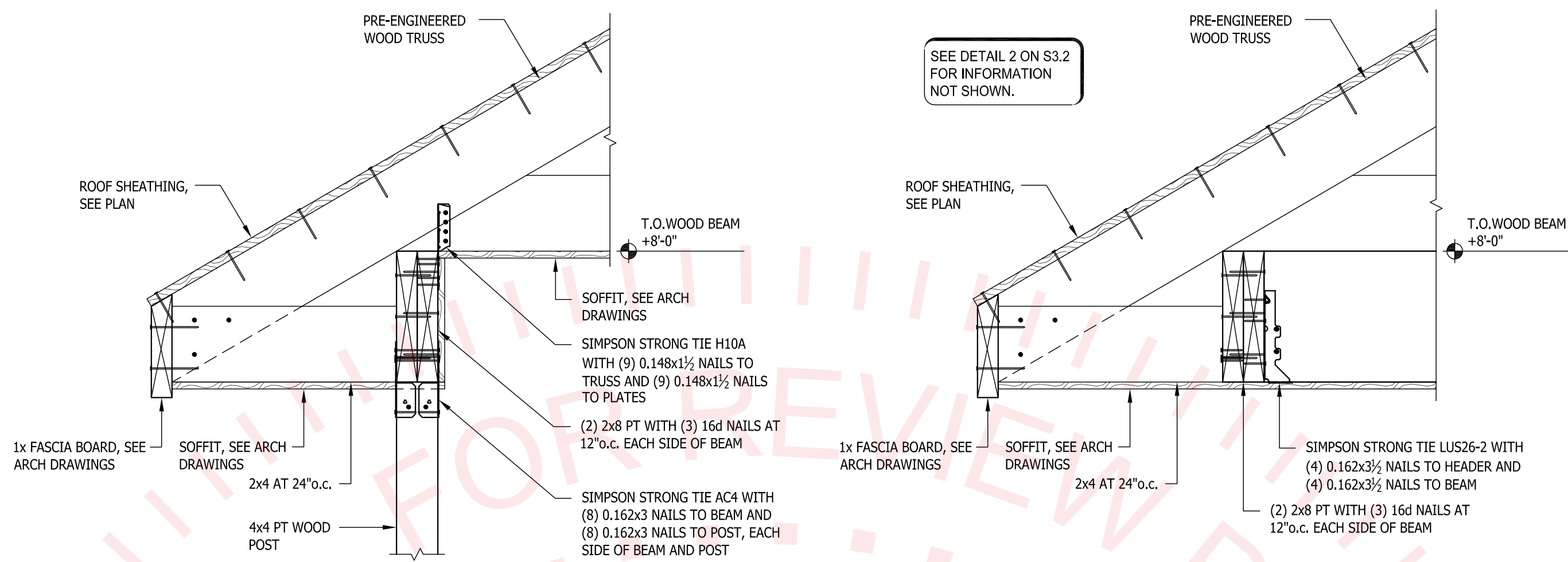
DATE: 11-5-2021
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PROJ. NO. 38022

SHEET NO.
S3.1



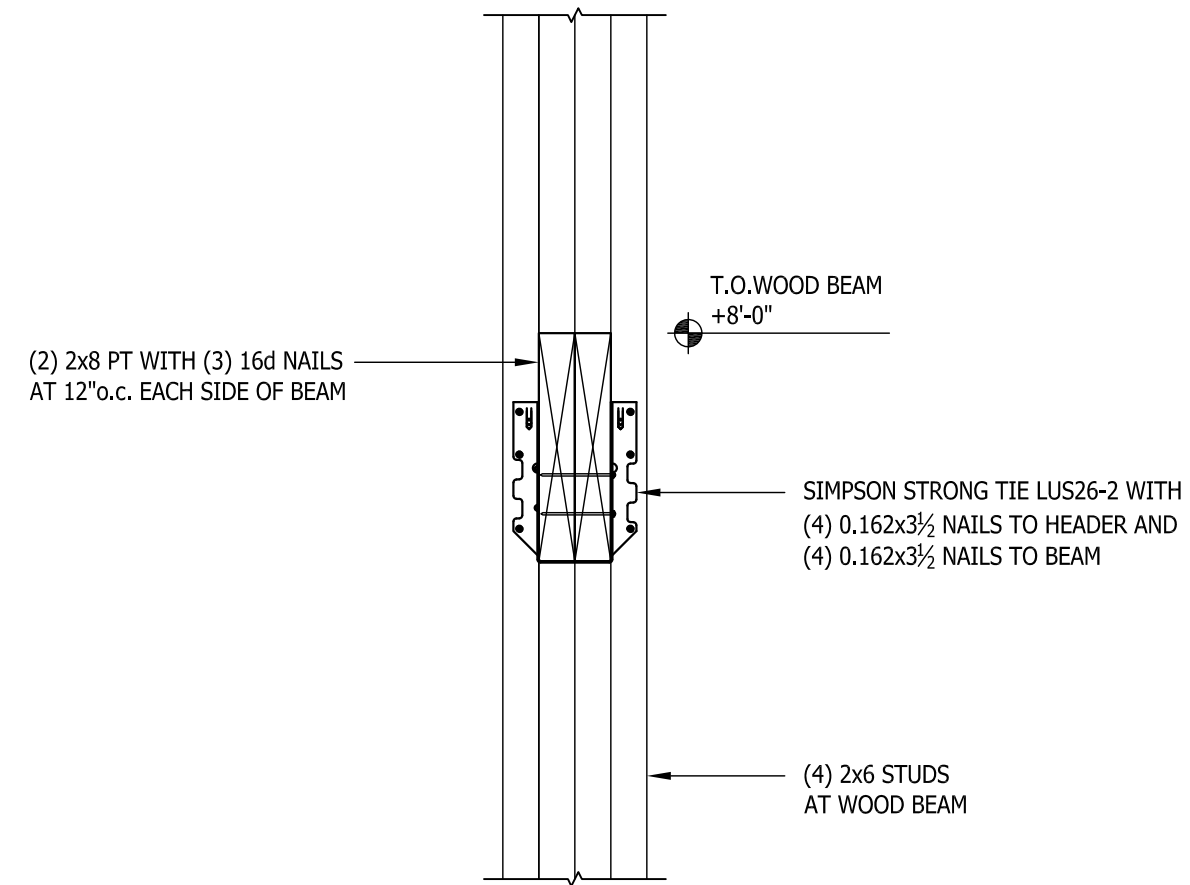
BRIDGING TO BE LOCATED AS REQUIRED BY TRUSS MANUFACTURER'S SPECIALTY ENGINEER.

1 TYPICAL TRUSS BRIDGING DETAIL

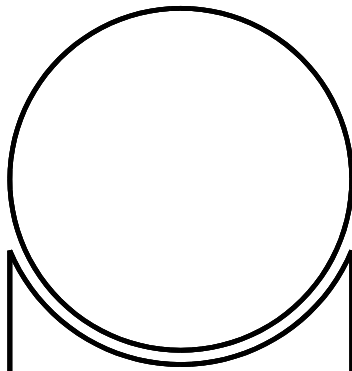


2 TYPICAL ROOF TRUSS / POST CONN. AT COVERED PORCH

3 TYPICAL BEAM TO BEAM CONN. AT COVERED PORCH



4 TYPICAL WOOD BEAM TO WALL CONN. AT COVERED DRIVEWAY



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NO.	DATE



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FL CA #32577
SEG Job No. 20T025

To best of my knowledge, the plans & specifications comply with applicable minimum building codes.

PISGAH FELLOWSHIP HALL
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

JERRY L. HICKS, ARCHITECT
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CONSULTING ENGINEERS
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850-906-9501

DATE:	3-1-2021
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PROJ. NO.	38022

SHEET NO.
S3.2

FOR CONSTRUCTION BIDS

DRAWING SCALE: 1 1/2" = 1'-0"

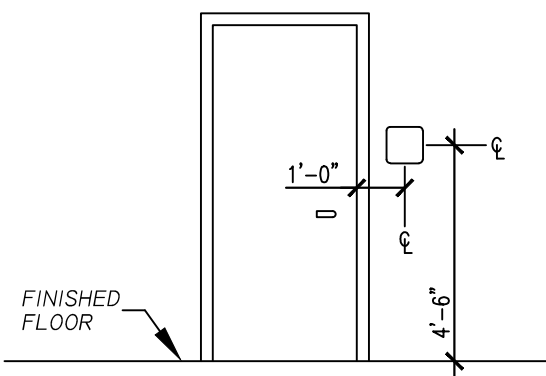
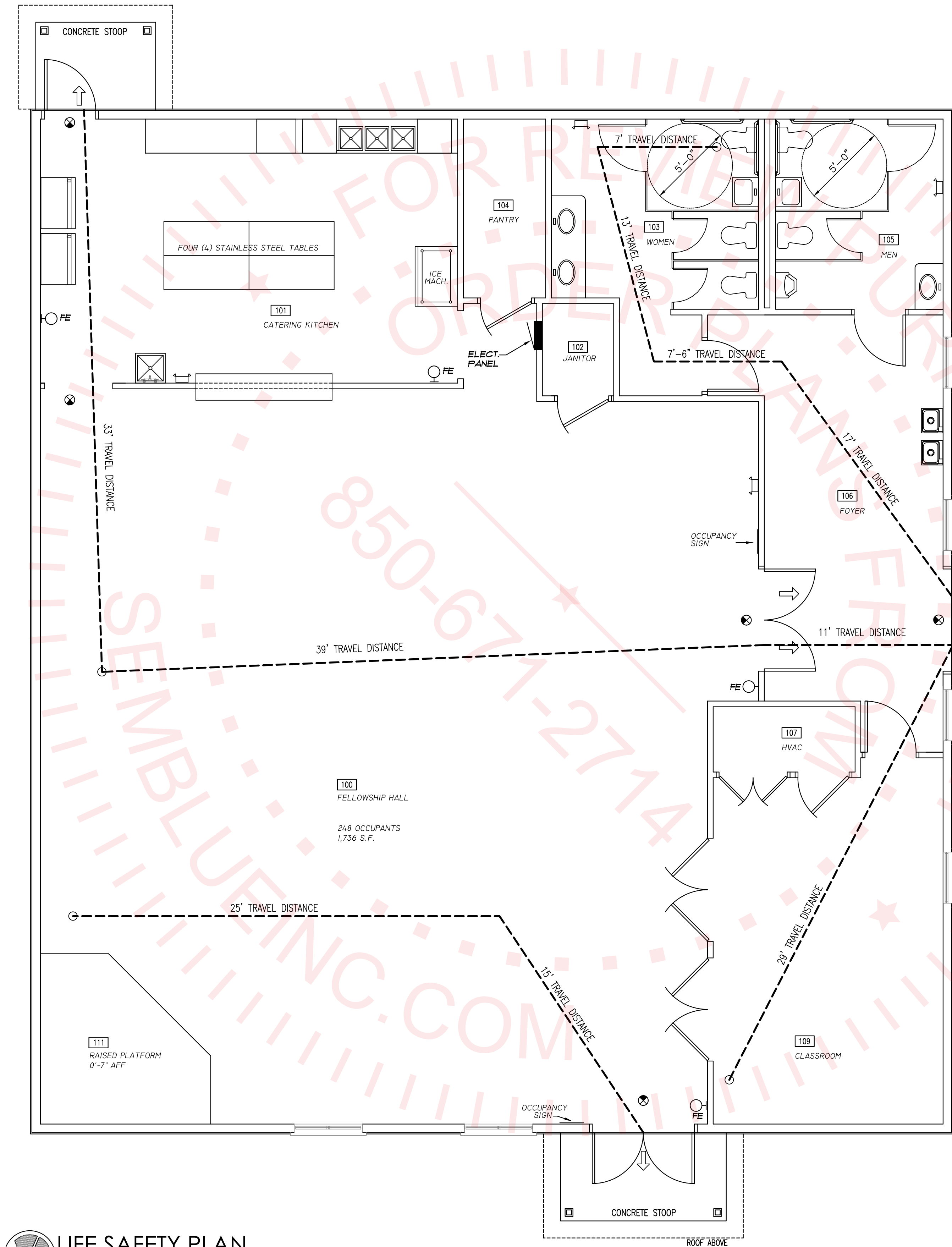
LIFE SAFETY NOTES:

1. Furniture layout is for reference only. Final furniture layout to be determined by Owner.
2. Exit discharge at 36" doors shall have 34" clear opening with a maximum 1/2" high threshold – ADA compliant.
3. Exit discharge 72" double doors shall have 68" clear opening width with a maximum 1/2" high threshold – ADA compliant.
4. All door hardware including closure, handles, pulls, latches, locks and panic hardware shall be ADA compliant.
5. Egress door exit shall be to dedicated path of travel, accessible route to public transportation stops, access, parking and passenger loading zones.
6. All egress/ingress paths shall be a minimum of 44" clear.
7. Fire extinguishers shall be 2A-10B:C minimum, bracket-mounted at 40" to bracket AFF.

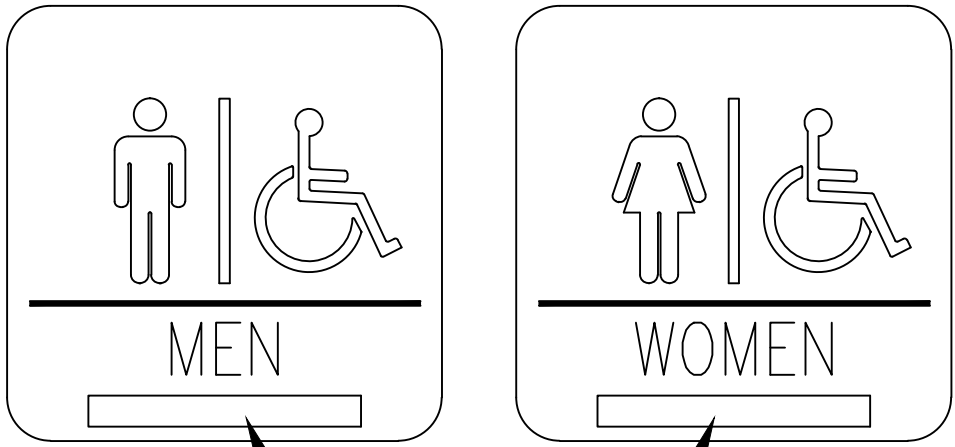
LIFE SAFETY SYMBOL LEGEND	
	EGRESS DIRECTION
	EGRESS TRAVEL PATH
	FIRE EXTINGUISHER
	ILLUMINATED EXIT SIGN
	DIRECTIONAL EXIT SIGN, ONE-SIDED
	DIRECTIONAL EXIT SIGN, TWO-SIDED
	EMERGENCY WALL LIGHT w/ BATTERY PACK

LEGEND NOTES:

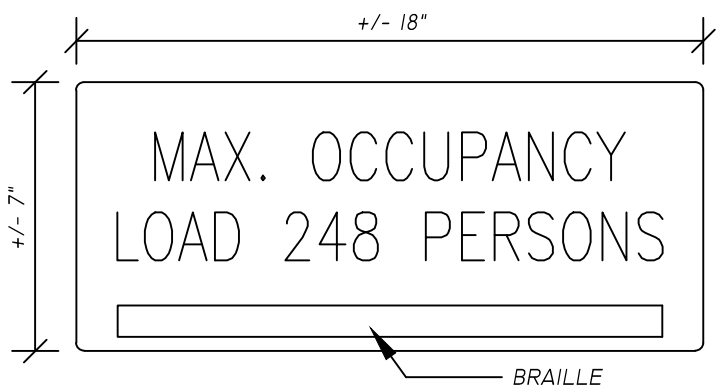
Fire Extinguisher shall be 2A-10B:C minimum size. Wall bracket shall be wall-mounted at a height of 40" AFF.



DOOR SIGN LOCATION



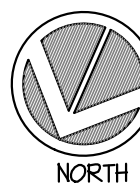
TOILET ROOM DOOR SIGN DETAIL



OCCUPANCY SIGN DETAIL

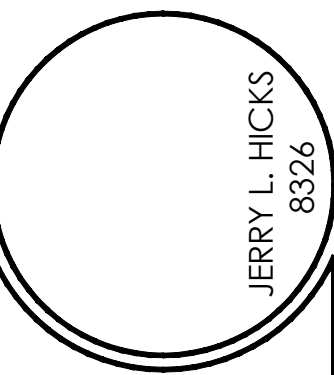
NOTES:

1. The engraved plastic sign shall have minimum 1/8"-high black letters on a white background.
2. Post signs in Fellowship Hall where indicated.
3. Mount sign 48" AFF to top.



LIFE SAFETY PLAN

SCALE: 1/4" = 1'-0"



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LIFE SAFETY PLAN

FELLOWSHIP HALL
PISGAH UNITED METHODIST CHURCH
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

JERRY L. HICKS, ARCHITECT
AR NO. 8326 (FLORIDA)
5383 PEMBRIDGE PLACE
TALLAHASSEE, FLORIDA 32309
850-544-0540

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OOTEN AND ASSOCIATES, LLC
CONSULTING ENGINEERS
HOMER A. OOTEN, P.E., LEED AP
PE 131 68 EB 5043
6700 TREASURE OAKS CIRCLE
TALLAHASSEE, FLORIDA 32309
850-906-9501

DATE: 11-05-2021
DRAWN BY: JLH/GJ
CHECKED BY: JLH
PROJ. NO. 38022

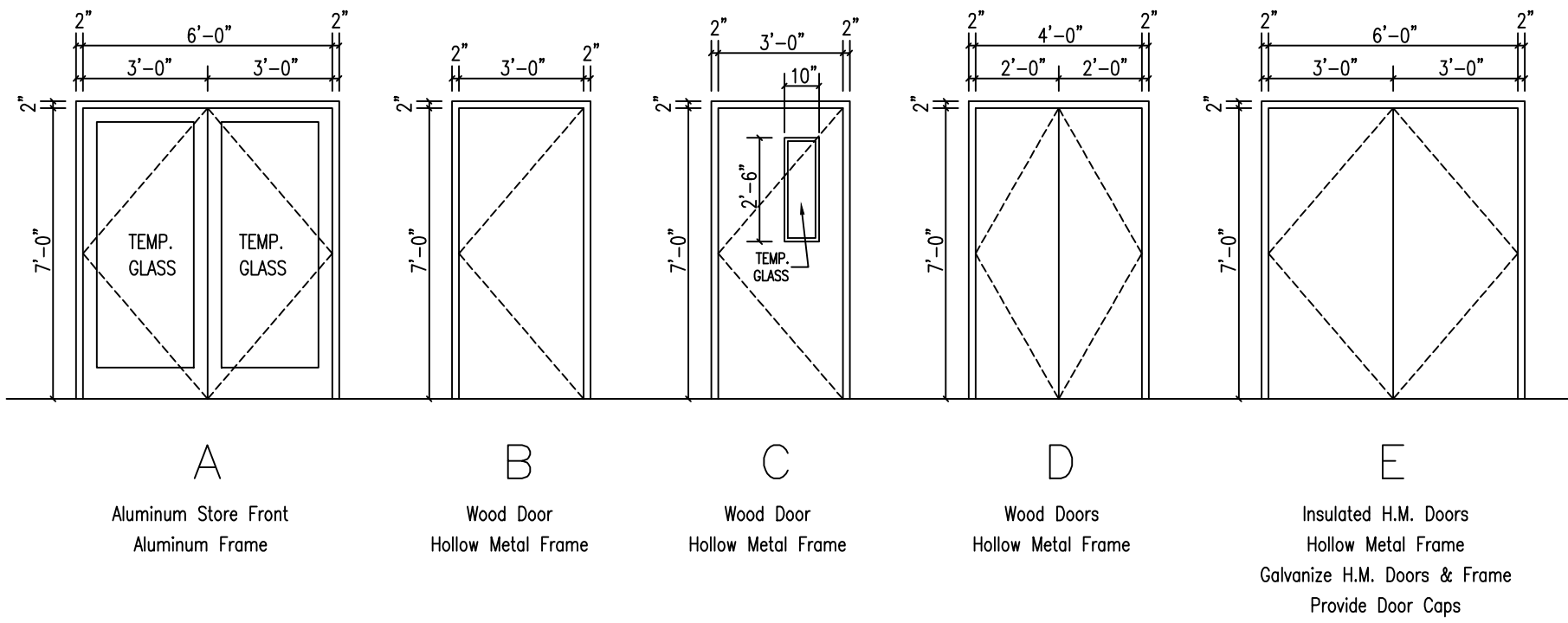


DOOR SCHEDULE											
DOOR DESCRIPTION					MATERIALS			DETAIL			HARDWARE
NO.	TYPE	W	H	T	DOOR	FRAME	FINISH	HEAD	JAMB	SILL	SET #
①	E	6'-0"	7'-0"	1-3/4"	Metal	Hollow Metal	Paint	6	5	4	10
②	B	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	2
③	B	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	5
④	B	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	2
⑤	B	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	5
⑥	A	6'-0"	7'-0"	1-3/4"	Alum/Glass	Aluminum	Anodized	3	2	1	9
⑦	A	6'-0"	7'-0"	1-3/4"	Alum/Glass	Aluminum	Anodized	6	5	4	8
⑧	C	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	7
⑨	B	3'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	2
⑩	D	4'-0"	7'-0"	1-3/4"	Wood	Hollow Metal	Stain	3	2	1	4
⑪	B	3'-0"	7'-0"	1-3/4"	Metal	Hollow Metal	Paint	6	5	4	1
⑫	A	6'-0"	7'-0"	1-3/4"	Alum/Glass	Aluminum	Anodized	3	2	1	9
⑬	A	6'-0"	7'-0"	1-3/4"	Alum/Glass	Aluminum	Anodized	3	2	1	9

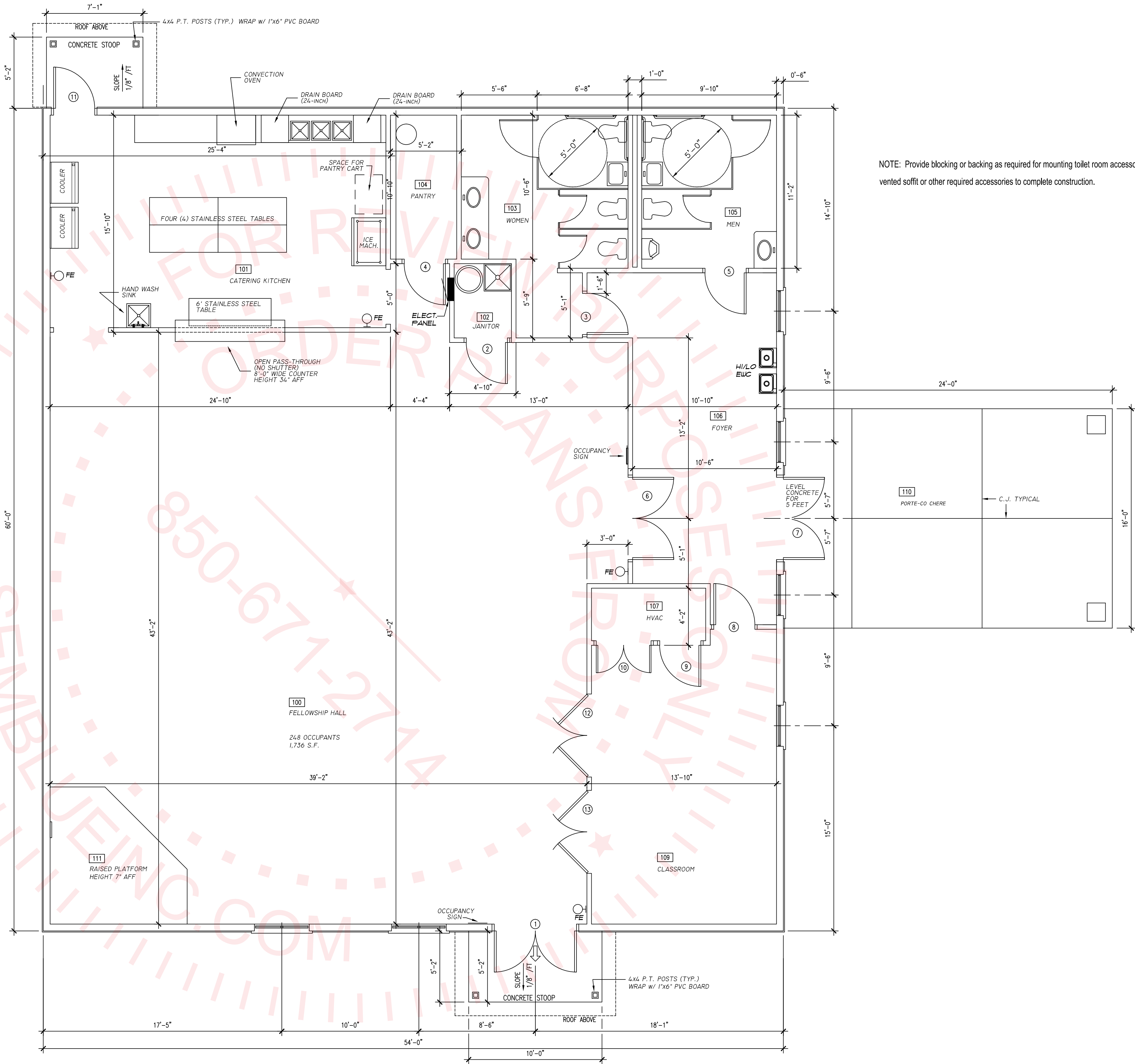
- NOTES:
- All wood doors shall be solid core birch or oak with stained factory finish prepared for pair of 4-1/2" x 1/2" hinges to fit hollow metal door frame. Coordinate lock set preparation of wood door with hollow metal frame.
 - Doors shall be prepared to receive scheduled door hardware.
 - Coordinate aluminum store front doors and frames with manufacturer's door hardware.
 - Seal top & bottom of all doors.
 - All exterior hollow metal frames shall be galvanized w/ accessible threshold and weather seal.
 - Where lever handle latch & locksets are indicated, provide 2-3/4" backset.
 - Provide door stops for all doors. (See Door Hardware Schedule.)
 - All hollow metal doors & frames shall receive two (2) coats paint.
 - Top of hollow metal doors shall be flush with no recess.
 - Exterior hollow metal doors & frames (Doors #1 & #11) shall have galvanized H.M. door frames and insulated H.M. galvanized doors.

ROOM FINISH SCHEDULE						
NO.	ROOM NAME	FLOOR FIN.	BASE	WALLS	CLG. FIN.	CLG. HT.
100	Fellowship Hall	CPT	Vinyl	PT	ACT	CPT
101	Catering Kitchen	VCT	Vinyl	PT	GWB-PT	9'-0"
102	Janitor	VCT	Vinyl	PT	GWB-PT	VCT
103	Women's Restroom	VCT	Vinyl	PT	GWB-PT	VCT
104	Pantry	CPT	Vinyl	PT	GWB-PT	CPT
105	Men's Restroom	VCT	Vinyl	PT	GWB-PT	VCT
106	Foyer	VCT	Vinyl	PT	ACT	VCT
107	HVAC	CPT	Vinyl	PT	GWB-PT	CPT
108	(Not Used)	--	--	--	--	--
109	Classroom	CPT	Vinyl	PT	ACT	10'-0"
110	Porte-Co Chere	SC	--	PT	Vinyl Soffit	12'-0"
111	Raised Platform	CPT	Vinyl	PT	ACT	CPT

- NOTES:
- Provide rubber transition strip where carpet abuts other flooring.
 - Provide manufactured interior and exterior premolded corners where vinyl base has been scheduled.
 - Color selections will be made from submitted samples.
 - Paint colors shall be selected by Owner. Provide two coats minimum on all interior walls and GWB ceilings.
 - ACT-1 shall be Armstrong 24"x24" & 5/8" thick tegular edge, Zoro white fussed.
 - Broom finish concrete at porte-co chere. Saw cut control joints at locations indicated. Seal concrete after curing is complete.
 - Exterior paint shall be Sherwin-Williams Duration Exterior Extra White Gloss -- 2 coats.



DOOR & FRAME ELEVATIONS
SCALE: 1/4" = 1'-0"



NOTE: Provide blocking or backing as required for mounting toilet room accessories, vented soffit or other required accessories to complete construction.

FLOOR PLAN
SCALE: 1/4" = 1'-0"
NORTH

JERRY L. HICKS
8326

REVISIONS
NO. DATE:

FLOOR PLAN AND NOTES;
DOOR SCHEDULE

FELLOWSHIP HALL
PISGAH UNITED METHODIST CHURCH
7000 PISGAH CHURCH RD.
TALLAHASSEE, FLORIDA 32309

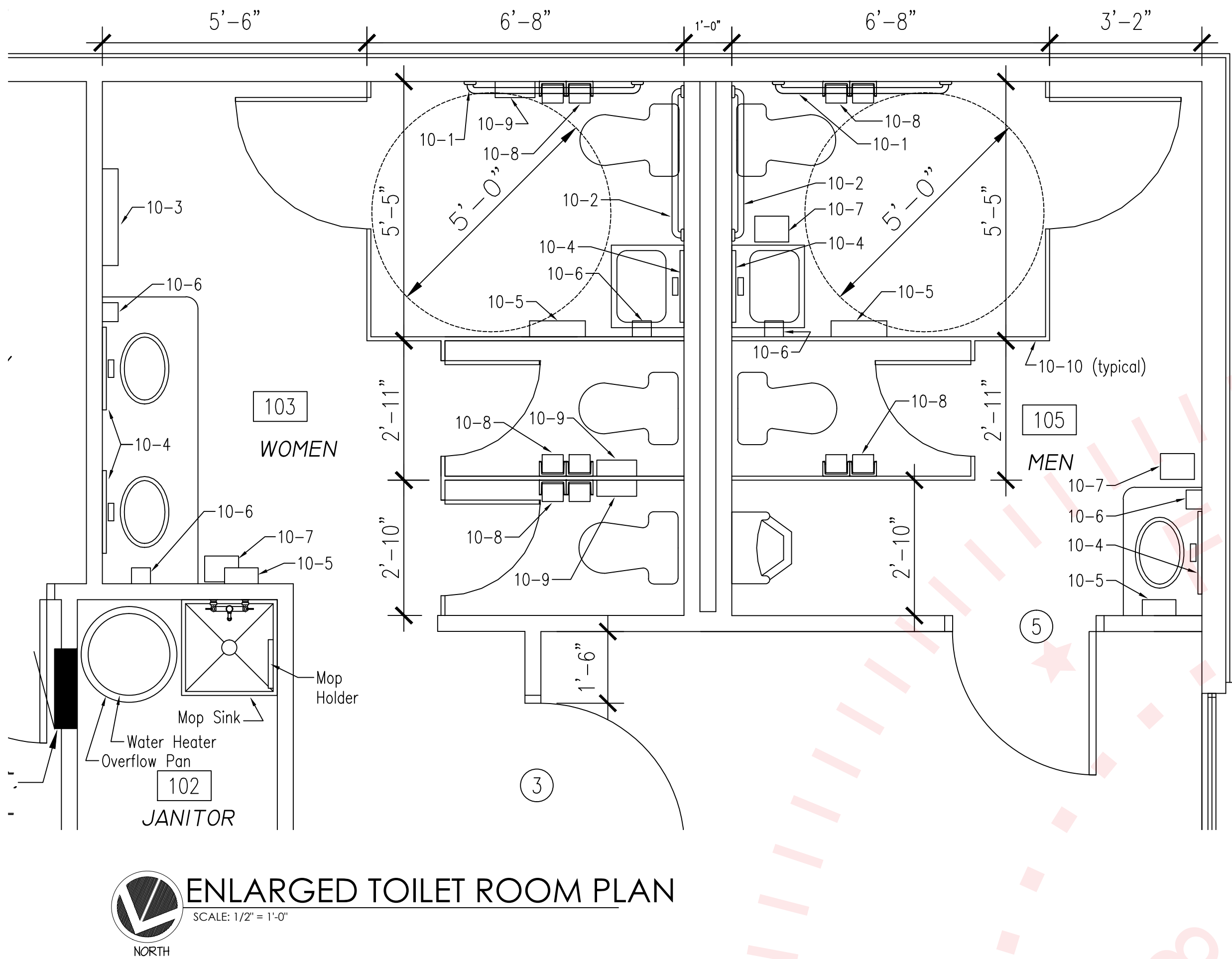
JERRY L. HICKS, ARCHITECT
AR NO. 8326 (FLORIDA)
5383 PEMBRIDGE PLACE
TALLAHASSEE, FLORIDA 32309
850-544-0540

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CONSULTING ENGINEERS
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PE 13148 EB 5043
6700 TREASURE OAKS CIRCLE
TALLAHASSEE, FLORIDA 32309
850-906-9501

DATE: 11-05-2021
DRAWN BY: JLH/GJ
CHECKED BY: JLH
PROJ. NO. 38022

SHEET NO.
A1



ENLARGED TOILET ROOM PLAN

TOILET ROOM ACCESSORIES SCHEDULE

REF.	ITEM	DESCRIPTION	QTY.	PART #	WALL MOUNTING INSTRUCTIONS
10-1	1-1/2" dia. Grab Bar w/ concealed mounting	Length 42"	2	812	Mount 33" AFF to centerline & 12" from back wall
10-2	1-1/2" dia. Grab Bar w/ concealed mounting	Length 36"	2	812	Mount 33" AFF to centerline & 12" from back wall
10-3	Surface mounted Baby Changing Station	Color Putty	1	904	Mount 34" AFF to bottom
10-4	ADA fixed-angle Mirror (stainless stl. frame)	20"W x 30"H	5	0535	Mount 40" to bottom edge AFF
10-5	Paper Towel Dispenser (hands-free)	--	4	T245	Mount 48" to bottom AFF
10-6	Liquid Soap Dispenser	Plastic	4	pllisodi	Mount above backsplash on wall
10-7	Waste Receptacle by Rubbermaid	Tall Kitchen	4	Rubbermaid	Set on floor (movable) as shown
10-8	Double Toilet Tissue Dispenser	--	5	A0965	Mount 17" to bottom AFF
10-9	Feminine Hygiene Disposal Receptacle	--	3	20852	Mount 17" to bottom AFF
10-10	Toilet Room Partitions (solid polymer plastic)	Color selected by the Owner	See Plan	--	As shown - floor mounted overhead braced; Use continuous mounting brackets at walls
10-11	Mop Holder (by Uline, 800-295-5510) (inside adjacent Room 102, "Janitor")	Length 36"	1	H-6090	Wall-Mount 60" AFF to centerline

NOTES FOR TOILET ROOM ACCESSORIES:

- Parts numbers listed are for AllPartitions.com items, Phone 1-866-255-9645. Provide catalog cut sheets for accessories prior to ordering. Provide shop drawings for toilet partitions w/ aluminum hardware including mounting screws and accessories. Provide pillar shoes, continuous wall mounting brackets, coat hook/door stop.
- Other manufacturers for accessories will be considered.
- Refer to enlargement toilet room plan, elevations, and Toilet Room Accessories Schedule, for mounting heights and locations. Comply with ADA.
- Provide wood blocking (backing) in wall partitions for mounting toilet room accessories including partitions.
- Water heater is A.O. Smith Signature 50-gal. electric 21½" diameter, installed in 24" overflow pan.
- Janitor Mop Sink is Riverbend 24"x24" (floor mounted).

DOOR HARDWARE SCHEDULE

SET #1 (Door #11) (Coordinate door hardware for pair of doors)
Each set to have...
6 Hinges CB1961 4½x4½-NRP 626 Stanley
2 Exit Device 99TP Grooved Mech. Case 626 Von Duprin
2 Exterior Lever Handles 626 Corbin-Russwin
2 Closers DC2210XA3XM54 689 Corbin-Russwin
1 Lock Guard 180S 630
2 Kickplates 8400(18"x2"L DWx0.050) 630 Rockwood
1 Gasket 315AR x Length Black Pemko
1 Stop 441UC 626 Rockwood
1 Threshold 2005AVxNDW Alum Pemko
2 Flush Bolts

SET #2 (Door #2, #4)
Each set to have...
3 Hinges CB1901 4½x4½ 626 Stanley
1 Lockset CL 3155 626 Corbin-Russwin
1 Kickplate 8400(18"x2"L DWx0.050) 630 Rockwood
1 Stop (Floor) 440 626 Rockwood
3 Silencers 608 Gray Rockwood

SET #3 (Not Used)

SET #4 (Door #10) (Coordinate door hardware for pair of doors)
Each set to have...
6 Hinges CB1961 4½x4½ 626 Stanley
1 Astragal or 2 Flush Bolts
1 Lockset CL 3155 626 Corbin-Russwin
6 Silencers 608 Gray Rockwood
1 Stop (Wall) 400 626 Rockwood

SET #5 (Door #3, #5)
Each set to have...
3 Hinges CB1961 4½x4½ 626 Stanley
1 Push Plate 70G (4"x20" Push Plate) 626 Rockwood
1 Pull / Plate 110x70C 626 Rockwood
1 Kickplate 8400(18"x2"L DWx0.050) 630 Rockwood
1 Stop (Wall) 440 626 Rockwood

SET #6 (Door #6)
Each set to have...
6 Hinges CB1901 4½x4½ 626 Stanley
1 Lockset CL 3110 626 Corbin-Russwin
2 Overhead Stop DH5400 x Size Required 626
2 Flush Bolts 458½ 626
1 Astragal Provide by Door Mfr.
2 Silencers 608 Gray Rockwood

SET #7 (Door #8)
Each set to have...
6 Hinges CB1961 4½x4½ 626 Stanley
2 Push Plates 70G (4"x20" Push Plate) 626 Rockwood
2 Pulls / Plates 110x70C 626 Rockwood
2 Closers DC2210XA3XM54 689 Corbin-Russwin
2 Kickplate 8400(18"x2"L DWx0.050) 630 Rockwood
2 Silencers 608 Gray Rockwood

SET #8 (Door #7)
Each set to have...
NOTE: Hardware set for aluminum storefront pair of doors at entrance into building shall be provided by storefront manufacturer. Cylinder shall match project cylinders. Manufacturer shall provide the following:
6 Hinges - Ball Bearing for Exterior Use
2 Push Bars
2 Pull Bars
2 Closers
1 Pair Flush Bolts
1 Gasket Set for Complete Opening
1 Threshold - Handicapped Accessible
2 Stops

SET #9 (Door #12, #13)
Each set to have...
NOTE: Hardware set for aluminum storefront pair of doors (interior) shall be provided by storefront manufacturer. Cylinder shall match project cylinders. Manufacturer shall provide the following:
6 Hinges - Ball Bearing
2 Push Bars
2 Pull Bars
2 Closers (provide closers on Door 6 only)
1 Pair Flush Bolts
2 Stops

SET #10 (Door #1)
Each set to have...
6 Hinges CB1961 4½x4½-NPP For Exterior Use
1 Exit Device 99TP Grooved Mech. Case 626 Von Duprin
1 Exterior Lever Handle 626 Corbin-Russwin
2 Closers
1 Lock Guard
2 Flush Bolts For Exterior Use
2 Kickplates 8400(18"x2"LDW) 630 Rockwood
1 Gasket 315AR x Length Black Pemko
2 Stops
1 Threshold 2005AVxNDW Alum. Pemko

HARDWARE NOTE: No storage function locksets are to be used. Hardware supplier shall verify that door hardware is adequate for all doors before supplying. Notify Architect if modifications are needed. Coordinate all door hardware with User/Architect.

ENLARGED TOILET ROOM PLAN
TOILET ROOM
ACCESSORIES SCHEDULE
DOOR HARDWARE SCHEDULE

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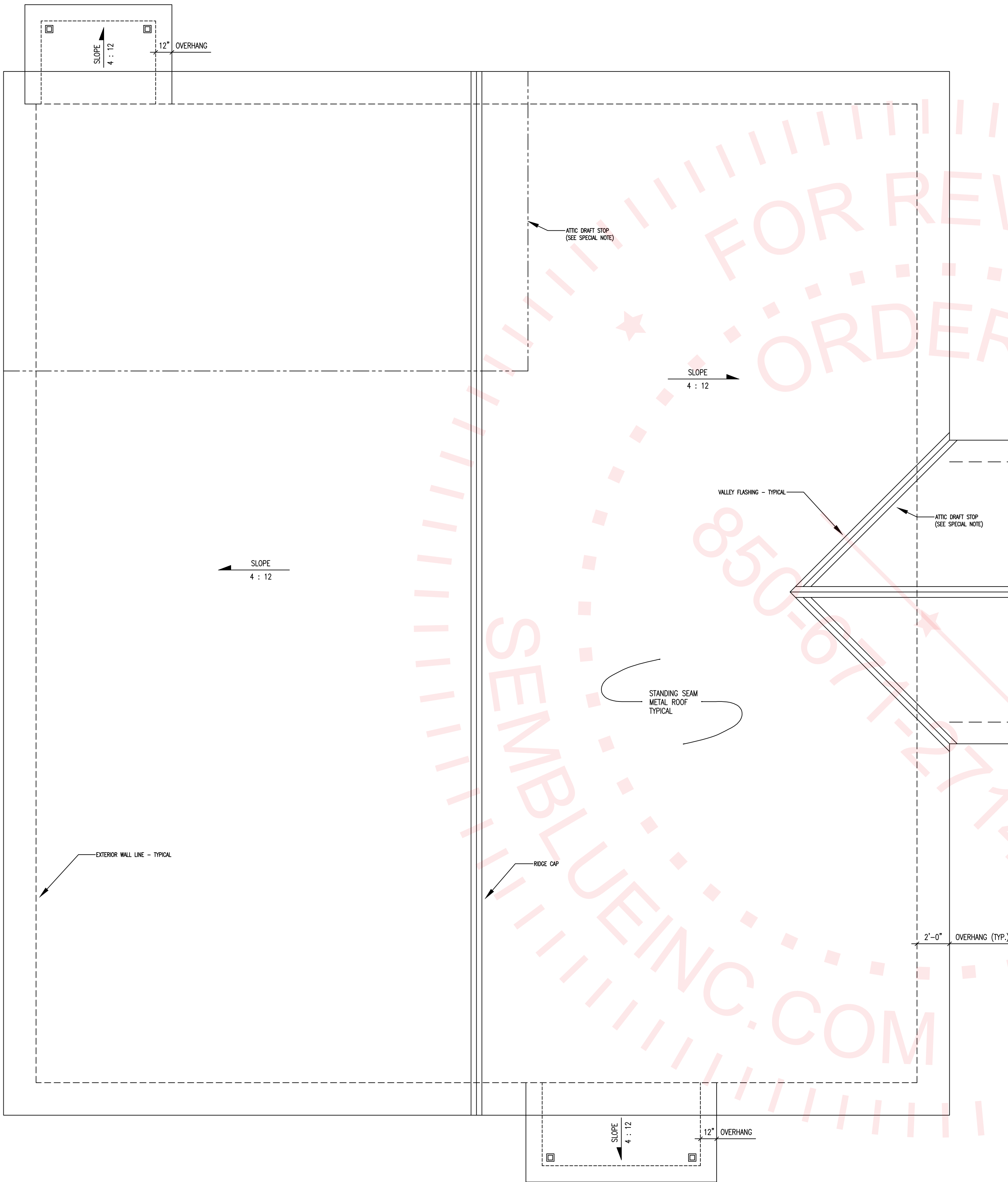
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850-906-9501

DATE: 11-05-2021
DRAWN BY: JLH/GJ
CHECKED BY: JLH
PROJ. NO. 38022

SHEET NO.

A2



ROOF PLAN NOTES:

1. Submit shop drawings/roof manufacturer's standard details for the standing seam painted baked enamel over 24 Gauge Galvalume – 40 year paint warranty and 30 year fade/chalk warranty. Submit concealed fastener details. All roofing installation shall comply with the Florida Building Code – Building (7th edition) 2020. Continuous drip edge and valley metal shall match selected roof color.
2. The concealed fastener standing seam metal roof shall be installed over a 60 mil peel & stick self healing membrane such as GRACE ICE & WATER SHIELD. Submit membrane sample for review.
3. Refer to the structural drawings for the required wind load design for the roof.
4. Submit standard details for all roof penetrations to assure penetrations will be properly flashed to prevent moisture infiltration.
5. The metal roof over combustible materials is considered a Class "A" roof.

SPECIAL NOTES:

1. The attic shall be divided by a DRAFT STOP to limit the attic square footage not to exceed 3,000 S.F. Doors or access in the attic smoke partition shall comply with Section 710.5.2.1 through 710.5.2.3. This may be achieved by screwing one layer of ½" GWB to one side of the roof truss, extending the GWB down to the GWB ceiling in the catering kitchen attic.
2. Doors or access in the attic smoke partition shall comply with Section 710.5.2.1 through 710.5.2.3.
3. The attic space at the Porti-Co-Chere shall be separated from the main main building roof system. This may be accomplished by extending the roof sheathing on the Fellowship Hall Building to the eave prior to installing the Porti-Co-Chere roof system.
4. Refer to structural drawings for mechanical connections (Simpson) and structural construction.

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ROOF PLAN

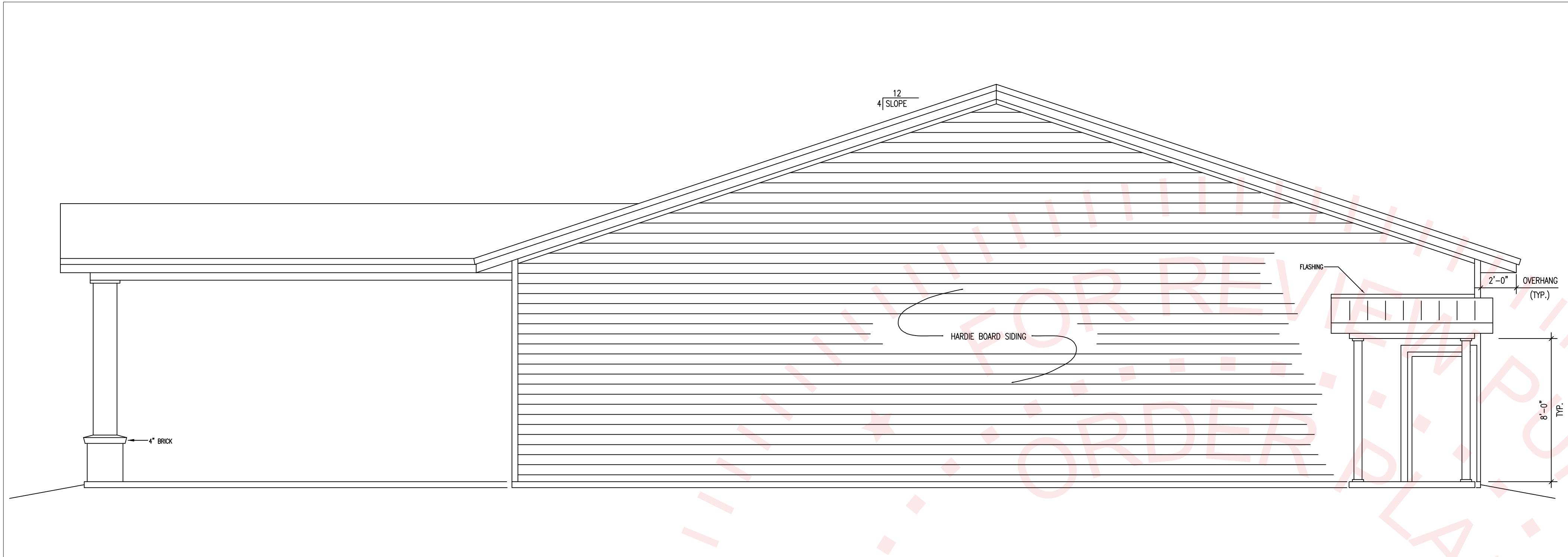
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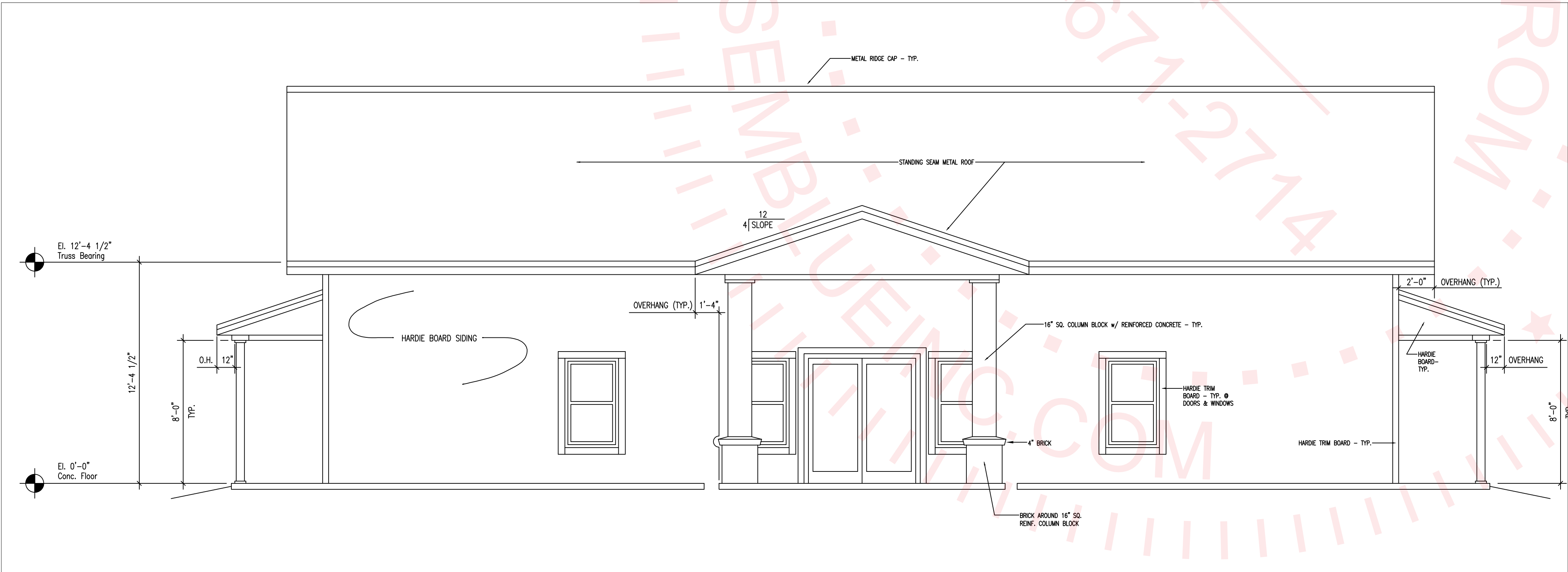
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RIGHT ELEVATION
SCALE: 1/4" = 1'-0"

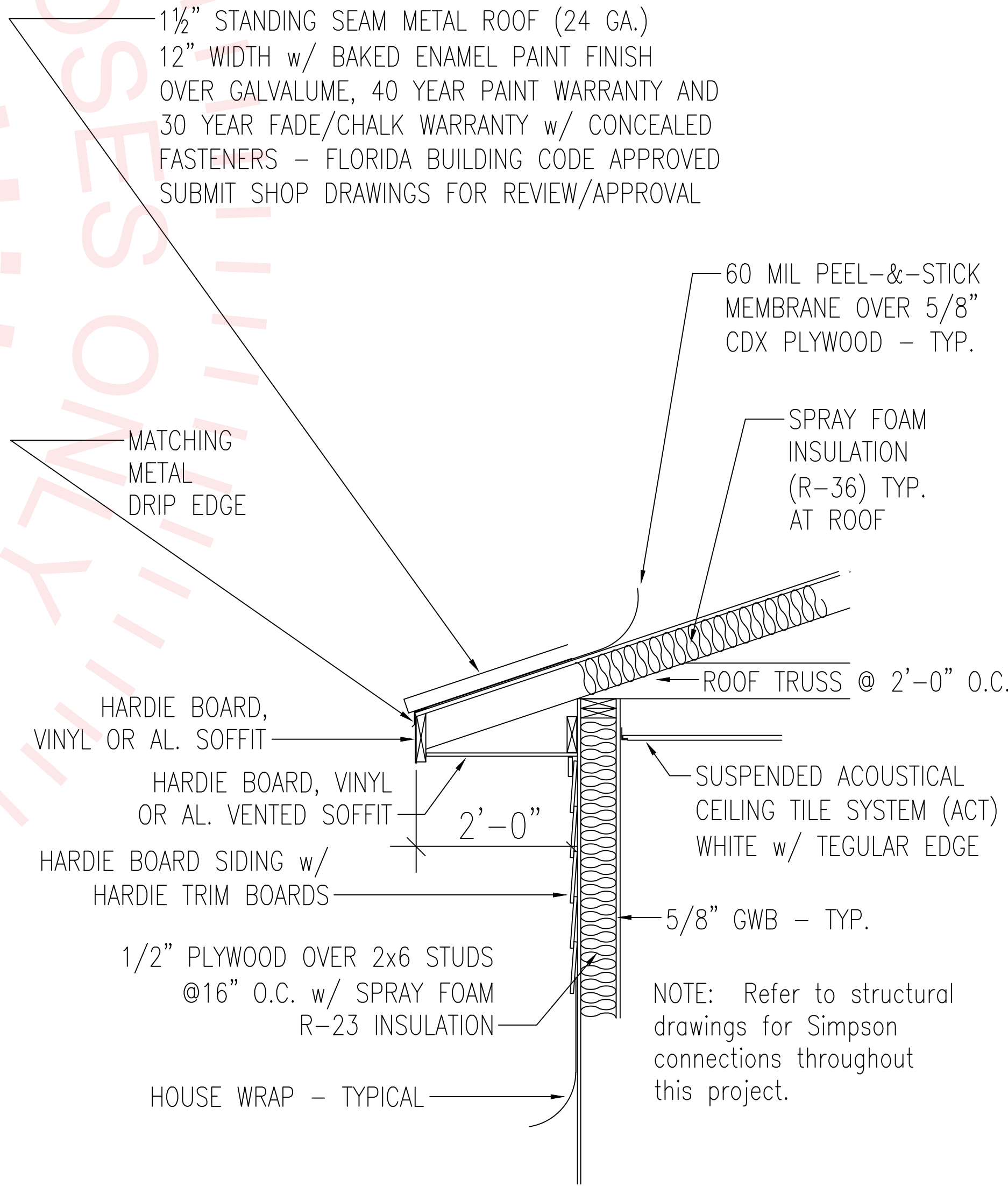


FRONT ELEVATION
SCALE: 1/4" = 1'-0"

NOTE:
Double or single hung windows shall be 3'-0" wide X 5'-0" high units vinyl clad with High Performance Low-E4 glass w/ 20 year warranty. Provide jamb extensions for 2x6 exterior walls.

NOTE:
Refer to structural drawings for Simpson connections throughout this project.

NOTE:
Spray foam building insulation shall conform to the requirements for icynene closed cell spray polyurethane foam meeting the fire-resistance characteristics to comply with ASTM E119 or NFPA 285 testing. Surface burning characteristics shall comply with ASTM E84 having a flame spread index of 25 or less and smoke-development index of 450 or less.

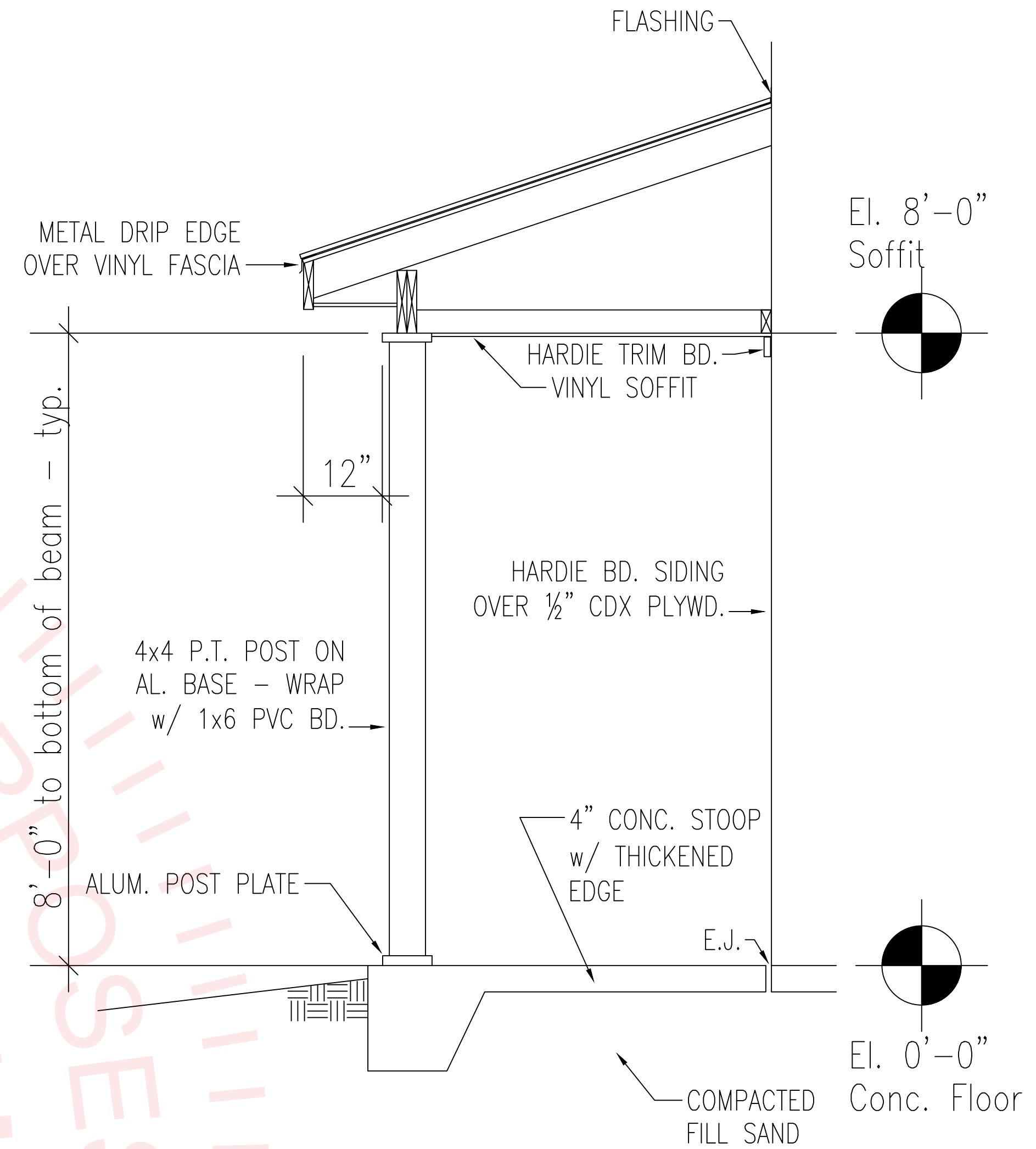


TYPICAL SOFFIT SECTION
SCALE: 3/4" = 1'-0"



REAR ELEVATION

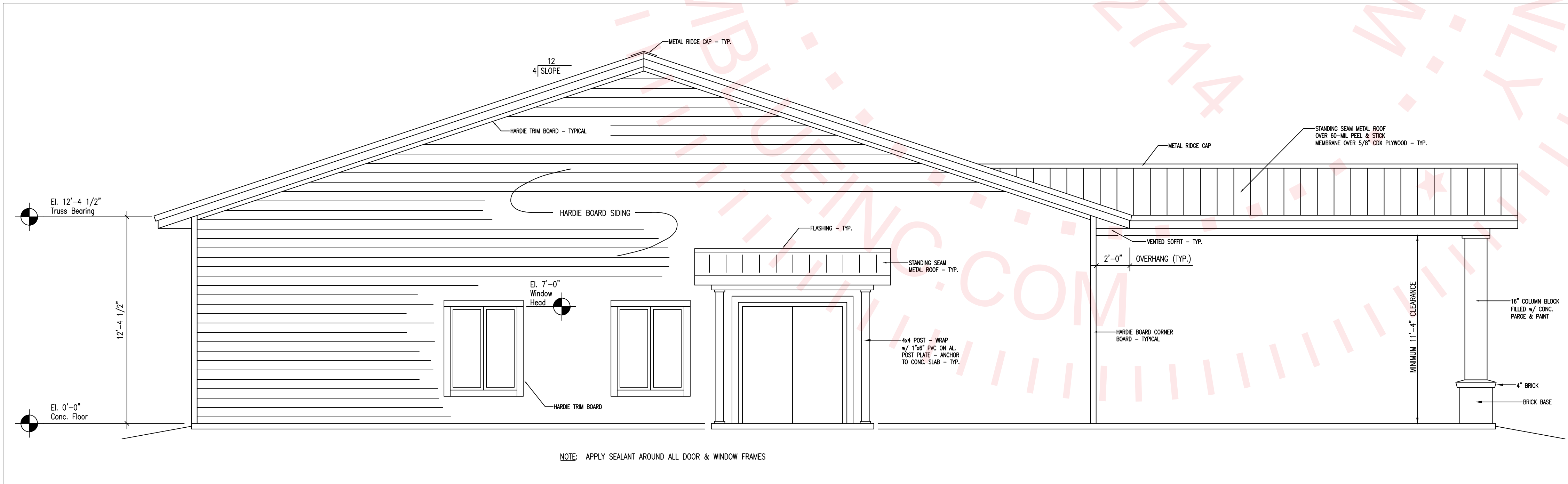
SCALE: 1/4" = 1'-0"



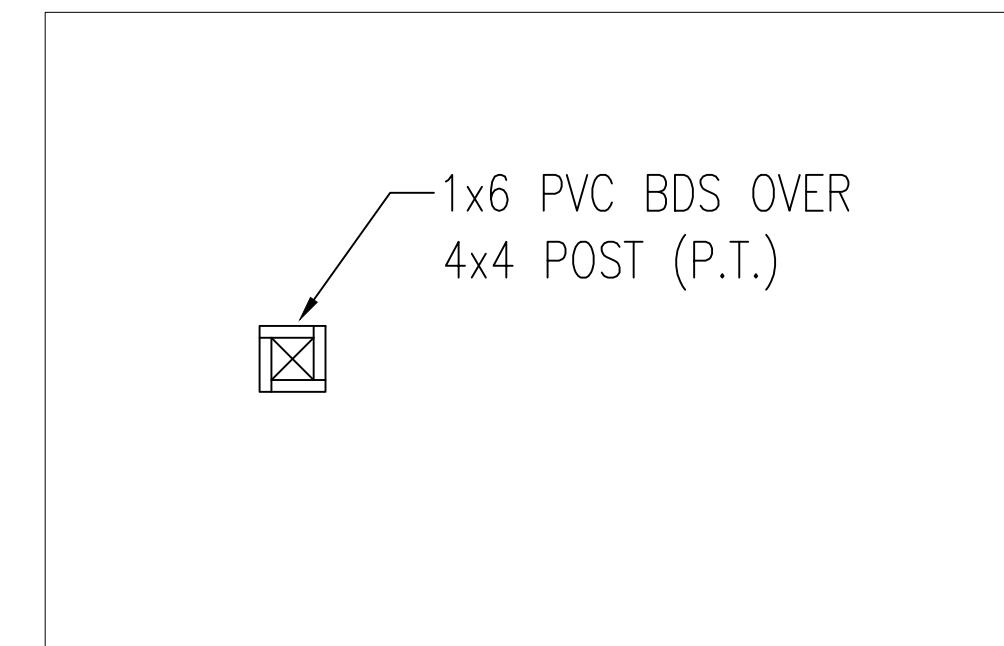
NOTE:
Refer to structural drawings for construction / Simpson connections & reinforcing.

COVERED STOOP SECTION

SCALE: 3/4" = 1'-0"



NOTE: APPLY SEALANT AROUND ALL DOOR & WINDOW FRAMES



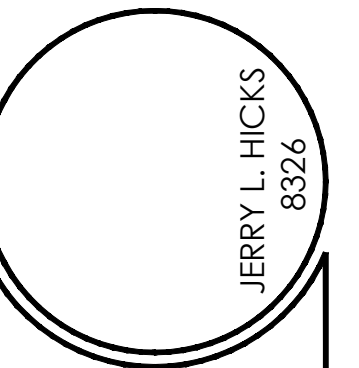
WRAPPED POST SECTION

SCALE: 3/4" = 1'-0"

NOTE:
Casement windows shall be pairs of 2'-0" wide X 5'-0" high units vinyl clad w/ High Performance Low-E4 glass w/ 20 year warranty. Provide jamb extensions for all windows for 2x6 exterior walls.

LEFT SIDE ELEVATION

SCALE: 1/4" = 1'-0"



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BUILDING ELEVATIONS;
COVERED STOOP SECTION

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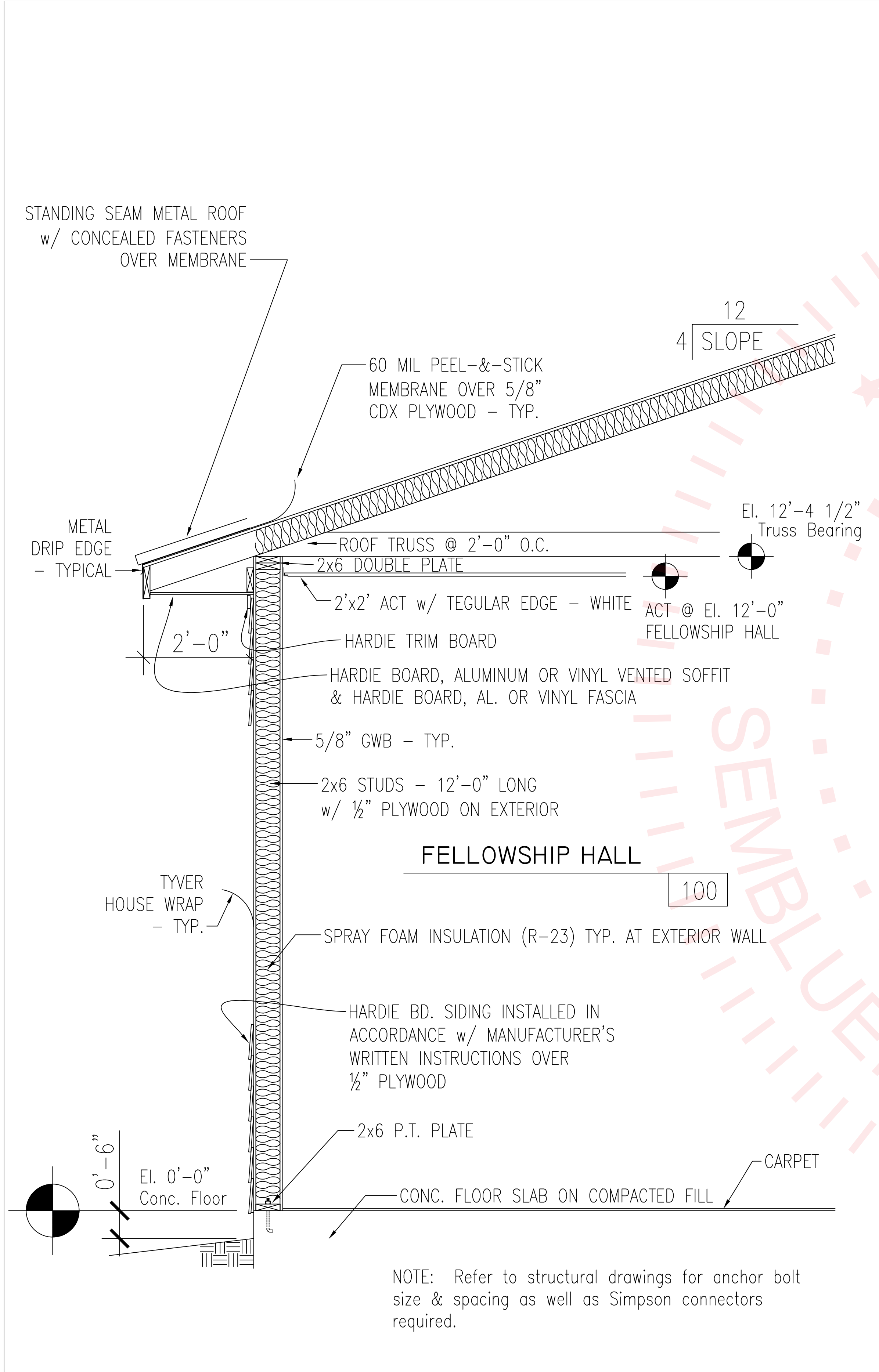
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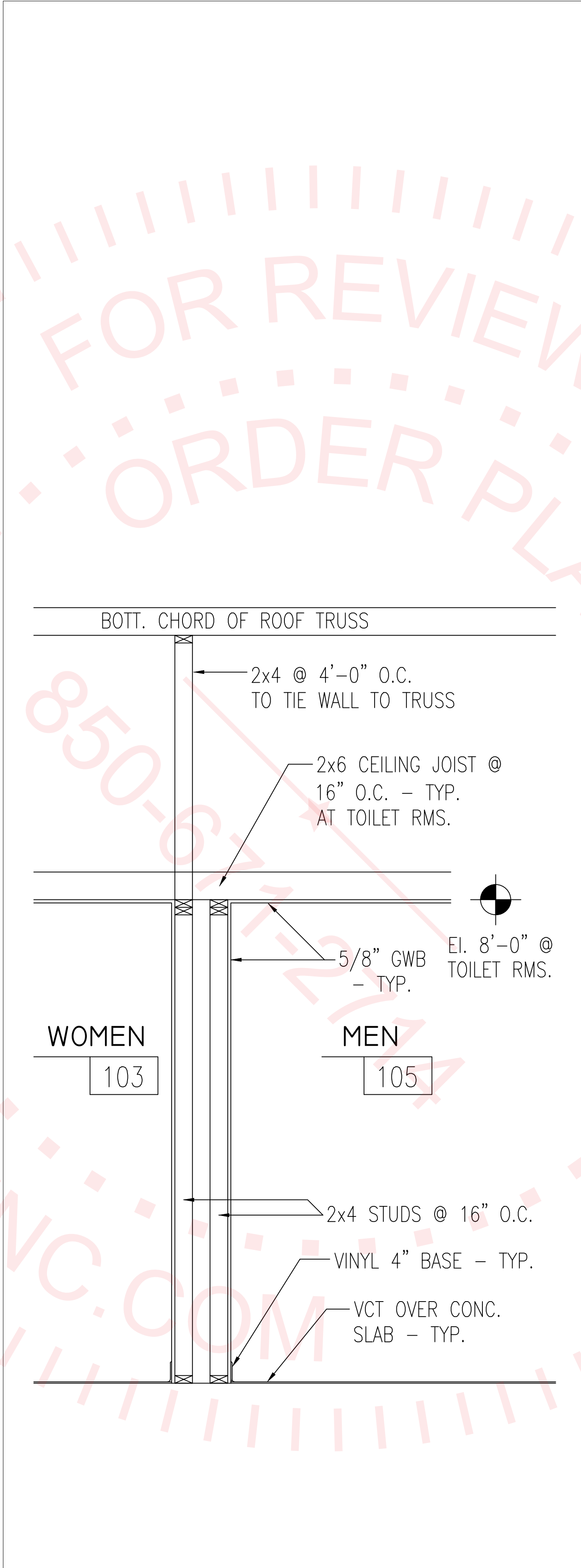
SHEET NO.

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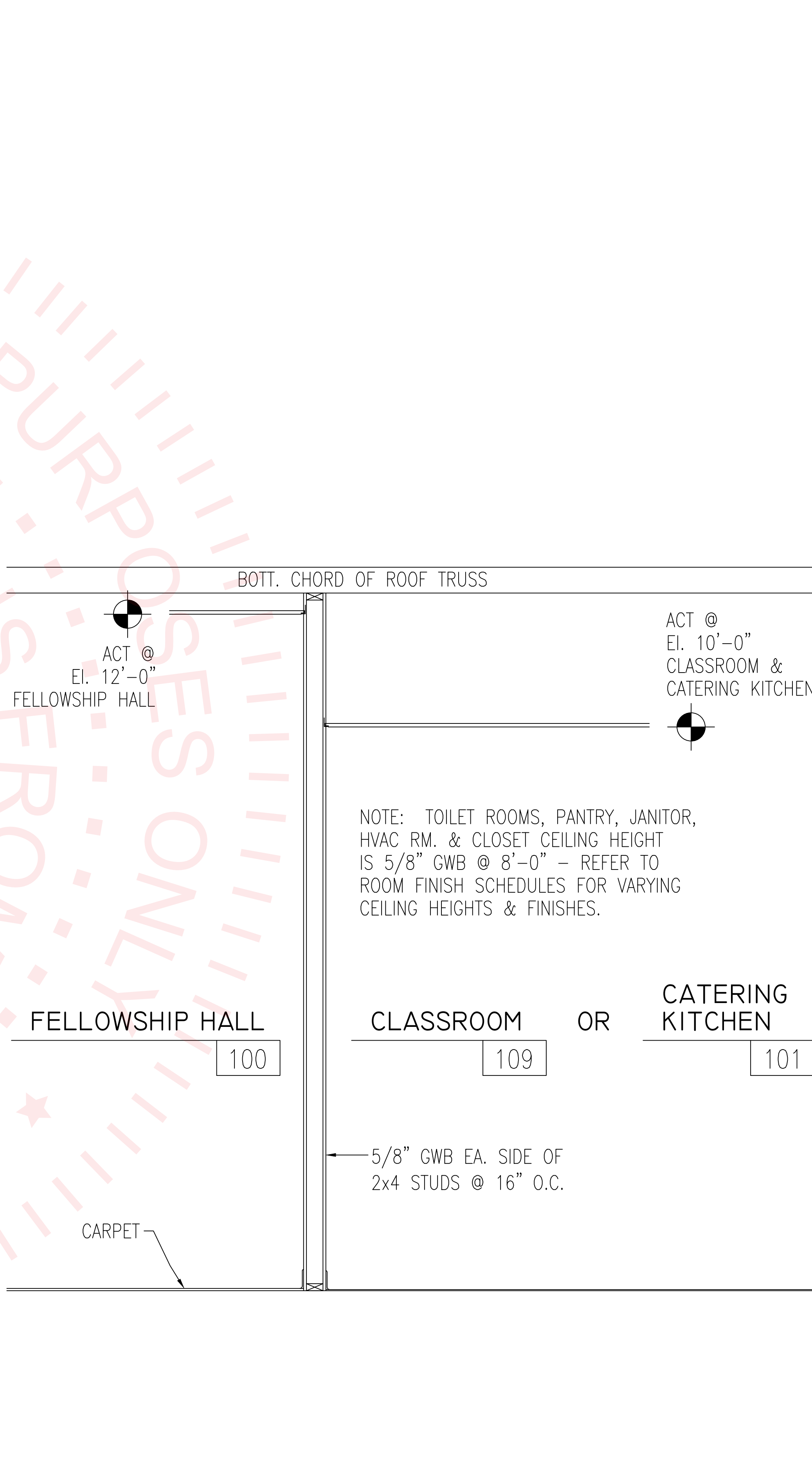
EXTERIOR WALL SECTION

SCALE: 3/4" = 1'-0"



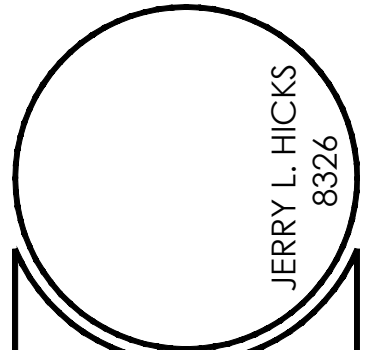
PLUMBING WALL SECTION @ TOILET ROOMS

SCALE: 3/4" = 1'-0"



SECTION @ CLASSROOM WALL

SCALE: 3/4" = 1'-0"



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WALL SECTIONS

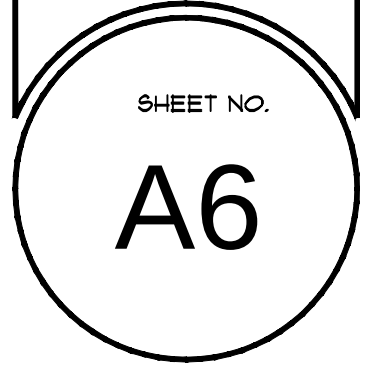
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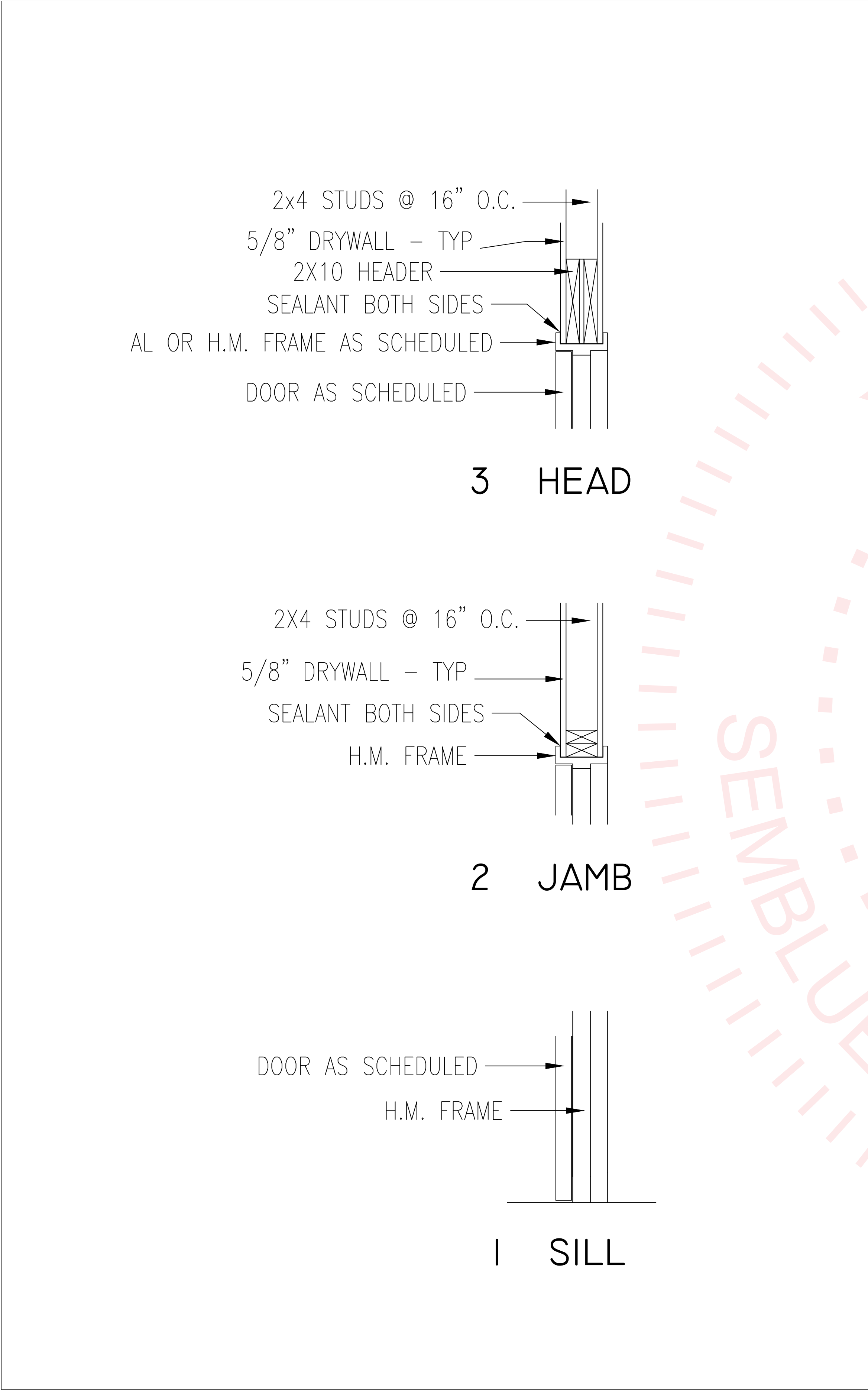
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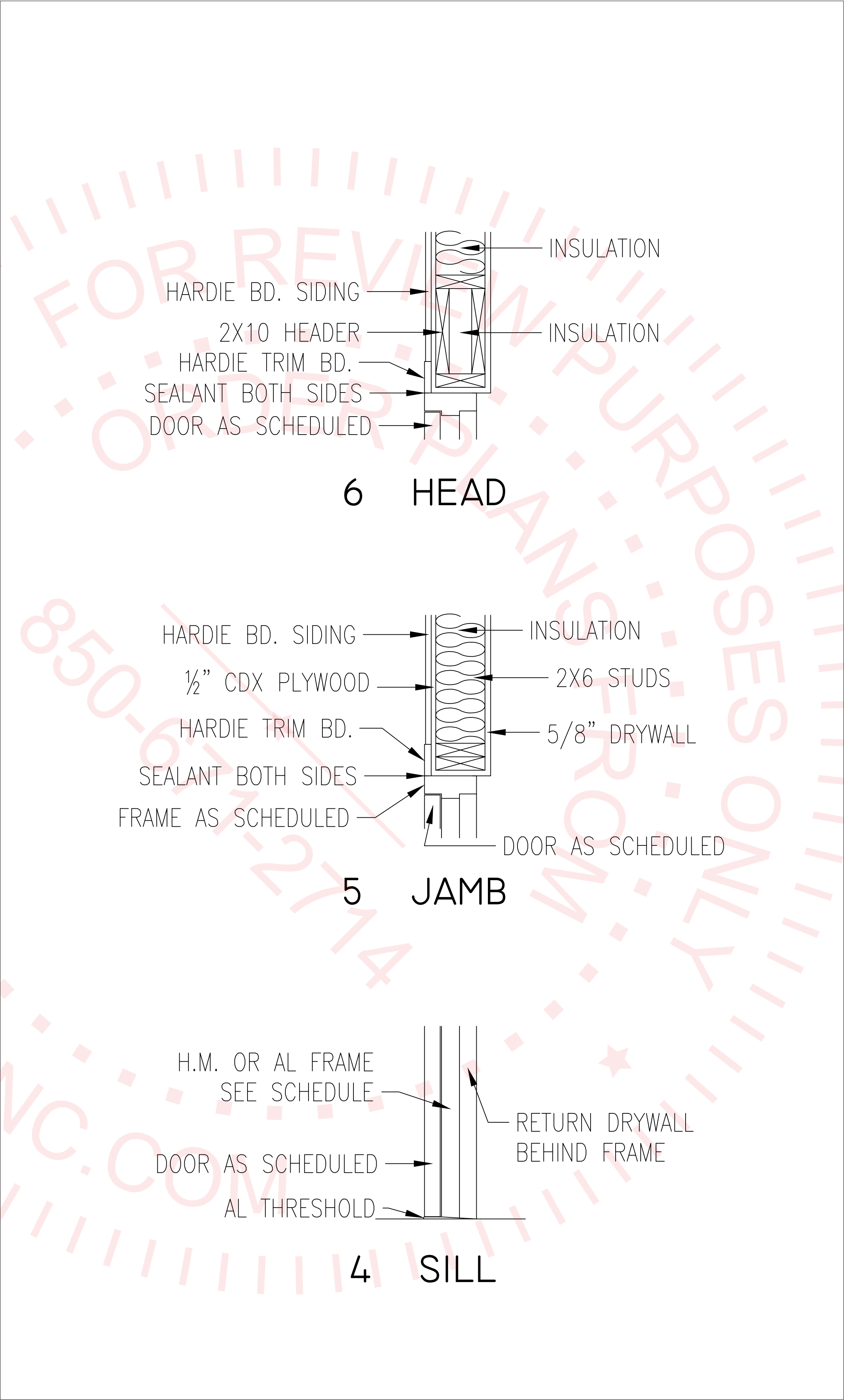
DATE:	11-05-2021
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INTERIOR DOOR FRAME DETAILS

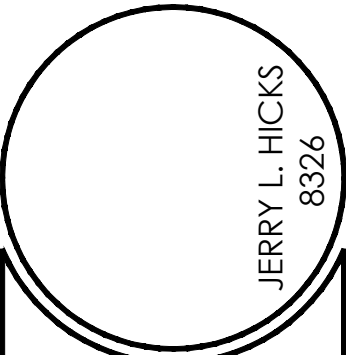
SCALE: 1 1/2" = 1'-0"



EXTERIOR DOOR FRAME DETAILS

SCALE: 1 1/2" = 1'-0"

- INTERIOR DOOR FRAME NOTE:
1. Provide horizontal blocking between 2x4 studs @ 4'-0" OC horizontally since studs are 12'-0" long - coordinate with drywall sheets.
- EXTERIOR DOOR FRAME NOTES:
1. Provide & install Vycor Plus by Grace self-adhering exterior door & window flashing at all exterior doors & windows in strict accordance with manufacturer's written instructions. Flashing comes in rolls 12" wide x 75' long.
 2. Coordinate door & window flashing in conjunction with DuPont Tyvek house wrap installation to insure proper installation of both Tyvek & flashing.
 3. All exterior H.M. frames shall be galvanized. All exterior H.M. doors shall be galvanized and insulated with flush door caps.



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DOOR FRAME DETAILS
AND NOTES

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MECHANICAL GENERAL PROVISIONS

GENERAL:

Applicable provisions of the General Conditions of the contract and special requirements govern work under this section.

CODES, ORDINANCES AND PERMITS:

General: Where requirements of these specifications exceed specified codes and ordinances, conform to these specifications. Materials and equipment included in Underwriters' Label Service shall bear that label.

Codes: The work covered under this division of specifications shall conform to the following codes as applicable:

- Florida fire Prevention Code, 6th ED
- Florida Building Code - Building, 6TH ED
- Florida Building Code - Plumbing, 6TH ED
- Florida Building Code - Mechanical, 6TH ED

HVAC SYSTEMS

SUBMITTALS:

General: All equipment and materials shall be new and of best grade. Review of all equipment and materials from the Engineer shall be obtained by the Contractor before any equipment is ordered.

Submittal Data: The Contractor shall check data carefully to insure compliance with these specifications prior to submitting. Submittal data shall be submitted at one time in one Portable Data Format (pdf) electronic file. Data shall include complete installation instructions, dimensional data, and electrical data, as applicable.

The file shall contain a cover sheet with project title, Architect, Engineer, and Contractor's names and contact information. The submittal shall include an index sheet listing the tab numbers. Submittals not prepared as specified herein will be returned to the Contractor for resubmittal.

- Grilles, Registers, and Diffusers
- Thermostats
- Duct Insulation
- Split System Heat Pumps

MANUFACTURER'S RECOMMENDATIONS:

The assembly, erection, testing, and adjusting of equipment shall be in strict accordance with the manufacturer's published instructions shipped with the equipment.

GUARANTEE, WARRANTIES AND INSTRUCTIONS:

Guarantee: All equipment and materials furnished and all work performed under this section of specifications shall be guaranteed to be free of defective materials and workmanship for a period of one year, five years for compressors, after final acceptance of the work by the Owner. During the guarantee period, the affected part or parts shall be promptly replaced with new parts by the Contractor at no additional cost to the Owner. All labor required to perform guarantee shall be included as part of the complete guarantee warranty.

DRAWINGS:

General: Both the drawings and specifications shall be considered supplemental to one another so that materials and labor required by one but not by the other shall be supplied and installed as though specifically called for by both. Where drawings and specifications conflict, Contractor shall conform to the more stringent requirements.

Scaling: The drawings are diagrammatic only and show generally the location of the equipment, ducts and pipes, but are not to be scaled. All dimensions shall be verified at the building site. Prefabrication of work from the drawings shall be at the Contractor's risk.

NAMEPLATES:

General: Provide nameplate for each piece of root mounted equipment.

Designation: The unit tag as shown on the plans of each device and electrical panel and circuit number shall be etched in 1/4" maximum, 1/8" minimum height letters attached to the item of equipment. Nameplate shall be white core black bakelite secured with brass screws, adhesive attachment will not be acceptable.

OPERATING AND MAINTENANCE INSTRUCTIONS:

Three bound and indexed Operating and Maintenance Manuals shall be prepared by the Contractor and be submitted to the Architect for approval prior to delivery to operating personnel.

Each manual shall contain the following information, data and drawings:

- List of contents. Insert under front cover.
- Copy of approved submittals, equipment and materials.
- Installation, operating and maintenance instructions for each item of equipment.
- Manufacturer's list of renewal parts for each item of equipment with recommended stock items and quantities indicated.
- Manufacturer's equipment warranties.
- Copy of shop drawings showing layouts and construction details.
- Copy of Test and Balance Reports including list of instruments and description of methods employed.

TESTING, ADJUSTING, AND BALANCING

Scope:

Testing and balancing shall be performed by an independent testing and balancing agency certified by either Associated Air Balance Council (AABC) or National Environmental Balancing Bureau (NEBB). Testing and balancing shall be performed for each item of heating, ventilating, and air conditioning equipment individually and then further testing and balancing shall be done to each system including accessories. Testing and balancing shall include but not be limited to equipment, air systems, control systems, and space temperature conditions.

Methods:

Testing and balancing shall be performed in complete accordance with AABC National Standards for Field Measurements and Instrumentation, or in complete accordance with Procedural Standards for Testing, Adjusting, Balancing of Environmental Systems, as published by the National Environmental Balancing Bureau when the testing and balancing work is performed by an NEBB certified firm. T & B shall satisfy all of the requirements of Section C408 of the Florida Building Code-Energy Conservation, 5th Edition.

DUCTWORK:

Concealed Low Pressure Supply and Return Ductwork:

General: Sheet metal low pressure ductwork shall be fabricated and installed in accordance with SMACNA HVAC Duct Construction Standards, Metal and Flexible, Current Edition, unless otherwise indicated on the drawings or specified herein.

Construction: Low pressure ductwork shall be constructed from lock forming quality galvanized steel sheets having a galvanized coating of 1-1/4 ounces total for both sides per one square foot of sheet. Sheet steel gauges and construction methods shall be as scheduled in SMACNA HVAC Duct Construction Standards.

Flexible Duct: Flexible duct shall not exceed 10 feet in length and shall be insulated with a minimum R-6 fibrous glass insulation.

INSULATION:

Concealed Ductwork: Insulate all supply, return, and outside air duct with R-6 fibrous glass batt insulation with FSK jacket and taped joints.

HEAT PUMP SCHEDULE			AIR HANDLING UNIT SCHEDULE		
TAG		HP-1 HP-2 HP-3	TAG		AHU-1 AHU-2 AHU-3
DESIGN MANUFACTURER		CARRIER	DESIGN MANUFACTURER		CARRIER
MODEL NUMBER		25HCE	MODEL NUMBER		FX4D
SIZE		5.0 TONS	ARRANGEMENT		AS SHOWN
REFRIGERATION CAPACITY	BTU/HR.	60,000	AIR FLOW RATE	CFM	2000
REFRIGERATION CIRCUITS	#	1	O/A FLOW RATE	CFM	AS NOTED
COMPRESSOR	#	1	EXT. STATIC PRESSURE	INCHES W.G.	0.50
			COIL ENTERING AIR	DB/WB	80/67
			COIL LEAVING AIR	DB/WB	58.3/58.1
REFRIGERANT		410A	DRIVE		DIRECT
LOCKED ROTOR AMPS		144.2	SPEED	RPM	—
RUN LOAD AMPS		24.4	MOTOR	HP	3/4
CONDENSER FAN			ELEC. CHARACTERISTICS	V/PH/Hz	208/230-1
NUMBER		1		MCA/MOCP	53.8/60
DISCHARGE		VERTICAL			
DRIVE		DIRECT	COOLING COIL CAPACITY , SH/TH	MBTUH	44.2/54.1
FULL LOAD AMPS (TOTAL)		1.52			
ELECTRICAL CHARACTERISTICS	V/Ø	208/230-1			
	MCA/MOCP	32.0/50			
ENERGY EFFICIENCY RATIO	SEER	14.0	HEATING COIL – ELEC.	KW	10
HEATING CAPACITY (ARI) 47 F BTU/HR		54,860			
HEATING COP		3.70			
NOTES:			NOTES		

- REFRIGERANT PIPING SHALL BE SIZED BY UNIT MANUFACTURER FOR LIFT AND EQUIVALENT LENGTH OF REFRIGERANT PIPING RUN.
- PROVIDE SINGLE POINT ELECTRICAL CONNECTION WITH INTEGRAL ELECTRICAL DISCONNECT ON INDOOR UNIT.
- PROVIDE NEW HONEYWELL MODEL TB8220 COMMERCIAL VISIONPRO 8000 TOUCHSCREEN PROGRAMMABLE THERMOSTAT FOR EACH SYSTEM.
- REFER TO NEW WORK NOTE 2, SHEET M2.

MINIMUM MECHANICAL VENTILATION

PER FBC-M 403 MECHANICAL VENTILATION:
{ [Rp (Dz Az/1000)]+ (Ra Az) } (1-N) = Vbz

Rp = O.A. CFM/PERSON
Dz = OCCUPANT DENSITY
Az = AREA OF ZONE
Ra = O.A. CFM/SF
N = FRACTION OF VENTILATION MET BY NATURAL VENTILATION
Vbz = MIN. O.A. VENTILATION TO BREATHING ZONE

FOYER

CORRIDOR: FLOOR AREA: 273 SF
4% FLOOR AREA = 11 SF
OPERABLE AREA = 51 SF

MINIMUM MECHANICAL VENTILATION = 0 CFM

FELLOWSHIP HALL

MULTIUSE SSSEMBLY: FLOOR AREA: 1736 SF
4% FLOOR AREA = 69 SF
OPERABLE AREA = 69 SF

MINIMUM MECHANICAL VENTILATIONVENTILATION = 0 CFM

CLASSROOM

CLASSROOM: FLOOR AREA: 283 SF
4% FLOOR AREA = 11 SF
OPERABLE AREA = 3 SF => 27% REQUIRED VENTILATION

Rp Dz Ra Az Vbz
5.0 70 0.06 283 85

MINIMUM MECHANICAL VENTILATION = 85 CFM

TOILET ROOMS

TOILET ROOMS EXHAUST: 6 FIXTURES X 50 CFM/FIXTURE = 300 CFM

TOTAL VENTILATION PROVIDED BY DESIGN = 300 CFM

GRILLE SCHEDULE

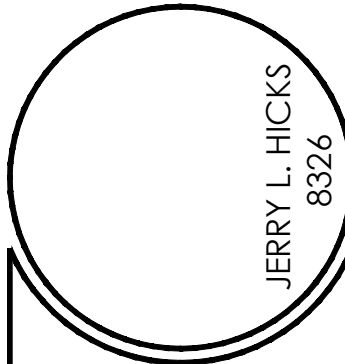
MARK	PRICE MODEL	SIZE	THROW	FINISH	NOTES.
A	9X9/24X24/AMDE/3P/4A/B12	9X9	4-WAY	WHITE	1,2
B	6X6/AMDE/1/4A/B12	6X6	4-WAY	WHITE	1,2
C	6X6/AMDE/1/1S/B12	6X6	4-WAY	WHITE	1,2
D	8X8/630DAL/F/S/A/B12	8X8	EXHAUST	WHITE	1,2
E	14X8/630DAL/F/L/A/B12	14X8	RETURN	WHITE	4
F	22X72/630DAL/F/S/A/B12	22X72	RETURN	WHITE	3

- NOTES:
- PROVIDE TRANSITION TO DUCT SIZE INDICATED ON THE PLANS.
 - WITH OPPOSED BLADE DAMPER
 - POSITION GRILLE 8" ABOVE FLOOR.
 - CENTER GRILLE ON WALL CENTERED BETWEEN DOOR HEADER HEIGHT AND CEILING.

EXHAUST FANS

MARK	CFM	APPROX EXT. SP IN W.G.	MOTOR HP	FAN RPM	MAX NOISE SONES	BASIS OF DESIGN	NOTES
EF-1	350	0.5	0.167	1351	8.1	COOK MODEL 100ACWB OR80	1,2,3

- NOTES:
- FAN TO BE INTERLOCKED TO RUN WHEN AHU-3 IS IN THE OCCUPIED MODE.
 - INTEGRAL DISCONNECT.
 - 120V/60HZ



REVISIONS

NO. DATE:

ELECTRICAL SPECIFICATIONS AND SCOPE OF WORK

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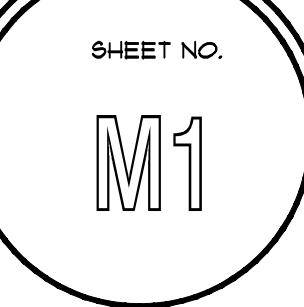
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DATE: 11-05-2021

DRAWN BY: HAO/YMO

CHECKED BY: HAO

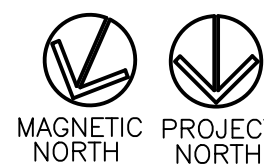
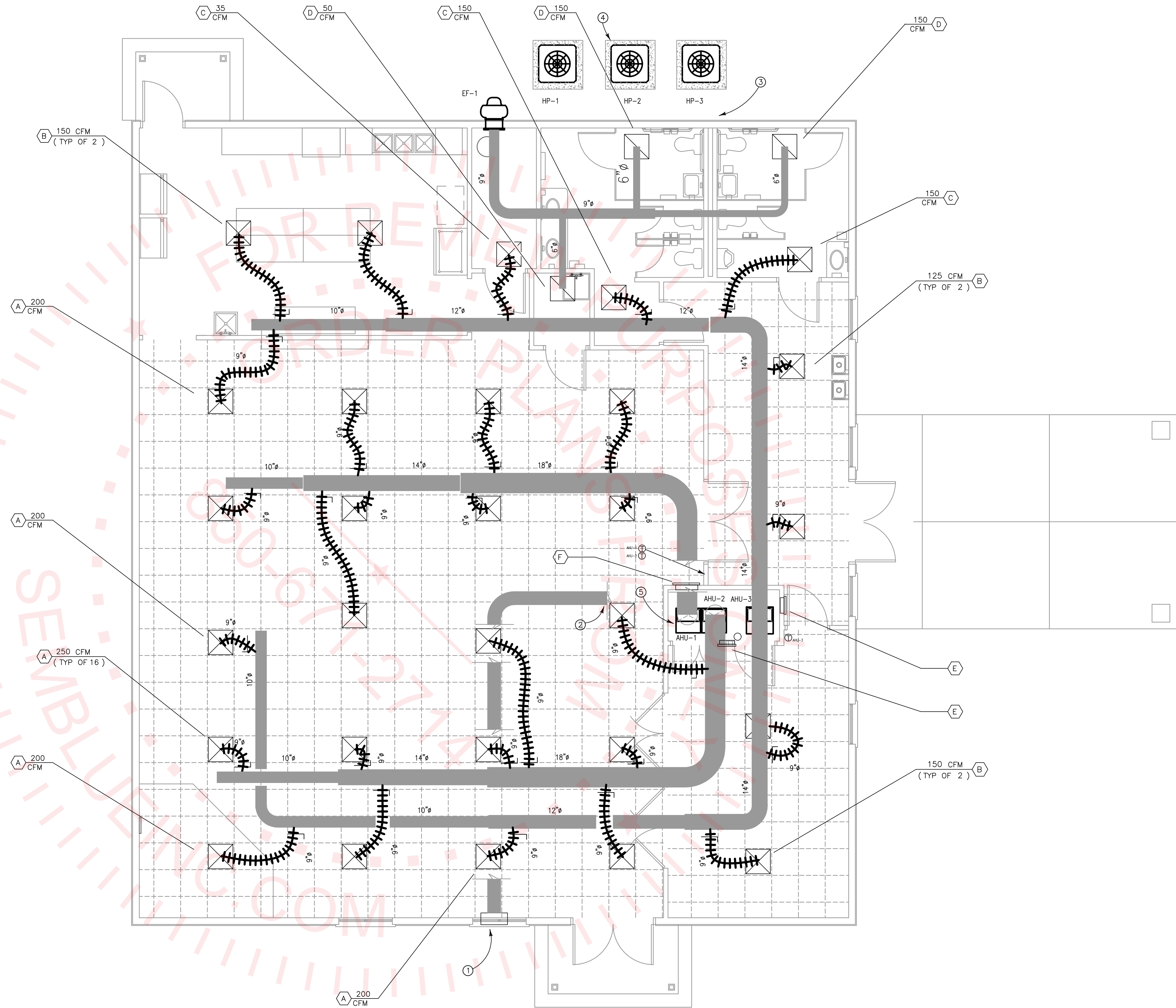
PROJ. NO. 38022



NEW WORK NOTES

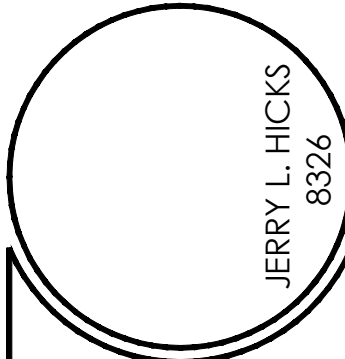
THIS SHEET ONLY

- ① PROVIDE 18X18 GABLE VENT CENTERED ON RIDGE LINE. EXTEND 12"Ø INSULATED DUCT FROM GABLE VENT THROUGH ATTIC AND TURN DOWN INTO AHU CLOSET.
- ② EXTEND 12"Ø INSULATED DUCT FROM GABLE VENT THROUGH ATTIC AND TURN DOWN INTO AHU CLOSET. PROVIDE MANUAL DAMPER AND MOTORIZED DAMPER. INTERLOCK MOTORIZED DAMPER WITH COMPRESSOR ON HP-1 TO OPEN ONLY WHEN HP-1 COMPRESSOR IS ENERGIZED. ADJUST OUTSIDE AIR FLOW NOT TO EXCEED 300 CFM WITH ALL AIR HANDLING UNITS ENERGIZED.
- ③ EXTEND REFRIGERANT PIPING FROM AHU'S THROUGH ATTIC TO SOUTH WALL. TURN DOWN ON WALL AND THEN EXIT WALL AT 12" ABOVE GRADE THROUGH AN AIREX TITAN OUTLET (TGS SERIES) ENGINEERED WALL SEAL WITH THE EXPOSED REFRIGERANT PIPING PROTECTED BY THE AIREX EFLEX GUARD SYSTEM.
- ④ PROVIDE A 4" THICK CONCRETE PAD WITH BRUSH FINISH AND BULL NOSED EDGES. REINFORCE PAD WITH 6X6X#10 WWM. ANCHOR OUTDOOR UNIT TO PAD TO STAISFY FBC WIND LOAD REQUIREMENTS.
- ⑤ EXTEND CONDENSATE DRAIN WITH 3" P-TRAP FROM EACH AHU TO FLOOR DRAIN AS DETAILED. PROVIDE SMOKE DETECTORS IN THE SUPPLY DUCT OF EACH AHU IN ACCORDANCE WITH FBC-MECHANICAL 606.2.1. PROVIDE ALARM AND RESET STATIONS IN THE CEILING OF FOYER,106, NEAR DOOR TO CLASSROOM,109.



MECHANICAL PLAN

SCALE: 1/4" = 1'-0"



REVISIONS	
NO.	DATE:

ELECTRICAL
SPECIFICATIONS
AND SCOPE OF WORK

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property of the Engineer
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consent of the Engineer.

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HOWARD A. OOTEN, P.E., LEED AP
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DATE:	11-05-2021
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DIVISION 15 - MECHANICAL

CODES, ORDINANCES AND PERMITS:

General: Where requirements of these specifications exceed specified codes and ordinances, conform to these specifications. Materials and equipment included in Underwriters' Label Service shall bear that label. Electrical equipment shall be UL approved as installed, unless noted otherwise herein.

Codes: The work covered under this division of specifications shall conform to the following codes as applicable:

- Florida Fire Prevention Code, 6th Ed.
- Florida Building Code - Building, 6TH Ed.
- Florida Building Code - Plumbing, 6th Ed.
- Florida Building Code - Mechanical, 6th Ed.

Permits: The Contractor shall obtain all necessary permits and inspections required for the installation of his work and pay all charges incidental thereto. He shall deliver to the Architect all certificates of said inspections issued by the authorities having jurisdiction.

PLUMBING

SCOPE

This section of specifications covers plumbing systems complete as shown on the drawings and as specified herein. All systems shall be tested and made ready for operation.

PART I - GENERAL

Plans are diagrammatic and show generally the location of piping and accessories and are not to be scaled for construction measurements. All dimensions shall be verified at the building site. Both the drawings and specifications shall be considered supplemental to one another so that materials and labor required by one but not by the other shall be supplied and installed as though specifically called for by both. Exact locations of rough-in connections to all equipment shall be determined before any rough-in is started. All work shall be coordinated with the work of other trades to avoid conflicts.

EXCAVATION AND BACKFILL; CUTTING AND PATCHING:

The excavation and backfill for the installation of the plumbing system shall be done in accordance with the requirements of these specifications. All junk metal shall be removed from the site before filling. The bottoms of all pipe trenches in rock or very hard strata shall be excavated a miniil-mum of 6" below the pipe bottoms and refilled with 7" or more of sand and gravel, tamped in place. When ditches are excavated in cinders, organic soils or other injurious materials, these materials shall be excavated a minimum of 6" below the bottom of the pipe line and replaced with inorganic soil. The backfill shall be tamped in 4" layers along the sides of the pipe and for at least 1" above the pipe. The remainder shall be tamped in 6" layers except beneath walks and roads where 4" layers should be used. Contractor shall provide shoring for installation of piping over six (6) feet deep. Contractor shall provide dewatering to keep trenches dry during installation of piping.

Existing slabs shall be saw cut where necessary for trenching and shall be repaired to match existing finish.

Cutting and patching in connection with the installation shall be done by the trade whose work is to be cut. Contractor shall lay out and install his work ahead of the work of other trades where possible.

PIPING SYSTEM:

Piping:

- Water Piping; Domestic Hot and Cold: Schedule 40 CPVC.
- Building Soil, Drain, Waste, Condensate, and Vent Piping: DWV-PVC.

CLEANOUTS:

Schedule:

Location	Finish	Zurn
Interior Areas	Bronze	Z-1400
Wall	Stainless Steel	Z-1446

Other Accepted Manufacturers: Josam, Ancon, Smith, Wade.

FLOOR DRAINS/SINKS:

General: Coordinate with equipment layout drawings for specific floor drain and floor sink location.

Schedule:

Type	Zurn	Notes
FS-1	Z-1901	2,3,6,7
FS-2	Z-1901	1,3,6,7
FD-1	Z-415-6B	1,4,6,7
FD-2	Z-415-5B	1,5,6,7

Notes:

- 2" connection
- 3" connection
- half grate
- 6" Type B strainer
- 5" Type B strainer
- Sewer gas seal
- Other Accepted Manufacturers: Ancon, Smith, Wade, Josam or Jonespec

Sewer Gas Seals:

General: Provide waterless sewer gas trap seals in lieu of trap primers. Accepted manufacturers include, SureSeal, Mifab., or Trap Guard

PRESSURE TESTS:

Plumbing and associate systems shall be subjected to constant inspection and final approval of the Architect/Engineer and code authorities having jurisdiction. Tests, in addition to those included in this section required to show compliance, shall be performed as directed.

Test shall be conducted as indicated in the following schedule. The specified test pressure shall be maintained for four hours. There shall be no drop in pressure during the test.

Pipe System	Test Pressure	Type Test
Soil, Waste and Vent	10 Ft.	Water Head
Potable Water	150 PSI	Hydrostatic

All work found to be defective or indicating leakage shall be removed and replaced with new materials. Tests shall be repeated until systems are proved tight. Hydrostatic test shall not be conducted in freezing weather. Caulking of screwed and welded joints will not be acceptable.

CLEANING; PROTECTION AND ADJUSTING:

The Contractor shall remove from the job site all debris and leftover materials for which he is responsible, clean all fixtures and equipment and repair any blemishes in the finish. The Contractor shall be held responsible for replacing fixtures where damage results from failure to provide protection.

Adjusting: All fixtures shall be tested to insure proper operation of all working parts and adjusted to provide proper water flow.

GUARANTEE:

All equipment and materials furnished and all work performed under this section shall be guaranteed to be free of defective materials and workmanship for a period of one year after final acceptance of the work by the Owner. Upon notice from the Owner of failure of any part of the guarantee equipment during the guarantee period, the affected part or parts shall be promptly replaced with new parts by the Contractor at no additional cost to the Owner. All labor required to perform guarantee shall be included as part of the complete guarantee warranty.

SUBMITTALS:

General: All equipment and materials shall be new and of best grade. Review of all equipment and materials from the Architect and Engineer shall be obtained by the Contractor before any equipment is ordered.

Substitutions: Approval of equipment or systems other than that shown must be obtained at least 10 working days prior to bid date.

Submittals: Submittal data shall be submitted for review and approval in one electronic file in pdf format. Submittal shall include the following:

- Pipe and fittings
- Floor sinks and drains
- Insulation
- Water heaters
- Tempering valve
- Plumbing Fixtures

INSULATION:

Insulation shall be applied in a workmanlike manner by experienced workmen.

Insulation shall not be applied until all pressure testing has been completed, inspected, and released for insulation application. Surfaces to be insulated shall be clean and dry.

All insulation joints shall be butted firmly together and all jackets and tapes shall be smoothly and securely installed.

Hot water piping throughout shall be insulated with 1" thick preformed fiberglass pipe insulation, minimum 4 lb. density, K factor of 0.23 at 75°F mean. Jacket shall be factory applied all service jacket with self-sealing lap (ASJ-SSL). Provide ASJ tape where insulation butts. ASJ jacket shall be continuous. Provide precut fiberglass insulation fitting inserts covered with PVC fitting covers. PVC fitting covers shall have a flame spread of 25 or less and a smoke development of 50 or less.

FIXTURE SCHEDULE:

Furnish and install all plumbing fixtures complete with all equipment, fittings, trim and accessories as required for a complete installation. All fixtures shall be Grade A. The name and trademark of the manufacturer shall be printed or pressed on all closets and lavatories. All fixtures shall be thoroughly cleaned after installation. All fixtures listed as handicap shall meet the requirements of ADA.

P1 ADA Water Closet: Sterling Model S402315, white in color, 1.28 gpf, Proflo seat Model PFTSOFE2000WH, Brasscraft BOCR19C angle stop service

P2 Water Closet: Sterling Model S403081, white in color, 1.28 gpf, Proflo seat Model PFTS2000WH, Brasscraft BOCR19C angle stop service

P3 Urinal: Proflo Model 1815, white vitreous china urinal, Zurn Model ZTR6203-WS1 flush valve, 1.0 gpf.

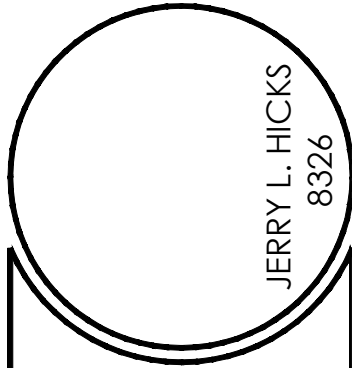
P4 ADA Lavatory: Proflo PF5511, vitreous china lavatory, faucet Moen Model L4605 with chrome grid strainer, chrome. I "x3/8" chrome angle stops and supplies, insulated drain and supply fittings with PVC jacket.

P5 Lavatory: Proflo PF19164, vitreous china lavatory, faucet Moen Model L4605 with chrome grid strainer, chrome. I "x3/8" chrome angle stops and supplies, insulated drain and supply fittings with PVC jacket.

P6 Mop Sink: Fiat Model MSBID2424 Service Basin, 24"x24"x10" one piece molded fiberglass basin with 3" ABS, drain and tail piece, 830AA wall faucet with support, 889CC mop hanger, 832AA hose and bracket, MSG2424 wall guard and E77AA vinyl bumperguard.

GENERAL NOTES	
1.	SEE ARCHITECTURAL DRAWINGS FOR BUILDING DIMENSIONS AND DETAILS. THESE PLUMBING DRAWINGS ARE DIAGRAMMATIC ONLY AND ARE NOT INTENDED TO BE SCALED.
2.	REFER TO EQUIPMENT LAYOUT PLANS FOR SPECIFIC PLACEMENT OF EQUIPMENT AND DRAINS.
3.	COORDINATE PIPE ROUTING AND EQUIPMENT LOCATIONS WITH HVAC, FIRE SPRINKLER AND ELECTRICAL INSTALLATIONS AND WITH BUILDING STRUCTURAL MEMBERS. CONFLICTING WORK NOT COORDINATED WILL BE REMOVED TO THE EXTENT NECESSARY TO ALLOW PROPER INSTALLATION OF ALL SYSTEMS.
4.	REFER TO ELECTRICAL DRAWINGS FOR VOLTAGE AND PHASE REQUIREMENTS OF ALL EQUIPMENT.
5.	SUPPORT ALL PIPING AND EQUIPMENT FROM PRIMARY BUILDING STRUCTURAL MEMBERS ONLY. ALL WORK SHALL TAKE CARE TO MINIMIZE DAMAGE TO BAR JOIST FIRE PROOFING. DISTURBED FIRE PROOFING SHALL REPAIRED AT THE CONTRACTORS EXPENSE.
6.	WHERE EXISTING SLAB MUST BE DEMOLISHED TO ACCOMPLISH REQUIRED WORK SLAB SHALL BE SAW CUT AS NECESSARY. WHERE SOIL IS DISTURBED NECESSARY CLEAN FILL SHALL BE PROVIDED AND COMPACTED AND THEN THE SLAB SHALL BE REPAIRED TO MATCH.

PLUMBING LEGEND	
	NEW BUILDING DRAIN
	NEW GREASE DRAIN
	VENT PIPING
	COLD WATER PIPING
	HOT WATER PIPING - 120°F
	TEMPERED WATER PIPING - 105°F
FI	PLUMBING FIXTURE TAG
FC	PLUMBING CONTRACTOR
DFU	DRAINAGE FIXTURE UNITS
WSFU	WATER SUPPLY FIXTURE UNITS
VTR	VENT THRU ROOF
V	VENT
W.C.O.	WALL CLEAN OUT
C.O.	CLEAN OUT



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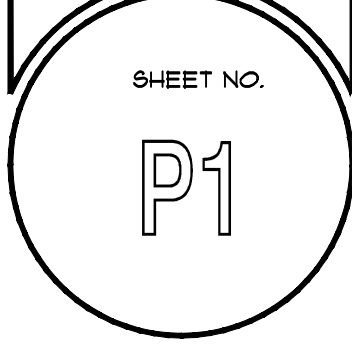
ELECTRICAL SPECIFICATIONS AND SCOPE OF WORK	
FELLOWSHIP HALL PISGAH UNITED METHODIST CHURCH	7000 PISGAH CHURCH RD. TALLAHASSEE, FLORIDA 32309

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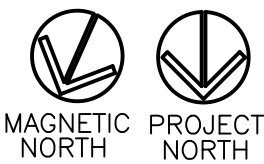
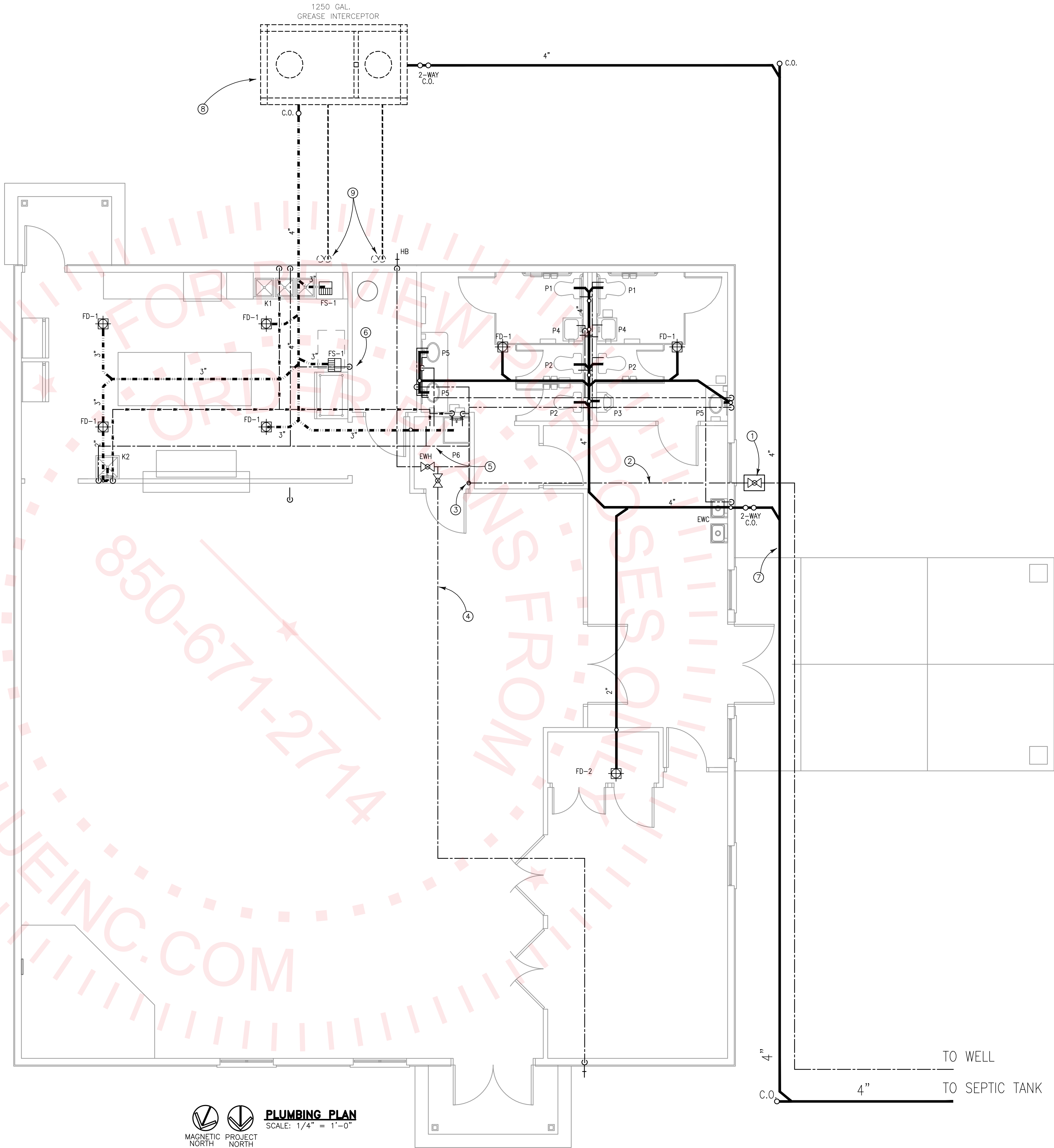
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NEW WORK NOTES

THIS SHEET ONLY

- ① BUILDING WATER SERVICE. PROVIDE VALVE BOX WITH FULL BORE BALL VALVE. EXTEND TO WELL PUMP.
- ② PIPING BELOW SLAB.
- ③ TURN UP EXPOSED IN UTILITY SPACE. PROVIDE ISOLATION VALVE AT BASE OF ENTRANCE RISER. PROVIDE DOUBLE CHECK REDUCED PRESSURE ZONE BACK FLOW PREVENTER EXPOSED ON WALL.
- ④ PIPING CONCEALED ABOVE CEILING.
- ⑤ PROVIDE 50 GALLON LOWBOY STYLE ELECTRIC WATER HEATER WITH DUAL NON-SIMULTANEOUS 4.5KW ELEMENTS. MOUNT UNIT AS HIGH AS POSSIBLE ABOVE THE UTILITY SINK USING A HOLDRITE MODEL #50-SWP-W-C EQUIPMENT PLATFORM WITH INTEGRAL PAN ASSEMBLY. DISCHARGE PAN TO UTILITY SINK. PROVIDE SUPPLEMENTAL R-6 INSULATION WITH REINFORCED REFLECTIVE SERVICE JACKET. PROVIDE EXPANSION TANK WITH APPURTENANCES AS DETAILED.
- ⑥ EXTEND ½" PIPING THROUGH ASEE 1024 BACKFLOW PREVENTER AND ISOLATION VALVE TO WATER FILTER. EXTEND FROM WATER FILTER ½" FOOD GRADE REINFORCED PVC TUBING TO ICE MAKER. EXTEND ICE MAKER DRAIN TO FLOOR SINK.
- ⑦ EXTEND BUILDING DRAIN TO EXISTING SEPTIC TANK/DRAIN FIELD SYSTEM.
- ⑧ PROVIDE GREASE INTERCEPTOR WITH EXTENSION RINGS AND MAN HOLE COVERS.
- ⑨ EXTEND 2" VENT FROM EACH CHAMBER OF THE GREASE TRAP TO THE WALL THEN TURN UP WITH GOOSE NECK DISCHARGE 12" ABOVE GRADE.



PLUMBING PLAN
SCALE: 1/4" = 1'-0"

JERRY L. HICKS
8326

REVISIONS
NO. DATE:

ELECTRICAL
SPECIFICATIONS
AND SCOPE OF WORK

FELLOWSHIP HALL
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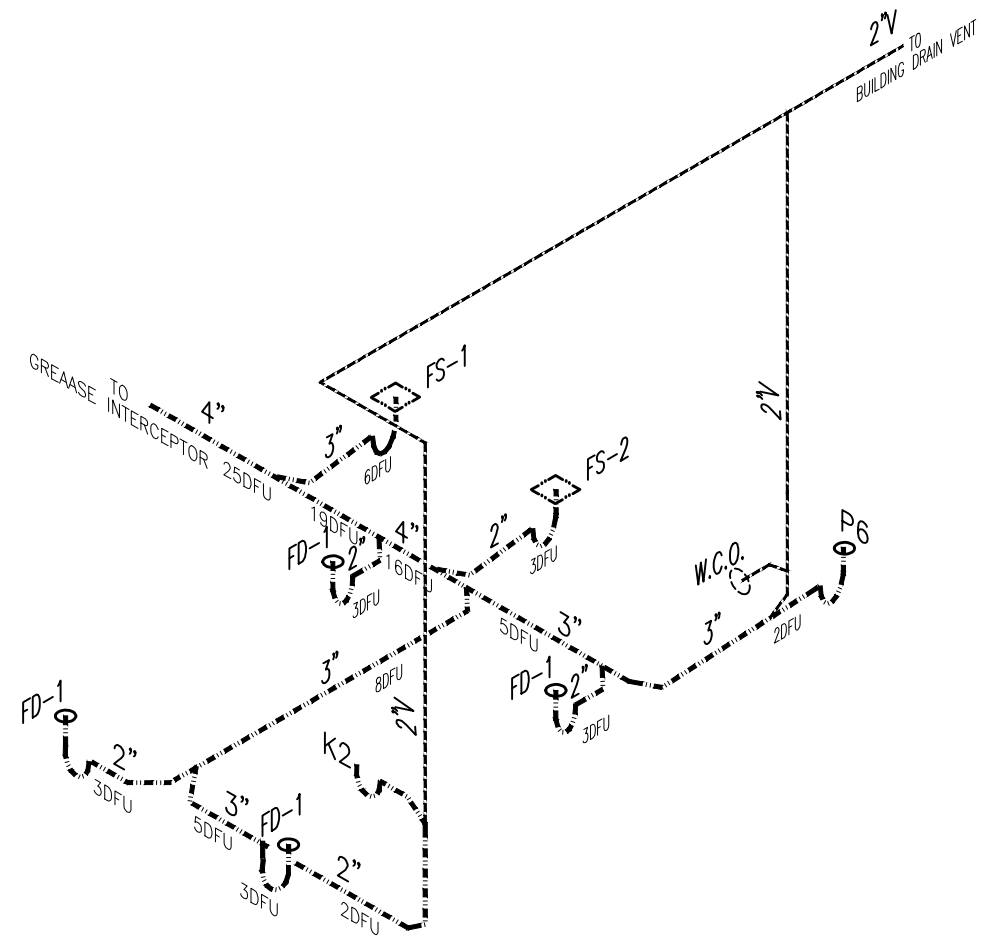
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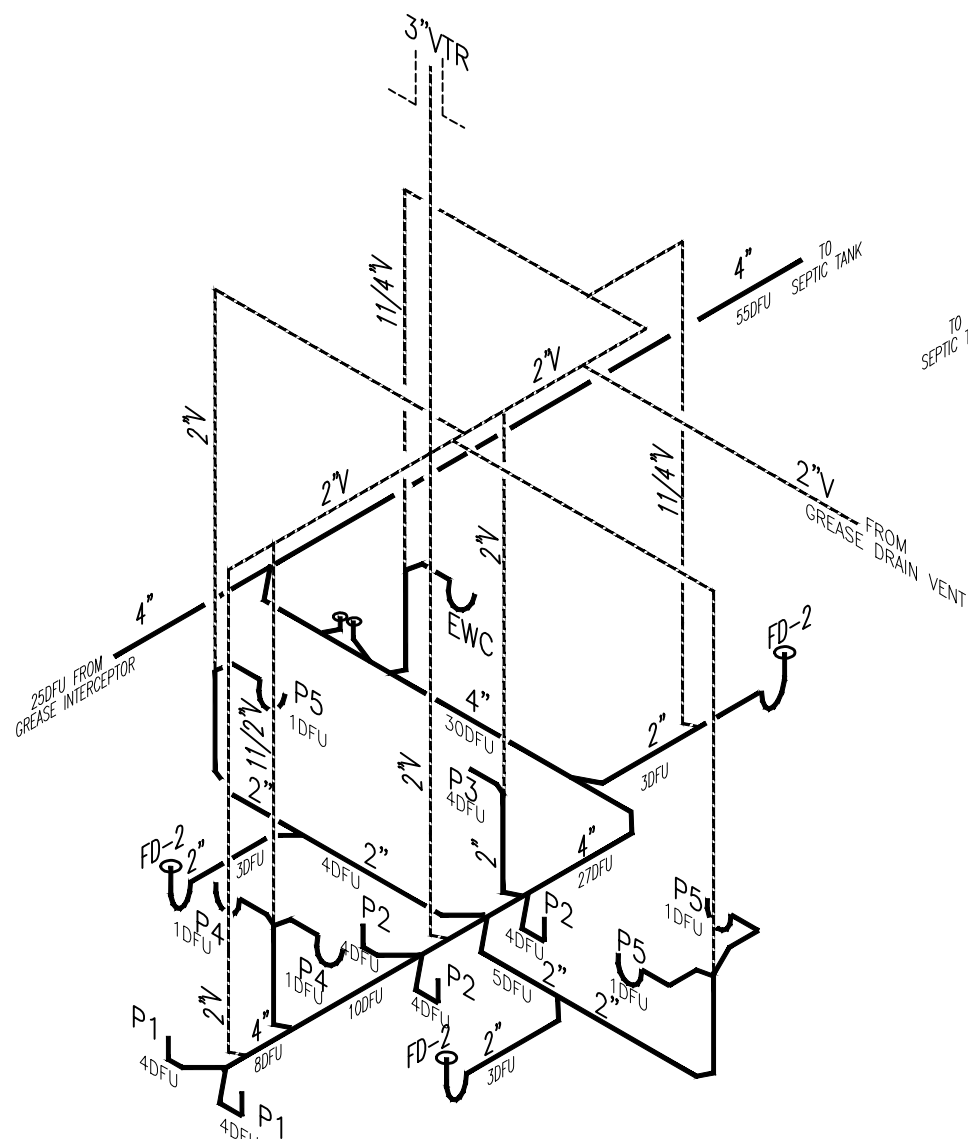
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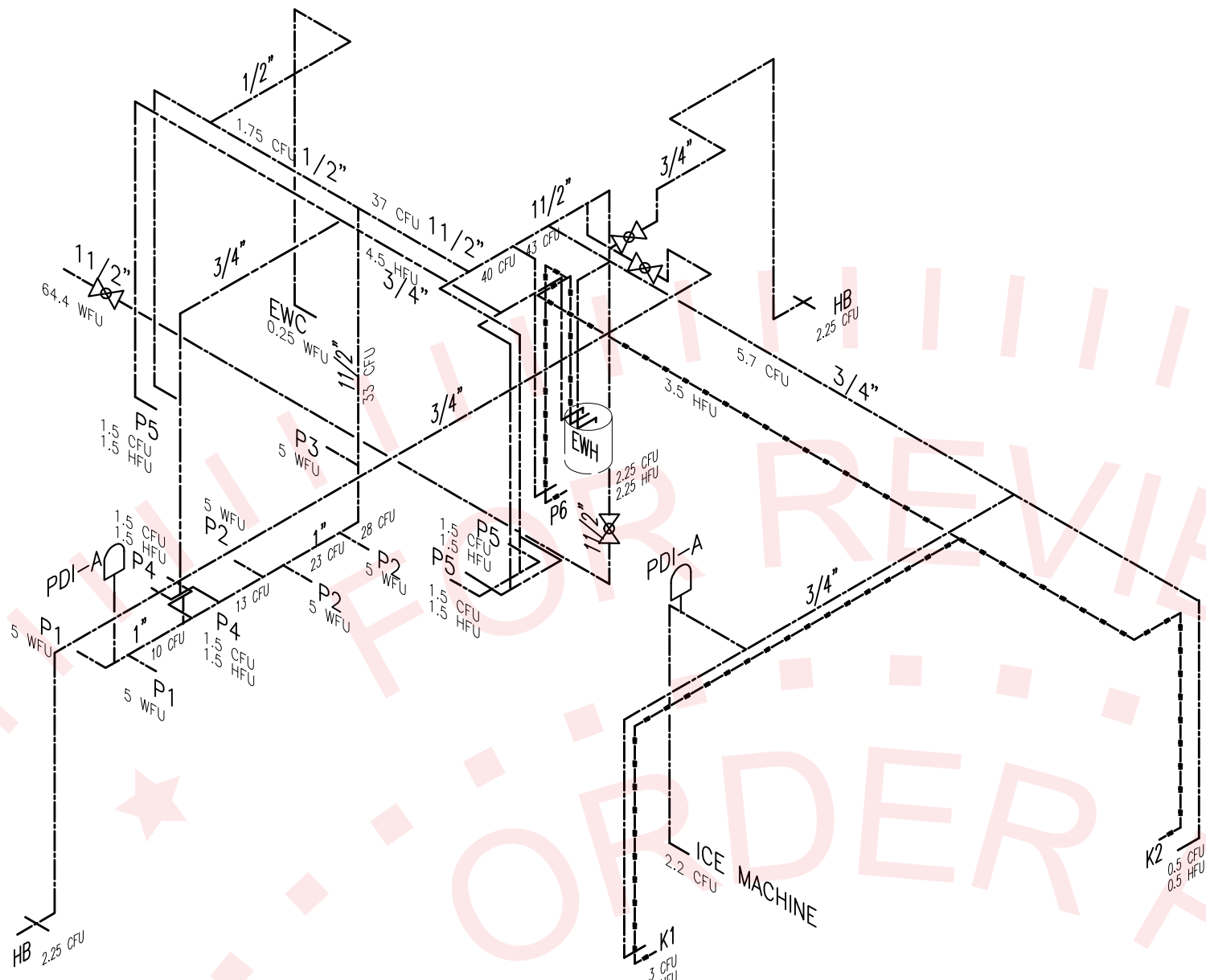
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P2



GREASE DRAIN RISER
SCALE: NONE

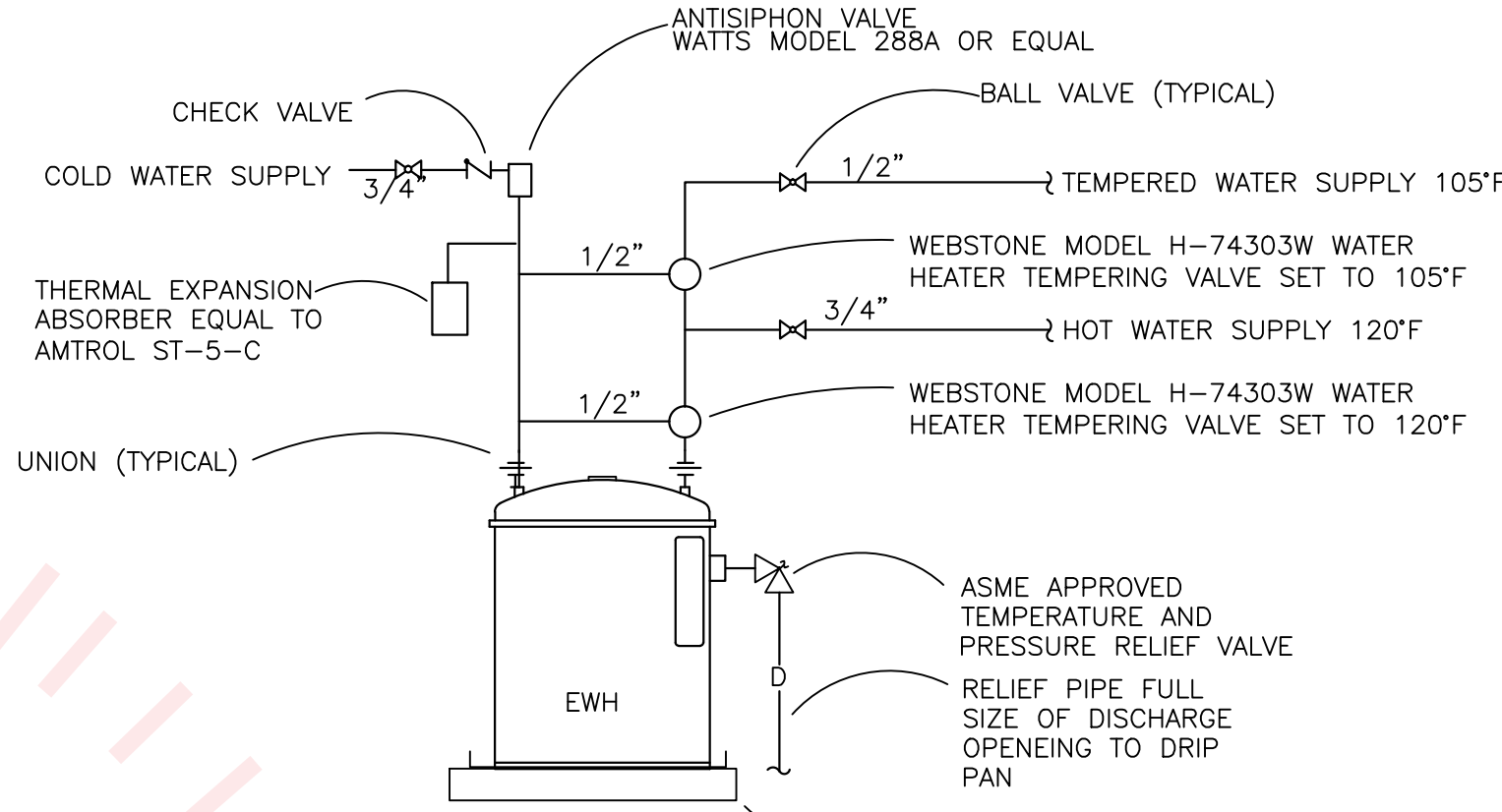


BUILDING DRAIN RISER
SCALE: NONE



POTABLE WATER RISER
SCALE: NONE

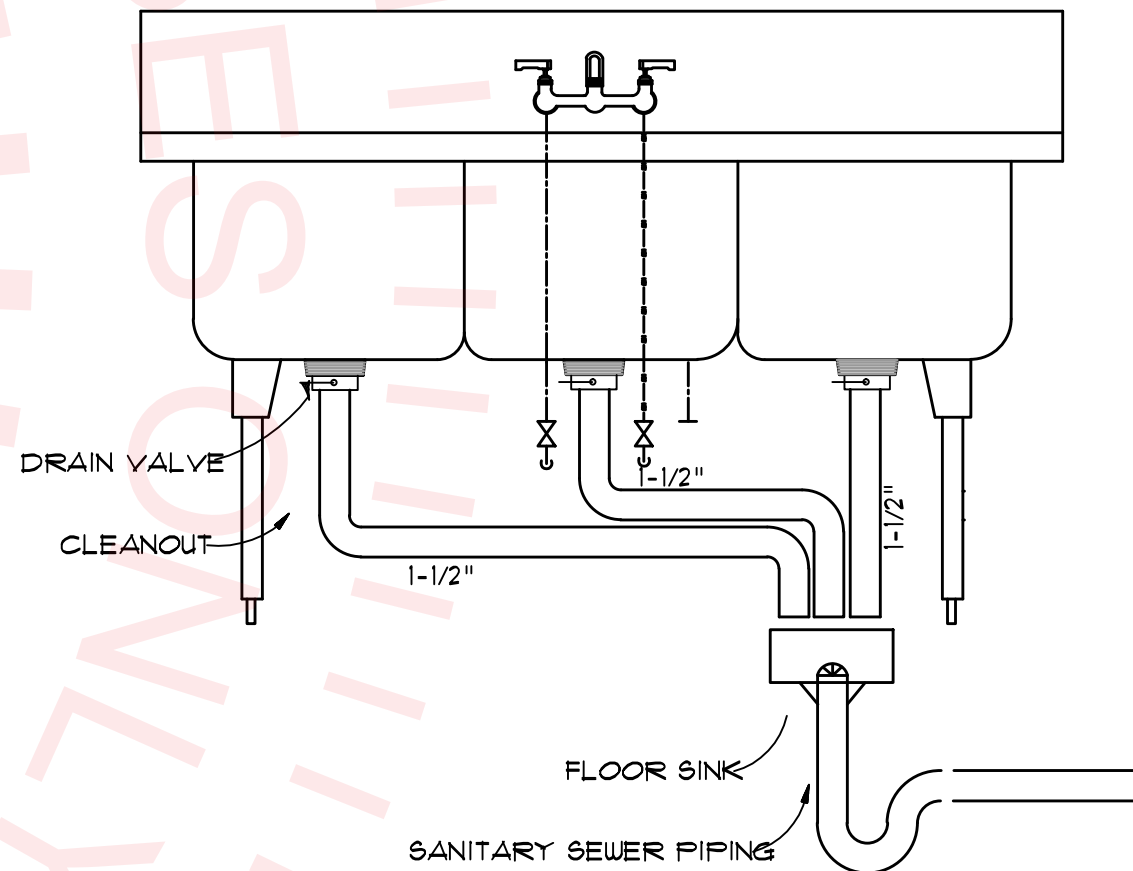
FIXTURE CONNECTION SCHEDULE							
MARK	FIXTURE	RIM HEIGHT	COLD WATER		HOT WATER		SOIL / WASTE
			BRANCH	CONN.	BRANCH	CONN.	BRANCH
P1	WATER CLOSET (ADA)	18"	1/2"	3/8"	--	--	4"
P2	WATER CLOSET	15"	1/2"	3/8"	--	--	4"
P3	URINAL	17"	3/4"	3/4"	--	--	2"
P4	LAVATORY (H.C.)	34"	1/2"	3/8"	1/2"	3/8"	2"
P5	LAVATORY (H.C.)	C.I.	1/2"	3/8"	1/2"	3/8"	2"
P6	MOP SINK	--	1/2"	3/8"	1/2"	3/8"	3"
K1	3 COMP SINK	--	1 1/2"	3/8"	1 1/2"	3/8"	2"
K2	HAND WASH SINK	34"	1 1/2"	1 1/2"	1 1/2"	2"	11/4"



PROVIDE NOMINAL 50 GAL LOW PROFILE WATER HEATER WITH DUAL NON-SIMULTANEOUS 4.5KW @ 240V HEAT ELEMENTS. PROVIDE SUPPLEMENTAL R-6 INSULATION WITH REINFORCED REFLECTIVE SERVICE JACKET. UNIT MUST BE COMPLIANT WITH NAECA 2015, 0.95 EF. MOUNT WITH WALL RACK EQUAL TO HOLDRITE MODEL 50-SWHP-W-C. HOLD HIGH AS POSSIBLE.

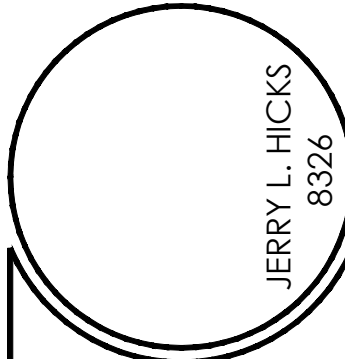
ELECTRIC WATER HEATER DETAIL

SCALE: NONE



3-COMPARTMENT SINK DETAIL

SCALE: NONE



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ELECTRICAL
SPECIFICATIONS
AND SCOPE OF WORK

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ELECTRICAL SPECIFICATIONS

1. GENERAL

A. SCOPE OF WORK

1. THE ELECTRICAL WORK SHALL COVER THE FURNISHING OF ALL EQUIPMENT AND MATERIALS, THE PERFORMANCE OF ALL LABOR AND SUPERVISION, AS NECESSARY FOR THE SATISFACTORY COMPLETION OF ALL WORK REQUIRED FOR OPERATION AS DESCRIBED HEREIN AND/OR AS SHOWN ON THE DRAWINGS. PAY FOR ALL REQUIRED LICENSE FEES AND INSPECTIONS. INCLUDE ALL AUXILIARIES AND ACCESSORIES FOR A COMPLETE ELECTRICAL SYSTEM WHETHER SPECIFIED HEREIN OR SHOWN ON DRAWINGS OR NOT. ALL WORK SHALL BE DONE IN A NEAT AND WORKERLIKE MANNER.

2. THE CONTRACTOR SHALL VISIT THE PREMISES TO THOROUGHLY FAMILIARIZE HIMSELF/HERSELF WITH ALL THE DETAILS OF THE WORK AND WORKING CONDITIONS, TO VERIFY ALL DIMENSIONS IN THE FIELD, AND SHALL ADVISE THE ARCHITECT/ENGINEER IN WRITING OF ANY DISCREPANCY BEFORE PERFORMING ANY WORK. NO CHANGES WILL BE ALLOWED IN CONTRACT PRICE FOR WORK REQUIRED TO COORDINATE WITH EXISTING CONDITIONS. CONTRACTOR SHALL IDENTIFY ANY AND ALL DEFICIENCIES, DISCREPANCIES, QUESTIONS OR EXCEPTIONS IN HIS/HER BID. ALL WORK REQUIRED TO CORRECT DEFICIENCIES, DISCREPANCIES OR QUESTIONS NOT INCLUDED WITH THE BID SHALL BE REQUIRED AS WORK INCLUDED, AND NO CHANGE IN CONTRACT PRICE WILL BE ALLOWED.

3. THE DRAWINGS ARE TO BE CONSIDERED DIAGRAMMATIC, NOT NECESSARILY SHOWING IN DETAIL OR TO SCALE ALL ACCESSORIES OR OTHER MINOR ITEMS. UNLESS SPECIFIC DIMENSIONS ARE SHOWN, THE STRUCTURAL, ARCHITECTURAL AND SITE CONDITIONS SHALL GOVERN THE EXACT LOCATIONS. THE CONTRACTOR SHALL FOLLOW DRAWINGS IN LAYING OUT WORK. CHECK DRAWINGS OF ALL TRADES TO VERIFY SPACES IN WHICH WORK WILL BE INSTALLED AND SHALL MAINTAIN MAXIMUM HEAD ROOM, OR SPACE CONDITIONS AT ALL POINTS. WHERE HEAD ROOM OR SPACE CONDITIONS APPEAR INADEQUATE, ARCHITECT/ENGINEER SHALL BE NOTIFIED BEFORE THE CONTRACTOR PROCEEDS WITH INSTALLATION. THIS CONTRACTOR SHALL, WITHOUT EXTRA CHARGE, MAKE FIELD MODIFICATION IN LAYOUT AS NEEDED TO PREVENT CONFLICT WITH WORK OF OTHER TRADES OR FOR PROPER EXECUTION OF THE WORK.

4. EXAMINE ALL DRAWINGS CAREFULLY PRIOR TO SUBMITTING A BID. THE CONTRACTOR WILL BE REQUIRED TO FURNISH, INSTALL AND MAINTAIN ALL ELECTRICAL EQUIPMENT, MATERIALS AND ITEMS SHOWN ON ANY OF THE ARCHITECTURAL, PLUMBING, OR AIR CONDITIONING DRAWINGS WITHOUT ADDITIONAL EXPENSE TO THE OWNER. IF DISCREPANCIES, CONFLICTS, INTERFERENCE OR OMISSIONS OCCUR BETWEEN DRAWINGS, NOTIFY THE ENGINEER IN WRITING PRIOR TO FINAL PRICE NEGOTIATION.

5. ALL MATERIALS SHALL BE NEW AND FREE FROM DEFECTS UNLESS NOTED OTHERWISE OR AS DIRECTED BY THE OWNER'S REPRESENTATIVE. INSTALL MATERIALS AND EQUIPMENT IN A NEAT AND FIRST CLASS WORKERLIKE MANNER.

B. CODES:

1. THE INSTALLATION SHALL COMPLY WITH CODES LISTED IN ELECTRICAL NOTE 1, SHEET E102.

C. PRIOR APPROVAL:

1. REQUESTS FOR PRIOR APPROVAL OF EQUAL PRODUCTS OR MATERIALS SHALL BE MADE TO THE ENGINEER AT LEAST 10 DAYS PRIOR TO FINAL BID OR PRICE NEGOTIATION TO ASSURE ADEQUATE TIME FOR EVALUATION OF THE REQUEST AND TO NOTIFY ALL BIDDERS OF THE SUBSTITUTIONS. ALL SUBSTITUTED EQUIPMENT THAT CANNOT MEET SPACE REQUIREMENTS SHALL BE REPLACED AT THE CONTRACTOR'S EXPENSE. ANY MODIFICATIONS RESULTING FROM SUBSTITUTIONS SHALL BE MADE AT THE CONTRACTOR'S EXPENSE. SUBSTITUTIONS DURING CONSTRUCTION WILL NOT BE ALLOWED.

D. SUBMITTALS:

1. ALL EQUIPMENT AND MATERIALS SHALL BE NEW AND OF BEST GRADE. REVIEW OF ALL EQUIPMENT AND MATERIALS FROM THE ARCHITECT AND ENGINEER SHALL BE OBTAINED BY THE CONTRACTOR BEFORE ANY EQUIPMENT IS ORDERED. THE CONTRACTOR SHALL CHECK DATA CAREFULLY TO INSURE COMPLIANCE WITH THESE SPECIFICATIONS PRIOR TO SUBMITTING. SUBMITTAL DATA SHALL BE SUBMITTED AT ONE TIME IN ONE PORTABLE DATA FORMAT (PDF) ELECTRONIC FILE. DATA SHALL INCLUDE COMPLETE INSTALLATION INSTRUCTIONS, DIMENSIONAL DATA, AND ELECTRICAL DATA, AS APPLICABLE. THE FILE SHALL CONTAIN A COVER SHEET WITH PROJECT TITLE, ARCHITECT, ENGINEER, AND CONTRACTOR'S NAMES AND CONTACT INFORMATION. THE SUBMITTAL SHALL INCLUDE AN INDEX SHEET LISTING THE TAB NUMBERS. SUBMITTALS NOT PREPARED AS SPECIFIED HEREIN WILL BE RETURNED TO THE CONTRACTOR FOR RESUBMITTALS. SUBMITTAL DATA SHALL BE PREPARED TO INCLUDE THE FOLLOWING: LIGHTING FIXTURES, POWER PANELS, CIRCUIT BREAKERS, TVSS'S, FIRE ALARM SYSTEM COMPONENTS, LIGHTNING PROTECTION SYSTEM AND COMPONENTS, RECEPTACLES, SWITCHES, COVER PLATES.

E. TESTS:

1. PRIOR TO ENERGIZATION, CHECK CONTINUITY OF EACH POWER AND LIGHTING CONDUCTOR. PRIOR TO ENERGIZATION OF EACH POWER AND LIGHTING CONDUCTOR, CHECK PHASING AND INSULATION RESISTANCE OF UNGROUNDED CONDUCTORS. INSULATION OPEN CIRCUIT RESISTANCE SHALL BE MINIMUM OF 20 MEGOHMS (CONDUCTOR TO CONDUCTOR AND CONDUCTOR TO GROUND) WHEN MEASURED WITH 500 VDC MEGGER. REPLACE ANY WIRE/CABLE MEASURED BELOW THAT LEVEL.

2. THE CONTRACTOR SHALL TEST ALL WORK AND EQUIPMENT AS DIRECTED BY THE ENGINEER AND BY AUTHORITIES HAVING JURISDICTION. THE ENTIRE INSTALLATION SHALL BE TESTED FOR SHORTS, GROUNDS AND OPEN CIRCUITS. ALL DEFECTS SHALL BE DEMONSTRATED TO BE IN PROPER WORKING AND OPERATING CONDITION TO THE COMPLETE SATISFACTION OF THE ENGINEER.

3. UPON COMPLETION OF THE WORK, THE SYSTEM SHALL BE TESTED FOR CONTINUITY AND GROUND (10 OHMS MAXIMUM) AND SHALL BE PROVEN TO PROVIDE COMPLETE SATISFACTORY OPERATION IN ACCORDANCE WITH THE REQUIREMENTS OF THE DRAWINGS AND SPECIFICATIONS.

F. GUARANTEE:

1. THE CONTRACTOR SHALL GUARANTEE THE ENTIRE INSTALLED ELECTRICAL SYSTEMS AND SHALL REPLACE WITHOUT ADDITIONAL CHARGE ALL WORK WHICH MAY BE DEFECTIVE OR DAMAGED OTHER THAN ORDINARY WEAR AND TEAR FOR A PERIOD OF ONE YEAR FROM THE DATE OF FINAL ACCEPTANCE.

2. MATERIALS:

A. WIRE AND CABLES:

1. CONDUCTORS SHALL BE COPPER, TYPE THWN OR THHN, UNLESS NOTED OTHERWISE ON THE DRAWINGS.

2. THE NEUTRAL CONDUCTORS SHALL HAVE AN IDENTIFYING WHITE COVERING. EACH CIRCUIT SHALL HAVE A SEPARATE NEUTRAL CONDUCTOR AND A SEPARATE GROUND CONDUCTOR. CONDUCTORS SHALL BE COLOR CODED CONSISTENT WITH EXISTING COLOR CODES IN THE FACILITY. MAINTAIN THE SAME COLORS THROUGHOUT THE JOB.

3. MINIMUM WIRE SIZE FOR CONTROL WIRING SHALL BE #14 OR AS APPROVED BY ENGINEER. MINIMUM WIRE SIZE FOR LIGHTING AND RECEPTACLE CIRCUITS SHALL BE #12, UNON.

4. ALL TAPS AND SPLICES IN NO. 8 OR SMALLER BRANCH CIRCUITS SHALL BE FASTENED TOGETHER BY MEANS OF "WIRENUT" CONNECTORS (IDEAL OR APPROVED SUBSTITUTION). ALL TAPS AND SPLICES IN BRANCH CIRCUITS LARGER THAN NO. 8 SHALL BE MADE WITH SCREW TYPE FITTINGS AND TAPED TO PROVIDE INSULATION EQUAL TO WIRE INSULATION.

B. RACEWAYS:

1. ALL ELECTRICAL CONDUCTORS (LIGHTING AND POWER) SHALL BE INSTALLED IN RACEWAYS, UNLESS OTHERWISE NOTED. MC CABLE, INCLUDING SOUTHWIRE LOW VOLT CABLE, IS ALLOWED WHERE IT CAN BE CONCEALED WITHIN WALLS AND/OR ABOVE CEILING. LOW VOLTAGE CONDUCTORS (CONTROL, FIRE ALARM SYSTEM, AND COMMUNICATIONS WIRING) MAY BE FREE WIRED (NOT IN CONDUIT).

2. MINIMUM TRADE SIZE: 1/2" EXCEPT FOR MC CABLE.

3. ALL CONDUIT UNDER THE SLAB OR IN SLAB SHALL BE GALVANIZED RIGID METAL CONDUIT (RMC) OR SCHEDULE 40 PVC. OTHER CONDUITS RUN ABOVE THE CONCRETE SLAB, AND IN DRY LOCATIONS, MAY BE THIN WALL (EMT) CONDUIT.

4. INSTALL EACH ENTIRE CONDUIT SYSTEM COMPLETE BEFORE PULLING IN ANY CONDUCTORS. CLEAN THE INTERIOR OF EVERY RUN OF CONDUIT BEFORE PULLING IN ANY CONDUCTORS TO GUARD AGAINST OBSTRUCTION AND OMISSIONS.

5. CUT ALL JOINTS SQUARE AND REAM SMOOTH. COAT CUTS, THREADS OR SCRATCHES ON STEEL CONDUIT WITH AN APPROVED ZINC CHROMATE OR WITH A 90 PERCENT ZINC PAINT. WHEN DRY, DRAW UP TIGHT.

6. MAKE BENDS WITH STANDARD ELLS OR CONDUIT BENT TO NOT LESS THAN SAME RADIUS. BENDS MUST BE FREE FROM DENTS OR FLATTERINGS. USE NO MORE THAN THE EQUIVALENT OF FOUR 90 DEGREE BENDS IN ANY RUN BETWEEN TERMINALS AND CABINETS, OR BETWEEN OUTLETS AND JUNCTION BOXES OR PULL BOXES.

7. CONDUIT SHALL BE SECURELY FASTENED TO ALL OUTLET BOXES OR CABINETS WITH DOUBLE LOCKNUTS AND BUSHINGS OF APPROVED MAKE. INSULATED BUSHINGS SHALL BE USED FOR ALL CONDUITS 1-1/4" OR LARGER.

8. GROUND CONNECTIONS AND BONDING SHALL BE INSTALLED AS REQUIRED BY NEC 250.50.

C. OUTLETS:

1. PROVIDE GALVANIZED STEEL BOXES OF SUFFICIENT SIZE TO ACCOMMODATE WIRING DEVICES TO BE INSTALLED. IF NEEDED, PROVIDE AN EXTENSION RING FOR THE DEVICE TO BE INSTALLED. SQUARE OR RECTANGULAR BOXES MAY BE SUPPLIED. UNLESS OTHERWISE NOTED, PROVIDE 1 1/2" x 4" BOX.

2. CEILING OUTLETS SHALL BE INSTALLED AS SHOWN ON THE DRAWINGS AND SHALL BE PROPERLY CENTERED AND SPACED. CEILING OUTLET BOXES, UNLESS OTHERWISE INDICATED ON THE DRAWINGS, SHALL BE 4" OCTAGON, OR AS REQUIRED, GALVANIZED AND NOT LESS THAN 1-5/8" DEEP.

3. SYMBOLS ON DRAWINGS AND MOUNTING HEIGHTS AS INDICATED ON DRAWINGS AND SPECIFICATIONS ARE APPROXIMATE ONLY. THE EXACT LOCATIONS AND MOUNTING HEIGHTS MUST BE DETERMINED ON THE JOB AND IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO COORDINATE WITH THE ARCHITECT AND ARCHITECTURAL DRAWINGS, AS WELL AS THESE DRAWINGS, TO SECURE THE CORRECT INSTALLATION.

D. DEVICES:

1. GENERAL PURPOSE WALL SWITCHES AND RECEPTACLES SHALL BE INSTALLED SINGLE OR GANGED FOR THE OUTLET CONTROL INDICATED ON THE DRAWINGS. SWITCHES AND RECEPTACLES SHALL BE HEAVY DUTY COMMERCIAL GRADE WIRING DEVICES, AND SHALL BE HIGH-IMPACT WHITE NYLON OF THE ENCLOSED FLUSH QUIET TYPE. FOR 120 VOLT SERVICE AND EQUAL TO LEVITON, HUNT, OR HUBBELL.

2. LIGHTING SWITCHES SHALL BE AS SHOWN ON SHEET E201, COMPLYING WITH FBC-EC C405.2.1 OCCUPANT SENSOR CONTROLS AND FBC-EC C405.2.2 TIME-SWITCH CONTROLS.

3. OCCUPANCY SENSOR SWITCHES TO BE LEGRAND WATTSOPPER TECHNOLOGY, WALL OR CEILING-MOUNTED SWITCHES DW-100/DW-100-347/DW-200, OCCUPANCY SENSORS DT-355, AND UNIVERSAL VOLTAGE POWER PACK BZ-50. ALTERNATES WILL BE REVIEWED BY THE ENGINEER IAW SPEC I.C.

E. WALL PLATES:

1. ALL SURFACE AND FLUSH MOUNTED SWITCH AND RECEPTACLE OUTLET PLATES SHALL BE HIGH-IMPACT WHITE NYLON.

F. LIGHTING FIXTURES:

1. THE ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL LIGHTING FIXTURES FOR ALL OUTLETS AT LOCATIONS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH THE LIGHTING FIXTURE SCHEDULE.

G. LEDS:

1. THE ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL NEW LED'S AS SHOWN ON THE LIGHTING FIXTURE SCHEDULE, AS REQUIRED FOR ALL FIXTURE TYPES, AND FOR COMPLETE INSTALLATION READY FOR OPERATION.

H. GROUNDING:

1. PROVIDE GROUNDING OF THE NEW BUILDING IN ACCORDANCE WITH ELECTRICAL NOTE 5, SHEET E102. GROUNDING SHALL COMPLY WITH NEC 250.50. ALL CIRCUITS SHALL CONTAIN A SEPARATE GROUNDING CONDUCTOR.

2. EC SHALL AUGMENT/SUPPLEMENT THE EXISTING GROUNDS AT THE EXISTING SANCTUARY AND ROBERTS HALL, IAW NEW WORK NOTES 2, 4, 9 AND 10 ON SHEET E103.

3. ALL STRUCTURAL STEEL, FRAMING AND MISCELLANEOUS STEEL SHALL BE MADE ELECTRICALLY CONTINUOUS THROUGH CONSTRUCTION. THE ELECTRICAL CONTRACTOR SHALL ASSURE THIS CONTINUOUS GROUNDING.

I. COMMUNICATIONS/DATA (C/D) SERVICE:

1. PROVIDE DEMARK INTERFACES AT NITS, ONE FOR A SATELLITE TV PROVIDER AND ONE FOR A TELEPHONE PROVIDER.

2. PROVIDE C/D OUTLETS WHERE SHOWN IN THE DRAWINGS. C/D OUTLET IS TO BE A STANDARD DEVICE BOX WITH TWO PORTS AS SHOWN ON THE LEGEND SYMBOL WITH 3/4" EMT, WITH PULL LINE TO 6" ABOVE CEILING. TURN TOWARD NIT. PROVIDE NYLON BUSHING ON END OF CONDUIT TO PROTECT WIRING. CAT 6 CONDUCTORS ARE TO BE INSTALLED BY THIS EC FROM C/D OUTLET TO NIT.

3. PROVIDE TV OUTLETS WHERE SHOWN ON DRAWING E202. OUTLET IS TO BE IAW LEGEND SYMBOL. R56 CONDUCTORS ARE TO BE INSTALLED BY THIS EC FROM TV OUTLET TO NIT.

J. PANEL BOARDS AND CIRCUIT BREAKERS:

1. POWER PANEL F SHALL BE CIRCUIT BREAKER TYPE, SQUARE D, SIEMENS OR CUTLER-HAMMER FOR 240/120 VOLTS, SINGLE PHASE. PANEL F SHALL HAVE COPPER BUS. PANEL BOARD SHALL BE LOCKABLE.

2. REPLACE FAULTY SURGE ARRESTOR (TVSS) AT THE ELECTRICAL UTILITY SERVICE AT THE EXISTING SANCTUARY.

3. PANEL BOARDS AND CIRCUIT BREAKERS SHALL BE BY THE SAME MANUFACTURER THROUGHOUT AND SHALL BE QUICK-MAKE, QUICK-BREAK, TRIP-FREE, THERMAL MAGNETIC TRIP, AUTOMATIC TRIPPING INDICATED BY HANDLE AT MID-POINT POSITION. MULTI-POLE BREAKERS TO HAVE COMMON TRIP.

4. ALL CIRCUIT BREAKERS SERVING HVAC EQUIPMENT SHALL BE HACR RATED.

K. MOTORS:

1. PROVIDE DISCONNECTING MEANS FOR EACH MOTOR, FUSIBLE OR NON-FUSIBLE, AS REQUIRED ON PLANS.

L. HVAC EQUIPMENT:

1. WIRE UP COMPLETE ALL THE POWER REQUIREMENTS FOR A/C EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS. CONTROLS ARE TO BE INSTALLED BY THE MECHANICAL CONTRACTOR. ELECTRICAL CONTRACTOR SHALL COORDINATE WITH THE MECHANICAL CONTRACTOR TO ASSURE THAT ALL ELECTRICAL POWER IS PROVIDED AS REQUIRED FOR HVAC CONTROLS.

2. THE ELECTRICAL CONTRACTOR SHALL FURNISH AND INSTALL THE NECESSARY CONDUIT AND DISCONNECTING MEANS FOR HVAC EQUIPMENT. COORDINATE WITH MC TO ASSURE CORRECT SIZE AND NUMBER OF POLES BEFORE ORDERING CIRCUIT BREAKERS.

3. FINAL CONNECTIONS SHALL BE BY HVAC CONTRACTOR. COORDINATE REQUIREMENTS WITH HVAC CONTRACTOR.

M. FIRE ALARM SYSTEM:

1. PROVIDE/INSTALL AN EXTENSION OF THE EXISTING ALARM SYSTEM WITH REQUIRED INITIATING DEVICES, AHU SHUTDOWNS, SIGNALING DEVICES, AUTOMATIC TELEPHONE DIALER FOR REMOTE NOTIFICATION, BACK-UP BATTERIES AND CONTROL PANEL IN COMPLIANCE WITH NFPA 72, NATIONAL FIRE ALARM CODE. SEE RISER DIAGRAM ON SHEET E301.

2. PROVIDE CEILING MOUNTED SMOKE DETECTORS IN FRONT OF FIRE ALARM CONTROL PANEL (FCP), FIRE ALARM ANNUNCIATOR PANEL (FAAP) AND FIRE ALARM TERMINAL CABINET (FATC) IF LOGIC COMPONENTS ARE HOUSED WITHIN THESE ENCLOSURES.

3. PROVIDE CERTIFICATION AND SMOKE DETECTOR SENSITIVITY TEST AND REPORT, PER MANUFACTURER.

4. SMOKE DETECTORS SHALL BE INSTALLED A MINIMUM OF 36" AWAY FROM AIR DIFFUSERS, AND SHALL NOT BE PLACED IN LOCATIONS WHERE THE AIR FLOW WILL EXCEED THE LISTED AIR VELOCITY OF THE SMOKE DETECTOR.

5. ACTIVATION OF ANY MANUAL OR AUTOMATIC FIRE ALARM INITIATION DEVICE SHALL INITIATE A GENERAL ALARM THROUGHOUT ALL THREE BUILDINGS.

6. PROVIDE TVSS ON THE FIRE CONTROL PANEL AND ON ANY FIRE ALARM INITIATING OR SIGNALING CIRCUIT ENTERING/LEAVING THE BUILDING.

N. LIGHTNING PROTECTION SYSTEM

1. ADD ALTERNATE: INSTALL COMPLETE LIGHTNING PROTECTION SYSTEM PER SHEETS E401 AND E402.

3. EXECUTION

A. COORDINATE ELECTRICAL WORK WITH WORK OF OTHER TRADES TO AVOID ALL CONFLICTS.

B. ALL EQUIPMENT SHALL BE LABELED BY A NATIONALLY-RECOGNIZED ELECTRICAL TESTING LABORATORY.

C. RUN ALL RACEWAYS PARALLEL TO THE BUILDING STRUCTURE.

D. RACEWAYS IN DAMP OR WET LOCATIONS SHALL BE RIGID METAL CONDUIT (RMC).

E. SIZE AND INSTALL RACEWAYS PER NEC EXCEPT AS NOTED HEREIN. ALL CONDUCTORS SHALL BE INSTALLED IN APPROVED RACEWAYS, UNON.

F. SIZE ALL BOXES IN ACCORDANCE WITH NEC.

G. ON CONCEALED CONDUIT SYSTEMS, SET BOXES FLUSH WITH FINISHED SURFACES.

H. FASTEN ALL BOXES SECURE TO BUILDING STRUCTURE.

I. BOXES EXPOSED TO WEATHER OR MOISTURE SHALL BE CAST METAL WITH THREADED HUBS AND RATED NEMA 3R, MINIMUM.

J. INSTALL A NYLON PULL LINE 200 LB TEST IN ALL EMPTY CONDUITS.

K. UNLESS LOW-OR-NO FRICTION CONDUCTORS, SUCH AS SOUTHWIRE SIMPLU ARE USED, LUBRICATE ALL CONDUCTORS WITH LUBRICANT (IDEAL "YELLOW" OR EQUAL) PRIOR TO PULLING WIRE.

L. IDENTIFY CLEARLY ON A TYPE WRITTEN SCHEDULE FOR ALL CIRCUITS AND EQUIPMENT TO CORRESPOND WITH THE PLANS. ATTACH INSIDE THE PANEL INSTALLED OR AFFECTED BY THIS PROJECT.

M. INSTALL NO WIRING SMALLER THAN NO. 12 AWG EXCEPT FOR SIGNAL OR CONTROL WIRING.

N. RUN ALL WIRING CONTINUOUS THE ENTIRE LENGTH. CONDUCTORS SHALL CONTAIN NO WELDS OR SPLICES.

O. TERMINATE STRANDED CONDUCTORS WITH CODE-APPROVED BOLTED PRESSURE CONNECTORS, T&B SERIES S4000.

P. TERMINATE SOLID CONDUCTORS WITH BOLTED PRESSURE CONNECTORS, OR CODE-APPROVED SPRING CONNECTORS "SCOTCH-LOK", IDEAL, OR T&B "PIGGY".

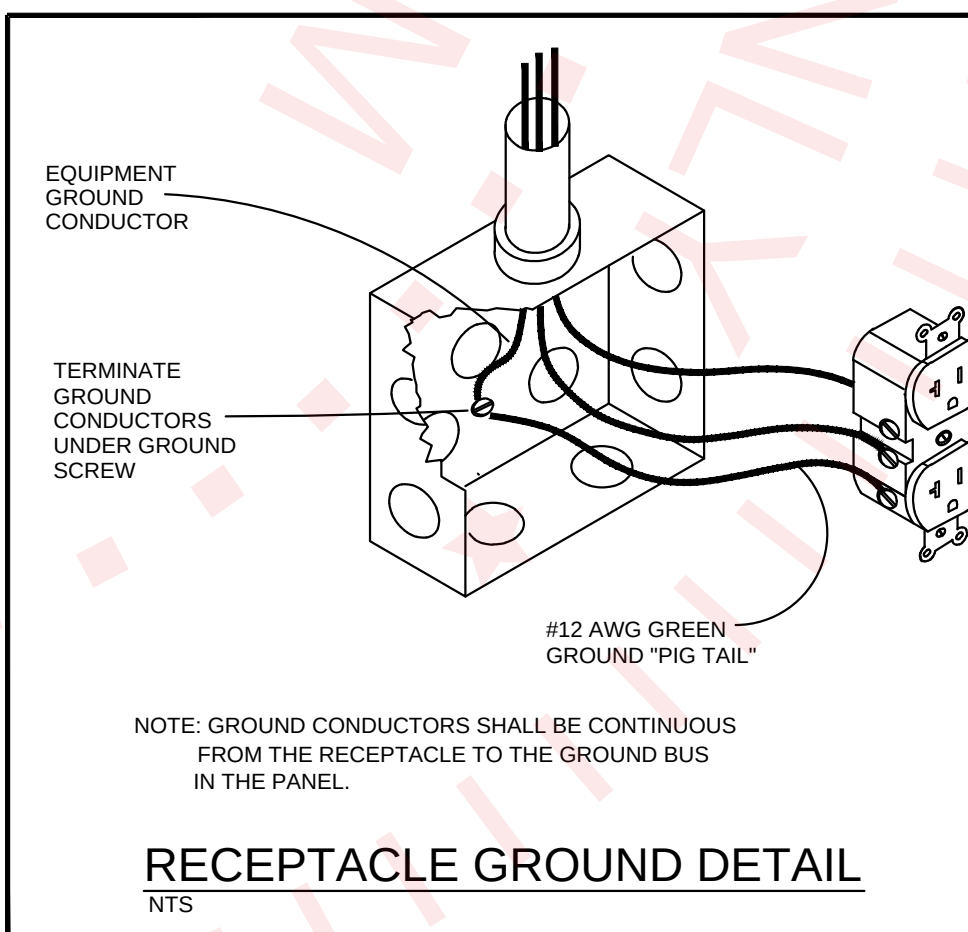
Q. CONNECT ALL CONTROLS, DISCONNECT SWITCHES, CIRCUIT BREAKERS, ETC., WHETHER FURNISHED UNDER THIS CONTRACT BY THE GENERAL CONTRACTOR, OTHER SUBCONTRACTORS, OR THE OWNER.

R. SUPPORT CONDUITS AS FOLLOWS:

1/2" AND 3/4" 5'-0" ON CENTERS
1" AND 1-1/4" 6'-0" ON CENTERS
1-1/2" AND LARGER 8'-0" ON CENTERS

S. RUN A GREEN GROUND WIRE FOR EACH CIRCUIT. BOND AT GROUND BUS AND TO EACH OUTLET BOX, APPLETON NO. SCR-1032 PLT1 GREEN HEAD GROUND SCREW WITH PIGTAIL.

T. AT COMPLETION OF JOB, THE ELECTRICAL CONTRACTOR SHALL GIVE THE OWNER AN AS-BUILT SET OF REPRODUCIBLE DRAWINGS SHOWING THE EXACT ELECTRICAL INSTALLATION OR ACCEPTABLE ELECTRONIC COPY OF "AS-BUILT" DRAWINGS IN "AUTOCAD 2004" OR LATER FORMAT.



ELECTRICAL SCOPE OF WORK

A. SEE ARCHITECTURAL DRAWINGS FOR BUILDING DIMENSIONS AND DETAILS. THESE ELECTRICAL DRAWINGS ARE DIAGRAMMATIC ONLY AND ARE NOT INTENDED TO BE SCALED.

B. PROVIDE DISTRIBUTION PANELS IAW ELECTRICAL RISER DIAGRAM, SHEET E301. PROVIDE POWER SERVICE TO POWER LOADS IAW SHEETS E103 AND E203.

C. COMPLETE LIGHTING INSTALLATIONS, IAW SHEETS E201 AND E202.

D. INSTALL C/D OUTLETS AND SATELLITE TV OUTLETS IN LOCATIONS SHOWN ON SHEET E203. CONDUCTORS ARE TO BE INSTALLED BY THIS EC FROM NITS.

E. EXTEND EXISTING SECURITY AND FIRE ALARM SYSTEM IAW FIA RISER ON SHEET E301 AND SPEC. 2.M. INSTALL FIRE ALARM DEVICES AS SHOWN ON SHEET E203. FIRE ALARM CONTRACTOR SHALL SUBMIT FOR A SEPARATE FIRE ALARM SYSTEM PERMIT.

F. ADD ALTERNATE: INSTALL COMPLETE LIGHTNING PROTECTIONS SYSTEM IN ACCORDANCE WITH PLAN ON SHEET E401 AND SPECIFICATIONS AND DETAILS ON SHEET E402. CONTRACTOR SHALL SUBMIT FOR A SEPARATE LIGHTNING PROTECTION SYSTEM PERMIT.

G. MODIFY AND UPGRADE GROUND SYSTEM IAW NEW WORK NOTES 2, 4, 9 AND 10 ON SHEET E103. REPLACE TVSS AT EXISTING SANCTUARY ELECTRIC UTILITY SERVICE.

H. FURNISH ALL DISTRIBUTION CONDUCTORS, RACEWAYS, CIRCUIT BREAKERS, DEVICES, AND OTHER ELECTRICAL HARDWARE TO PROVIDE COMPLETE ELECTRICAL SYSTEMS TO SERVE ALL LIGHTING, RECEPTACLE/POWER LOADS, AND SYSTEMS REQUIRED WITHIN THE SCOPE OF THIS PROJECT.

I. PROVIDE COMPLETE TESTING OF ALL ELECTRICAL SYSTEMS ON THIS PROJECT AND ASSURANCE THAT ALL EQUIPMENT AND SYSTEMS ARE FUNCTIONING PROPERLY. PROVIDE COMPLETE TEST OF FIRE ALARM SYSTEM. TAG FIRE ALARM SYSTEM PER FIRE MARSHAL AND NFPA 72 (2016), NATIONAL FIRE ALARM AND SIGNALING CODE.

ABBREVIATIONS

AC	AIR CONDITIONING UNIT	IMC	INTERMEDIATE METALLIC CONDUIT INCLUDED
A/E	ARCHITECT/ENGINEER	INCL	JUNCTION BOX
ADA	AMERICANS WITH DISABILITIES ACT	JB	THOUSAND CIRCULAR MILS
AFF	ABOVE FINISHED FLOOR	KCMIL	KILOVOLT AMPERE
AFG	ABOVE FINISH GRADE	KVA	KILOVOLT AMPERE
AHJ	AUTHORITY HAVING JURISDICTION	KW	KILOWATT
AHU	AIR HANDLING UNIT	LCD	LIQUID CRYSTAL DISPLAY
AIC	AMPS INTERRUPTING CAPACITY	LED	LIGHT EMITTING DIODE
AL	ALUMINUM	LP	LIGHTNING PROTECTION INSTITUTE
ARCH	ARCHITECT	MBS	MAINTENANCE BYPASS SWITCH
ATS	AUTOMATIC TRANSFER SWITCH	MC	MECHANICAL CONTRACTOR, OR
AV	AUDIO VISUAL	MCA	MINIMUM CIRCUIT AMPS
AWG	AMERICAN WIRE GAUGE	MDP	MAIN DISTRIBUTION PANEL
C/D	COMMUNICATIONS/DATA	MGB	MAIN GROUND BUS
C	CONDUIT	MH	METAL HALIDE
CB	CIRCUIT BREAKER	MLO	MAIN LUGS ONLY
CKT	CIRCUIT	MOCP	MAXIMUM OVERCURRENT PROTECTION
CL	CENTER LINE	MPE	MECHANICAL PLUMBING AND ELECTRICAL
C.O.T.	CITY OF TALLAHASSEE	MTS	MANUAL TRANSFER SWITCH
CJ	COPPER OR CONDENSING UNIT	NEUTRAL	NEUTRAL
DACT	DISTAL ALARM COMMUNICATIONS	NEC	NATIONAL ELECTRICAL CODE
	TRANSMITTER	NEMA	NATIONAL ELECTRICAL CODE
DC	DOOR CONTACT		
DWG	DRAWING		
EC	ELECTRICAL CONTRACTOR	NF	NON-FUSED
EF	EXHAUST FAN	NFPA	NATIONAL FIRE PROTECTION
EGS	EMERGENCY GENERATOR SYSTEM		ASSOCIATION
EM	EMERGENCY	NIC	NOT IN CONTRACT
EMT	ELECTRICAL METALLIC TUBING	NIT	NETWORK INTERFACE TERMINAL
ENT	ELECTRICAL NONMETALLIC TUBING	OAU	OUTSIDE AIR UNIT
E.P.	ELECTRONIC PROGRAM	OHE	OVERHEAD ELECTRIC
ERD	ELECTRICAL RISER DIAGRAM	OPCI	OWNER PROVIDED CONTRACTOR
EXIST	EXISTING		INSTALLED
FA	FIRE ALARM	PH	PHASE
FAAP	FIRE ALARM ANNUNCIATOR PANEL	PE	PHOTOELECTRIC CELL
FATC	FIRE ALARM TERMINAL CABINET	PVC	POLYVINYL CHLORIDE CONDUIT
FBC	FLORIDA BUILDING CODE	RMC	RIGID METAL CONDUIT
FBC-EC	FLORIDA BUILDING CODE- ENERGY CONSERVATION	RTU	ROOF TOP UNIT
FCP	FIRE ALARM CONTROL PANEL	SCH	SCHEDULE
FCU	FAN COIL UNIT	SP	SQUARE FEET
FFPC	FLORIDA FIRE PREVENTION CODE	SLC	SIGNALING LINE CIRCUIT
GF1	GROUND FAULT INTERRUPTER	SPD	SURGE PROTECTION DEVICE
GND, G	GROUND	TEC	TALQUIN ELECTRIC COOP
HACR	HEATING/AIR CONDITIONING-RATED	TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSER
HP/HPU	HEAT PUMP UNIT	TYP	TYPICAL
HP	HORSEPOWER	UGE	UNDERGROUND ELECTRIC
HZ	HERTZ	UL	UNDERWRITERS LABORATORY
IAW	IN ACCORDANCE WITH	UNON	UNLESS OTHERWISE NOTED
IBC	INTERNATIONAL BUILDING CODE	UPS	UNINTERRUPTIBLE POWER SUPPLY
IES	ILLUMINATED EXIT SIGN	V	VOLT (S)
		W	WATT
		WH	WATER HEATER
		WP	WEATHER PROOF
		XFMR	TRANSFORMER

INDEX OF ELECTRICAL SHEETS

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E102	ELECTRICAL NOTES, LEGEND AND FIXTURE SCHEDULE
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E201	ELECTRICAL LIGHTING PLAN
E202	DIMMING SYSTEM SCHEMATIC
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E301	ELECTRICAL RISER DIAGRAMS
E401	LIGHTNING PROTECTION SYSTEM
E402	LIGHTNING PROTECTION SPECIFICATIONS AND DETAILS

JERRY L. HICKS
8326

REVISIONS

NO. DATE:

ELECTRICAL
SPECIFICATIONS
AND SCOPE OF WORK

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PROJ. NO. 38022

SHEET NO.

E101

ELECTRICAL NOTES

1. ALL WIRING SHALL BE IN ACCORDANCE WITH THE 7TH EDITION (2020) OF THE FLORIDA BUILDING CODE BUILDING (FBC), THE 2017 EDITION OF THE FLORIDA ELECTRICAL CODE (FEC), THE 7TH EDITION (2018) OF THE FLORIDA FIRE PREVENTION CODE (FFPC), THE 2016 EDITION OF NFPA 72, NATIONAL FIRE ALARM AND SIGNALING CODE, ELECTRICAL SPECIFICATIONS ON SHEET E101, AND THE FOLLOWING GENERAL NOTES:
- A. ALL CONDUCTORS ARE TO BE COPPER. MC CABLE IS ALLOWED ON THIS PROJECT WHERE IT CAN BE CONCEALED INSIDE WALLS AND/OR ABOVE CEILINGS. FIRE ALARM SYSTEM CONDUCTORS, AND OTHER LOW VOLTAGE CONDUCTORS MAY BE RUN AS FREE-WIRED (NOT IN CONDUIT) . LIGHTING AND POWER CONDUCTORS SHALL BE INSTALLED IN RACEWAYS, UNON.
- B. ALL WIRING SHALL BE CONCEALED WITHIN WALLS OR ABOVE CEILINGS. SECURE RACEWAYS TO OVERHEAD WHEN INSTALLED ABOVE CEILINGS. DO NOT ALLOW RACEWAYS OR FREE WIRE TO BE LOOSE ABOVE CEILINGS. PROVIDE STRAP WITHIN 12" OF BEND AND AT SPACING PER SPECIFICATIONS SECTION 3.0E.
- C. ALL CONDUIT EXPOSED TO WEATHER SHALL BE RIGID METAL CONDUIT (RMC) OR PVC SCH. 40.
- D. NO CONDUIT IS TO BE INSTALLED ON EXTERIOR OF THE BUILDING.
- E. SETSCREW DIE CAST FITTINGS ARE ACCEPTABLE FOR INDOOR APPLICATIONS. COMPRESSION DIE CAST FITTINGS ARE REQUIRED FOR EXTERIOR SITUATIONS.
2. THE ELECTRICAL CONTRACTOR IS RESPONSIBLE FOR ACQUIRING PERMITS FOR THIS CONSTRUCTION AND FOR SCHEDULING APPROPRIATE INSPECTIONS DURING CONSTRUCTION WITH ALL AUTHORITIES HAVING JURISDICTION (AHJ).
3. THE ELECTRICAL CONTRACTOR SHALL FURNISH ALL MATERIALS AND LABOR TO COMPLETE THE ELECTRICAL LIGHTING, AND POWER SYSTEMS. GREEN SCHEDULING. ALL WORK SHALL BE CAREFULLY DONE BY CAPABLE, EXPERIENCED ELECTRICIANS. ALL ITEMS SHOWN ARE TO BE NEW AND SHALL BE PROVIDED BY THE CONTRACTOR, UNLESS SPECIFICALLY STATED OTHERWISE.
4. IT IS THE INTENTION OF THE DRAWINGS TO FULLY COVER ALL WORK AND MATERIALS FOR A COMPLETE FIRST CLASS ELECTRICAL INSTALLATION. ANY DEVICES SUCH AS PULL BOXES USUALLY EMPLOYED IN THIS CLASS OF WORK, THOUGH NOT SPECIFICALLY MENTIONED OR SHOWN ON THE DRAWINGS, MAY BE REQUIRED FOR THE SATISFACTORY COMPLETION OF THE WORK. SHALL BE FURNISHED AND INSTALLED AS PART OF THE CONTRACTOR'S TOTAL WORK.
5. CIRCUITS SHALL BE BONDED AND SHALL HAVE A GROUND CONDUCTOR TO PROVIDE ELECTRICAL CONTINUITY. PROVIDE INSULATED GROUND CONNECTION FOR THE ELECTRICAL PANEL F TO THE EXISTING GROUND BUS AT PANEL H. GROUND CONDUCTOR SIZE SHALL BE AS NOTED ON RISER DIAGRAM. GROUND SYSTEM SHALL BE CONNECTED TO CONCRETE-ENCASED ELECTRODES OF AT LEAST 20 FEET OF MINIMUM OF #5 REBAR OR #4 AWG BARE COPPER CONDUCTOR IN THE BUILDING'S CONCRETE FOUNDATION, IAW NEC 250.52(3). GROUND SYSTEMS SHALL BE BONDED TOGETHER. THIS INCLUDES THE REBAR OR COPPER CONDUCTOR IN THE FOUNDATION. THE GROUND BUS IN EXISTING PANEL H, UNDERGROUND WATER SERVICE PIPE (CONNECT WITHIN 5 FEET OF ENTRY OF PIPE INTO BUILDING), LIGHTNING PROTECTION GROUND RODS, AND OTHER GROUNDING MEDIUMS.
6. ANY CONDUIT PENETRATING THE ROOF OR WALLS SHALL HAVE WATER-TIGHT FLASHING, AND PAINT TO RESTORE PENETRATION TO ORIGINAL CONDITION AND APPEARANCE. ALL EXTERIOR ELECTRICAL EQUIPMENT SHALL BE WEATHERPROOF. ENCLOSURES FOR EXTERIOR APPLICATIONS ON ITEMS SUCH AS DISCONNECT SWITCHES, STARTERS, AND EXHAUST FANS SHALL BE NEMA 3R.
7. ELECTRICAL CONTRACTOR SHALL COORDINATE WITH MECHANICAL CONTRACTOR TO AVOID CONFLICTS BETWEEN LOCATIONS OF DIFFUSERS, FIRE ALARM DEVICES, AND LIGHTING FIXTURES. COORDINATE WITH OTHER TRADES TO PROVIDE ALL NEEDED ELECTRICAL SERVICE.
8. THE MECHANICAL CONTRACTOR SHALL PROVIDE ALL LOW VOLTAGE CONTROL WIRING FOR ALL HVAC SYSTEMS. ELECTRICAL CONTRACTOR SHALL PROVIDE POWER SOURCE FOR ALL HVAC POWER AND CONTROL SYSTEMS.
9. ELECTRICAL CONTRACTOR SHALL VERIFY ALL DOOR SWINGS WITH ARCHITECTURAL DRAWINGS BEFORE ROUGHING IN SWITCHES AND SHALL VERIFY EXACT LOCATION OF ALL MOTORS AND EQUIPMENT BEFORE ROUGHING IN.
10. ELECTRICAL CONTRACTOR SHALL VERIFY THE EXACT LOCATION OF COUNTER TOPS, BACKSPASHES AND EQUIPMENT LOCATIONS ON ARCHITECTURAL DETAIL AND/OR EQUIPMENT AND CASEWORK SHOP DRAWINGS AND SHALL ADJUST SPECIFIED MOUNTING HEIGHTS OF WALL OUTLETS & J-BOXES AS REQUIRED TO BEST SERVE EQUIPMENT. VERIFY WIRING REQUIREMENTS FOR ALL EQUIPMENT.
11. CONTRACTOR SHALL CHECK ALL LIGHTING FIXTURES INSTALLED ON THIS PROJECT FOR EXACT TYPE, MOUNTING AND SPACE REQUIRED BEFORE ROUGHING IN. SUPPORT OF ALL LIGHTING FIXTURES SHALL BE THE RESPONSIBILITY OF THIS ELECTRICAL CONTRACTOR. FIXTURES SHALL BE SUPPORTED FROM STRUCTURAL MEMBERS OF THE BUILDING, INDEPENDENT OF CEILING. PROVIDE PLASTER GRIPS AS NEEDED FOR LIGHTING FIXTURES.
12. ALL ELECTRICAL PANELS INSTALLED OR CHANGED BY THIS PROJECT ARE TO BE FURNISHED WITH TYPED DIRECTORIES, IN ACCORDANCE WITH THE SCOPE OF ELECTRICAL WORK OF THIS PROJECT. LABEL ALL ELECTRICAL EQUIPMENT (PANELS, DISCONNECT SWITCHES, AND SIMILAR) AS TO TITLE, ELECTRICAL CHARACTERISTICS (I.E. 240V/120 V, SINGLE PHASE), AND SOURCE OF POWER (I.E. FED FROM PANEL F, CIRCUIT 1, 3). (LABELS SHALL BE ON RIGID ACRYLIC, WHITE LETTERS ON BLACK BACKGROUND, MINIMUM 1/4" LETTERS).
13. ALL CONDUIT, DEVICES, AND FIXTURES SHALL BE SUPPORTED BY CLIPS OR STRAPS APPROVED FOR THE PURPOSE, AND IN ACCORDANCE WITH THE N.E.C.
14. ANY DAMAGE TO THE BUILDING OR EQUIPMENT BY ELECTRICAL WORKERS SHALL BE REPAIRED BY WORKERS SKILLED IN THIS TYPE OF WORK AT NO EXPENSE TO THE OWNER.
15. ARC FLASH WARNING LABELS REQUIRED ON ALL EQUIPMENT DESIGNATED IN THE 2011 NEC, ART. 110.16 ELECTRICAL EQUIPMENT, SUCH AS SWITCHBOARDS, PANEL BOARDS, INDUSTRIAL CONTROL PANELS, METER SOCKETS ENCLOSURES, AND MOTOR CONTROL CENTERS THAT ARE IN OR WITHIN DWELLING UNITS AND ARE LIKELY TO REQUIRE EXAMINATION, ADJUSTMENT, SERVICING OR MAINTENANCE WHILE ENERGIZED SHALL BE FIELD MARKED TO WARN QUALIFIED PERSONS OF POTENTIAL ELECTRIC ARC FLASH HAZARDS. THE MARKING SHALL BE LOCATED SO AS TO BE CLEARLY VISIBLE TO QUALIFIED PERSONS BEFORE EXAMINATION, ADJUSTMENT, SERVICING, OR MAINTENANCE OF THE EQUIPMENT. NEC 110.16.

ELECTRICAL LEGEND	
SYMBOL	DESCRIPTION
	RECESSED, SURFACE, OR WALL-MOUNTED LED FIXTURE. SEE LIGHTING FIXTURE SCHEDULE FOR TYPE AND MOUNTING. DIAGONAL CROSS-HATCH INDICATES INTEGRAL BATTERY TO PROVIDE 90 MINUTES OF BACK-UP LIGHTING FOR EGRESS.
	ILLUMINATED EXIT SIGN (IES). SEE LIGHTING FIXTURE SCHEDULE.
	FLUSH-MOUNTED SINGLE POLE SWITCH. COLOR PER SPEC. 2,D.1 AND 2,E.1. MOUNT AT 48" AFF TO BOTTOM. LEVITON 5521-21, 20A, 120V. INSTALL WITHIN 4" OF DOOR FRAME, LOCK-SET SIDE. SM = OCCUPANCY SENSOR CONTROL SWITCH. SDM = COMBINATION DIMMER AND MOTION SENSOR SWITCH. ST = TIME-SWITCH CONTROLS. SOT = COMBINATION DIMMER AND TIME-SWITCH CONTROLS.
	CEILING-MOUNTED OCCUPANCY SENSOR TO SWITCH LIGHTS ON WHEN THE SPACE IS OCCUPIED AND OFF WHEN THE PRE-SET TIMER COMPLETES ITS TIME.
	FLUSH-MOUNTED DUPLEX CONVENIENCE RECEPTACLE WITH GROUND. NEMA 5-20. COLOR PER SPEC. 2,D.1 AND 2,E.1. MOUNT AT 18" AFF TO BOTTOM OR AS NOTED. G = CONNECTED TO GROUND FAULT PROTECTED CIRCUIT. WP = WEATHERPROOF.
	DUPLEX RECEPTACLE, 44" AFF TO BOTTOM OR 6" ABOVE COUNTER. COLOR PER SPEC. 2,D.1 AND 2,E.1. TURN HORIZONTAL. WP = WEATHERPROOF. G = GFCI CIRCUIT.
	DOUBLE-DUPLEX RECEPTACLE, COLOR PER SPEC. 2,D.1 AND 2,E.1., 18" AFF TO BOTTOM OR AS NOTED. C = 6" ABOVE COUNTER.
	J-BOX TO SERVE PURPOSE SHOWN ON PLANS.
	DISCONNECT SWITCH. SIZED AND NUMBER OF POLES AS SHOWN OR TO CONTROL CONNECTED LOAD IN ACCORDANCE WITH NEC. SWITCH TO BE SQUARE D QUICK-MAKE, QUICK-BREAK TYPE, NEMA 3R FOR EXTERIOR.
	PHOTOCELL CONTROL FOR LIGHTING, ALR-AT-15. INSTALL AT ROOF LEVEL, FACING NORTH.
	ELECTRONIC TIMER, 7-DAY PROGRAMMABLE WITH BATTERY BACK-UP. INTERMATIC MODEL. INSTALL AT 5'-6" AFF, ADJACENT TO PANEL SERVING LOAD.
	HVAC UNIT. SEE MECHANICAL SHEETS FOR SPECIFICATIONS. COMPLY WITH INSTALLATION AND SPACING REQUIREMENTS OF NEC 110-26.
	EXHAUST FAN. SEE MECHANICAL SHEETS FOR SPECIFICATIONS.
	COMBINATION TV OUTLET AND POWER RECEPTACLE SERVED FROM CKT. AS SHOWN ON DRAWINGS. PROVIDE R6G COAX CONDUCTOR IN CONDUIT TO 6" ABOVE SUSPENDED CEILING. RECESS AT 60" AFF TO BOTTOM UNLESS NOTED OTHERWISE. RECEPTACLE TO BE RECESSED TYPE, SIMILAR TO LEVITON TYPE 689 OR 690, TO ALLOW FOR FUTURE SURFACE-MOUNT OF FLAT SCREEN TV. EC TO PRE-WIRE ALL TV AND CD OUTLETS WITH CAT6 CONDUCTORS FROM THE CID NIT. EC TO PRE-WIRE SATELLITE TV OUTLETS WITH R6G COAX CONDUCTORS FROM THE CABLE TV INTERFACE AT THE SATELLITE TV NIT TO EACH CABLE TV OUTLET. CAT 6 AND R6G CONDUCTORS MAY BE FINE WIRED (NOT IN CONDUIT) ABOVE SUSPENDED CEILINGS.
	COMMUNICATIONS/DATA (C/D) OUTLET IAW SPEC. 2.1.1. MOUNT AT 18" AFF TO BOTTOM, UON. PROVIDE 2" x 4" DEVICE BOX FOR TWO C/D PORTS. PROVIDE 3/4" C. TO 6" ABOVE CEILING IAW SPEC 2.1.1. EC TO PRE-WIRE ALL C/D OUTLETS WITH CAT 6 CONDUCTORS FROM THE C/D NIT.
	FIRE ALARM CONTROL PANEL (FCP). EXISTING IN MAIN BLDG. ELECTRICAL ROOM. SEE FIRE ALARM RISER DIAGRAM, SHEET E301. AND SPECIFICATION 2.M.
	FIRE ALARM ANNUNCIATOR PANEL. RECESS AT 5' AFF TO TOP.
	RECESSED FIRE ALARM MANUAL PULL STATION, 44" AFF TO BOTTOM.
	FIRE ALARM HORN/STROBE COMBINATION. RECESS AT 80" AFF TO BOTTOM OF DEVICE. MINIMUM 70 CANDELA OR AS NOTED. STANDARD RED COLOR. C EQUALS CEILING MOUNTED.
	FIRE ALARM STROBE ONLY. MINIMUM 70 CANDELA. RECESS AT 80" AFF TO BOTTOM OF DEVICE OR AS NOTED. STANDARD RED COLOR.
	ADDRESSABLE CEILING MOUNTED HEAT DETECTOR (HD).
	CEILING MOUNTED ADDRESSABLE SMOKE DETECTOR (SD).
	ELECTRICAL PANEL. NQOD, BOLT-ON, COPPER BUS. MOUNT AT 5' AFF TO TOP, OR AS NOTED. PER SPECIFICATION 2.J. BRANCH CIRCUITS MAY BE 10,000 AMPS RMS. SEE PANEL SCHEDULES.
	TRANSIENT VOLTAGE SURGE SUPPRESSOR (TVSS). PER DETAIL, SHEET E301.
	NETWORK INTERFACE TERMINAL (NIT) FOR SATELLITE TV OR TELEPHONE INTERFACE.

LIGHTING FIXTURE SCHEDULE							
SYMBOL	MFGR. & MODEL OR APPROVED EQUAL (SEE NOTE 3)	LAMPS (SEE NOTE 1)				MOUNTING	REMARKS
		TYPE	NO.	WATTS	VOLTS		
A	WILLIAMS 6DR-TL-L20-835-DIM-UINV-AW-OF- CS-WHT-N-F1	LED	1	20	120	RECESSED	RECESSED CAN 6" RAISED PLATFORM
B	WILLIAMS A72-24L55/835-D-DIM-UNV	LED	1	44	120	RECESSED IN GRID 2' x 2'	100, 106, 109
C	WILLIAMS 39-4-L52/830-A-DIM-120	LED	1	37	120	SURFACE	SEE NOTE 5
C2	WILLIAMS 39-2-L52/830-A-DIM-120	LED	1	20	120	SURFACE	RESTROOMS 103, 105
E	LITON WLRD-G-L22/830-BZ-UNV	LED	1	20	120	WALL AT 7'-6" AFF TO CENTER, OR ABOVE DOOR	OUTDOOR WALL MOUNT
F	WILLIAMS 75R-4-L50/835-DRV-UNV	LED	1	22	120		HVAC, JANITOR 107, 102
G	SCOTT ARCHITECTURAL LIGHTING 53875-L18 + 30K + BA + FA1	LED	1	15	120	SEE NOTE 4	25 1/4" HIGH x 4 5/8" WIDE x 3" DEEP. DIMMABLE LED
H	WILLIAMS 6DR-TL-L30/830+DIM-UNV-OW-OF- CS-WET/CC-N-F1	LED	1	20	120	RECESSED	PORTE-CO-CHERE 110
	BEGHELLI XCL-R-U-W	LED	1	3	120	SEE NOTE 2	SEE NOTE 2. WITH BATTERY AND CHARGER

LIGHTING FIXTURE NOTES:

- LED COLOR SHALL BE 3500 K FOR ALL "INSIDE" FIXTURES AND 3000 K FOR ALL "EXTERIOR" FIXTURES.
- PROVIDE CEILING MOUNT, SURFACE MOUNT OR ABOVE DOOR. PROVIDE SINGLE OR DOUBLE FACE AS SHOWN, WITH APPROPRIATE L/R ARROW, AS SHOWN. EXIT LIGHTING FIXTURES MOUNTED ABOVE DOORS SHALL BE AT 4" ABOVE DOOR FRAME TO BOTTOM OF FIXTURE IAW DETAIL ON THIS SHEET.
- REQUESTS FOR APPROVAL OF SUBSTITUTES SHALL BE SUBMITTED TO THE ENGINEER A MINIMUM OF 10 DAYS PRIOR TO BID OPENING.
- TOILET ROOM 103 VERTICAL ON BOTH SIDES OF MIRROR. 48" AFF TO BOTTOM.
- DIMMABLE LED IN KITCHEN 101. NON-DIMMABLE IN PANTRY 104.

CONNECT AS NEAR THE SERVICE CONDUCTOR CONNECTIONS AS POSSIBLE. USE ASCO/EMERSON ACCU GUIDE COAX CABLE FOR INSTALLATION.

PANEL

VISUAL INDICATIONS OF PROTECTION STATUS IS REQUIRED.

SURGE SUPPRESSER (TVSS) DETAIL

N.T.S.

NOTES:

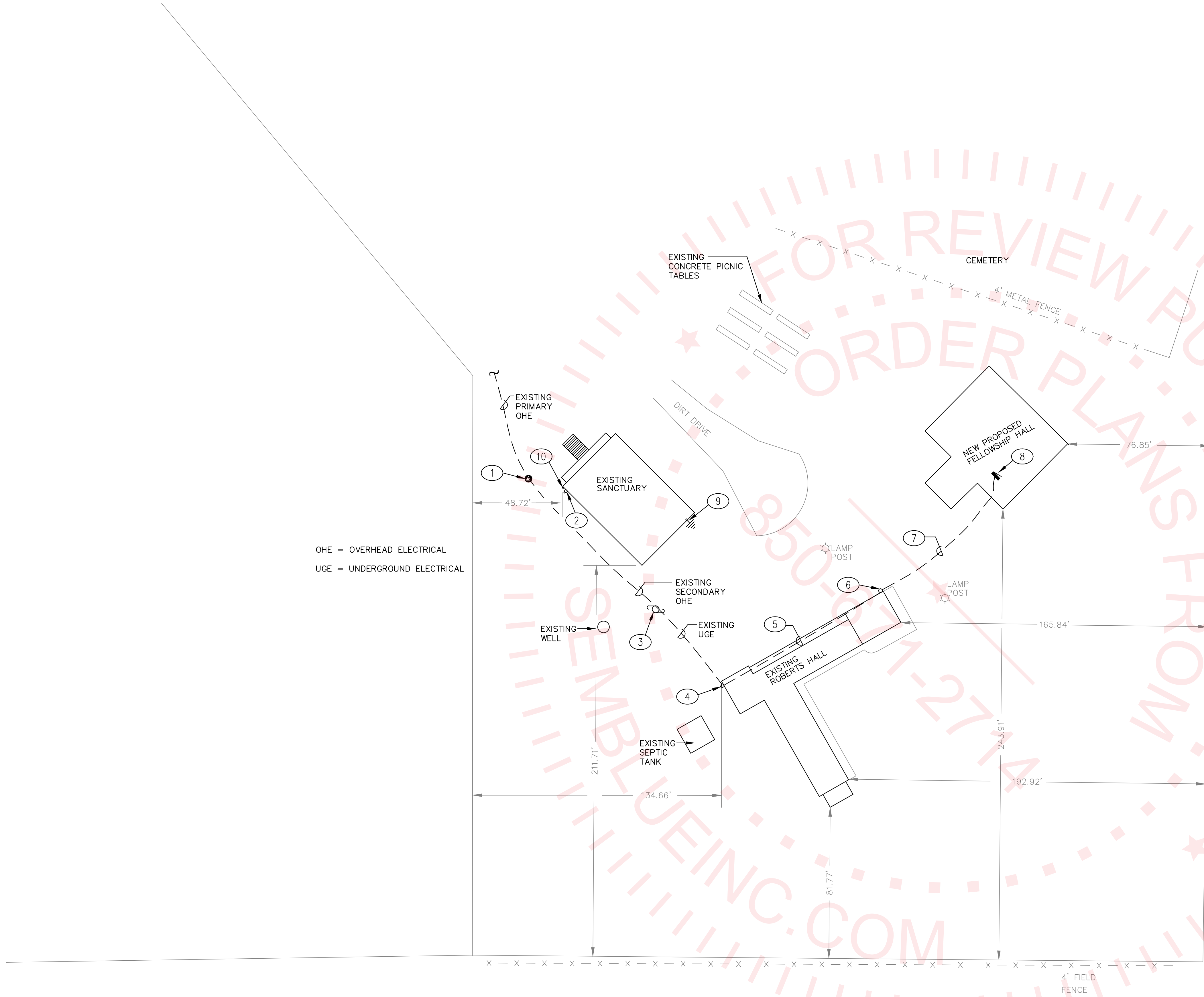
1. MOUNT SURGE SUPPRESSION ON PANEL DP AND F, AS SHOWN ON RISER DIAGRAM, SHEET E301.
2. SURGE SUPPRESSION SHALL BE BY ASCO/EMERSON.
3. CONDUCTORS BETWEEN THE SUPPRESSOR AND SWITCH BUSES SHALL BE CONTINUOUS WITHOUT LOOPS AND SHARP BENDS, AND NO SPLICES. CONDUCTOR LENGTHS SHALL BE AT A MINIMUM.

COORDINATE PLACEMENT OF LIGHTING FIXTURES, FIRE ALARM DEVICES, EXHAUST FANS AND HVAC DIFFUSERS WITH DIMENSIONS SHOWN ON THE ARCHITECT'S REFLECTED CEILING PLAN. CONTACT THE A/E IF DESIGN INTENT IS NOT CLEAR.

NOTE: FIRE ALARM CONTRACTOR SHALL SUBMIT FOR A SEPARATE FIRE ALARM SYSTEM PERMIT.

NOTE: ADD ALTERNATE; LIGHTNING PROTECTION CONTRACTOR SHALL
SUBMIT FOR A SEPARATE LIGHTNING PROTECTION SYSTEM PERMIT.

MOUNTING HEIGHTS DETAIL
N.T.S.



OHE = OVERHEAD ELECTRICAL
UGE = UNDERGROUND ELECTRICAL

PARTIAL ELECTRICAL SITE PLAN
SCALE: 1" = 30'-0"
NORTH

NEW WORK NOTES: (THIS SHEET ONLY)

- EXISTING TALQUIN ELECTRIC COOP (TEC) POWER POLE WITH POLE-MOUNTED TRANSFORMER (240/120 V., SINGLE PHASE) AND AREA LIGHT.
- EXISTING TEC METER 35 341 877, PANEL, AND TELEPHONE CO. NIT. R (METER & NIT) = 16.5 OHMS. R (PANEL) = 70.1 OHMS. IMPROVE GROUND RESISTANCE BY DRIVING ADDITIONAL GROUND RODS TO ACHIEVE 10 OHMS MAXIMUM. SEE PHOTO A. REPLACE FAULTY TVSS.
- EXISTING TEC POWER POLE, NO. L020867, WITH AREA LIGHT AND GROUND CONDUCTOR. R = 282 OHMS.
- EXISTING COMBINATION TEC METER 31 880 013, DISTRIBUTION PANEL (DP) AND TELEPHONE CO. NIT. R (PANEL DP) = 109 OHMS. R (NIT) = (INFINITE). IMPROVE GROUND RESISTANCE BY DRIVING ADDITIONAL GROUND RODS TO ACHIEVE 10 OHMS MAXIMUM. SEE PHOTO B
- 3 # 4/0, 1 # 2 G. XHHW, 2 1/2" C. SERVICE LINE TO SERVE NEW PROPOSED FELLOWSHIP HALL, RUN ABOVE PORCH CEILING (OR ATTIC) TO EAST END OF THE COVERED PICNIC AREA.
- CONDUIT DOWN TO 24" UNDERGROUND WITH 3 # 4/0, 1 # 2 G. XHHW, 2 1/2" C. SEE PHOTO C.
- UNDERGROUND SERVICE TO SERVE NEW PROPOSED FELLOWSHIP IAW ELECTRICAL RISER DIAGRAM ON SHEET E301.
- NEW PANEL F IN NEW PROPOSED FELLOWSHIP HALL OUTSIDE THE PANTRY 104 (SEE SHEET E203), IAW ELECTRICAL RISER DIAGRAM ON SHEET E301.
- LIGHTNING PROTECTION DOWN CONDUCTOR. R = 1,200 OHMS. IMPROVE GROUND RESISTANCE BY DRIVING ADDITIONAL GROUND RODS TO ACHIEVE 10 OHMS MAXIMUM.
- LIGHTNING PROTECTION DOWN CONDUCTOR. R = 1,190 OHMS. IMPROVE GROUND RESISTANCE BY DRIVING ADDITIONAL GROUND RODS TO ACHIEVE 10 OHMS MAXIMUM.



PHOTO A



PHOTO B



PHOTO C

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REVISIONS
NO. DATE

PARTIAL ELECTRICAL
SITE PLAN

FELLOWSHIP HALL
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DRAWN BY: HAO/YMO
CHECKED BY: HAO
PROJ. NO. 38022

SHEET NO.
E103

NOTES: (THIS SHEET ONLY)

1. SERVE LIGHTING FIXTURES IN ROOMS 101, 102, 103, 104, 105, 107 AND 109 FROM CKT. F-9.

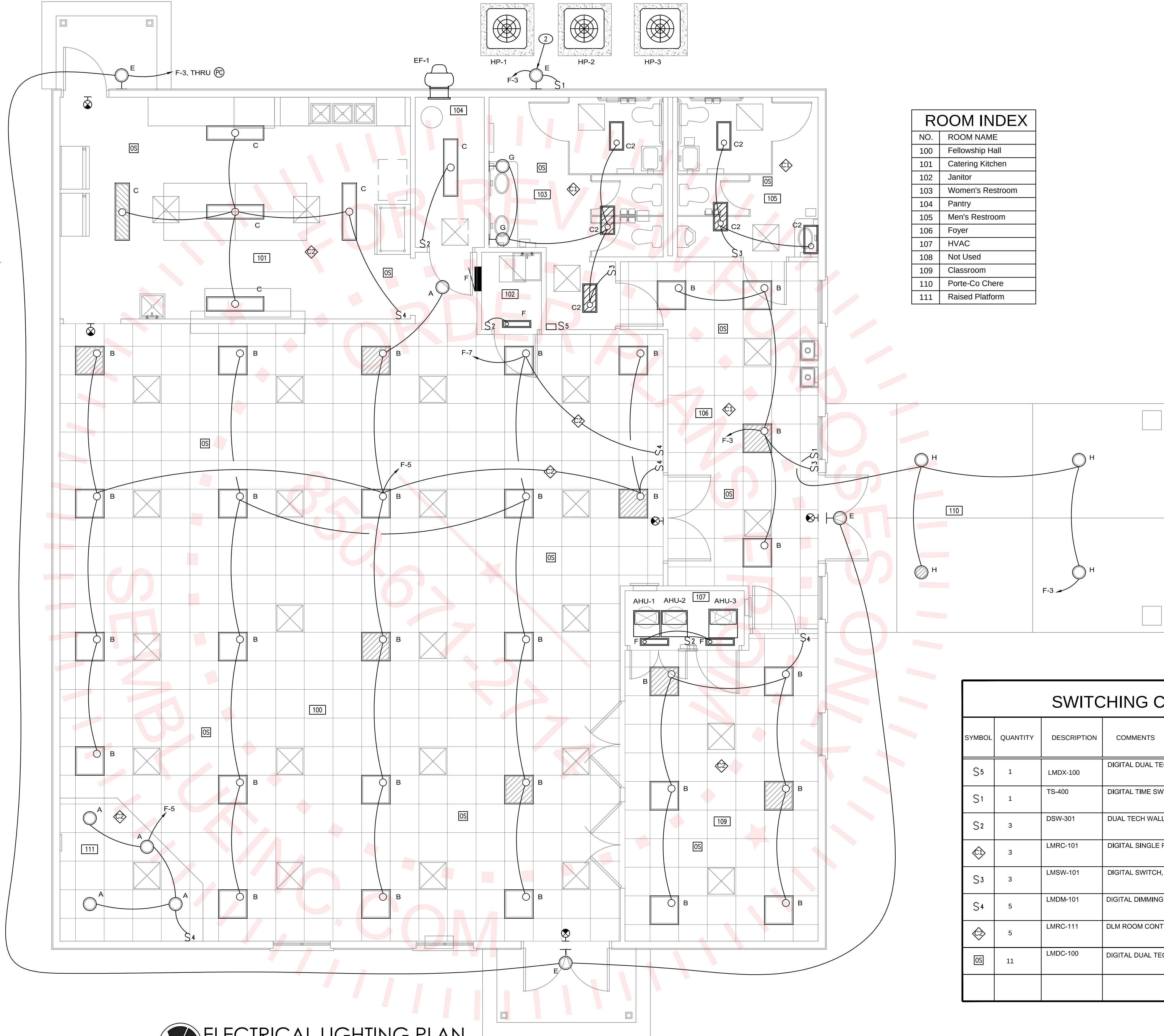
2. FIXTURE E FOR HVAC MAINTENANCE. INSTALL (WP)S1 @ 48" AFG.

3. SERVE IES'S FROM NEAREST LIGHTING CIRCUIT WITH NO SWITCH.

4. THIS LIGHTING DESIGN COMPLIES WITH FLORIDA BUILDING CODE - ENERGY CONSERVATION, 7TH EDITION (2020) TABLE C405.4.2(2), INTERIOR LIGHTING POWER ALLOWANCES, BUILDING AREA METHOD, WHICH ALLOWS 0.64 WATTS/S.F. FOR RELIGIOUS BLDG., FELLOWSHIP HALLS:

INTERIOR WATTS (2,018 W)
INTERIOR SQUARE FEET 3,240 SF)

LPD (LIGHTING POWER DENSITY)
 $\frac{2,018}{3,240} = 0.63 \text{ W/SF}$



ROOM INDEX	
NO.	ROOM NAME
100	Fellowship Hall
101	Catering Kitchen
102	Janitor
103	Women's Restroom
104	Pantry
105	Men's Restroom
106	Foyer
107	HVAC
108	Not Used
109	Classroom
110	Porte-Co Chere
111	Raised Platform

SWITCHING CONTROLS

SYMBOL	QUANTITY	DESCRIPTION	COMMENTS
S5	1	LMDX-100	DIGITAL DUAL TECH CORNER MOUNTSENSOR
S1	1	TS-400	DIGITAL TIME SWITCH 100-300 VAC, 0-800/1200 WATTS (WP)
S2	3	DSW-301	DUAL TECH WALL SWITCH OCCUPANCY SENSOR, 120/277 V.
◇	3	LMRC-101	DIGITAL SINGLE RELAY ROOM CONTROLLER, ON/OFF, 120/277 V.
S3	3	LMSW-101	DIGITAL SWITCH, 1-BUTTON W/ INFRARED
S4	5	LMDM-101	DIGITAL DIMMING WALL SWITCH, 1 PADDLE, W/ IR.
◇	5	LMRC-111	DLM ROOM CONTROLLER, 1 RELAY, KO, 0-10 V, 10 A
OS	11	LMDC-100	DIGITAL DUAL TECH CEILING MOUNT SENSOR



ELECTRICAL LIGHTING PLAN

SCALE: 1/4" = 1'-0"

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ELECTRICAL
LIGHTING PLAN

FELLOWSHIP HALL
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SHEET NO.

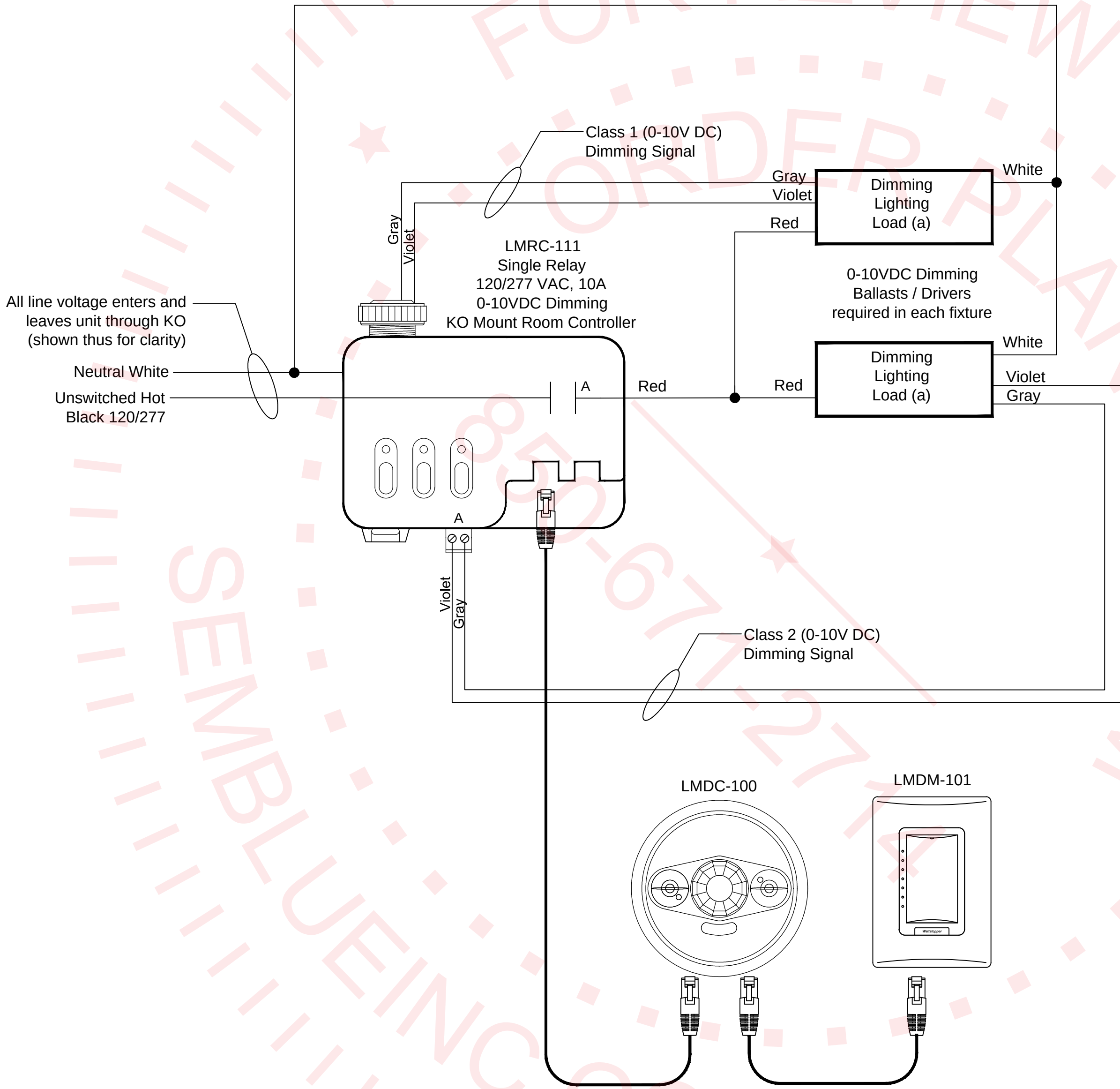
E201

LMRC-111

1-RELAY 0-10V KO MOUNT DIMMING CONTROLLER

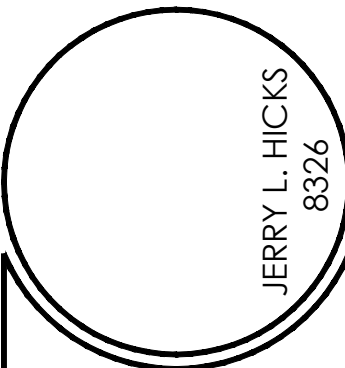
One (1) 0-10V dimming controllable switch-legs, 10A max

Note: Class 1 and Class 2 wires can be used individually or together.



FULL DLM 0-10 V DIMMING SYSTEM

N.T.S.



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NO.	DATE:

DIMMING SYSTEM
SCHEMATIC

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NEW WORK NOTES: (THIS SHEET ONLY)

- ① NIT FOR TELEPHONE AT 18" AFG.
- ①.1 NIT FOR SATELLITE TV (FUTURE) AT 11'-8" AFG.
- ② 3 # 4/0, 1 # 2 G. XHHW, 2 1/2" C. FROM EXISTING 200/2 CB IN EXISTING PANEL H AT THE WEST END OF ROBERTS HALL TO SERVE NEW PANEL F. SEE NEW WORK NOTES 5, 6 AND 7 SHEET E103.
- ③ POST AN ACRYLIC SIGN BELOW THE HORN STROBE AT 48" AFG WITH 2" LETTERS, WHITE ON RED WITH THE WORDING "FIRE ALARM-CALL 911".
- ④ FIRE ALARM TERMINAL CABINET. SERVE FROM CKT. F-1.
- ⑤ THE EC SHALL OBTAIN SHOP DRAWINGS FOR THE EWC'S AND SHALL LOCATE THE RECEPTACLE (GFI) ADJACENT TO THE EWC AND ACCESSIBLE. SERVE FROM CKT. F-13.
- ⑥ SERVE SEVEN WP/GFCI EXTERIOR RECEPTACLES FROM CKT. F-11. INSTALL TWO RECEPTACLES AT PORTE-CO CHERE BENEATH THE WINDOWS. INSTALL ONE RECEPTACLE AT 11'-8" AFG ADJACENT TO NIT FOR SATELLITE TV.
- ⑦ EF-1. SERVE FROM CKT. F-19.
- ⑧ SERVE RECEPTACLE BELOW S2 AND EF-1 FROM CKT. F-19.
- ⑨ NF DISCONNECT SWITCH AS NOTED BELOW. SERVE FROM CKT. BREAKER LISTED BELOW. AHU'S AND HP'S ARE CONTROLLED BY THE NF (NON-FUSED) DISCONNECT SWITCHES AS FOLLOWS, AND ARE SERVED FROM CIRCUIT BREAKERS AS FOLLOWS. (SEE MECHANICAL SHEETS FOR LOCATION OF UNITS). COMPLY WITH WORKING CLEARANCE REQUIREMENTS OF NEC 110-26.

AHU'S (INDOOR)

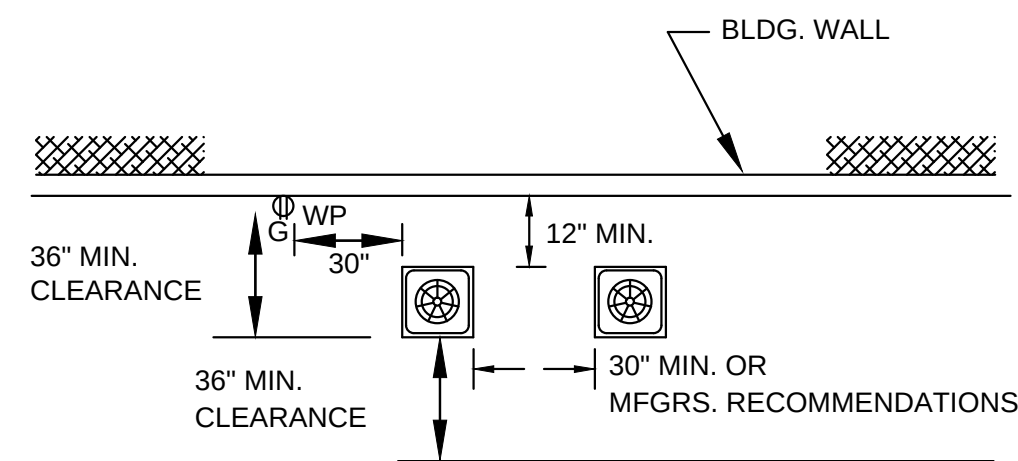
UNIT #	DISC. SW	BKR. AMPS	CKT.
1	*	60/2	F-2, 4
2	*	60/2	F-6, 8
3	*	60/2	F-10, 12

* INDOOR UNIT HAS INTEGRAL DISCONNECTING MEANS.

HP'S (OUTDOOR)

UNIT #	DISC. SW	BKR. AMPS	CKT.
1		60/2	50/2
2		60/2	50/2
3		60/2	50/2

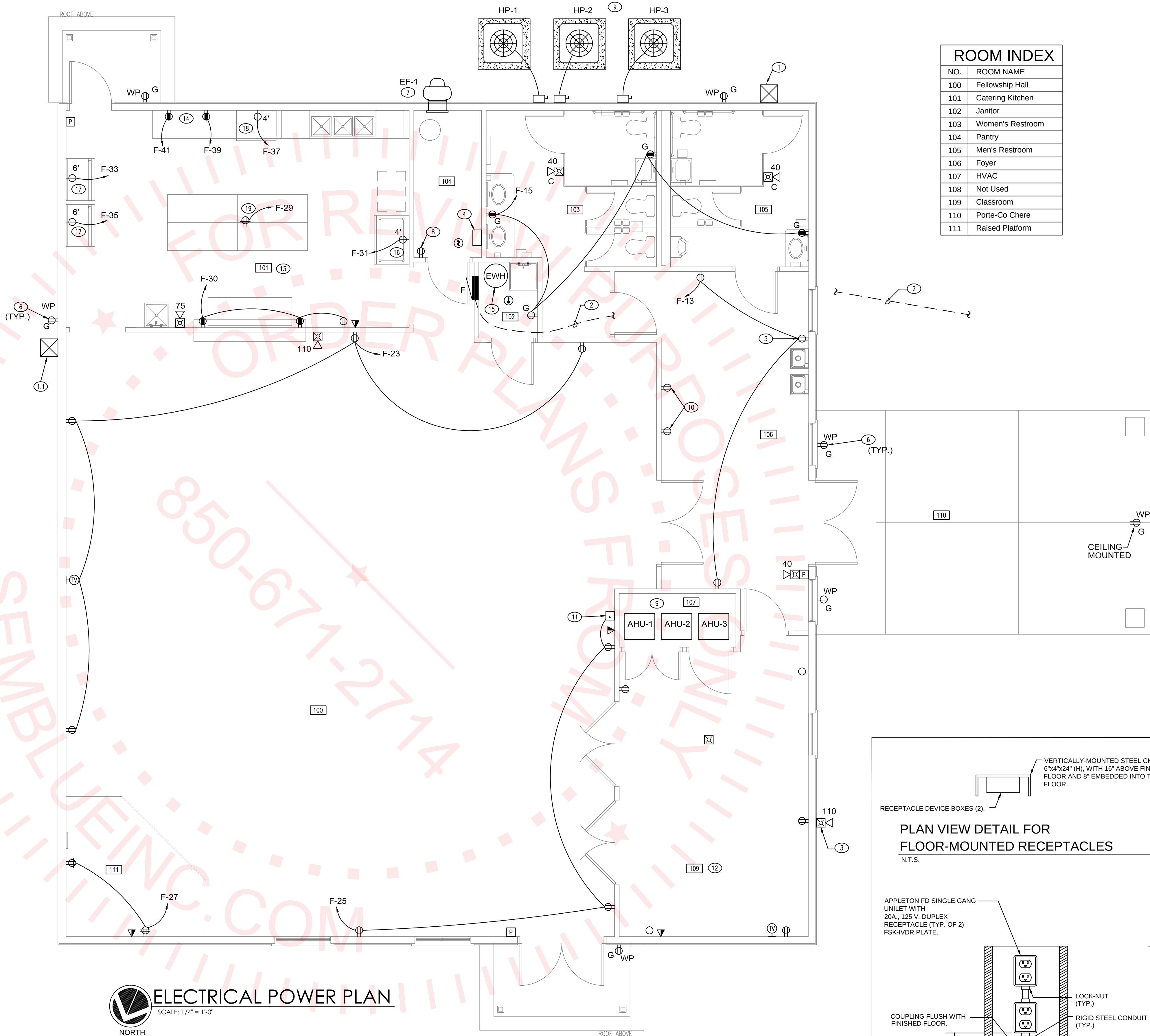
- ⑩ PROVIDE RECESSED RECEPTACLES (CLOCK TYPE) AT 7'-2" AFF. SERVE FROM CKT. F-17
- ⑪ OVERHEAD 4" x 4" J-BOX (RECESSED AT 9' - 0" AFF) FOR PROJECTOR TO BE INSTALLED (BY 'OTHERS' IN FUTURE) FOR VERTICAL PROJECTOR, OR EQUIVALENT. SERVE FROM RECEPTACLE CKT F-25.
- ⑫ SERVE RECEPTACLES IN CLASSROOM 109 FROM CKT. F-21.
- ⑬ ALL RECEPTACLES IN CATERING KITCHEN 101 SHALL BE GFCI PROTECTED.
- ⑭ DEDICATED RECEPTACLES AT COUNTER FOR MICROWAVE OVENS.
- ⑮ EWH. PROVIDE 30/2/NF. SERVE FROM CKT. F-26, 28.
- ⑯ ICE MACHINE
- ⑰ COOLER (2)
- ⑱ CONVECTION OVEN
- ⑲ FLOOR-MOUNTED RECEPTACLE IAW DETAIL, THIS SHEET.



1. INSTALL A/C SERVICE PANELS TO PROVIDE 36" DEEP MINIMUM WORKING CLEARANCE, PER NEC 110-26.
2. INSTALL A/C UNITS TO MANUFACTURER'S MINIMUM HORIZONTAL CLEARANCE AND 36" DEEP MINIMUM UNRESTRICTED FRONT CLEARANCE, PER NEC 110-26.

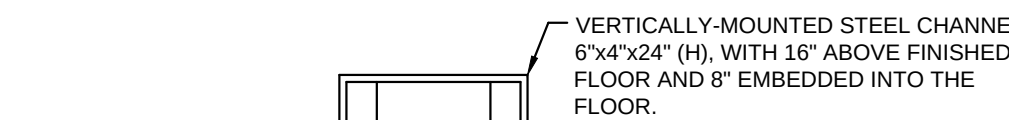
A/C INSTALLATION DETAIL

N.T.S.



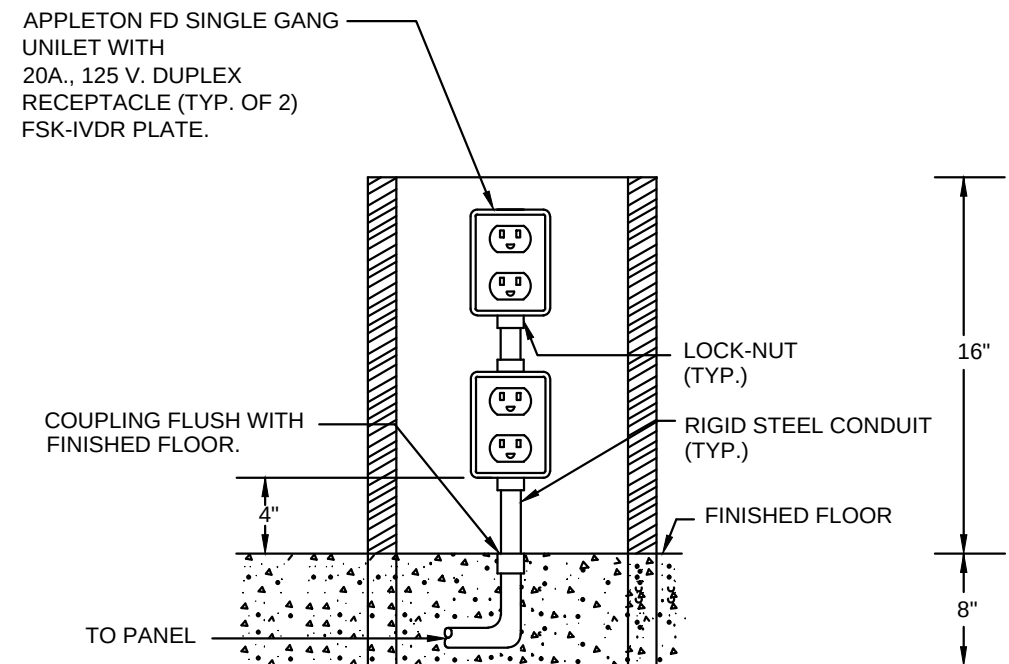
ROOM INDEX

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110	Porte-Co Chere
111	Raised Platform



PLAN VIEW DETAIL FOR FLOOR-MOUNTED RECEPTACLES

N.T.S.



ELEVATION DETAIL FOR FLOOR-MOUNTED RECEPTACLES

N.T.S.

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E203

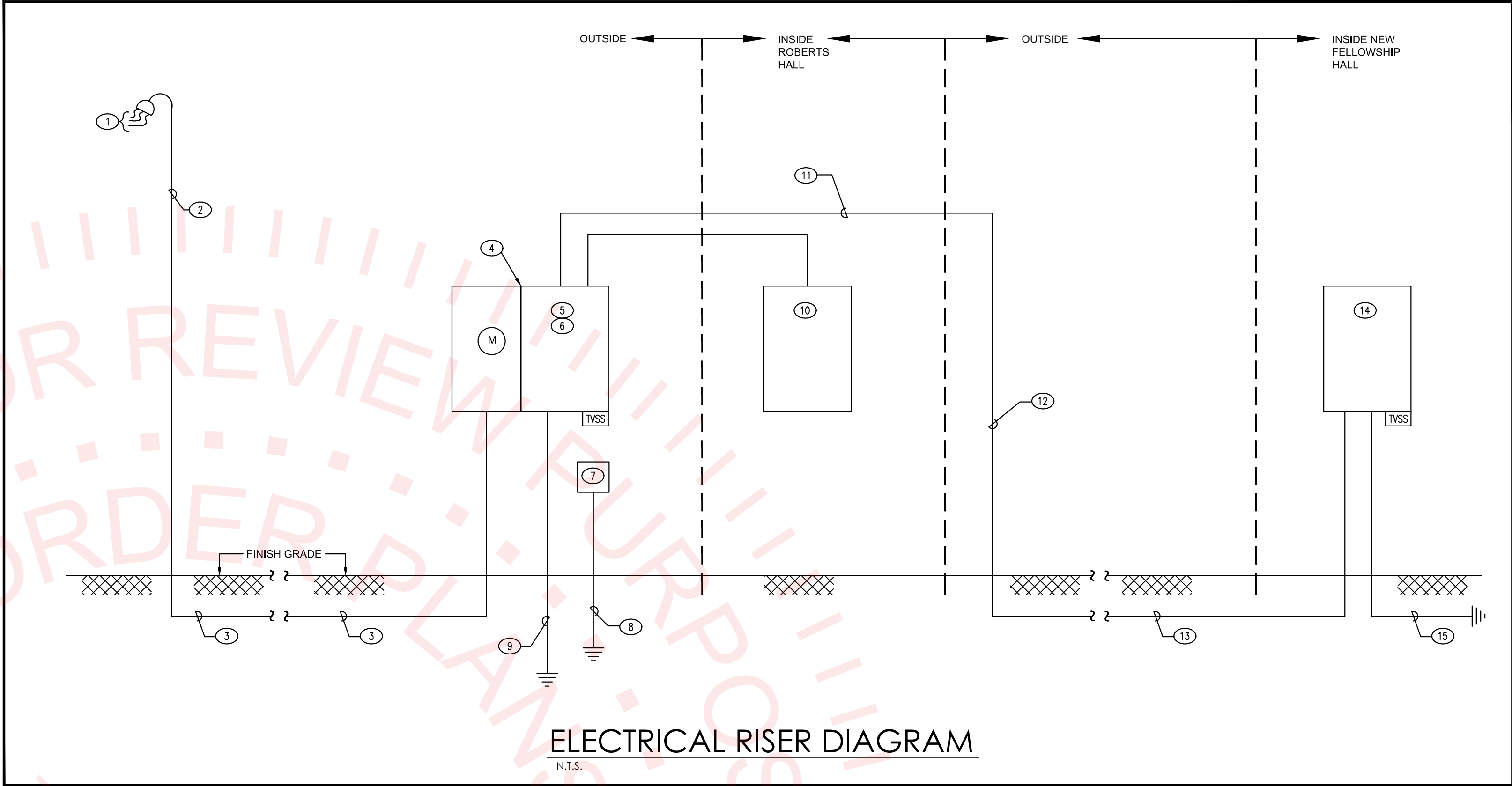
JERRY L. HICKS
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NEW PANEL		PANEL SCHEDULE										22,000 A.I.C.											
F		240/120 V., 1 PH., 3 W.			200 AMP MAIN			225 AMP FRAME															
EQUIPMENT SERVED		CKT. BREAKER			KVA/PHASE			CKT. NO.			KVA/PHASE			CKT. BREAKER			EQUIPMENT SERVED						
		TRIP	POLE	WIRE	A	B	C				A	B	C				TRIP	POLE	WIRE				
FIRE ALARM TERM. CAB. *		20	1	2#12	.4			1	2	4.1					60	2	3#6				AHU-1		
LGTS., OUTSIDE 101-107, 109						1.1		3	4		4.1												
↓ 100, 111						1.1		5	6	4.1					60	2	3#6				AHU-2		
SPARE								7	8		4.1												
↓								9	10	4.1					60	2	3#6				AHU-3		
RCPTS., EXTERIOR (6)				2#12		1.1		11	12		4.1												
EWC, FOYER 106 (5)					.6			13	14	3.1					50	2	3#8				HP-1		
GFT'S RESTROOMS, JAN					.7			15	16		3.1												
FOYER 106 (10)					.4			17	18	3.1					50	2	3#8				HP-2		
PANTRY 104 AND EF-1 (7 X 8)					.5			19	20		3.1												
CLASSROOM 108					.9			21	22	3.1					50	2	3#8				HP-3		
FELLOWSHIP 100					.9			23	24		3.1												
↓					.7			25	26	2.3					30	2	3#10				EWB		
RAISED PLATFORM 111					.7			27	28		2.3												
KITCHEN FLOOR 101					.4			29	30	.6					20	1	2#12				RCPTS., KITCHEN 101		
ICE MACHINE					.6			31	32														
COOLER					.7			33	34														
COOLER					.7			35	36														
CONVECTION OVEN					.6			37	38														
COUNTER					.2			39	40						30	2	4#10				TVSS		
↓					.2			41	42														
* PROVIDE LOCKABLE C.B. PAINT CB RED		CONNECTED KVA										A B C											
												30.6 28.5											
TOTAL CONNECTED LOAD:		59.1 KVA		246 AMPS		LOCATION: HALLWAY OUTSIDE PANTRY 104																	
DEMAND LOAD:		40.1 KVA		167 AMPS		TYPE: RECESSED, NEMA 1, BOTTOM FED																	

EXISTING PANEL				PANEL SCHEDULE										22,000 A.I.C.			
DP				240/120 V.,			1 PH., 3 W.			M.L.O.			400 AMP FRAME				
EQUIPMENT SERVED		CKT. BREAKER			KVA/PHASE			CKT. NO.		KVA/PHASE			CKT. BREAKER			EQUIPMENT SERVED	
		TRIP	POLE	WIRE	A	B	C			A	B	C	TRIP	POLE	WIRE		
PANEL A (EXISTING)		200	2	3#30	17.3			1	2							SPACE	
		---	---	---		17.4		3	4								
PANEL F (NEW)		* 200	2	ERD	30.6			5	6								
		---	---	---		28.5		7	8								
SPACE								9	10								
								11	12								
								13	14								
								15	16								
								17	18								
								19	20								
								21	22								
								23	24								
								25	26								
TVSS (NEW)		** 30	2	4#10				27	28								
		---	---	---				29	30								
* EXISTING 200/2 CB TO SERVE NEW FELLOWSHIP HALL PANEL F																	
** NEW TVSS IAW DETAIL ON SHEET E301																	
CONNECTED KVA										A	B	C					
										47.9	45.9						
TOTAL CONNECTED LOAD:										93.8	KVA	391	AMPS	LOCATION: WEST END, ROBERTS HALL			
DEMAND LOAD:										61.9	KVA	258	AMPS	TYPE: EXISTING			

NEW WORK NOTES (ERD ONLY):

- EXISTING TALQUIN ELECTRIC (TEC) OVERHEAD ELECTRICAL (OHE) SECONDARY ELECTRICAL SERVICE (240/120 VOLTS, SINGLE PHASE).
- EXISTING DOWN CONDUCTOR ON EXISTING TEC POWER POLE NO. L020867 (SEE NEW WORK NOTE 3) ON SHEET E103).
- EXISTING UNDERGROUND ELECTRICAL (UGE) SECONDARY ELECTRICAL SERVICE (240/120 VOLTS, SINGLE PHASE, 300 AMP CAPACITY).
- EXISTING COMBINATION TEC METER AND (DISTRIBUTION PANEL) PANEL DP (SEE NEW WORK NOTE 4) ON SHEET E103).
- EXISTING PANEL DP, 240/120 VOLTS, SINGLE PHASE, 3 WIRE, 400 AMP FRAME/BUS, M.L.O.
- PANEL DP CONTAINS TWO 200 AMP, 2 POLE C.B.'S. ONE OF THE TWO SERVES EXISTING PANEL A INSIDE EXISTING ROBERTS HALL. THE SECOND 200/2 CB IS TO SERVE NEW PANEL F IN THE NEW FELLOWSHIP HALL THROUGH 3 # 4/0, 1 # 2 G XHHW CU, 2 1/2" C. INSTALL 300/2 CB IN EXISTING PANEL DP AND SERVE A NEW TVSS THROUGH 4 # 10, TVSS TO BE ASCO/EMERSON MODEL ACV-120S-111 R.
- EXISTING TELEPHONE COMPANY NETWORK INTERFACE TERMINAL (NIT), EXTEND EXISTING SERVICE (CAT 6 CONDUCTORS) TO NEW FELLOWSHIP HALL IN 3/4" C. PARALLEL TO THE 2 1/2" C. ELECTRICAL SERVICE, IAW NEW WORK NOTES 11, 12, 13. THIS SHEET.
- EXISTING NIT GROUND CONDUCTOR. RESISTANCE-TO-GROUND MEASURED TO BE INFINITE, COMPLETELY "OPEN CIRCUIT."
- EXISTING GROUND CONDUCTOR FOR EXISTING PANEL DP. RESISTANCE-TO-GROUND MEASURED TO BE 109 OHMS. EC SHALL COMBINE THE TWO GROUND CONDUCTORS 8 AND 9 AND DRIVE ADDITIONAL GROUND RODS TO IMPROVE GROUND RESISTANCE TO R < 10 OHMS.
- EXISTING PANEL INSIDE EXISTING ROBERTS HALL. NO WORK BY THIS EC.
- 12, 13 PROPOSED ROUTE OF 2 1/2" C. ELECTRICAL SERVICE AND 3/4" C. TELEPHONE SERVICE FROM WEST END OF ROBERTS HALL (SEE PHOTO B ON SHEET E103) TO THE NEW FELLOWSHIP HALL. SEE NEW WORK NOTES 5, 6 AND 7 ON SHEET E103. SEE PHOTO C ON SHEET E103 FOR PROPOSED LOCATION OF DOWN CONDUCTOR TO TRANSITION FROM ATTIC TO UNDERGROUND.
- NEW PANEL F IN NEW FELLOWSHIP HALL. 240/120 VOLTS, SINGLE PHASE, 3 WIRE, 225 AMP FRAME/BUS, 200/2 MAIN BREAKER, IAW PANEL SCHEDULE F, THIS SHEET.
- CONNECT PANEL F GROUND BUS TO GROUND TRIPPOD, IAW DETAIL E ON SHEET E402.



ELECTRICAL RISER DIAGRAM

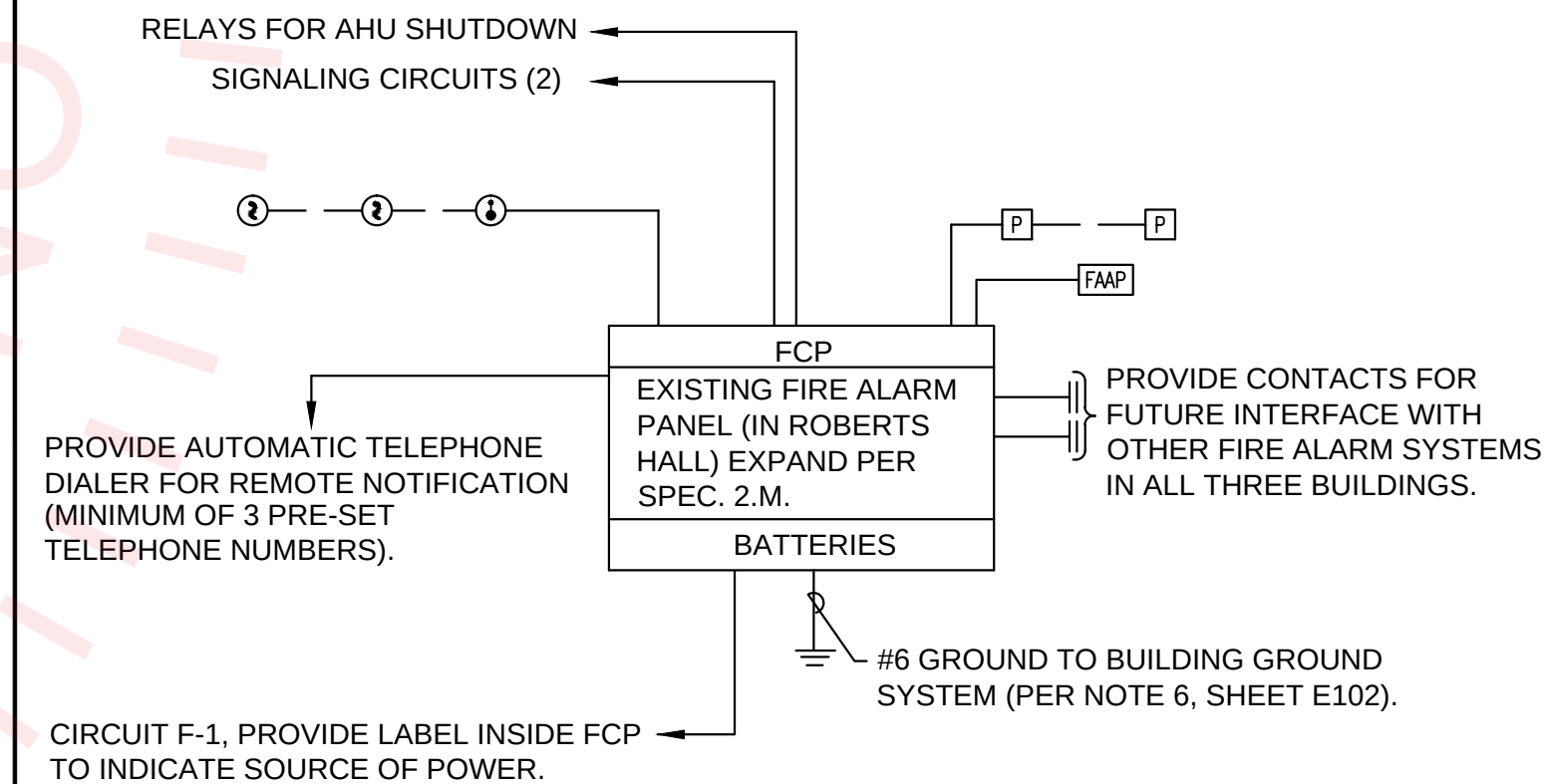
N.T.S.

LOAD SUMMARY - PANEL F

	CONNECTED KVA	LOAD FACTOR	DEMAND KVA
LIGHTING	2.2	1.25	2.8
RECEPTACLES (1ST 10 KVA)	10.0	1.0	10.0
RECEPTACLES (REMAINING)	0.4	0.5	0.2
HEAT	24.6	1.0	24.6
A/C	18.6	-0-	-0-
MISC.	5.0	0.5	2.5
			40.1

1 @ 240/120 V., 1 PH = 167 AMPS

FIRE ALARM SYSTEM RISER



- NOTE:
- FIRE ALARM SYSTEM CONTROL PANEL SHALL BE OF SUFFICIENT CAPACITY FOR 20 PERCENT EXPANSION.

FIRE ALARM SEQUENCE OF OPERATION

- THE ACTIVATION OF A MANUAL PULL STATION OR AUTOMATIC DETECTOR SHALL CAUSE:
 - ALL EVACUATION HORNS TO SOUND AND STROBES TO FLASH.
 - REMOTE REPORTING WILL OCCUR BY PRE-SET AUTOMATIC DIALER.
 - INDICATION OF THE ZONE OF THE ALARM DEVICE AT THE FIRE ALARM CONTROL PANEL AND THE ANNUNCIATOR PANEL
- WHEN THE ALARM SILENCE IS PRESSED, THE HORNS AND STROBES SHALL CEASE TO FUNCTION. THE SYSTEM SHALL REMAIN IN ALARM UNTIL THE CAUSE OF THE ALARM IS CORRECTED.
- THE AIR HANDLING UNITS SHALL CEASE TO OPERATE. THEY SHALL SEQUENTIALLY START WHEN THE SYSTEM IS RESET.

NOTE: FIRE ALARM CONTRACTOR SHALL SUBMIT FOR A SEPARATE FIRE ALARM SYSTEM PERMIT.

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REVISIONS

NO. DATE:

ELECTRICAL RISER DIAGRAM

FELLOWSHIP HALL
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7000 PISGAH CHURCH RD.
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DATE: 11-05-2021
DRAWN BY: HAO/YMO
CHECKED BY: HAO
PROJ. NO. 38022

SHEET NO.

E301

NEW WORK NOTES: (THIS SHEET ONLY)

- ① 1/2" x 18" BLUNT-TIPPED SOLID ALUMINUM AIR TERMINAL (TYP.). SEE SHEET A402 FOR INSTALLATION DETAILS, IAW DETAIL A OR C, AND DETAIL B FOR BOTH DETAILS A AND C.
- ② DOWNLEAD CONNECTION TO GROUND ROD, IAW DETAIL D ON SHEET E402.
- ③ GROUND ROD TRIPOD, PER DETAIL E ON SHEET E402.

NOTES:

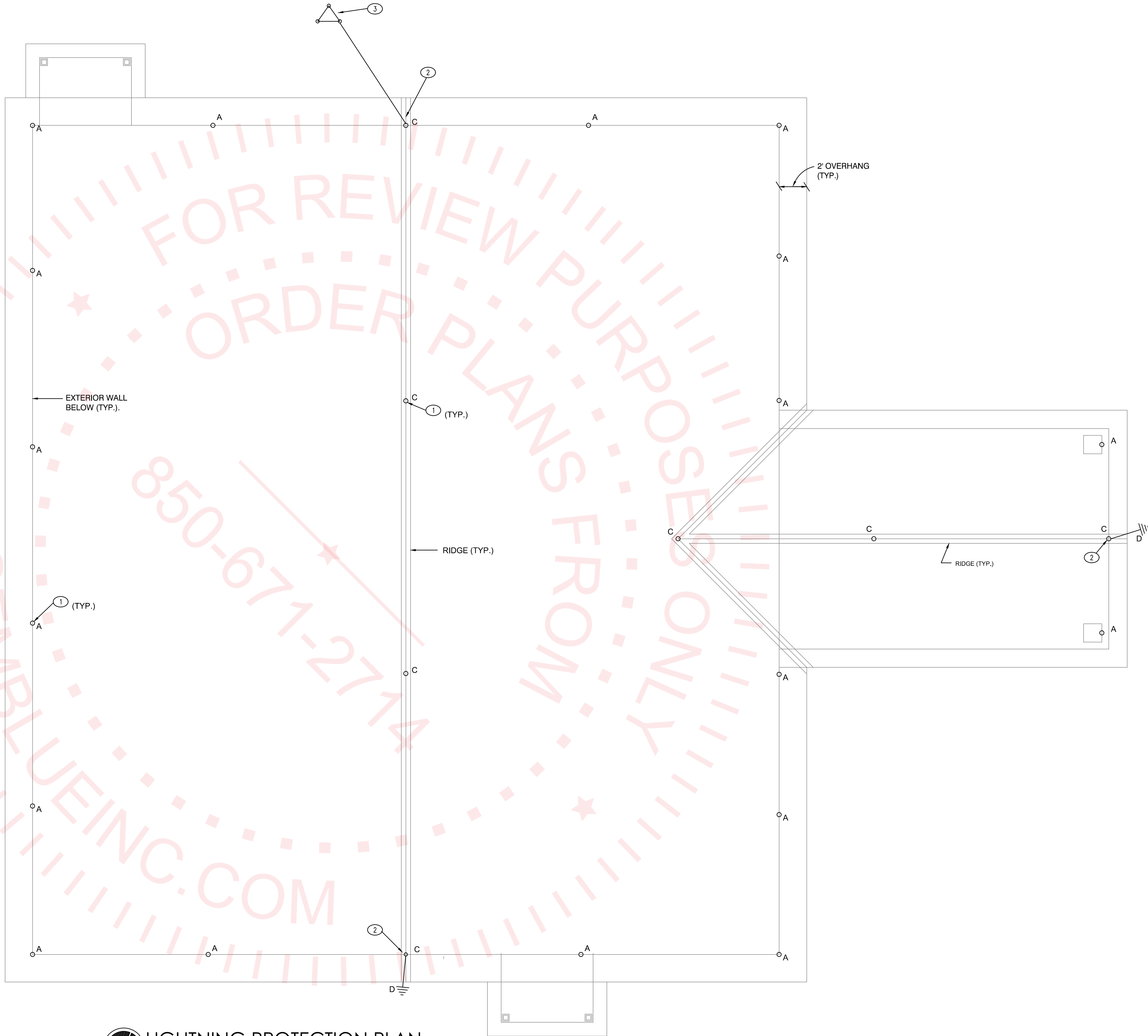
1. INSTALL LIGHTNING PROTECTION SYSTEM IN ACCORDANCE WITH SPECIFICATIONS ON SHEET E402.
2. UPON INSTALLATION OF THE LIGHTNING PROTECTION SYSTEM, TEST RESISTANCE-TO-GROUND (EARTH CONNECTION) WITH RESISTANCE TESTER. THE ENGINEER SHALL BE NOTIFIED IN ADVANCE TO WITNESS THE RESISTANCE-TO-GROUND TEST FOR OHM RESISTANCE. MAXIMUM ALLOWABLE RESISTANCE ON DOWN CONDUCTORS IS 10 OHMS, IAW SPEC 1.E.3 ON SHEET E101.

LIGHTNING RISK ASSESSMENT

A RISK ASSESSMENT CALCULATOR FROM MAXWELL LIGHTNING PROTECTION CO. WAS USED WITH INPUTS RELATED TO THIS PROJECT AS FOLLOWS:

- LIGHTNING FLASH DENSITY:
FLASHES/SQUARE MILE/YEAR FOR TALLAHASSEE: 8
- STRUCTURE COLLECTIVE DATA:
RECTANGULAR STRUCTURE : 60' x 54' x 22.67' HIGH.
- RELATIVE STRUCTURE LOCATION:
STRUCTURE ROOF HEIGHT IS HIGHER THAN HEIGHT OF STRUCTURE WITHIN A DISTANCE OF 68 FEET (3 H).
- TYPE OF CONSTRUCTION:
NONMETALLIC FRAMEWORK AND METAL ROOF.
- STRUCTURE CONTENTS:
STANDARD VALUE AND NONFLAMMABLE.
- OCCUPANCY:
NORMALLY UNOCCUPIED.
- LIGHTNING CONSEQUENCE:
CONTINUITY OF FACILITIES SERVICES NOT REQUIRED, NO ENVIRONMENTAL IMPACT.

THIS OPERATION RESULTED IN THE CONCLUSION FROM THE MAXWELL LIGHTNING PROTECTION CO. ALGORITHM THAT THE PISGAH UNITED METHODIST CHURCH FELLOWSHIP HALL IS "AT HIGH RISK OF A LIGHTNING STRIKE."



 **LIGHTNING PROTECTION PLAN**
SCALE: 1/4" = 1'-0"

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LIGHTNING PROTECTION
SYSTEM

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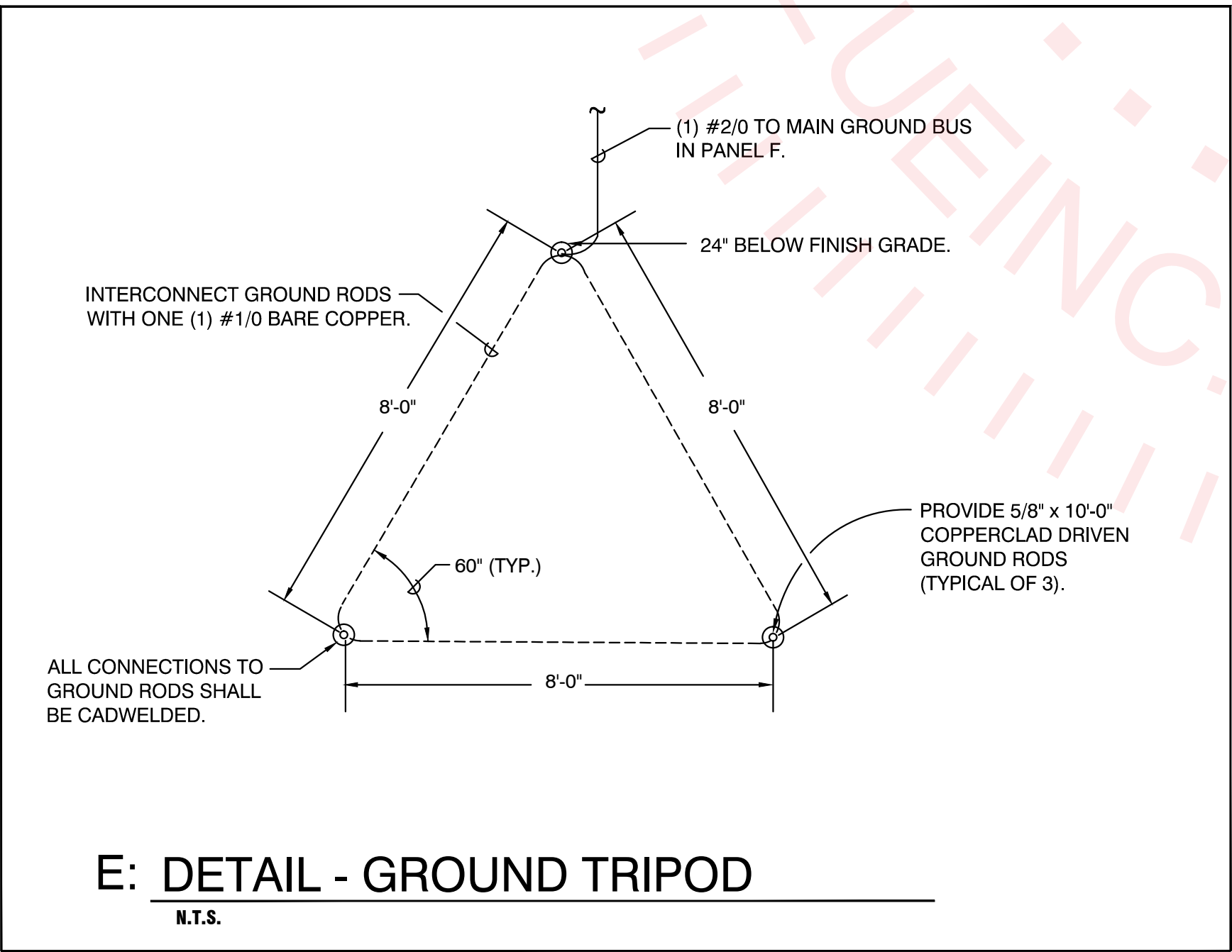
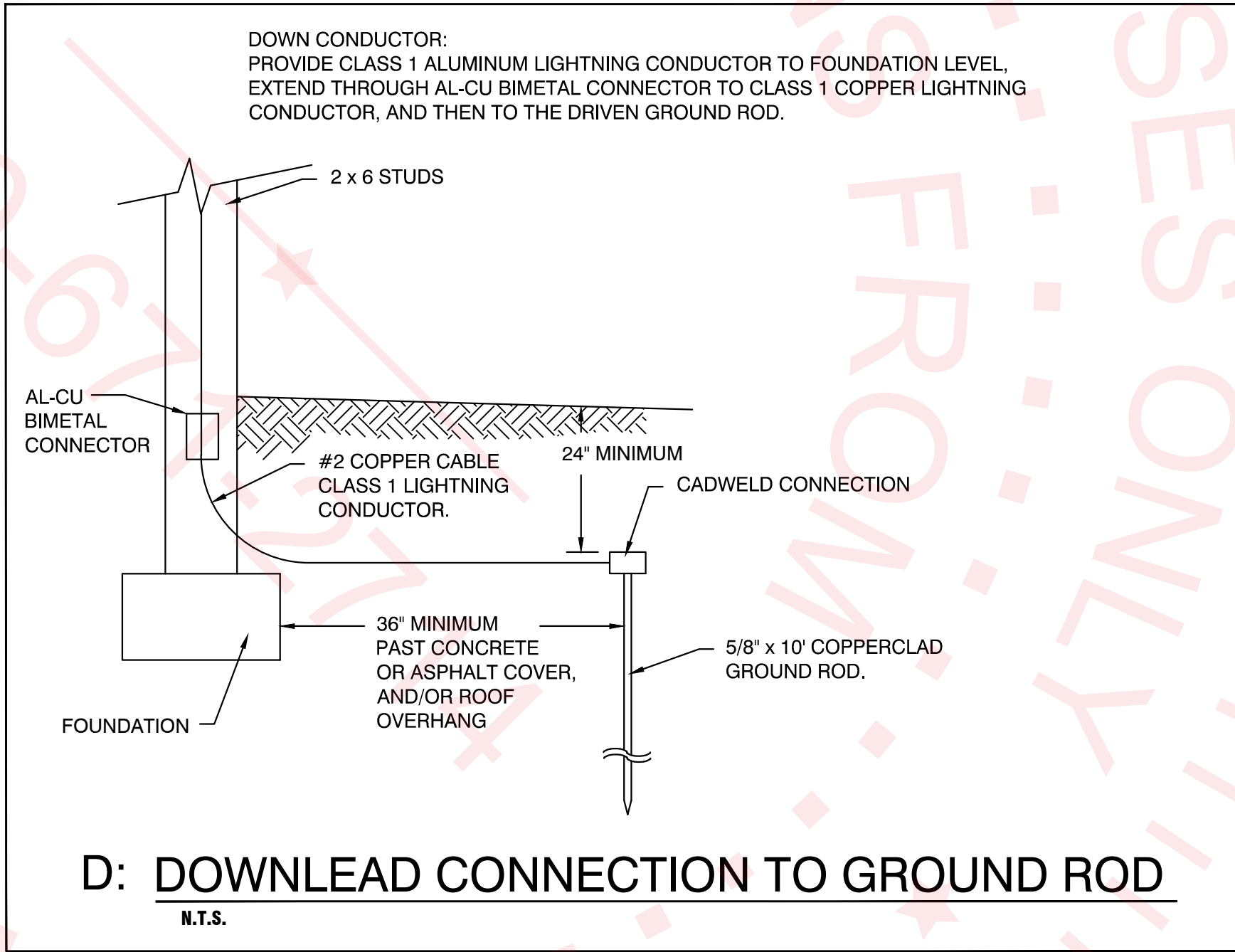
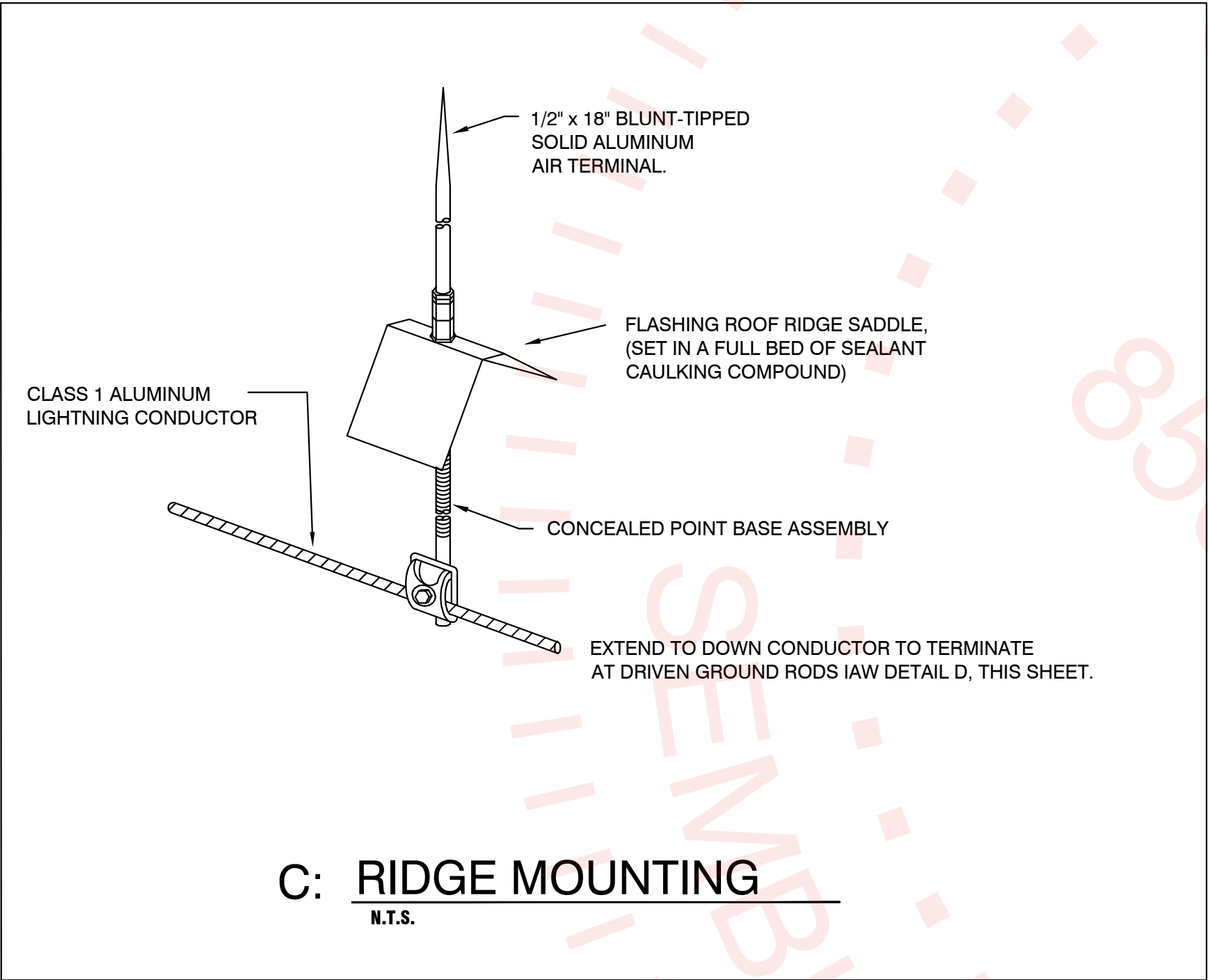
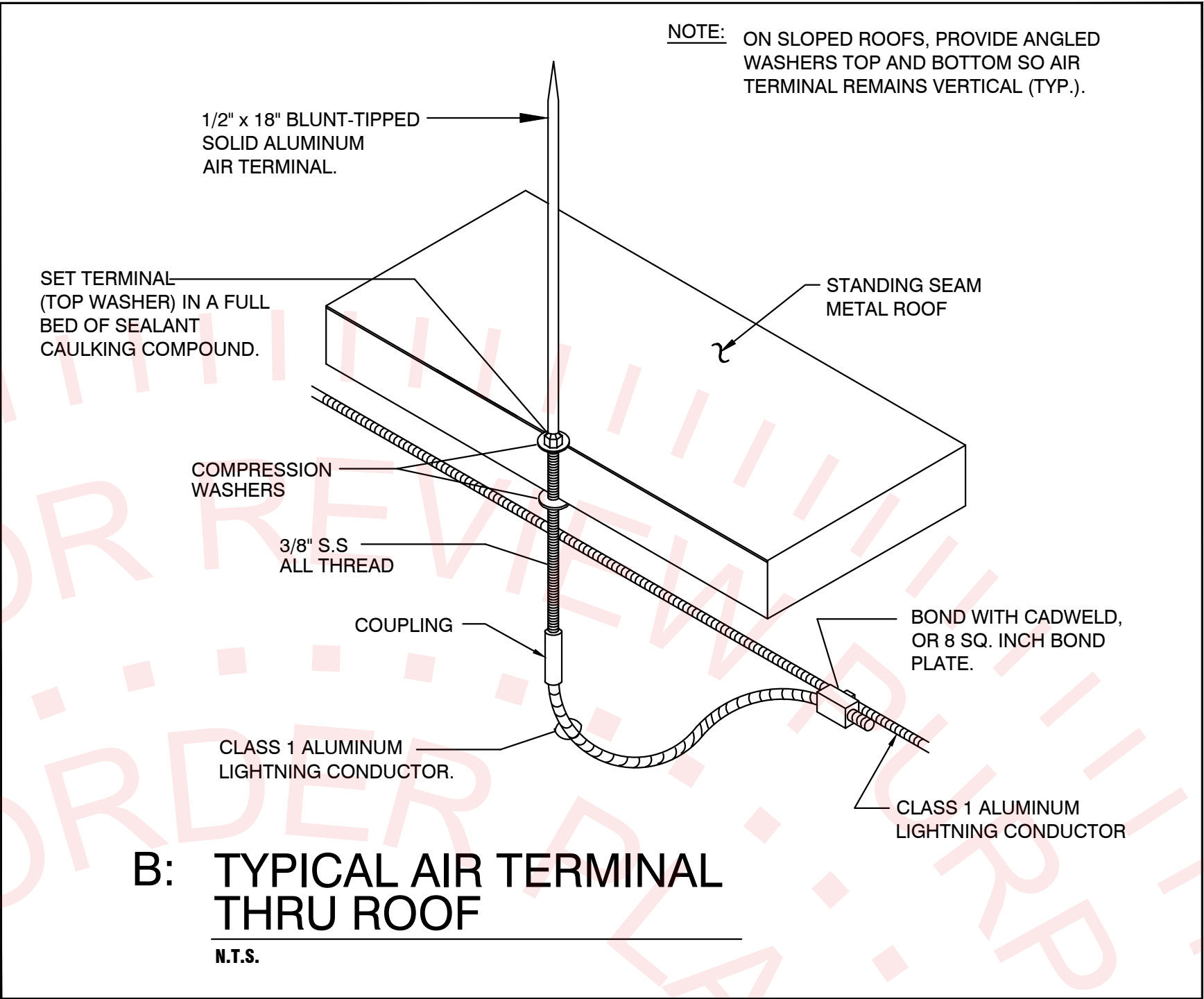
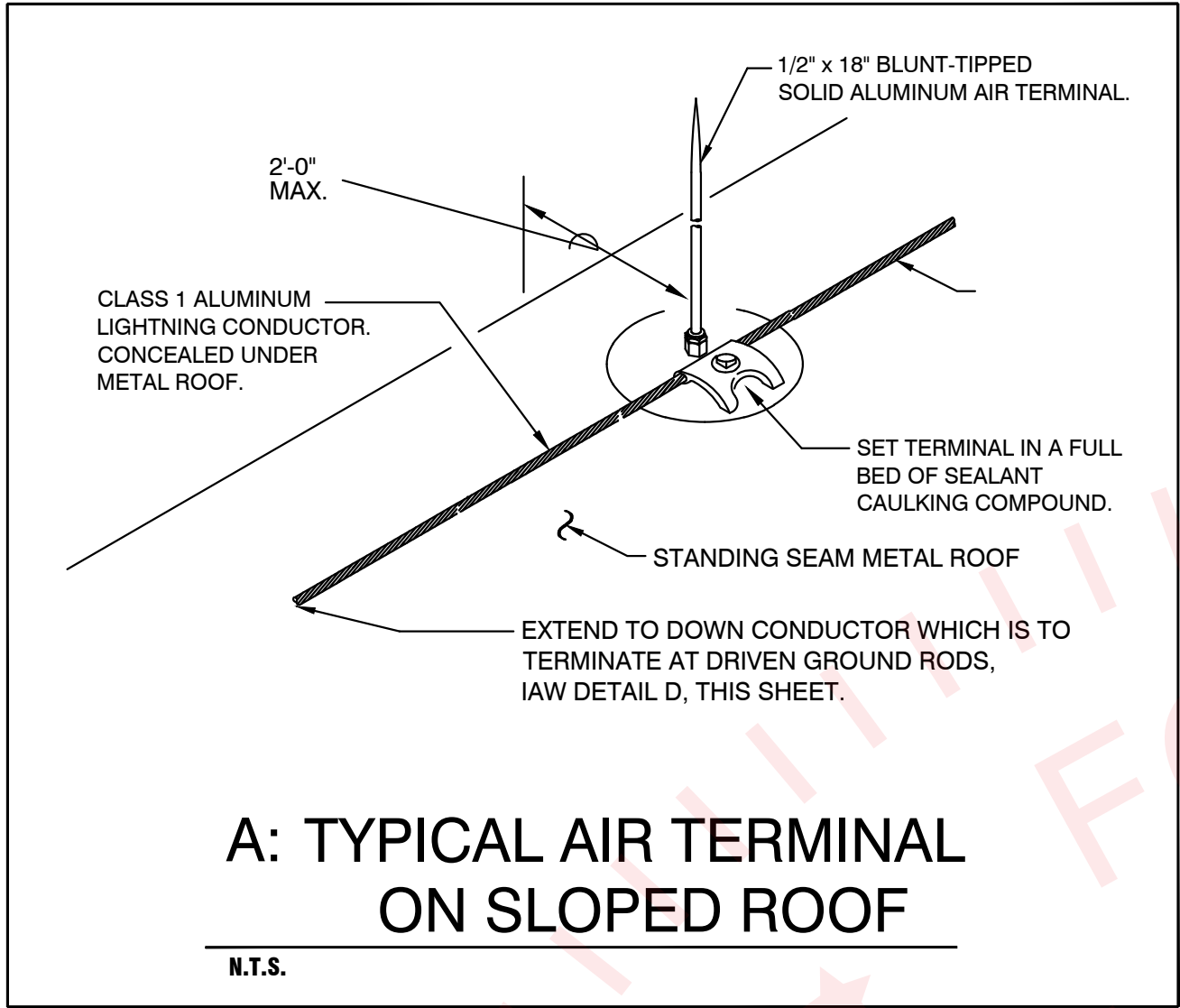
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SHEET NO.
E401



LIGHTNING PROTECTION SYSTEM SPECIFICATIONS

- FURNISH AND INSTALL ALL MATERIALS AND LABOR REQUIRED TO PROVIDE A COMPLETE FUNCTIONAL LIGHTNING PROTECTION AND COMMON GROUNDING SYSTEM FOR THE BUILDING, IN STRICT ACCORDANCE WITH THESE SPECIFICATIONS. SUBMIT SHOP DRAWINGS FOR APPROVAL, IAW SPEC 1.D, SHEET E101.
- THE FOLLOWING SPECIFICATIONS AND STANDARDS FORM A PART OF THIS SPECIFICATION:
 - LIGHTNING PROTECTION INSTITUTE - INSTALLATION CODE LPI-175.
 - UNDERWRITERS LABORATORIES (UL) CODE 96A, MASTER LABEL, OR EQUAL FROM A NATIONALLY-RECOGNIZED ELECTRICAL TESTING LAB.
 - NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) #780.
- EQUIPMENT MANUFACTURER SHALL BE CERTIFIED BY THE LIGHTNING PROTECTION INSTITUTE (LPI), AND PRODUCTS APPROVED FOR UL LISTING. ACCEPTABLE INSTALLERS: MAXWELL LIGHTNING PROTECTION OF FLORIDA, THOMPSON LIGHTNING PROTECTION, INC., ROBBINS LIGHTNING PROTECTION, OR APPROVED EQUAL.
- IN ORDER TO INSURE INTEGRITY OF INSTALLATION, THE SYSTEM SHALL BE INSTALLED UNDER THE DIRECT JOB SITE SUPERVISION OF A CERTIFIED MASTER INSTALLER, WHO HAS QUALIFIED UNDER THE LPI'S CERTIFICATION PROGRAM AS A MASTER INSTALLER.
- SYSTEM COMPONENTS SHALL BE CONCEALED WHEREVER PRACTICAL, WITH ROOF PERIMETER CABLES AND MID-ROOF CABLES CONCEALED BENEATH ROOF. PRIMARY AND SECONDARY BONDING OF ROOF METALS AND EQUIPMENT SHALL ALSO BE BENEATH ROOF. ROOF PENETRATIONS SHALL BE MADE WITH SOLID BAR THRU-ROOF CONNECTORS WITH PITCH POCKETS. DOWN CONDUCTORS SHALL BE CONCEALED WITHIN THE STRUCTURE AND/OR BUILDING STEEL SHALL BE USED AS DOWN CONDUCTORS.
- SYSTEM MATERIALS IN GENERAL SHALL BE ALUMINUM AND ALUMINUM ALLOY CASTINGS, AND SHALL COMPLY IN WEIGHT, SIZE, AND COMPOSITION FOR THE CLASS OF STRUCTURE TO BE PROTECTED, AS SPECIFIED IN CODES MENTIONED IN 2 ABOVE. THE SYSTEM SHALL CONSIST OF ALL NECESSARY CABLES, AIR TERMINALS (POINTS), MOUNTING BASES, FITTINGS, COUPLINGS, CONNECTORS, GROUND RODS, AND FASTENERS, AS REQUIRED FOR A COMPLETE AND COORDINATED SYSTEM. ALL CABLES AND ALL AIR TERMINALS SHALL BEAR UL LABELS (OR EQUAL).
- ALL SYSTEM FITTINGS EXCEPT CABLE HOLDERS, REGARDLESS OF STRUCTURE CLASSIFICATION, SHALL BE HEAVY-DUTY TYPE MADE FROM ALUMINUM ALLOY CASTINGS AND SECURED WITH BOLTED-PRESSURE CLAMPS. PRESSURE PLATES MADE FROM STAMPED OR PRESSED METAL PARTS, OR FITTINGS UTILIZING CRIMP-TYPE PRESSURE DEVICES ARE NOT ALLOWED. ALL BOLTS, SCREWS, AND RELATED TYPE HARDWARE SHALL BE STAINLESS STEEL.
- THE ALUMINUM LIGHTNING PROTECTION MATERIALS ARE REQUIRED FOR UL 96-96A/NFPA 780 COMPLIANCE AND CERTIFICATIONS, AND TO MAINTAIN THE INTEGRITY OF ROOFING MATERIALS AGAINST DAMAGE AND POTENTIAL ROOF SYSTEM FAILURES.
- COPPER LIGHTNING PROTECTION MATERIALS SHALL NOT BE PLACED ON ALUMINUM, GALVANIZED STEEL, GALVALUME SURFACES OR ANY OTHER DISSIMILAR METAL SURFACES, PAINTED OR NOT, NOR SHALL ALUMINUM MATERIALS BE PLACED ON COPPER SURFACES.
- EXPOSED FITTINGS AND AIR TERMINALS SHALL HAVE ANODIZED FINISH (DECORAL) TO MATCH COLOR OF ROOFING. REFER TO ARCHITECTURAL SPECIFICATIONS.
- METAL BODIES OF INDUCTANCE LOCATED ABOUT THE ROOF SUCH AS: HVAC EQUIPMENT, EXHAUST FANS, ACCESS HATCH, METAL FLASHINGS, ROOF DRAINS, AND SOIL PIPE VENTS SITUATED WITHIN 6'-0" OF A LIGHTNING CONDUCTOR OR BONDED METAL BODY SHALL BE BONDED TO THE LIGHTNING PROTECTION SYSTEM.
- AIR TERMINALS SHALL BE PLACED NOT MORE THAN 2'-0" FROM THE ENDS OF RIDGES, OUTSIDE CORNERS, OR OUTSIDE EDGES OF MAIN ROOFS AND HIGH ROOF, AND MUST EXTEND A MINIMUM OF 10" ABOVE OBJECT TO BE PROTECTED.
- NO BEND OF A CONDUCTOR SHALL FORM A FINAL INCLUDED ANGLE OF LESS THAN 90 DEGREES NOR SHALL HAVE A RADIUS OR BEND OF LESS THAN 8". CONDUCTORS SHALL MAINTAIN A HORIZONTAL OR DOWNWARD PATH FREE FROM "U" AND "V" POCKETS. ANY RISE IN HORIZONTAL CONDUCTOR SHALL NOT EXCEED 8 INCHES. CABLE MAY RISE UPWARDS AT A RATE OF 3" OF RISE PER 1'-0" RUN MAXIMUM.
- AIR TERMINALS SHALL BE PLACED ON 20'-0" MAXIMUM SPACING AROUND THE ROOF PERIMETER. MIDROOF AIR TERMINALS SHALL BE SPACED AT 50'-0" O.C. MAXIMUM SPACING. ALL LIGHTNING PROTECTION CABLES SHALL BE FASTENED NOT MORE THAN 3'-0" MAXIMUM SPACING.
- ALL ADHESIVE FIXTURES SHALL BE SET WITH AN ADHESIVE COMPOUND COMPATIBLE WITH THE ROOFING MATERIAL. ADHESIVES SHALL BE APPROVED IN ADVANCE BY THE ROOFING CONTRACTOR.
- CONNECTIONS TO GROUND SHALL BE MADE IAW DETAILS ON THIS SHEET. BOND ALL METAL PIPES (SUCH AS WATER, STORM, SEWER WHICH ENTER THE STRUCTURE) TO THE NEAREST DOWNLEAD.
- ACTUAL JOB SITE CONDITIONS MAY NECESSITATE SLIGHT ALTERATIONS IN AIR TERMINAL AND GROUND ROD LOCATIONS. LIGHTNING PROTECTION INSTALLER MUST FIELD VERIFY THE ACTUAL CONDITIONS OF THE BUILDING(S) TO DETERMINE THE PROTECTION LEVEL NEEDED AND MAKE ANY ADJUSTMENTS, UPWARD OR DOWNWARD, AS REQUIRED FOR U.L. 96-96A/NFPA 780 COMPLIANCE AND CERTIFICATION.
- THE LIGHTNING PROTECTION SYSTEM SHALL BE INSTALLED IN A NEAT AND INCONSPICUOUS MANNER SO THAT ALL COMPONENTS WILL BLEND IN WITH THE APPEARANCE OF THE BUILDING.
- UPON JOB COMPLETION, CONTRACTOR SHALL FURNISH ARCHITECT WITH WRITTEN CERTIFICATION ON LPI FORMS 175-A AND 175-B PLUS UL MASTER LABEL "C" (OR EQUAL), TO CERTIFY THAT THE SYSTEM IS INSTALLED IN COMPLIANCE WITH APPLICABLE CODES.

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REVISIONS	
NO.	DATE:

LIGHTNING PROTECTION
SPECIFICATIONS
AND DETAILS

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SHEET NO.
E402